

Coupole

Reference manual

Version 0.8.0 — 19 October 2026

Free toolbox for students of the « Explorer et Comprendre l'Univers » diploma (Observatoire de Paris).

OHP image bank · DU shared folders · image quality · spectra and series · sites and times · cosmology · animations · course calculators.

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Data: « OHP student observations » bank, Observatoire de Paris / PADC (Etalab 2.0 licence)

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CHAPTER 1

Overview

Coupole is a free application (GPL-3.0) written in Python 3.10+, running on Windows, macOS and Linux, with a graphical interface (PyQt6) and a complete command line. Every displayed text exists in French and English; the system language is detected and can be forced (*Language* menu, Preferences, or `-lang fr|en`).

Module	Purpose
OHP image bank	Inventory, catalogue, download, astrometric check, fixed headers, conversion, stackable sets, anomalies, sky map.
DU shared folders	Collaborative folders of the challenges and practicals: structure, previews, what's new, checked measurement tables, analysis tools (Moon and ISS parallax), upload of one's own measurement without overwriting anything.
Observatory archives	Public images from Hubble, JWST, ESO, Spitzer, WISE, 2MASS, NOIRLab, Keck, SDSS, Voyager, Cassini, Juno: search, volume, download, extraction, alignment, colour.
Image quality	Optional. FWHM, ellipticity, background, noise, gradient, saturation, trails, sampling.
Spectra and series	Reading and plotting 1D data (radio spectra, light curves, tables).
Cosmology	From redshift to distances, ages, volume, distance modulus, scale (Planck 2018 and others).
Sites and times	Site database, OpenStreetMap map, UTC and local time.
Course calculators	The course exercises and practicals step by step with your numbers, then the exact calculation: Parallax (course practical).
My computer	Hardware diagnosis; minimal example module.

Principles: all long work runs off the GUI thread (the interface stays responsive, with progress and cancellation); an interrupted run resumes where it stopped; nothing is deleted without a trace; no data is sent without consent; optional components (ASTAP, SEP, CuPy) may be missing.

CHAPTER 2

Installation

2.1 All-inclusive packages

The packages bundle Python and every library (PyQt6, numpy, scipy, astropy, zstandard, lz4): nothing else to install. They update themselves. ASTAP stays optional and is not included.

Optional libraries bundled (0.2.1). Optional for a pip installation (extras `qualite`, `alignement`), they are in *every* package (Windows installer, .dmg, tar.gz, .deb): `reproject` (exact and adaptive alignment in the Archives module, with the `astropy-healpix`, `dask` and `zarr` it needs; up to 0.2.0 it was missing from the packages, and alignment fell back on scipy bilinear resampling), `sep` (Image quality module), `psutil` (machine measurements) and `lxml` (XISF validation). Only CuPy (NVIDIA card, unused for now) stays out. Each built package imports and runs them before publishing; added weight: about 20 MB to the Linux archive (101 → 121 MB), 12 MB to the .deb (109 → 121 MB). The *My computer* module and *Help* > *About* list the optional libraries present and missing, with their version.

System	File	Installation
Windows 10/11 64-bit	`Coupole-Setup-x.y.z.exe`	Double-click. SmartScreen warning at first launch (unsigned installer): « More info », « Run anyway ». Installed in the user profile, no administrator rights.
macOS Apple Silicon	`Coupole-x.y.z-macos-arm64.dmg`	Drag Coupole into Applications. First launch: right click, « Open ».
macOS Intel	`Coupole-x.y.z-macos-x86_64.dmg`	Same.
Ubuntu / Debian x86_64, ARM64	`coupole_x.y.z_amd64.deb`	`sudo apt install ./coupole_x.y.z_amd64.deb` (`_arm64.deb` on ARM): apt installs the Qt system libraries; `coupole` command, menu entry. Ubuntu 22.04 or newer, Debian 12 or newer. Uninstall: `sudo apt remove coupole`.
Other Linux x86_64 / ARM64	`Coupole-x.y.z-linux-*.tar.gz`	Unpack, run `installer.sh` (copied into `~/.local/share/coupole`, menu entry, `coupole` command through a link `~/.local/bin/coupole`; if that folder is not in the `PATH`, the installer says so and gives the line to add). The `Coupole.sh` launcher follows symbolic links (links to links included).

2.2 With the computer's Python

The scripts in the source folder look for Python 3.10 or newer, create a virtual environment, install the dependencies and create launchers. One question only: Enter to confirm (`-oui` / `-Oui` to skip it).

Command

```
sh install.sh # Linux, macOS
install.bat # Windows (double-click; runs install.ps1)
```

Without Python, or without the `venv` module (Debian: `python3-venv` package), the script says so and explains how to get it. Locations: Linux `~/.local/share/coupole/venv`, launchers in

`~/.local/bin`; macOS `~/Library/Application Support/Coupole/venv` and `~/Applications/Coupole.app`; Windows `%LOCALAPPDATA%\Programs\CoupolePython`, Desktop and Start menu shortcuts.

2.3 pip and pipx

Command

```
pipx install git+https://github.com/ARP273-ROSE/coupole
pip install .  # from the sources; pip install ".[qualite]" adds SEP
python -m coupole # interface; coupole -help:  command line
```

Extras: `qualite` (SEP), `validation` (lxml, xisf: XSD validation and second reader), `gpu` (CuPy), `test` (pytest).

2.4 Updates, uninstallation, data

At startup (can be disabled) the packages check the latest release on GitHub and offer to install it: only the code archive (`coupole-app-x.y.z.zip`, about 1 MB) is downloaded and unpacked by the application, the previous version being restored on failure. A pip/pipx installation shows the command to run. The `.deb` package does not update through the archive: its files (`/opt/coupole`) belong to dpkg; Coupole announces the new version, gives the address of the new `.deb` and the `apt` command to run (*Download the package* button). Settings, cache and reports live outside the installation folder and survive updates as well as removal:

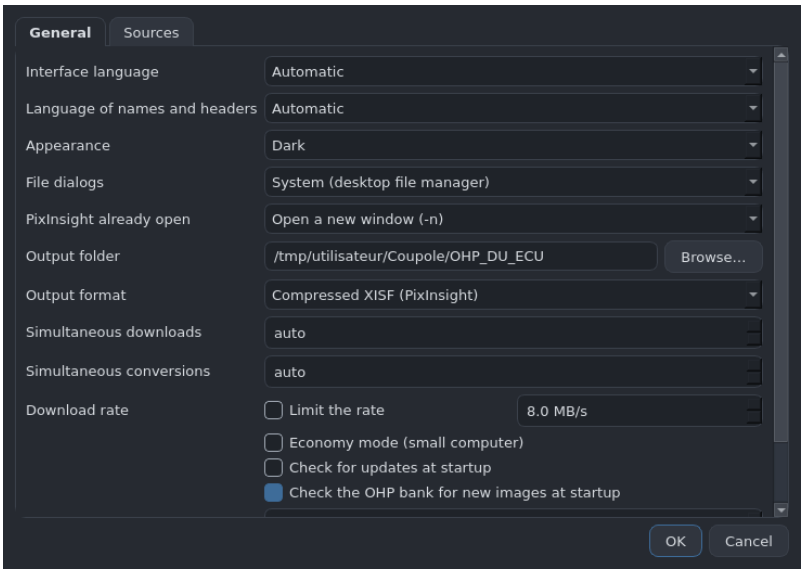
Windows	`%LOCALAPPDATA%\Coupole`
macOS	`~/Library/Application Support/Coupole` (cache: `~/Library/Caches/Coupole`)
Linux	`~/.config/coupole` (cache: `~/.cache/coupole`)
All	`COUPOLE_HOME` variable: replaces both (portable installations, tests)

CHAPTER 3

Getting started

The window has a module bar on the left (Ctrl+1 to Ctrl+9) and the current module on the right. Every interactive element has a tooltip. *Help* menu: help on the screen shown (F1), this manual (Shift+F1), keyboard shortcuts, report a problem, incident reports, check for updates, About (version, licence, detected configuration).

F1	help on the screen
Shift+F1	PDF manual
Ctrl+1 … Ctrl+9	module 1 to 9
Ctrl+,	Preferences
Ctrl+Shift+A	ASTAP assistant
Ctrl+Shift+D	dark / light theme
Ctrl+R	OHP bank: refresh the inventory
Ctrl+Q	quit



Preferences (Ctrl+,): interface language; language of folder names, object names and FITS comments; appearance (section 3.2); file dialogs (section 3.1); PixInsight already open (section 4.11); output folder; format; parallelism; download rate (*Limit the rate* box, unticked by default: no cap; ticked: cap from 0.1 to 1,000 MB/s, for the OHP bank and the Archives alike); economy mode; update check; check of the bank for new images at startup and its maximum frequency (section 4.7). The *Sources* tab lists the service addresses (chapter 17). An unreadable settings file is set aside (suffix `.corrompu-...`) and the defaults take over: the application always starts.

3.1 Choosing a file or a folder, folder on a NAS

The *Browse* buttons open the system file manager, with its places, bookmarks and network shares: Explorer on Windows, Finder (NSOpenPanel) on macOS, and on Linux the desktop's own through the **XDG portal** (`xdg-desktop-portal`): Dolphin on KDE Plasma, Files (Nautilus) on GNOME. The standalone package bundles its own Qt library, which cannot load the desktop plugin installed on the system: up to 0.1.7 it fell back to Qt's generic dialog, in English and without the system places. At startup, unless the user already set `QT_QPA_PLATFORMTHEME`, Coupole checks (waiting at most one second) whether the portal is installed or running with a backend able to choose files, and uses it; otherwise, on a GTK desktop (GNOME, Cinnamon, Xfce, MATE…), the GTK dialog; otherwise Qt's dialog, **in the interface language**, its sidebar completed with the home folders, `/media`, `/mnt`, `/run/media/$USER`, the shares opened in the file manager (`/run/user/<uid>/gvfs/...`) and the cifs, nfs and sshfs mounts found in `/proc/mounts`.

Preferences > *File dialogs*: *System* (default) or *Qt*, the latter only if the system dialog does not open or misbehaves (broken portal). On Linux, switching back to *System* takes effect at next start. All texts provided by Qt (standard buttons, context menus of text fields, fallback dialog) follow the interface language and change with it.

Output folder on a NAS.

- *Linux*: first open the share in Dolphin or Files (`smb://server/share`); it shows up under `/run/user/<uid>/gvfs/smb-share:server=server,share=share` (GNOME; KDE with the `kio-fuse` package, without which an unmounted `smb://` share cannot be picked by an application), or mount the share with cifs or nfs (`/mnt/partage`, advised for large volumes; default options, no need for `nobrl`: section 4.6);
- *macOS*: Finder > Go > Connect to Server (Cmd+K), `smb://server/share`, then pick the volume under `/Volumes`;
- *Windows*: network drive (`Z:`) or UNC path `\\server\share`, which can also be typed into the folder field.

These paths are accepted everywhere: state database read read-only (SQLite URI of a UNC path written with an empty authority), free space, write test, Windows long paths (`\\?\UNC\...`), wrapped display; a share is recognised as such (UNC, network drive, cifs, nfs, sshfs, gvfs, kio-fuse) and the Quality module limits its parallel reads there.

3.2 Appearance

Coupole has its own theme, independent of the computer's (a system in dark mode no longer mixes its colours with the application's: Fusion style, complete palette, fixed font size, WCAG contrast $\geq$ 4.5 for text, fields, buttons, tooltips and selection). The **dark** theme is the default; the light theme remains available. It is changed through the *View* > *Appearance* menu (checkable *Light* / *Dark* entries, Ctrl+Shift+D to toggle) or in the Preferences: the same setting, applied at once and saved; both stay in sync. A user who had chosen the light theme keeps it: the saved setting takes precedence over the default. The screenshots of this manual are taken in the default theme.

3.3 Saved settings

Each time it starts, Coupole shows the screen as it was left. Kept:

- **window**: size, position, screen, maximised state; size of every dialog (Preferences and its tab, ASTAP assistant, About, report, help, object correction, new site);
- **layout**: module shown, tab of the OHP bank, splitter positions, width and order of the columns, sort column and direction of every table;
- **OHP bank**: search, type, telescope, *New*, *To check*, *Missing only* and *Without doubtful dates* boxes, night and filter, chosen objects (restored as soon as the inventory is loaded), anomaly and sky-map filters; Download tab: output folder *as soon as it is changed*, format, language of names, *Keep duplicates*, *Keep FITS*, ASTAP mode, *Check quality*;
- **Image quality**: analysed folder, sample mode and number of images per batch;
- **Spectra and series**: folder of the open dialog and *Recent* menu (last 10 files; a file that has disappeared is greyed out); chosen axis;
- **Cosmology**: parameter set, custom parameters (H_0, Ω_m, kept even when going back to Planck), Ω_k, current z, SH0ES option, scale of the curves (logarithmic or linear);
- **Sites and times**: chosen site, zoom and centre of the world map, *Online map*;
- **file dialogs**: each one opens again in the last folder used (CSV export, reorganisation, ASTAP, report...).

The layout lives in `interface.json`, next to `reglages.json` (section 2.4); processing choices stay in `reglages.json`. Writing is *deferred and grouped*: at most once every 2 s and once on closing, atomic (temporary file then replacement); typing in a field never writes to disk. Tolerant reading: a missing value, a wrong type or an out-of-range value leaves the default; an unreadable file is set aside (`.corrompu-...`). Safeguards: an unplugged screen or a position outside the screens centres the window again, a size larger than the screen is brought back onto it (as at 150 or 200 %). A folder kept on a network share that is absent never blocks start-up: its existence is checked in the background, and the dialog opens in the default folder until it is confirmed. Nothing that would be dangerous to replay is kept: an interrupted processing never restarts on its own (it resumes when you run it again).

To go back to the original: Preferences > *Reset the layout* (immediate; language, theme, output folder and the other settings do not change), or `coupole -reinitialiser-interface` (`-reset-interface`).

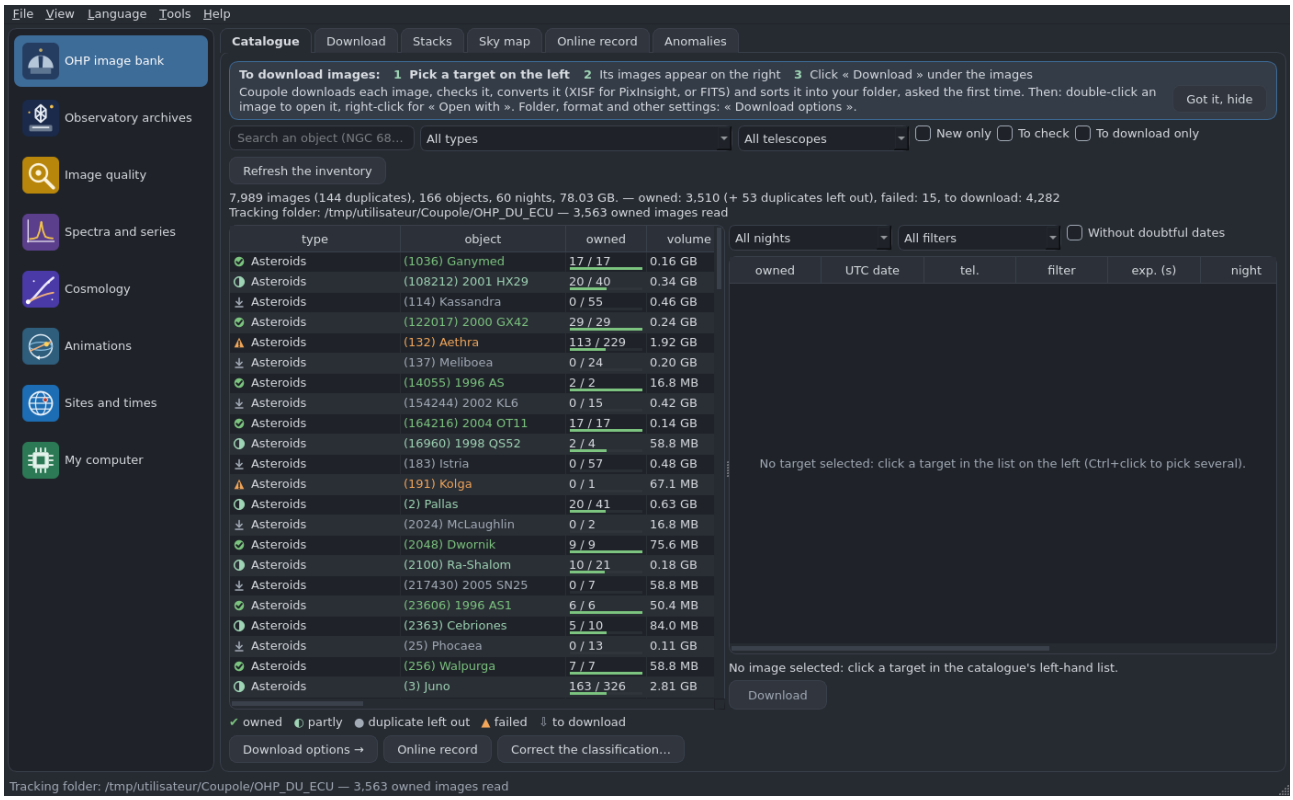
CHAPTER 4

OHP image bank module

4.1 The bank

« OHP student observations » gathers the images taken by the diploma's students at the Observatoire de Haute-Provence: 7,989 FITS images on 8 October 2026, 166 objects under 194 names, 60 nights from 2015 to 2026, 78.0 GB; 1.20 m telescope (T120, 7,419 images, 1024 pixels of 0.77″, 13.1′ field) and the 50 cm IRIS robotic telescope (570 images, 2048 pixels of 0.68″ or 4096 pixels of 0.46″, 23′ and 31′ fields). The images are already calibrated (dark, flat) and astrometrically solved: **they must not be recalibrated**.

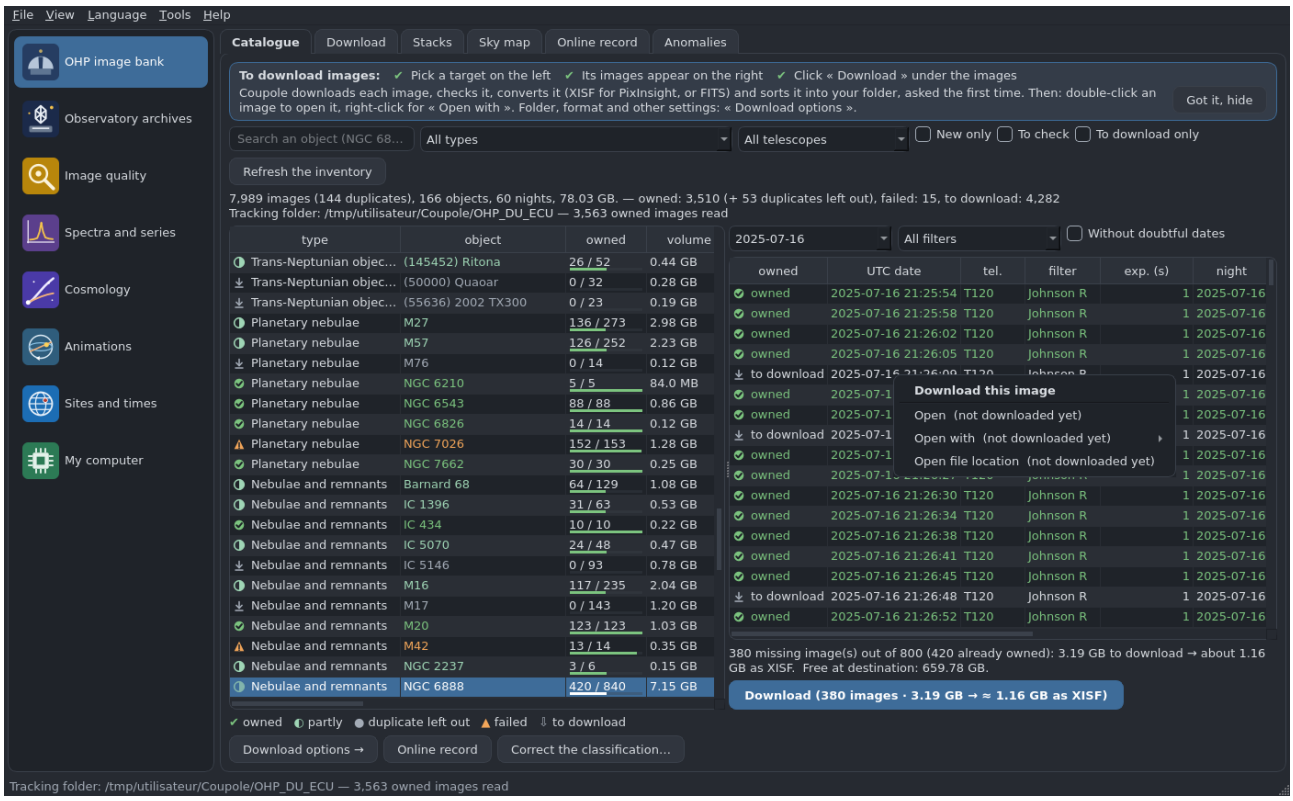
4.2 Download in three steps (0.7.2)



At first start, a banner at the top of the catalogue guides step by step; each step done is ticked.

1. **Pick a target**: click a row of the left-hand list (Ctrl+click or Shift+click for several; the search box and the type list help find it). While nothing is selected, the image list says so: « No target selected: click a target in the list on the left ».
2. **See its images**: they appear on the right, filterable by night, filter and doubtful dates.
3. **Download**: the blue button under the image list says what it will do, « Download (840 images · 7.15 GB → ≈ 2.64 GB as XISF) ». The first time, Coupole asks for the folder to put the images in (the system's full file dialog; it creates `OHP_DU_ECU` there); afterwards it reuses the same folder and format. Progress shows under the button, with *Stop*; at the end, each image's badge turns into a green check and the status bar says so (« Download finished: … Double-click an image to open it »).

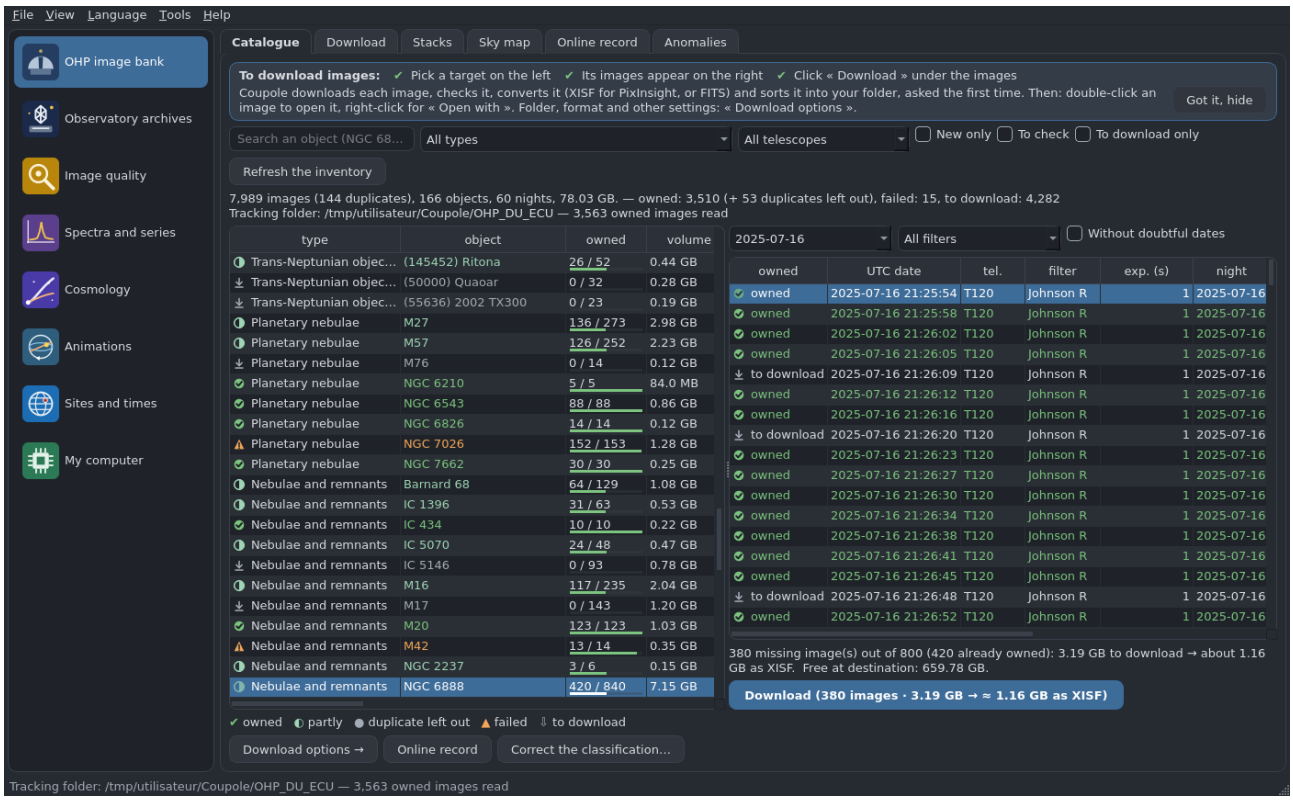
The button is greyed out, with the reason on hover, while no target is selected; it reads « Already downloaded ✓ » when everything is there, « Downloading… » during a download. *Got it, hide* closes the banner; *Help > Getting started (OHP bank)* shows it again. The settings (folder, XISF or FITS format, duplicates, original FITS, astrometry, parallelism) are in the *Download* tab, under *Options* (catalogue button *Download options →*).



Right-click and double-click. An image's menu starts with *Download this image* (or *Download these N images* when several rows are selected), then *Open*, *Open with*, *Open file location*, greyed out with their reason (« not downloaded yet ») while the file does not exist; a target's menu starts with *Download all the images of NGC 6888*. Double-clicking an image (or a target) not downloaded yet offers to download it; on a downloaded image, it opens it.

For students. « In Coupole, *OHP image bank* module: 1) click a target in the left-hand list; 2) its images appear on the right; 3) click the blue *Download* button under the image list and choose your folder the first time. Then double-click an image to open it. »

4.3 Catalogue

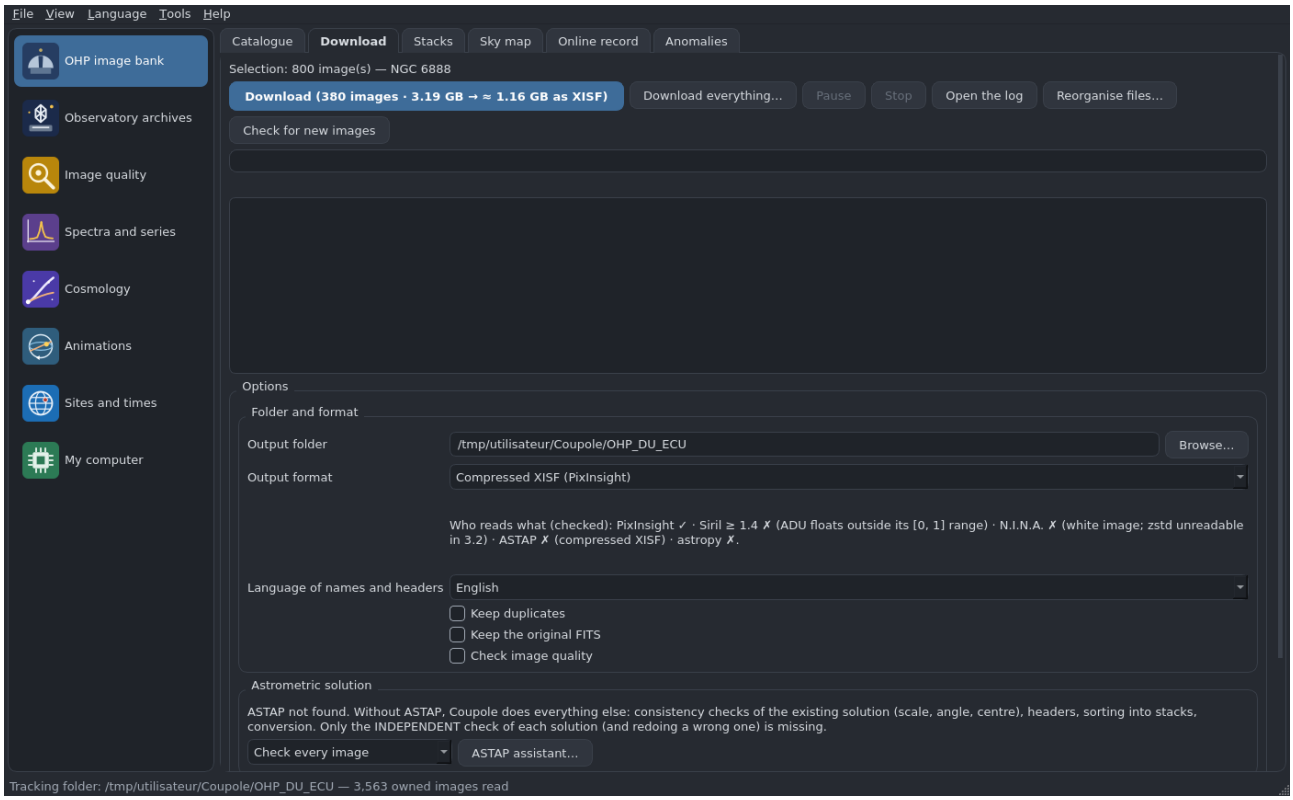


The inventory comes from the Observatory's public TAP service (`http://tap-ufe.obspm.fr/tap`, table `ivoa.obscore`, one synchronous ADQL query), kept in a cache; a snapshot is shipped for offline work. *Refresh the inventory* (Ctrl+R) queries the service again and flags new images and names (« new since … »). Database names are grouped into canonical objects; selecting one or several objects lists their images on the right, filtered by night, filter and doubtful dates; this list is what *Download* fetches. The estimate gives the volume to download, the expected output volume and the free space. Times are shown in UTC and in site time (hover: also the user's time). Each column header explains itself on hover; key columns (state, date, telescope, filter, exposure; object, owned, volume) come first and stay visible without scrolling at 1366 × 768.

What you already own. According to the output folder (`_traitement/etat.sqlite`, read again in the background when opening, after each run and whenever the folder changes, never blocking), the image list tells apart with a marker and a soft text colour (readable in both themes, contrast ≥ 4.5): **owned** (converted and sorted, green check), **duplicate left out** (same pixels as an image already converted, grey dot), **failed** (to retry, orange triangle) and **to download** (normal text, arrow); the marker's tooltip gives the status, the date and the local path of the file. The **object list** shows the same state, in the same colours: each row starts with a marker (green check: everything owned, left-out duplicates included; green half disc: partly; orange triangle: at least one failure; arrow: nothing downloaded), the object name takes the same colour, and the tooltip details it (« 12 owned / 12 · 0 to download · 0 failed · 3 duplicate(s) left out »). The *owned* column (« 120 / 300 », mini bar) comes right after the name (a column order chosen by hand is kept); sorting it orders objects by state (failed, nothing, partly, complete). The legend, below both lists, applies to both. The inventory summary line adds the overall ownership (« 7,989 images (144 duplicates), 166 objects, 60 nights, 78.03 GB — owned: 7,625 (+ 220 duplicates left out), failed: 0, to download: 0 »). The *To download only* box hides fully owned objects and, in the list, the images already processed; an emptied list explains itself in its centre (« Everything is already downloaded in … »), as does the image list while no object is selected (« No target selected: click a target in the list on the left »; at first start, no target is selected: the guide and this list say what to do). The estimate counts only what is missing (« 180 missing images out of 300: 1.5 GB »). Changing the output folder (Preferences, field or *Browse* in the Download tab) re-reads its ownership in the background at once: markers, column and summary follow, and the status bar says so (« Ownership recomputed: 7,625 images found in … »). Command line: `coupole ohp inventory`

`-missing [-dest FOLDER] [-json]`, and `possedee`, `statut_local`, `fichier_local` columns in the CSV of `coupole ohp list`.

4.4 Download (tab and options)



The *Download* tab (formerly *Processing*) shows the selection, the same *Download* button, *Download everything…*, *Pause*, *Stop*, progress and the log; it gathers under *Options* the folder and format, the language of names, *Keep duplicates*, *Keep the original FITS*, *Check image quality*, the astrometric check (ASTAP) and parallelism. Internal file and folder names do not change (`_traitement/etat.sqlite`…), nor does the command line (`coupole ohp process`): folders already filled stay compatible. For each image, duplicates excluded:

1. download: at most 4 at once (3 by default), full rate by default (total cap available in the Preferences), User-Agent naming Coupole, resume by HTTP Range requests, checks of size (server and inventory), multiple of 2,880 bytes and `SIMPLE` card;
2. reading of the FITS cards as written and of the pixels; SHA-1 fingerprint of the pixels (exact duplicates);
3. astrometric solution check (section 4.8);
4. fixed header (section 4.9);
5. writing in the chosen format, reading back and pixel-by-pixel comparison;
6. at the end: sorting into stacks, `LOT.txt`, `INDEX_LOTS.csv`, `_traitement/journal.csv`.

The state is kept in `_traitement/etat.sqlite`: *Stop* then *Download* (or running the same command again) resumes without redoing anything. A folder keeps the name language and format of its first run. *Keep duplicates* also processes what would be left out; *Keep the original FITS* stores them in `_traitement/fits_origine`.

The chain is a *pipeline*: downloads (threads) and conversions (separate processes, one per available core) overlap, the queue of waiting FITS is bounded (2× conversions + downloads; when free space at the destination is tight it is reduced automatically — never below conversions + 1 — and the log says so, rather than refusing the job), and the overall rate is that of the network or of the chosen cap

(measured: 24 images of 8 MB at 8 MB/s in 24.5 s, i.e. the duration of the download alone; without a cap, 86 MB/s of FITS converted on 6 cores). *Pause* suspends feeding the pipeline (images in progress finish) and *Resume* restores it; *Stop* interrupts downloads and conversions without waiting (processes are terminated, ASTAP included: by signal on Linux and macOS, by a *job object* on Windows). At the end, a report sums up images, duration, converted, duplicates, failures (with their reason), volume written and stacks.

On a network share (NAS over SMB), where every access is a round trip, the state is committed at most once per second and `JOURNAL.txt` is written in batches (a power cut only redoes the last second); the final sorting rewrites only the `LOT.txt` files that changed and no longer walks the whole tree: running a processing again for one image no longer costs a minute of sorting. The state database, however, is never written directly on a share: section 4.6.

4.5 Download everything, log, reorganise

Download everything… takes the whole bank (about 8,000 images, 78 GB of FITS, 30 GB as XISF). On first use (or when no output folder is set), Coupole asks for the destination folder through a native dialog and proposes `Documents/Coupole/OHP_DU_ECU`; then it shows, **before any confirmation**, the volume to download, the expected output volume, the estimated duration — at the rate cap if one is set (78 GB at 8 MB/s: 2 h 42), otherwise at the rate measured during previous downloads, otherwise "depends on the connection" — and the free space; it clearly refuses when space is short. Processing is the same as for any selection: resume after a cut or a close (per-image state in `_traitement/etat.sqlite`, already downloaded FITS found again), pause, end report. Command line: `coupole ohp all -yes`.

Open the log shows `_traitement/JOURNAL.txt`: one timestamped (UTC), bilingual line per event — session start and end (resume, pause), each image downloaded, converted or left out with its reason, each error, each name conflict; rotation at 5 MB (`JOURNAL-1.txt`, `JOURNAL-2.txt`). The tabular details stay in `journal.csv`.

Reorganise files… sorts into the stack tree images already converted by Coupole that live elsewhere or in an older layout: each file is recognised by its header (source address `OHP:Source:URL`, or `HISTORY` card « converted from … »), attached to the inventory and **moved** (never copied); a name already taken by a foreign file gets a suffix and a log line; nothing is ever overwritten. Headers are read by several processes, progress is shown (« 1,200 / 7,625 headers read ») and *Stop* interrupts the reading (whatever is already recognised is sorted). Command line: `coupole ohp reorganise FOLDER -dest DEST`.

4.6 Output folder on a network share

Why. The state database (`_traitement/etat.sqlite`) is an SQLite database, which locks its file on every write. On Linux, an SMB share mounted by `cifs` with default options passes these locks on to the server as SMB byte-range locks, and writing fails: « database is locked » after one minute, 0-byte database (checked on a real Samba server; true of every version up to 0.1.9: downloading new images, *Download everything* or reorganising into a NAS folder failed). Read-only access and file copies do work.

What Coupole does (0.1.10). When the output folder is recognised as a share (cifs, smb3, nfs, sshfs, gvfs, kio-fuse on Linux; smbfs, afpfs, nfs on macOS; UNC path or network drive on Windows) or when SQLite cannot write there within 3 s (unrecognised share, another program), it works on a **local working database** in the user's cache folder (`bases/<key>/etat.sqlite`, one per output folder, the key derived from the path), and says so in the log and the status bar: « Folder on a network share: local working database, copied back to the share every 30 s ».

- *At the start*: the database on the share is copied (plain file copy) into the working database.
- *During processing*, every 30 s, *at the end*, on *stop* and *cancel*: the working database is copied back to the share — consistent snapshot, `PRAGMA integrity_check`, copy into a temporary file in the same folder, read back and SHA-256 compared, then replaced in one go (`os.replace`). No lock is ever taken on the share: no orphan lock is possible after a crash, and a reader (another computer, the NAS) always sees a complete database.

- *At the next start*: if the database on the share is newer (processed elsewhere), it is copied in; if the working database holds writes that were not copied back (crash, power cut), they are copied at once.
- *Two writers* (two computers, or the NAS, processed into the same folder meanwhile): nothing is overwritten. Coupole says so clearly and offers a **merge**: for each image the most advanced status is kept (converted > duplicate left out > failed > in progress), the most recent on a tie; pixel fingerprints are pooled. Both original databases are kept next to the working database (`etat.sqlite.local-<date>`, `etat.sqlite.partage-<date>`). Command line: `coupole ohp merge -dest FOLDER`.

Ownership, new images and anomalies read the working database when it is up to date (faster than reading page by page over the network), otherwise the database on the share, read-only. `JOURNAL.txt`, the CSV files and the `LOT.txt` files are still written directly on the share (text files, replaced in one go). The Quality module's cache is only a cache: it stays in the user's cache folder, without copy back (section 9).

Per system. *Linux*: default cifs fails (cause above); mounting with `nobrl` is no longer needed. *macOS*: the smbfs client handles locks differently (depending on how SQLite was built); this could not be checked here, so Coupole does not rely on it and uses the same working database. *Windows*: SQLite can in principle write on a share (native SMB locks), but several writers risk damaging the database; same working database, checked in continuous integration on `\\localhost\C$`. If the database on the share is open in another program when it is replaced, Coupole retries, then waits for the next pass.

Windows: network drive or UNC path (0.2.1). An output folder on a network drive letter (`O:\OHP_DU_ECU`, mapped share) or on a UNC path (`\\server\share\OHP_DU_ECU`) is recognised as a share (`GetDriveTypeW` = remote drive). Up to 0.2.0, the state database of a folder on a network drive *letter* was unreadable: the path was « resolved » into a UNC path, and the resulting SQLite address (`file://server/...`) was rejected; the error went unnoticed and everything looked to be downloaded, while the same folder seen from Linux was fully owned. Since 0.2.1, the address is built without resolving (`file:///O:/...`, `file:////server/share/...`) and a database on a share is read through a **local copy** (user cache, redone only when the database changes); a database in WAL journal mode (old script) is copied with its `-wal`, which Windows cannot read on a share. The letter and the UNC path of the same share each have their working database (the key comes from the path typed): harmless, each follows the share. Checked in continuous integration on the machine's `C$` administrative share, under its name (not « localhost », which SQLite accepts in an address), as UNC and through a letter set by `net use`.

Tracking folder. The OHP bank summary line and the status bar give the folder really read and what was found there: « Tracking folder: <path> — N owned images read ». If the tracking database exists but cannot be read, an **error banner** says so with the cause (« The tracking database of this folder could not be read: … ») instead of showing « nothing owned »; `coupole.log` records each read (working database, copy or direct read, number of images read) and each synchronisation with the share. If `_traitement/` is missing while converted images are sorted (partial copy, restored backup), the banner offers **Recognise existing files**: each file's header (`OHP:Source:URL`, otherwise the `HISTORY` card « converted from … ») gives the original image, recorded as owned; in the background, with progress; nothing is downloaded, moved or rewritten (`coupole ohp recognise -dest FOLDER`). *Refresh the inventory* also re-reads ownership.

4.7 New images in the bank

At startup (at most once a day, adjustable, can be disabled in Preferences), when the Internet is available and a local copy exists in the output folder, Coupole compares a fresh TAP inventory with the copy. If there are new images, a banner appears in the OHP bank (and a message in the status bar): « N new images (M objects, X GB) since DD/MM — Download now? / Later / Show ». *Download* processes only the new images into the same tree; *Show* filters the catalogue; nothing is ever downloaded without explicit consent. The *Check for new images* button runs the comparison on demand. Command line: `coupole ohp new` (list) and `coupole ohp new -download`.

4.8 Astrometric solution

Consistency checks, always made: solution present and readable; scale within 2 % of the nominal one (`XPIXSZ/FOCALLEN`) or of the night median; angle within 5° of the night median (modulo 180°); centre within 1.5° of the object medoid (fixed) or 2° of the object-and-night medoid (moving). Medians and medoids come from the whole inventory.

With ASTAP (optional), each image is also solved independently:

Status	Meaning
confirmed	ASTAP finds the same centre within 10″ and the same scale within 1 %.
validated	solution kept: checks pass, ASTAP silent (poor or crowded field, short exposure) or absent.
redone	missing or inconsistent solution replaced by ASTAP's (with SIP); the old one goes to HISTORY.
doubtful	checks fail and no ASTAP to decide: image stored apart.
failed	nothing usable: image stored in `_sans_solution_astrometrique` (`_no_astrometric_solution` with English names).

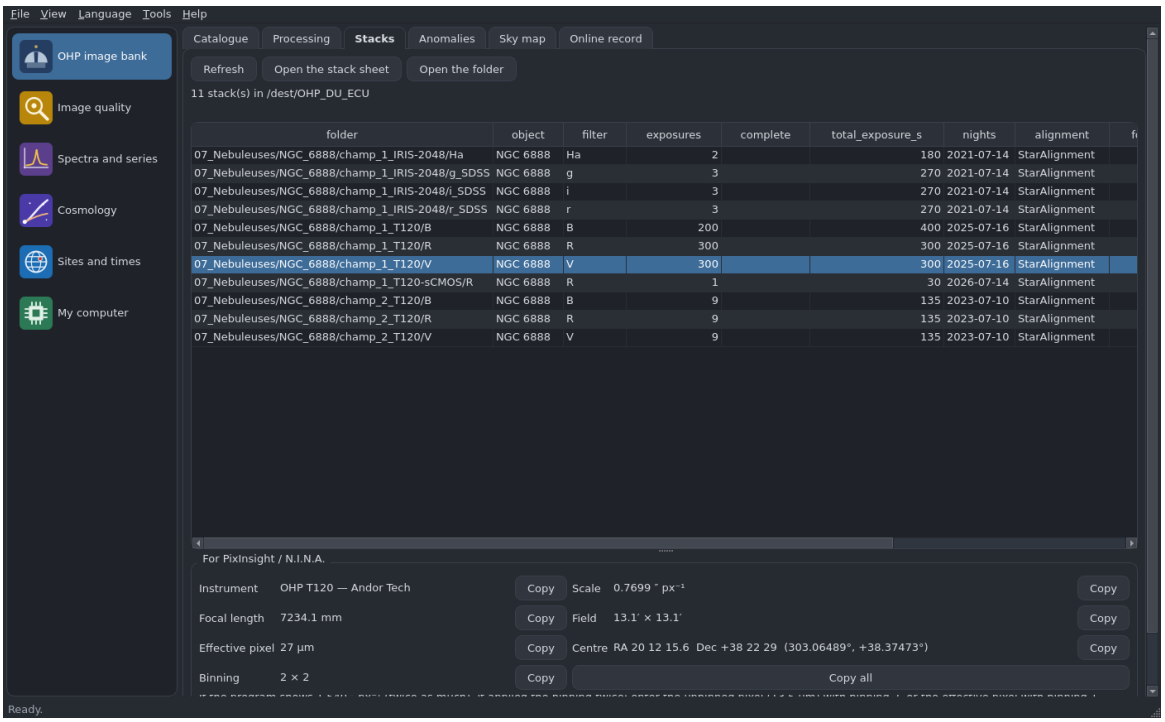
4.9 Fixed headers

All original cards are kept in order; every replaced value is copied into a `HISTORY original value: ...` card (French wording when names are in French).

Keyword	Fix
`OBJECT`	canonical name (database aliases merged).
`FILTER, FILTSYS, FILTORIG`	normalised filter (B, V, R, Ha, OIII, g, r, i, z), system (Johnson, Cousins, SDSS, Gunn, Narrowband), original value.
`LATITUDE, LONGITUD`	fixed **only when the swap is proven** (swapped values matching the site); correct: untouched; ambiguous: flagged, untouched. Idempotent.
`SITELAT, SITELONG, SITEELEV`	added when missing (MPC 511 site converted to WGS 84); present and consistent: kept; inconsistent: flagged.
`RA, DEC, OBJCTRA, OBJCTDEC`	true centre of the solution.
`PIXSCALE, ORIENTAT`	measured scale, position angle of the Y axis.
`FOCALLEN, XPIXSZ`	recomputed only when missing or inconsistent by more than 2 %.
`CTYPE1/2`	« -SIP » without coefficients (IRIS, PinPoint) brought back to TAN.
`DATE-OBS, DATEDOUT`	added when missing; `DATEDOUT = T` when the date is shared with another exposure.
`XISFCONV`	description of the conversion.

The `t_min` column of the database equals `DATE-OBS` − exposure/2; `DATE-OBS` (exposure start) prevails. A header date written in local time (whole-hour offset equal to the site time-zone offset against `MJD-OBS/JD`, or a « night » exposure dated in daylight) is flagged in the log, never silently fixed.

4.10 Stacks



A stack gathers what is stacked together: same object, same instrument (T120, T120-bin1, T120-sCMOS, IRIS-2048, IRIS-4096), same filter, and same field for a fixed object (all nights) or same night for a moving object.

Command

```
07_Nebulae/NGC_6888/field_1_T120/V/20250716-214427_NGC_6888_V_1s.xisf
01_Asteroids/914_Palisana/night_2023-08-15_T120/R/LOT.txt
```

`LOT.txt` (French then English) gives object, database names, instrument, filter, exposures and total exposure, nights, mean centre, angle, solution status, doubtful dates and stacking advice (StarAlignment or CometAlignment, rejection suited to the number of exposures). File names: UTC date and time of the exposure start, object, filter, exposure; sorting by name = sorting in time. Names are safe under Windows. The *Stacks* tab adds a *complete* column: converted exposures against the bank's exposures for the same key (object, instrument, filter, and night for a moving object); a fixed object split into several fields counts its stacks together.

Owned ≠ stackable together. « 840 / 840 » in the *owned* column says all the images of an object are there, not that they form one stack: the *stacks* column of the object list and its tooltip give the breakdown (« 840 images sorted into 9 stacks: field 1 T120: B 200 · V 300 · R 300; … — stack each stack separately »), and right-clicking an object offers *Show this object's stacks* (Stacks tab filtered, *All stacks* button) and *Open the target folder* (double-click: the same).

For PixInsight / N.I.N.A. Below the stack table, a box gives, for the selected stack, what to enter in PixInsight (ImageSolver, WBPP *Astrometric solution* panel) or N.I.N.A.: instrument, **focal length** (the one giving the measured scale: $f = 206.265 \times \mathrm{XPIXSZ/PIXSCALE}$, for instance 7,234.1 mm for the T120 instead of the original header's 7,200), **effective pixel** (binning included: T120 iKon-L in binning 2: 27 µm), binning, **scale** ($''\,\mathrm{px}^{-1}$), field and centre, each with *Copy*, plus *Copy all* (also on a stack's right-click: *Copy the astrometry parameters*). The values are read from the header of the stack's images, never hard-coded; the *focal length*, *pixel* and *scale* columns can be hidden by right-clicking the header. `LOT.txt` gives the same values. If a program shows twice the scale, it applied the binning twice: enter the unbinned pixel (13.5 µm) with binning 2, or the effective pixel with binning 1.

Checked in PixInsight's code (PJSR: `AstrometricMetadata.js`, `BPP-Solver.js`): ImageSolver reads the `Instrument:Telescope:FocalLength` (m) and `Instrument:Sensor:XPixelSize` (µm) prop-

erties, otherwise `FOCALLEN` and `XPIXSZ`, without multiplying by `XBINNING`; WBPP solves the integrated *master* with its metadata if they survive integration, otherwise with the values of its panel, which stay those of the last instrument used: enter focal length and pixel there (one WBPP settings file per instrument avoids typing them again). Coupole therefore writes `FOCALLEN` matched to the measured scale (old value in `HISTORY`) and the `Instrument:Sensor:XPixelSize/YPixelSize`, `Instrument:Camera:XBinning/YBinning`, `Instrument:Camera:Name`, `Instrument:Telescope:FocalLength` and `Aperture` (T120: 1.2 m) properties. For files already converted: `coupole ohp metadata FOLDER -rewrite` (without `-rewrite`: dry run) completes the headers *without touching the pixels* (block copied byte for byte, read back and compared; atomic writing; log `_traitement/metadonnees.csv`) and notes focal length and pixel in the state database; `coupole ohp sort` then rewrites the `LOT.txt` files. To check a master: open it, *FITSHeader* or *Image > Properties*, look for `XPIXSZ`, `FOCALLEN` and `Instrument:*`.

4.11 Opening an image

Double-click an owned image (image list, Quality table): the system's associated application. Right-click: *Open*, *Open with* and *Open file location* (file selected: Explorer `/select`, Finder `open -R`, freedesktop file manager through D-Bus `FileManager1.ShowItems` — Dolphin, Nautilus… —, otherwise the folder). *Open with* only offers the astronomy programs found on the computer (PixInsight, Siril, ASTAP, Aladin, N.I.N.A.: PATH, usual locations, flatpak, `/Applications`, Program Files) *and* able to read this file according to the checked table of section 8; the others are greyed with the reason (« ASTAP only reads uncompressed XISF: choose the FITS format in the download options »). An image not downloaded yet: the menu starts with *Download this image*, and double-click offers to download it (0.7.2). A disabled entry carries its reason in its label (« Open with (not downloaded yet) ») and the full reason shows in the status bar on hover: no more tooltip over the menu (on KDE it grabbed the mouse).

How each program is started (0.1.12). Always the detected program, with a *list* of arguments (never a shell); each launch is recorded in `coupole.log` (method, program, arguments, process id; if the program stops within the first 3 seconds: exit code and the beginning of its output) and its result shows in the status bar: « Opening in Siril… », or the error (program not found, immediate stop with its message). Never silent.

- **PixInsight** — Linux: the `bin/PixInsight.sh` script (the publisher states that the binary alone does not start on Linux). This script rebuilds its arguments through an `eval` that only quotes those containing a space: a path with `$`, `‘`, `’`, `"`, `\`, `(`, `*`… would be interpreted; Coupole then passes a *temporary symbolic link* with a safe name (to save: « Save As » into the original folder). Windows: `bin\PixInsight.exe`. macOS: `open -a PixInsight.app` *`file`*, or `open -n -a PixInsight.app -args -n` *`file`* for a new instance.
- **PixInsight already open.** Without options, PixInsight hands the file to the running instance (« Yielded execution to running application instance #1 », code 0); if that instance is busy (script running), *nothing opens*. Hence the setting *Preferences > PixInsight already open*: « Open a new window (-n) » (default, safe) or « Send to the open window (try) ». When an instance is running (process detected), the menu offers both: « PixInsight (new window) » and « PixInsight (open window, try) », the setting's one first; if the output says PixInsight yielded, the status bar says so.
- **Siril**: `siril` *`file`*; as a flatpak: `flatpak run -file-forwarding org.free_astro.siril @@` *`file`* `@@` (the document portal only forwards the file if the sandbox cannot see it). The Flathub package sees the whole system (`-filesystem=host`) except `/tmp`, `/run` (apart from `/run/media`), `/usr`…; if a `flatpak override` removes a folder, the status bar says so and gives the command: `flatpak override -user -filesystem=/mnt/partage org.free_astro.siril`.
- **ASTAP, Aladin**: *`program file`*.

- **N.I.N.A.** does not open an image given as an argument (its only options: profile, sequence, debug, graphics acceleration): it stays greyed in *Open with*, with the reason; open the image from within N.I.N.A.

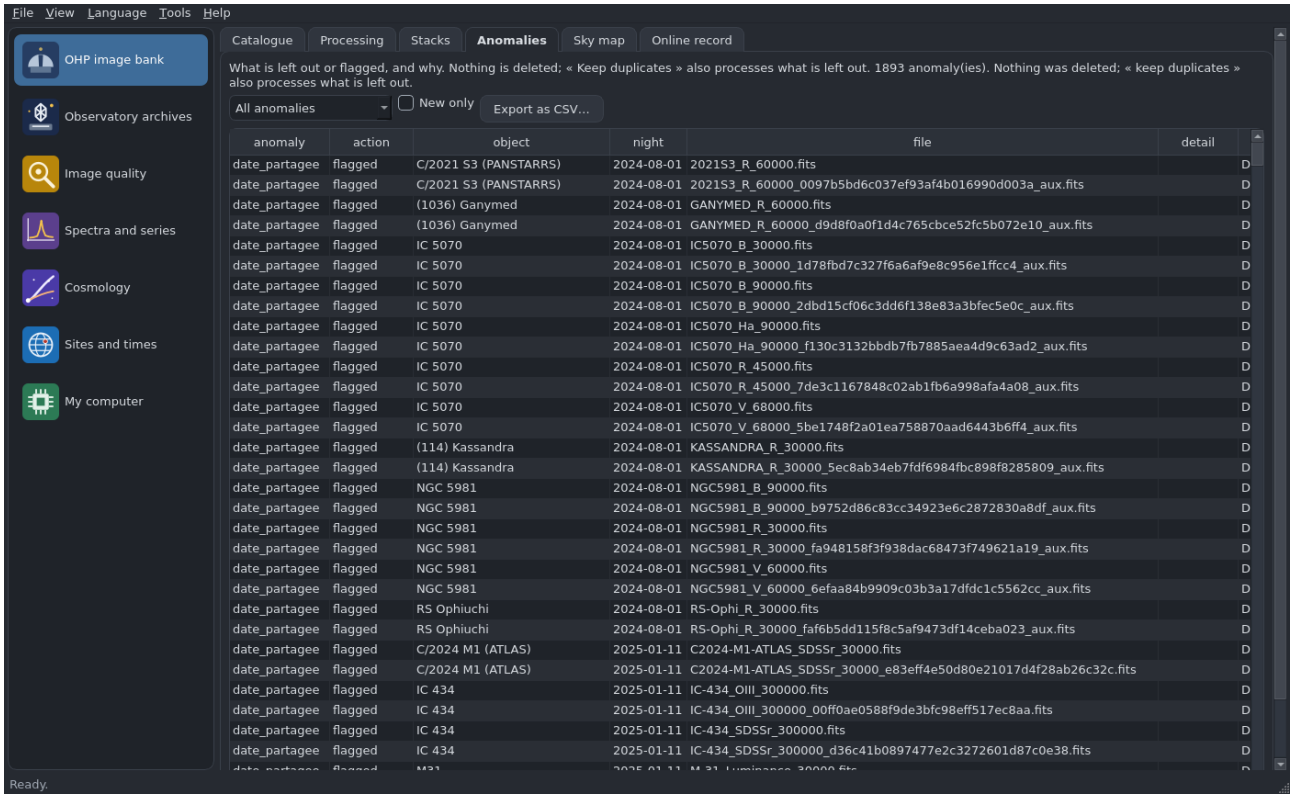
Double-click opens PixInsight without the image (Linux). `xdg-open` starts the application associated with the file type through its `.desktop` file. If its `Exec=` line has no `%F` code (`%f`, `%u`, `%U`), the application starts *without* the file — the case of a hand-written `pixinsight.desktop`. Coupole detects it (`xdg-mime query filetype`, then `xdg-mime query default`, reading the `.desktop`) and then starts the program directly (status bar: « PixInsight started directly… »); another application in that case is not started, a message says why. To fix the whole desktop (Coupole does not change this file), in `~/.local/share/applications/pixinsight.desktop`:

```
Exec=/opt/PixInsight/bin/PixInsight.sh %F
MimeType=image/x-xisf;image/fits;
```

then `update-desktop-database ~/.local/share/applications` and, if needed, `xdg-mime default pixinsight.desktop image/x-xisf`.

Copy from an older processing run. An output made before Coupole (script `ohp_xisf.py`) or on another computer records the absolute path of the time in its state database (« /srv/ancien/OHP_DU_ECU/… »). Coupole finds each file in the current output folder, in this order: relative path recorded in the database, `_traitement/journal.csv` (`destination` column), former root or type folder, then the expected name in the stack folder; paths found through the journal are recorded once and for all in the database (in the background, at first load; through the local working database on a share). Open, *Open with*, file location, target folder and stacks therefore work on a copy of the whole bank (7,625 images found in 0.34 s, recorded in the database in 0.8 s). `coupole ohp metadata FOLDER -rewrite` records these locations too.

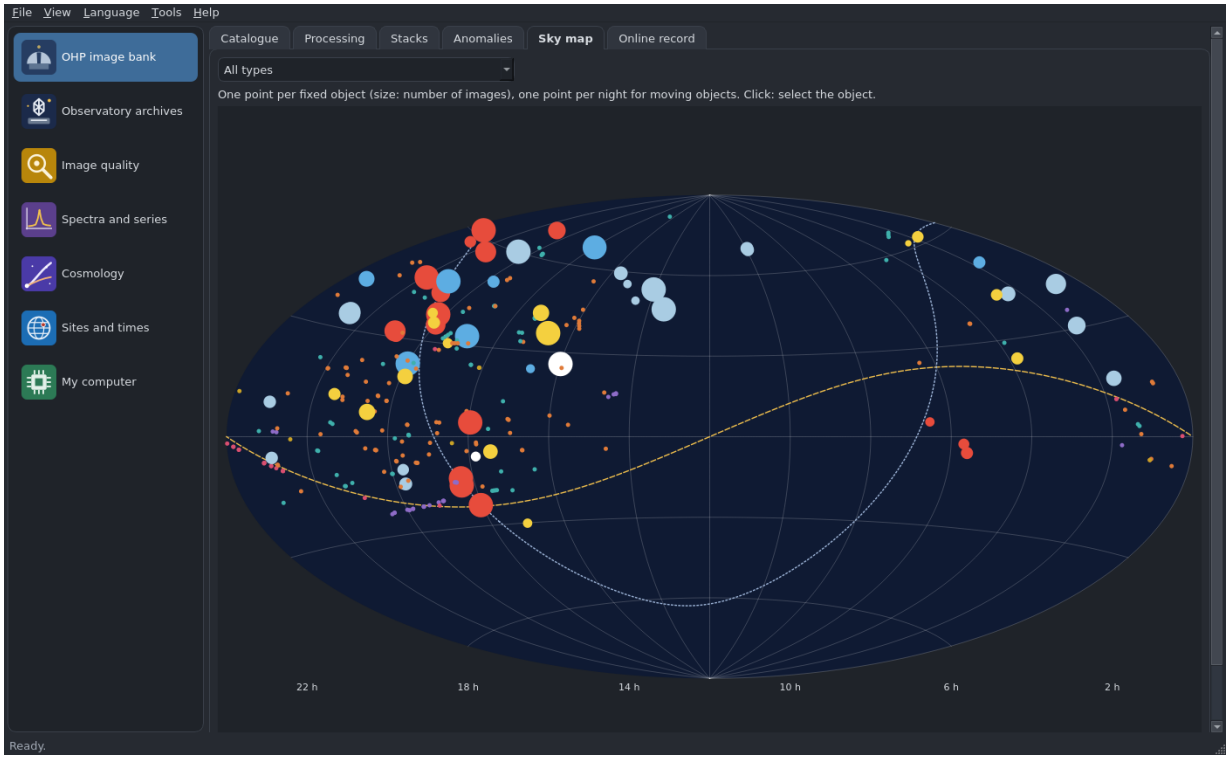
4.12 Anomalies



Kind	Action	Meaning
`meme_fichier`	left out	same exposure under another path or spelling
`copie_diurne`	left out	daytime-dated copy of a correctly dated image
`pixels_identiques`	left out	same pixels (SHA-1) as another image, at processing
`memes_meta`	flagged	same date, filter, exposure, size, different name
`date_partagee`	flagged	DATE-OBS identical to another exposure
`date_diurne`	flagged	Sun above the horizon at the site (never for the Sun or radio data)
`champ_incoherent`	flagged	centre far from the other exposures of the object
`classement`	flagged	name classified automatically or not at all
`instrument`	flagged	instrument missing from the table

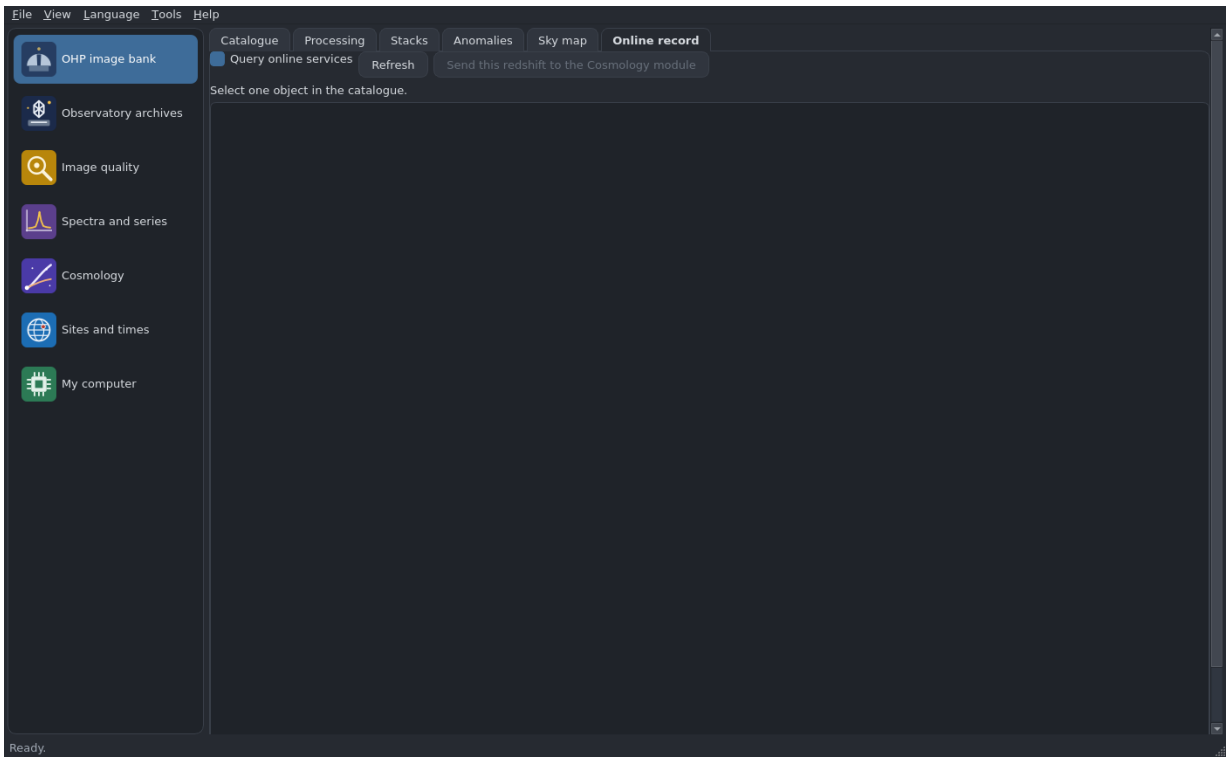
On 8 October 2026: 133 + 11 = 144 duplicates left out, 1,654 shared dates, 3 daytime exposures, 92 inconsistent fields. CSV export with bilingual columns; nothing is deleted.

4.13 Sky map



Aitoff projection, right ascension increasing to the left, 12 h at the centre; one point per fixed object (size: number of images) and per night for moving objects; ecliptic dashed, galactic plane dotted.

4.14 Online record (optional)



When the Internet is available, the *Online record* tab (or the button of the same name under the catalogue) shows public information on the selected object:

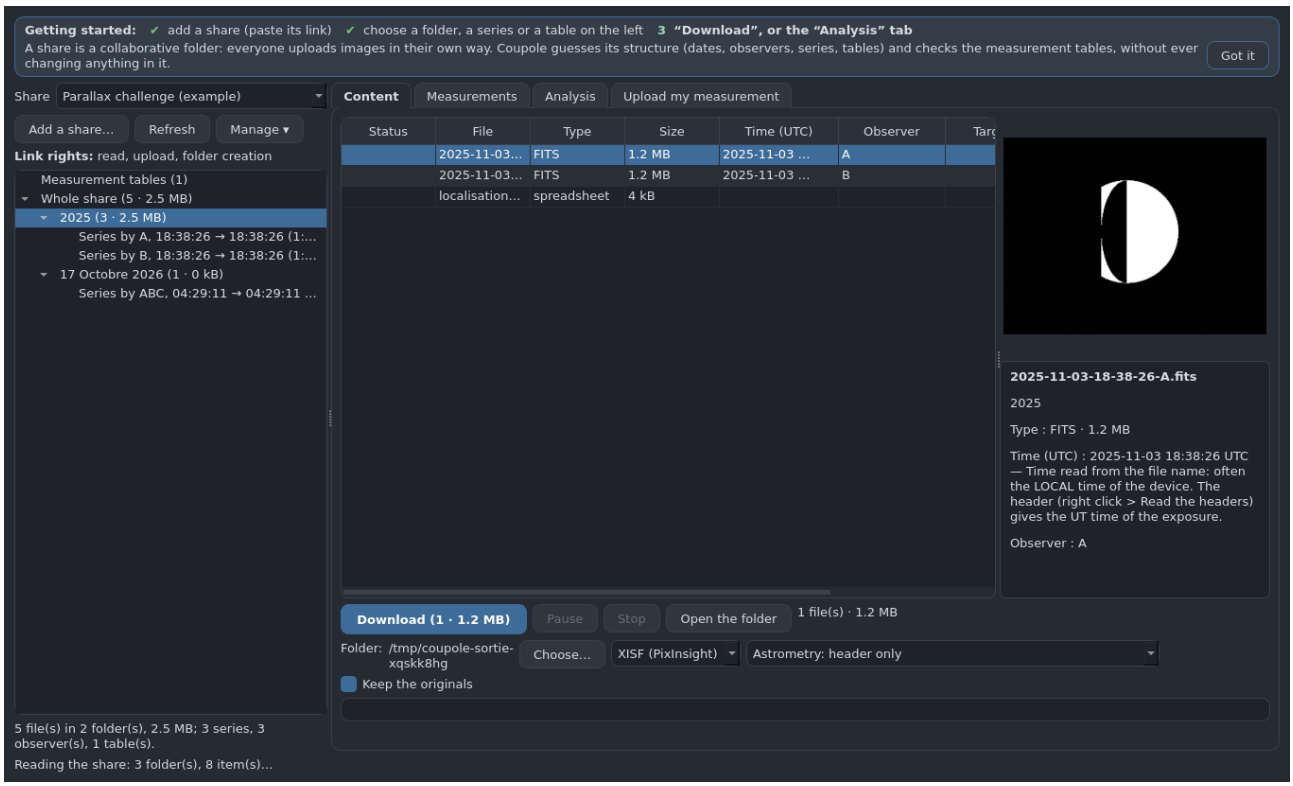
- deep sky and stars: **SIMBAD** (CDS, Strasbourg) — type, ICRS coordinates, magnitudes, parallax (and the distance it gives when precise enough), preferred measured distance with its method and reference, radial velocity and redshift, angular size, spectral and morphological types, other names; links to the SIMBAD page, **Aladin Lite** (sky image centred on the object) and, for galaxies, **NED**; when SIMBAD does not know the name, **Sesame** (CDS) also queries NED and VizieR;
- asteroids, comets, trans-Neptunian objects: **JPL** Small-Body Database — orbit class, near-Earth or not, elements (a, e, i, q, Q, Ω, ω, period, epoch), diameter, albedo, rotation period, absolute magnitude;
- planets and NEOCP designations: no record (explanatory message).

The identifier chosen by the service is always shown: SIMBAD, for instance, attaches « NGC 6888 » to its Wolf-Rayet star HD 192163; the record says so. A positive redshift can be sent to the Cosmology module (*Send this redshift to the Cosmology module* button).

Politeness and offline use. Nothing is sent while the tab is hidden; one request per selected object (after 0.6 s without another selection change), at most one per second and per service, 8 s timeout, Coupole's User-Agent. Answers are kept for 30 days in Coupole's cache (`enligne.json`, with their date; « not found » for one day). Offline, the last known record is shown with its date, otherwise a clear message; the rest of the application never depends on it. The *Query online services* box stops all requests. The addresses (`simbad.base`, `simbad.page`, `aladin.page`, `ned.page`, `sbdb.page`, `resolution.sbdb`, `resolution.sesame`) are set like the other sources.

CHAPTER 5

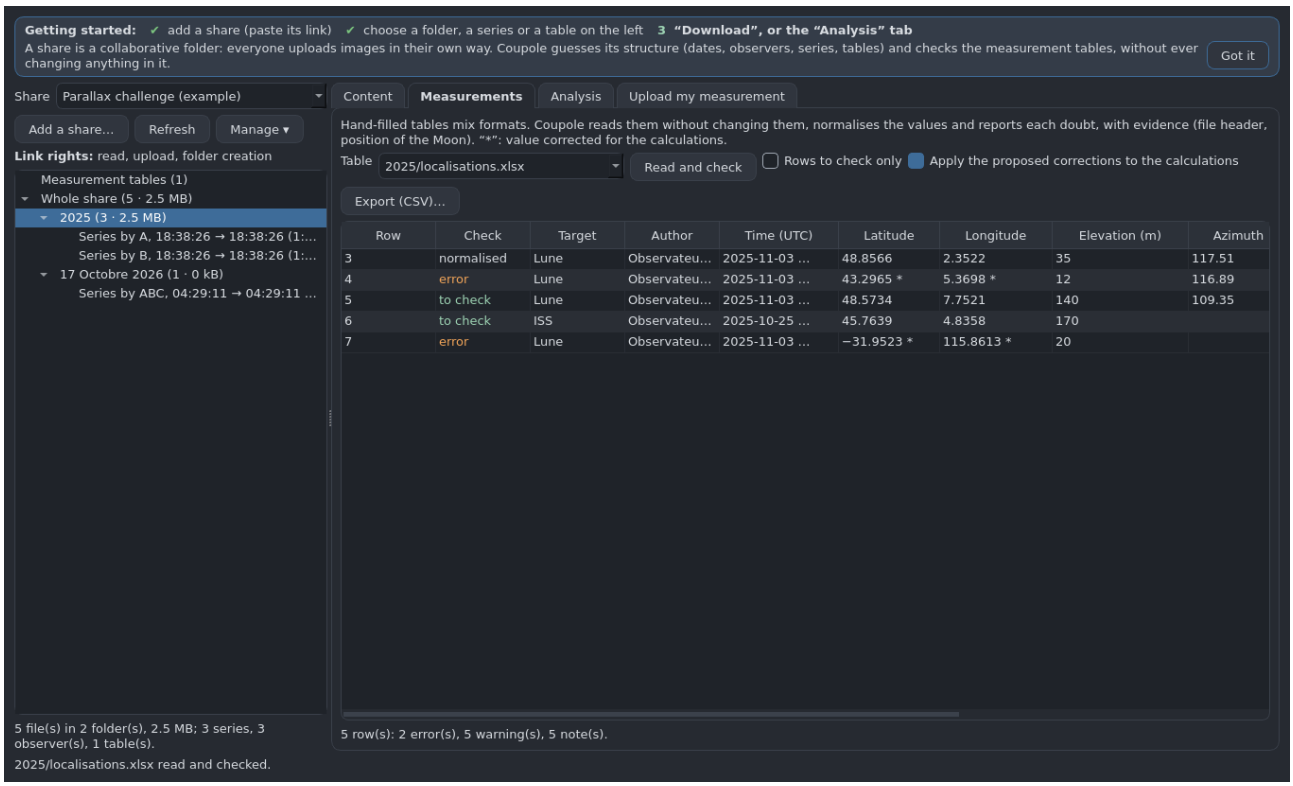
DU shared folders



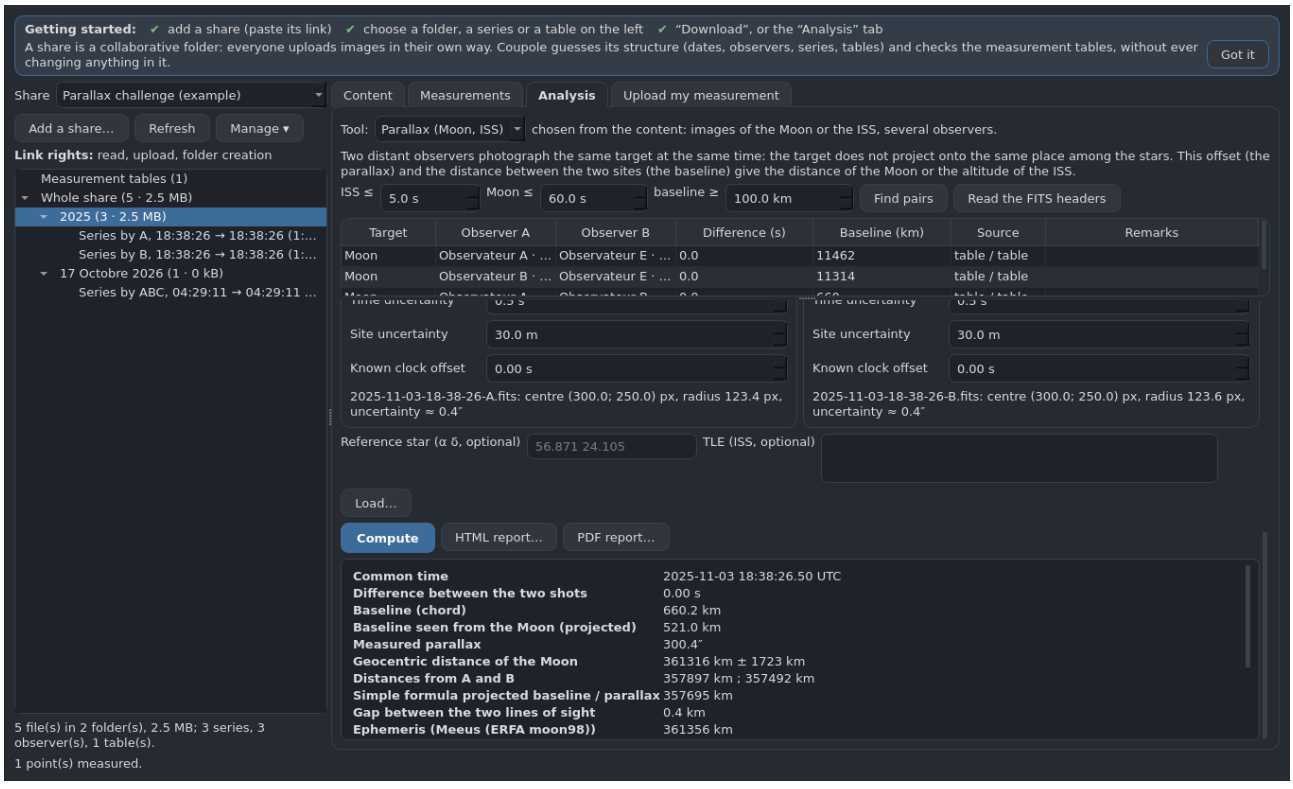
The challenges and practicals of the course often take place in a *shared folder* (the Observatory's Nextcloud): a "`https://.../s/...`" link where everyone uploads images, videos and sometimes a row in a table. This module gathers everything in one place: several remembered shares (free names), their structure, previews, what's new, download, checked measurement tables, analysis tools and the upload of your own measurement. **No link ships with Coupole**: you paste the one the course team gave you; it stays on your computer (`partages.json`, readable by you only) and never appears in a report.

OHP bank or share? The OHP bank is a professional archive (TAP service, normalised metadata, FITS from the T120 and IRIS telescopes). A share is a collaborative student folder: heterogeneous formats and metadata, typed by hand, to be validated. Coupole guesses the structure (dated folders, observers, series, tables), tells where each piece of information comes from (header, table, file name) and flags doubts. The layout is the same as the bank's: list on the left, content on the right, blue "Download (N · size)" button, right-click menu starting with "Download", three-step guide on first launch (*Help > Shares guide*), previews (FITS, JPEG, PNG, TIFF, first frame of a SER or AVI video) and a banner for what is new since your last visit.

What Coupole guesses. The kind of each file; the instant from the name ("`YYYY-MM-DD-HH-MM-SS`", the practical's "`DD-MM-YY-...`", "`IMG20260618223000`"...) — a name that reads two ways (17-10-26: 17 October 2026 or 26 October 2017) is settled by the upload date, and a time read from a name is marked "from the name" (often local time); the observer (end of the name, or a "date – place – name" folder); the target (ISS, Moon...); the shots (FITS/PNG/TIFF variants of one photo) and the series (exposures less than 2 min apart). FITS headers are read *without downloading the image* (partial request: the first 34 kB): site (SITELAT/SITELONG), start time and mid-exposure.



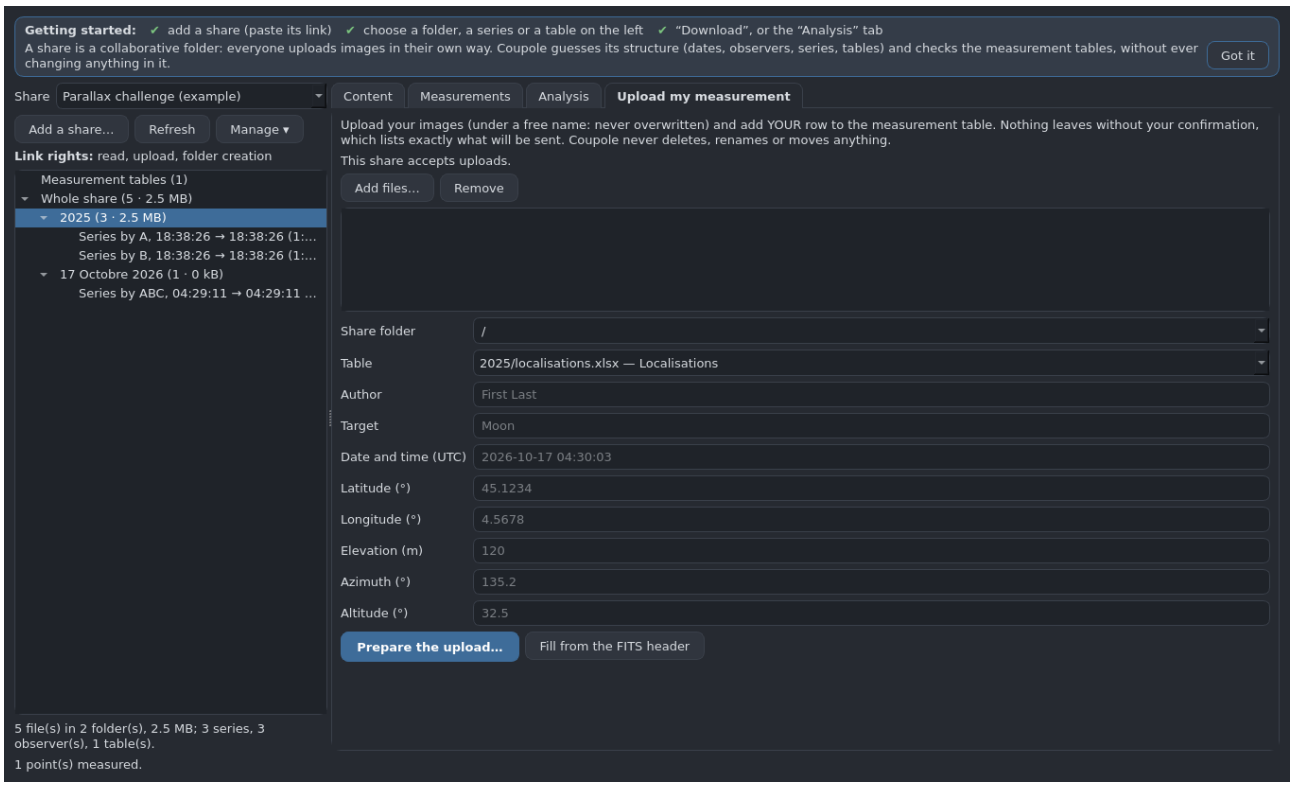
Measurement tables. Workbooks (xlsx, csv) are read *without being modified*. Normalised formats ("2,45°", "48°47′05″ N", "+44:13:51:00", Excel day numbers, US dates) and row-by-row checks, with the proof when there is one: latitude and longitude swapped (certain if $|\text{latitude}| > 90°$; proven by the file header, by the Moon's position — the noted altitude and azimuth only fit the swapped coordinates — or by comparison with reliable sites); local time instead of UT (the header or the Moon agree with the time minus 1 or 2 h); ambiguous date settled by the file name; missing fields, missing file. Proposed corrections (marked "*") are used in the calculations if you wish, never in the original table; "Export to CSV" gives the normalised table.



"Parallax (Moon, ISS)" tool. Analysis tools plug in according to the share's content (this is the first; others are added without touching the rest). *Find pairs*: same target, two distant observers (baseline ≥ 100 km), almost at the same instant (ISS ≤ 5 s, Moon ≤ 60 s). *Measure* (right-click an image): Moon centre fitted on the limb (Kåsa circle, the terminator is left out; a saturated Moon or a disc of abnormal size is flagged and the uncertainty widened), moving object of a series of short exposures or trail of a long exposure for the ISS; the direction (α, δ) comes from the image's astrometric solution. *Compute*, then the HTML or PDF report.

- **Moon**: topocentric vector method (WGS84 positions of both sites in the same frame, closest point of the two lines of sight). Directions measured on a photo are *astrometric*: the Moon where it was ≈ 1.2 s earlier, seen from an Earth that has moved since; since the light time differs for the two observers, Coupole turns each line of sight into the geometric direction before triangulating (otherwise ≈ 20 km of error for a 600 km baseline). Comparison with JPL DE440 (a 32 MB file downloaded once) or Meeus, error budget (direction, site, time).
- **ISS**: triangulation of both tracks at the common instant, estimate of the clock offset between the observers, plane method (no common clock), comparison with the TLE (SGP4).
- Single-observer methods: diurnal parallax of the Moon (several hours of measurements), angular speed or circular orbit of the ISS.

Verification. The calculations are compared with independent references (`outils/verif_parallaxe/`): skyfield and JPL DE440 (observer position within 2 m, Moon directions within $0.05''$ with DE440, Moon distance recovered within 0.5 km by triangulation, ISS altitude within 20 m, TLE prediction within 10 m) and SageMath at 200 bits (WGS84 ellipsoid both ways, closest point of two lines, refraction, Kåsa circle, zenith orbit).



Uploading your measurement. Only on your action, after a confirmation listing exactly what will be sent. Each file gets a free name ("_2" if the name exists) and is sent with the condition "the file must not exist"; your row is added in the first free row of the table (a filled cell is never replaced, styles and other sheets copied byte for byte) with the condition "the table has not changed since it was read"; a local copy of the table is kept before the addition; after three conflicts, the row goes into a small "`ligne_<name>_<date>.csv`" file next to it. Coupole's WebDAV client has *no* delete, rename or move method: even if the link allows it (Coupole tells you), it never touches other people's files.

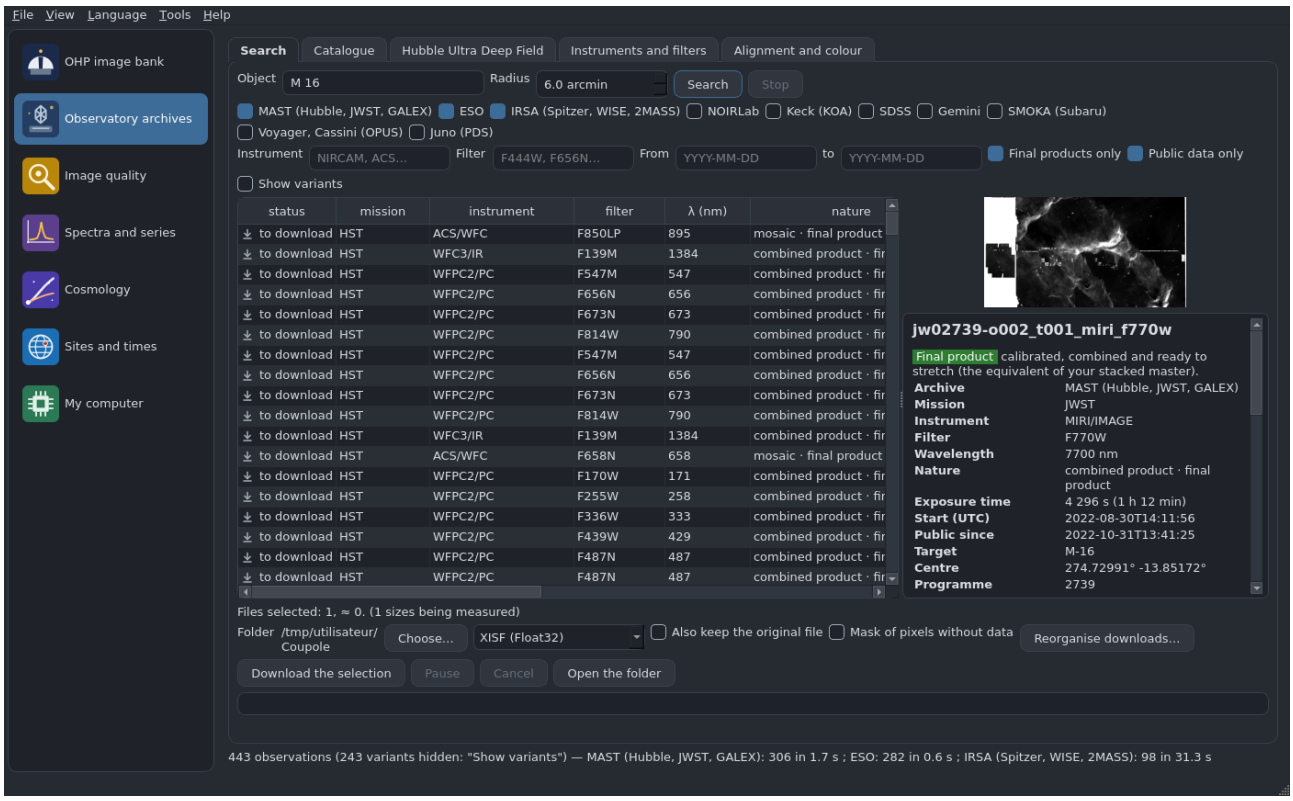
Astrometry of downloaded images. The header's solution is kept; otherwise, at your choice: ASTAP or astrometry.net installed on your computer, then, as a last resort, the online nova.astrometry.net service (section 7).

Command

```
coupole partages add "Parallax challenge" "https://.../s/..."
coupole partages explore "Parallax challenge"
coupole partages table "Parallax challenge" -json
coupole partages pairs "Parallax challenge" -headers
coupole partages measure image.fits -target moon
```

CHAPTER 6

Observatory archives



Major observatories publish their images once the proprietary period of the team that requested them is over (often twelve months). The module searches these public images, current and past, downloads them politely, extracts the science image and helps align and colour them, as the OHP bank does for the training nights. Detailed method and sources: `docs/archives_methode.md` (French, English summary).

Archive	Stage	What is offered
MAST (STScI)	1	Hubble (`_drz`, `_drc`), JWST (`_i2d`), GALEX (`-int`); official TAP service, `dbo.obspointing` view
ESO	1	reduced « Phase 3 » products (VISTA, VST, HAWK-I, FORS, MUSE…); TAP ObsCore
IRSA (NASA/IPAC)	1	Spitzer (SEIP mosaics), WISE (AllWISE Atlas), 2MASS (Atlas); SIA 2
NOIRLab	2	stacks and resampled images (DECam, Mosaic…)
Keck (KOA)	2	raw NIRC2, OSIRIS, MOSFIRE frames (no final imaging product)
SDSS	2	calibrated u, g, r, i, z images
Gemini	2	**account required** even for public data: reported, nothing downloaded
SMOKA (Subaru)	2	web form only: reported, not supported
OPUS (PDS)	3	Voyager ISS (`_GEOMED`, distortion corrected), Cassini ISS (`_CALIB`)
PDS Imaging Atlas	3	JunoCam (RDR, *framelet* strips)

6.1 What you download, compared with your telescope

With your telescope you take single exposures (lights), calibrate them (bias, dark, flat), then align and stack them into a *master*. Archives deliver the same steps, already done:

Nature (column)	Amateur equivalent
single exposure · calibrated	one of your calibrated lights (Cassini, Voyager, JunoCam, SDSS, Keck); blue *Calibrated exposure* badge
combined product · final product	your stacked master: exposures aligned, cosmic rays rejected, distortion corrected (Hubble `drz`/`drc`, JWST `i2d`, ESO Phase 3); green *Final product* badge
mosaic	several visits or fields merged on a common grid (Hubble MVM, JWST `c...`, Spitzer SEIP, WISE)
detection image	sum of all filters of a visit, made for catalogues: not a filter, left out by default

The *total exposure* column gives the sum of the exposures and their number when known ("8 × 650 s = 5,200 s (1 h 27 min)"): compare with your nights. The calibration level in the record is spelt out (2 = calibrated frame by frame, 3 = combined final product, 4 = product of a science team).

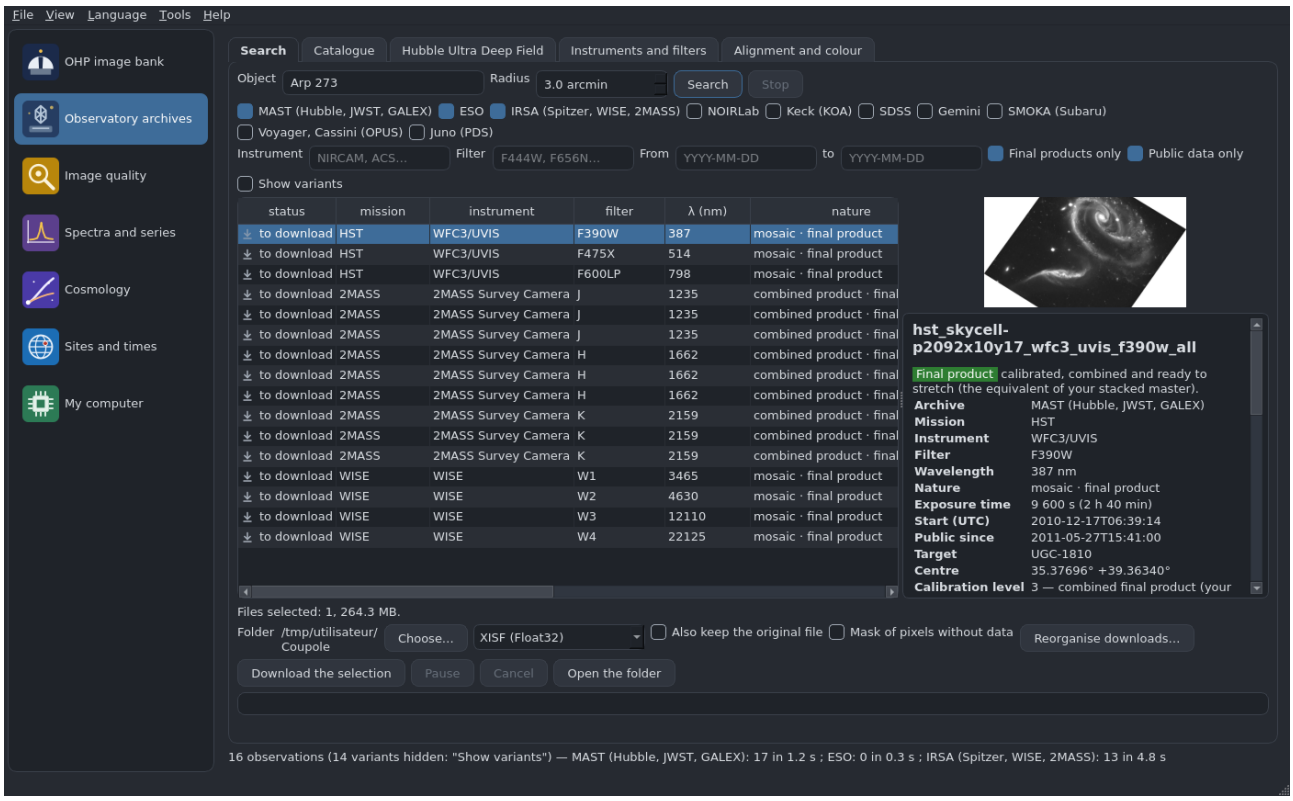
Units. Hubble gives electrons per second, JWST MJy/sr: not ADU. Values are tiny (0.01 to 10), so the image looks black until stretched (PixInsight STF, Siril autostretch). Nothing is lost: Coupole keeps the physical values and the photometric constants.

Hubble: drz, drc, SVM, MVM and "skycell". For the same exposures, MAST publishes up to four products (checked on Arp 273, programme 12326: 17 files for 3 filters):

- `ipppssoot_drz`: standard pipeline, an association of exposures of one visit and one filter; `_drc`: the same from exposures corrected for charge transfer efficiency (CTE, ACS/WFC and WFC3/UVIS) — *HST Data Handbook*, "Types of ACS files";
- `hst_<prog>_<visit>_..._drc`: **single-visit mosaic** (SVM); all filters of the visit aligned to each other and to Gaia, on the same grid; `..._total_...` is the detection image;
- `hst_skycell-pPPPPxXXyYY_..._all_drc`: **multi-visit mosaic** (MVM): every public exposure in the filter within a *sky cell*, a fixed grid 0.2° wide at 0.04″ per pixel, filled only where Hubble observed. The cell is trimmed to the smallest rectangle holding *all* exposures of the cell, every filter included (DrizzlePac, "Multi-visit Mosaic Processing"): one filter may fill only a corner. Real example: M 16, ACS F775W, 19,806 × 15,741 pixels (2.49 GB at MAST) for a data strip of 762 × 5,217 pixels.

One logical product per filter: Coupole picks the MVM when it exists (all exposures, longest total time, CTE corrected, grid shared by all filters of the cell: direct colour — MAST, "Multi-Visit Mosaics from Hubble Now Available", 2022); otherwise, per visit, the SVM; otherwise the standard pipeline product (WFPC2 for instance). For JWST: the mosaic of several observations (`c...`), else the observation with the longest exposure. The other products of the same exposures become *variants*, hidden by default (*Show variants* box); the record says why each product is chosen or set aside. Documented MVM limits: residual shift of a few pixels between visits, moving objects doubled, no WFPC2.

6.2 Searching



The *Object* field accepts:

- a name or catalogue number, resolved by SIMBAD then Sesame (which also queries NED and VizieR); common spellings are rewritten (“m16” → M 16, “ngc6611”, “arp273”, “sh2-155”) and usual French and English names recognised (“Pillars of Creation”, “Carina Nebula”, “Stephan’s Quintet”, “HUDF”);
- a **Solar System body**, in French or English, case and accents ignored (jupiter, Saturne, Enceladus, Iapetus, Phoebe…). SIMBAD does not know the planets and a planet moves: the body is searched **by target name** in MAST (Hubble, JWST: `target_name` “JUPITER”, “JUPITER-MAPS”… with the moving-target flag), Voyager/Cassini (OPUS) and Juno (PDS), even if those archives are not ticked;
- coordinates in degrees (`83.82 -5.39`) or sexagesimal (`05:35:17 -05:23:28`).

Suggestions appear as you type (bodies, usual names). An object not found gives proposals and advice. Radius: 3 arcminutes by default. Filters: instrument (part of the name: `NIRCAM`, `ACS`), filter (`F444W`), dates. Boxes: *Final products only*, *Public data only*, *Show variants*. Archives answer in parallel; one that is unreachable, requires an account or has no interface is reported without stopping the others. A reply truncated by the limit is reported; when an archive serves two missions (OPUS: Voyager and Cassini), the limit is shared between them (in 0.2.1, “saturne” returned only Voyager).

Size, record and preview. The size of each file shows up by itself: announced by the archive, or measured in the background by a one-byte request (at most 60 at once, the others extrapolated and marked “≈”); the selection total is under the table. The record gives mission, instrument, filter (with its wavelength and amateur equivalent), nature, total exposure, dates, programme, investigator, calibration level, file, credit and terms of use. The preview is the archive’s official thumbnail, read in full, reduced without being decompressed at full size and cropped to the useful area (MAST JPEGs are full resolution, up to 19,806 × 15,741 pixels, white where there are no data); failing that, a thumbnail made from the prepared file; otherwise the reason (“the archive gives no thumbnail…”).

6.3 Downloading and preparing

Select rows (Ctrl+click, Shift+click), then the blue *Download the selection (N · size)* button, right under the table (0.7.2: greyed out with the reason on hover while nothing is selected, « Already downloaded ✓ » when everything is there). Right-click starts with *Download this observation* (or *these N observations*), then *Open*, *Open with*, *Open file location* and the archive page; double-clicking a row not downloaded yet offers to download it. Beyond 2 GB (setting `archives_seuil_go`), confirmation. Full rate by default (cap: Preferences, or setting `archives_debit_mo_s`), two files at a time, an interrupted file resumes. The bar moves in **two phases**: download (up to 70 %, from bytes received), then preparation (up to 100 %); below it: files downloaded and prepared, bytes received out of the total, rate, time left in all and for the current file. At the end the state is spelt out: "Finished: N files, X downloaded, Y kept, Z errors — Open the folder".

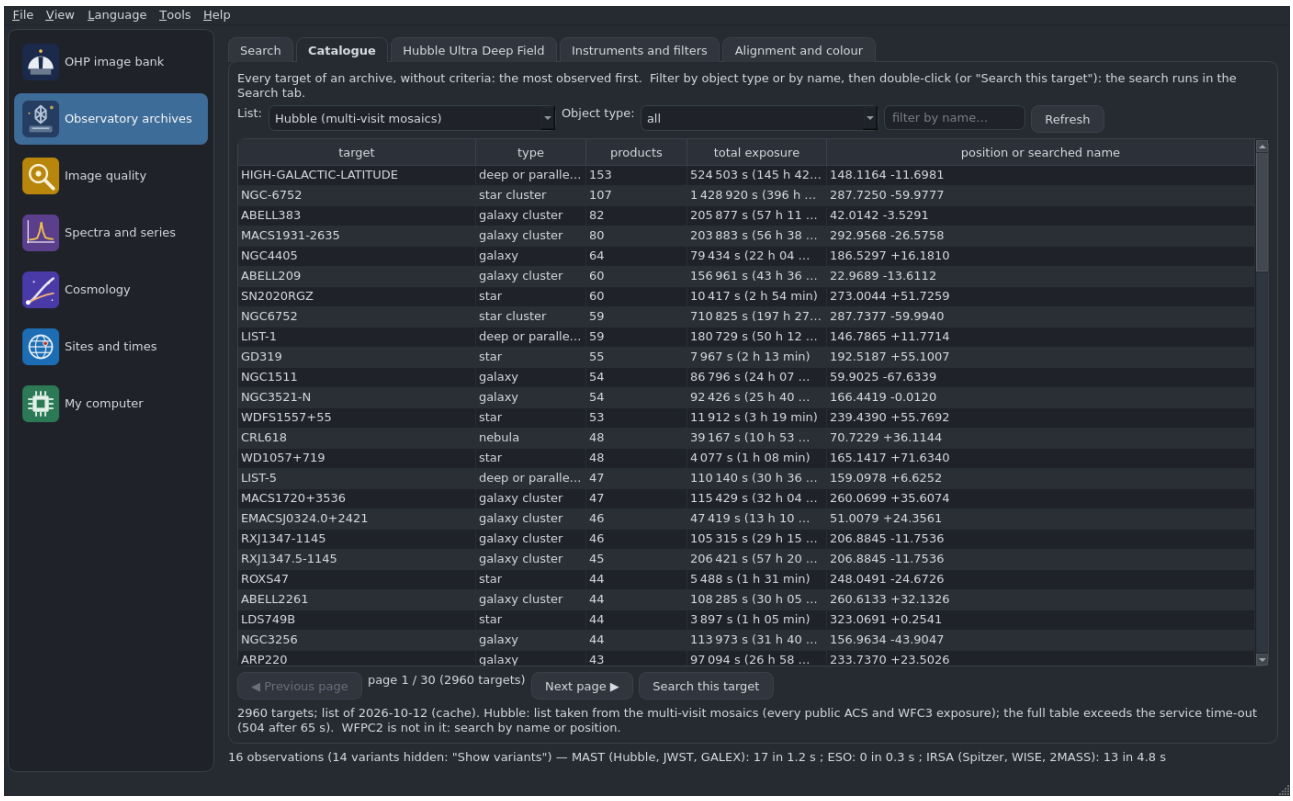
Each image is prepared then filed **by field**, as in the OHP image bank: `<folder>/Archives/<field>/<target>_` for example `Arp273_WFC3-UVIS_F390W_2010-12-17_MVM_8x1200s.xisf` or `HUDF_WFC3-IR_F160W_2009-11-03_HL` (type: MVM, SVM-v21 for visit 21, drz, drc, pose, i2d, HLSP…). The archive identifier stays in the header (`ARCHID`, `ORIGFILE`) and in the record. Preparation:

- science extension extracted (`SCI`; otherwise the first image), one file per detector for an ESO multi-detector exposure (`_d1`, `_d2`…);
- **cropping to the useful area**: smallest rectangle of valid pixels, the WCS follows (`CRPIX` shifted; `CROPX0`, `CROPY0`, `ORIGNAX1`, `ORIGNAX2` keep track); sizes before and after in the status bar, the record and the log;
- header: WCS (SIP included), `BUNIT`, photometric constants, telescope, instrument, filter, dates, exposure time, number of exposures (`NPOSES`), programme, investigator, `CREDIT` and terms of use;
- remaining pixels without data (NaN) set to 0 (`NANCOUNT`, `NANFILL`);
- XISF Float32 (physical values kept) or FITS Float32.

By default nothing but the prepared product: the original file (*Also keep the original file* box, filed in `<field>/_originaux`) and the mask of pixels without data (*Mask…* box, compressed 8-bit integers, written after cropping) are options. Without a mask, alignment treats exact zeros as "no data". The state is kept in `Archives/_etat/etat.sqlite` and the bilingual log in `Archives/_etat/JOURNAL.txt`.

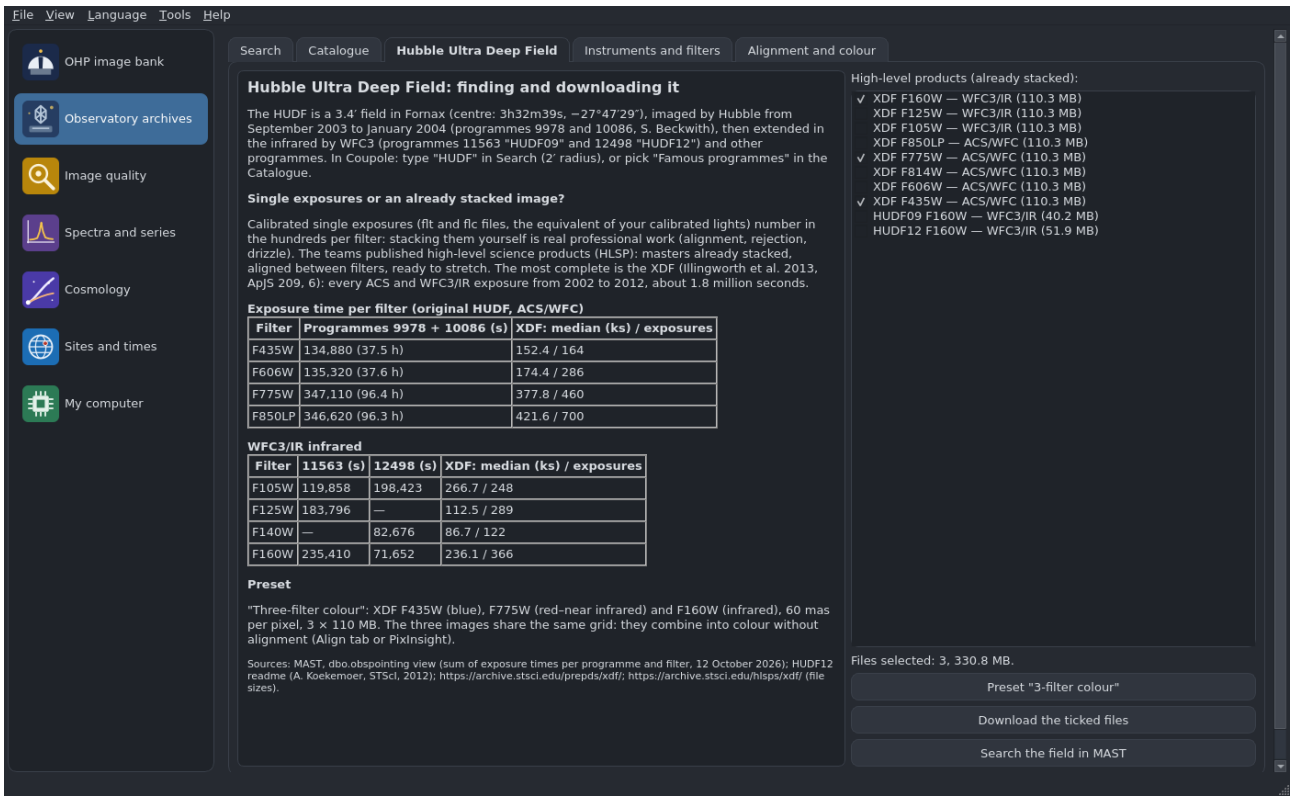
Reorganising downloads. The *Reorganise downloads…* button (and `coupole archives reorganise`) takes up what was downloaded with 0.2.0 or 0.2.1: readable names, filing by field, cropping, deletion of full-size masks, variants and detection images in `<field>/_variantes`. A confirmation shows what will change; nothing is ever overwritten (a taken name gets `_2`); every move is written in the log and the state database follows. Real test on 18 files of a database written under Windows: 18 filed, 11 cropped, 16 masks deleted, 1,641 → 910 MB.

6.4 Target catalogue



The *Catalogue* tab lists an archive's targets **without criteria**, most observed first, in pages of 100, filtered by object type (galaxy, cluster, nebula, star, Solar System, deep field) and by name; double-click: the search starts. Aggregate queries (`GROUP BY target_name`) measured: Hubble 12 to 17 s (restricted to multi-visit mosaics: the full table exceeds the service time-out), JWST 34 s, ESO 2 s, Voyager, Cassini (OPUS) and Juno (PDS) in under a second; lists are kept 30 days (*Refresh* to read them again). IRSA, NOIRLab, Keck and SDSS have no useful list (all-sky surveys, numbered fields, raw frames): the tab explains it. *Famous programmes*: Hubble Deep Field (6337), Hubble Ultra Deep Field (9978, 10086), HUDF09 (11563), HUDF12 (12498), Pillars of Creation 1995 (5773), 2014 (13926) and JWST 2022 (2739), Carina (10475; JWST 2731), Stephan's Quintet (2732), Southern Ring (2733), Arp 273 (12326): the search then keeps only that programme's products.

6.5 Hubble Ultra Deep Field



The tab leads to the HUDF (centre 3h32m39s, −27°47′29″): calibrated single exposures (`flt`, `flc`) number in the hundreds per filter; the teams published **high-level science products** (HLSP) already stacked and aligned between filters: XDF (Illingworth et al. 2013: every ACS and WFC3/IR exposure from 2002 to 2012, about 1.8 million seconds), HUDF09, HUDF12. Exposure times checked in MAST (sum of the pipeline products): programmes 9978 + 10086, ACS/WFC F435W 134,880 s, F606W 135,320 s, F775W 347,110 s, F850LP 346,620 s; HUDF12 (team readme) F105W 198,423 s, F140W 82,676 s, F160W 71,652 s. “3-filter colour” preset: XDF F435W, F775W and F160W at 60 mas, 3 × 110 MB, same grid (`coupole archives hudf -download`).

6.6 Instruments and filters

The *Instruments and filters* tab gives, for each instrument (Hubble ACS/WFC, WFC3 UVIS and IR, WFPC2; JWST NIRCam, MIRI, NIRISS; ESO HAWK-I, FORS2, MUSE, VIRCAM, OmegaCAM; Spitzer IRAC and MIPS; WISE; 2MASS; SDSS; Keck NIRC2; Voyager and Cassini ISS; JunoCam) its field, scale and range, and for each filter its wavelength, width, use and **amateur equivalent**. Hovering over a filter name in the results table gives the summary. The name encodes the filter: F = filter, number = wavelength (nm for Hubble, hundredths of a µm for JWST), letter = width (W wide, M medium, N narrow, LP long pass, X extra wide).

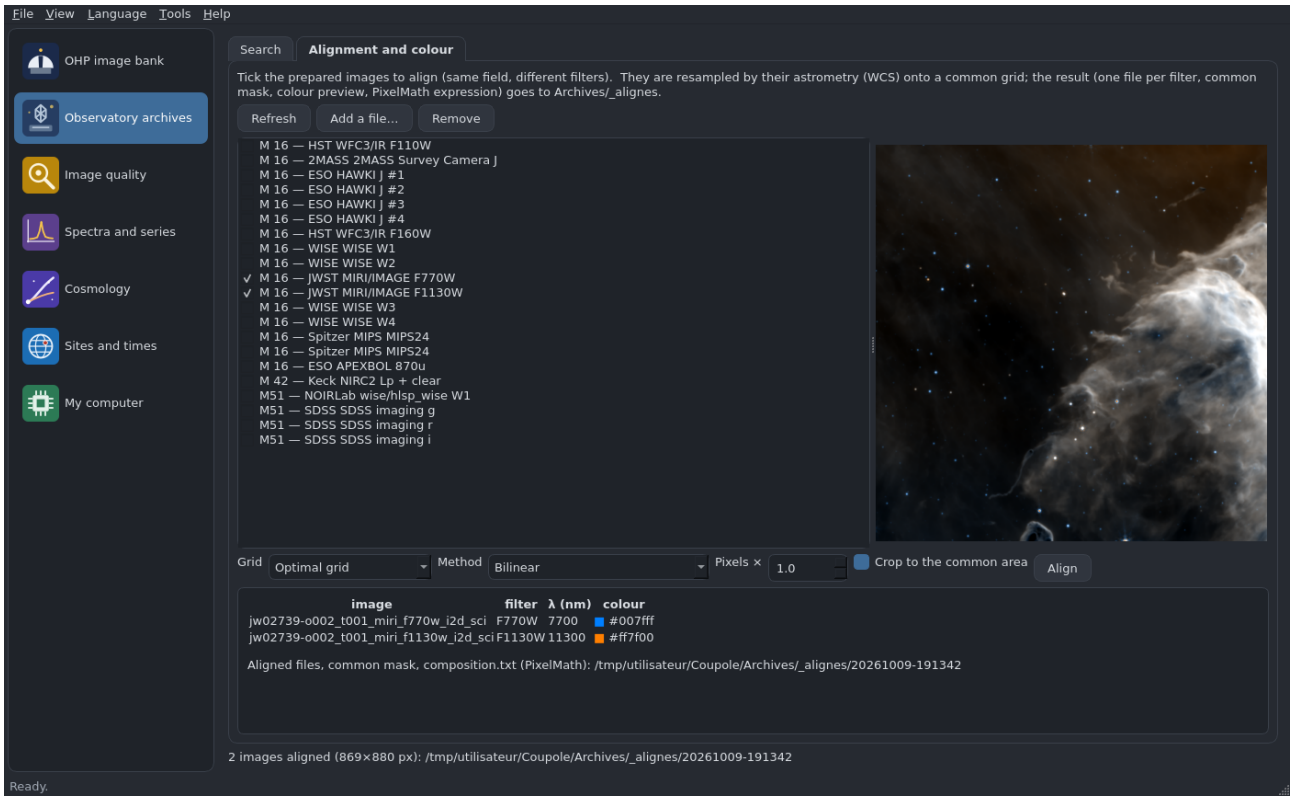
Filter	Instrument	λ (nm)	Width	Amateur equivalent
F435W	ACS/WFC	430	104	B
F555W, F606W	ACS/WFC	535, 591	119, 234	V; F606W, very wide, comes close to a luminance
F656N, F657N	WFC3/UVIS	656	2, 12	Hα (your 3–7 nm filters)
F502N	WFC3/UVIS	501	7	OIII
F673N	WFC3/UVIS	677	12	SII
F775W, F814W	ACS/WFC	776, 833	153, 251	near infrared (IR-pass filter)
F160W	WFC3/IR	1,537	268	none: beyond 1 µm a CMOS sensor sees nothing
F444W, F770W	NIRCam, MIRI	4,402, 7,639	1,024, 1,950	none: infrared, in false colour

Sources: WFC3 (tables 6.2 and 7.2, pivot wavelength) and ACS (table 5.1, central wavelength) instrument handbooks, JDox (NIRCam, MIRI, NIRISS), ESO instrument pages, Spitzer handbooks,

WISE and 2MASS explanatory supplements, SDSS, PDS Ring-Moon Systems Node, Hansen et al. 2017 (JunoCam); approximate values marked "≈" in the tab.

Planetary formats. Voyager and Cassini products are VICAR files with a PDS3 label; Juno products are PDS3 files; PDS4 (XML label) is read too. Coupole reads these formats itself (8 to 32-bit integers, IEEE reals, byte order, line prefixes, scaling), flips the rows so the image shows as in the archive and writes FITS or XISF. Peculiarity: the PDS3 label of the calibrated Cassini images places the image one line too early (a VICAR binary line precedes it); the data are read from the VICAR header. Not handled (clear message): compressed files (Voyager EDR), VAX reals, multi-band BIL/BIP interleaving, cubes; a JunoCam image is converted as is (strips of 128 lines per filter), without geometric reconstruction.

6.7 Aligning and colouring



Archive images have an exact astrometry: no star matching (stars change from one instrument and wavelength to another); each image is **resampled** onto a common grid: the grid of one image (often the sharpest filter) or an *optimal grid* covering them all, at the finest pixel scale, optionally enlarged (*Pixels* ×). Methods: bilinear (default, fast), adaptive (anti-aliased, better when the pixel scales differ a lot), exact (exact pixel overlap, slow; below 0.05″ per pixel, adaptive replaces it), nearest neighbour. Adaptive and exact need the optional `reproject` library (`pip install ".[alignement]"`); without it, bilinear is done by scipy. A **per-pixel** quantity (Hubble electrons/s, SDSS nanomaggies) is multiplied by the pixel-area ratio; a surface brightness (JWST MJy/sr) is kept as is. Optional crop to the largest rectangle covered by every image (a dead column in the middle of a mosaic does not shrink it). The accuracy is that of the archives' astrometry: a residual offset of about a pixel may remain between missions; star alignment in PixInsight or Siril on the resampled files then only has a translation left to correct.

Output in `Archives/_alignes/<date>/`: one file per filter (`_aligne`), the common mask, a colour preview (`apercu_couleur.png`, asinh stretch) and `composition.txt`. Filters are sorted by wavelength (*chromatic ordering* of the Hubble and JWST press images): three filters give blue, green and red; more give hues spread from blue to red. `composition.txt` gives the colour table and the PixelMath

expression (image identifiers as PixInsight creates them when opening); in Siril, the RGB composition uses the same, already aligned files.

6.8 Command line

```
coupole --lang en archives list
coupole --lang en archives search "M 16" --radius 6 --mission JWST --instrument MIRI
coupole --lang en archives search arp273                # one product per filter
coupole --lang en archives search arp273 --variants     # every variant
coupole --lang en archives search saturn                # MAST, Voyager, Cassini by target nam
coupole --lang en archives download "M 16" --mission JWST --instrument MIRI --dest ~/Sky
coupole --lang en archives download M51 --archive sdss --filter g --filter r --filter i --dest
coupole --lang en archives catalog                      # famous programmes
coupole --lang en archives catalog jwst --type nebuleuse --page 2
coupole --lang en archives search HUDF --radius 2 --program 9978 --program 10086
coupole --lang en archives hudf --download --dest ~/Sky
coupole --lang en archives instruments F656N
coupole --lang en archives reorganise --dest ~/Sky --dry-run
coupole --lang en archives prepare file_i2d.fits --format fits
coupole --lang en archives align A.xisf B.xisf C.xisf --dest ~/Sky/Pillars --optimal
coupole --lang en archives status --dest ~/Sky
```

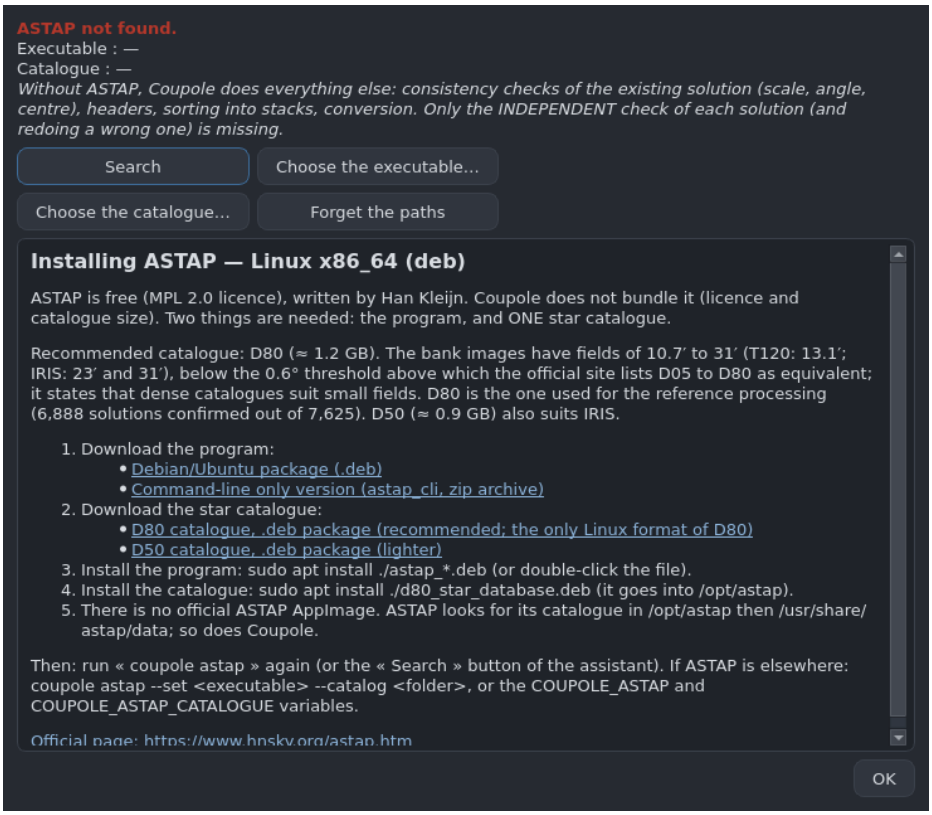
`-id` keeps precise observations (identifier, with or without the archive prefix, or beginning of the file name); `-all-levels` also shows calibrated and raw frames; `-keep-original` and `-mask` keep the original file and the mask; `-json` and `-csv` for scripts. Beyond the threshold, `download` needs `-yes`.

Credits. Any image published from these data must cite the archive and the mission: « NASA/ESA/CSA, STScI » (JWST), « NASA/ESA, STScI » (Hubble), « ESO » and the programme ID, « NASA/JPL-Caltech » (Spitzer), « NASA/JPL-Caltech/UCLA » (WISE), « 2MASS/UMass/IPAC-Caltech/NASA/NSF », NOIRLab, KOA, SDSS, and for the probes « NASA/JPL-Caltech » (Voyager), « NASA/JPL-Caltech/Space Science Institute » (Cassini), « NASA/JPL-Caltech/SwRI/MSSS » (Juno). Coupole writes it in the record and in the header of every file (`CREDIT`).

CHAPTER 7

ASTAP (optional)

Without ASTAP, Coupole does everything else. ASTAP (Han Kleijn, MPL 2.0 licence, https://www.hnsky.org/astap.htm) adds an independent check of each solution. It needs the program and one star catalogue.



Catalogue. The official site states that for a field of 0.6° or more, D05, D20, D50 and D80 all suit, and that dense catalogues serve small fields. The bank's fields range from 10.7′ (0.18°) to 31.4′ (0.52°): **D80** is recommended (about 1.2 GB; catalogue of the reference processing, 6,888 solutions confirmed out of 7,625); D50 (about 0.9 GB) also suits IRIS.

System	Program (links of the official page)
Windows x64	`astap_setup.exe` then `d80_star_database.exe` (same folder, `C:\Program Files\astap`); optional: `astap_command-line_version_win64.zip`
Windows ARM64	`astap_command-line_version_win11_aarch64.zip` in `C:\astap`; catalogue: `d80_star_database.exe` (x86 installer, emulated) into that folder, or `d50_star_database.zip` (no zipped D80)
macOS Apple Silicon	`astap_M1.pkg`, `d80_star_database.pkg`; once: `codesign -force -s - /Applications/ASTAP.app/Contents/MacOS/astap`
macOS Intel	`astap.pkg`, `d80_star_database.pkg`
Linux Debian/Ubuntu	`astap_amd64.deb` (or `astap_aarch64.deb`), `d80_star_database.deb` (into `/opt/astap`)
Linux Fedora/openSUSE	`astap_amd64.rpm`; D80 extracted from the `.deb` (below) or zipped D50 into `/opt/astap`
Linux Arch/Manjaro	`astap_amd64_gtk3.pkg.tar.zst` (or GTK2); D80: AUR package `d80-star-db-astap` or extracted from the `.deb`; D50: `d50_star_database.pkg.tar.zst`
Other Linux	`astap_amd64.tar.gz` or `astap_command-line_version_Linux_amd64.zip`; no official AppImage

No zipped D80. The « D80 (zipped) » link of the official page is dead (404, checked on 9/10/2026): D80 only exists as `.exe`, `.pkg` and `.deb`. Outside Debian, extract it from the `.deb` without touching the permissions of `/` (command checked on the real package: 1,213,392,756 bytes, 1,476 tiles):

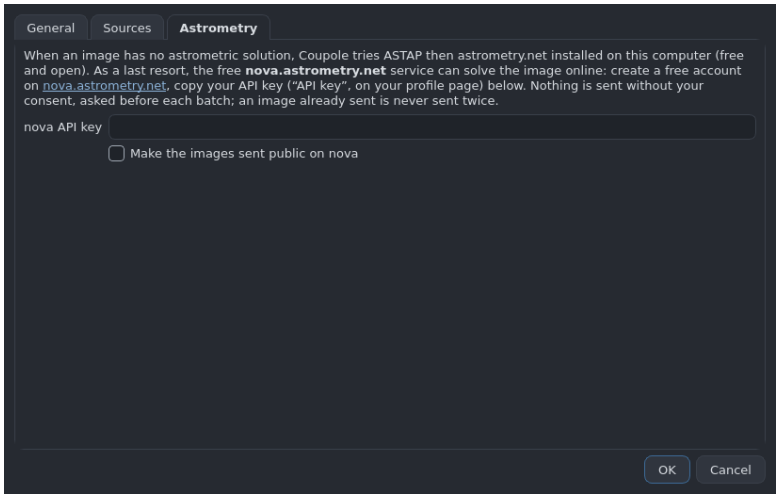
Command

```
bsdtar -xf d80_star_database.deb data.tar.xz
sudo mkdir -p /opt/astap
sudo tar -xJf data.tar.xz -C /opt/astap -strip-components=3 -no-same-owner
    -wildcards './opt/astap/d80_*'
```

(never `sudo tar -x ... -C /`: the archive contains `./` and `./opt/`). **Arch/Manjaro**: the official package puts `astap` and `astap_cli` in `/opt/astap` only (nothing on the PATH: Coupole finds them there); the GTK2 graphical `astap` needs `libgtk-x11-2.0`, missing from the official repositories (AUR only) and not declared; `astap_cli` is self-contained (statically linked) and is all Coupole needs; for ASTAP's interface, the GTK3 package. A continuous-integration test checks every link, for every system.

Detection. Order: chosen path (Preferences, `coupole astap -set`), `COUPOLE_ASTAP` variable, `astap_cli` everywhere (PATH then default locations), only then `astap`, default locations (Windows `Program Files\astap`, `%LOCALAPPDATA%\astap`; macOS `/Applications/ASTAP.app`, `/usr/local/opt/astap`; Linux `/opt/astap`, `/usr/bin`, `/usr/local/bin`, `~/astap`). Catalogue: chosen folder, `COUPOLE_ASTAP_CATALOGUE`, program folder, `/opt/astap`, `/usr/share/astap/data`, `/usr/local/opt/astap`. A catalogue is declared complete with its 1,476 tiles; the version is read from `astap_cli -h`.

Online astrometry (nova.astrometry.net)



When an image has no astrometric solution and no tool is installed, Coupole can hand it to the *free and open* nova.astrometry.net service (Lang, Hogg, Mierle, Blanton & Roweis 2010). Order followed everywhere (DU shared folders, OHP bank for images without a solution when ASTAP is missing): header solution, installed ASTAP, installed astrometry.net (`solve-field`), then nova. You need *your own* API key (free account on `https://nova.astrometry.net`, profile page) in *Preferences* > *Astrometry*; it is kept separately (`nova.json`, readable by you only) and only sent to nova. **Nothing is sent without your consent**, asked before each batch. What Coupole sends, following the official API documentation: a lightened image (16-bit greyscale, reduced above 4 million pixels; the solution is then brought back to the original image, SIP coefficients included), the known hints (scale in ″/px with ±20 % bounds, approximate centre and radius), images *not public* by default, without commercial-use or modification licence. One submission at a time, spaced polling (5 s then up to 30 s), maximum delay, clean abort; an image already sent (same fingerprint) is *never* sent again — its result, or the failure, is cached, and an interrupted submission resumes polling instead of re-sending the image.

CHAPTER 8

Output formats

Format	Content	Readers
XISF (default)	Float32 in ADU, `bounds="-1000:65535"`; UInt16 for integer IRIS FITS; zstd+sh level 9 compression; `Observation:*`, `Instrument:*` properties	PixInsight
Compatible XISF	UInt16 + 1,000 ADU pedestal (`PEDESTAL`), zlib+sh (what N.I.N.A. writes)	PixInsight, Siril ≥ 1.4, N.I.N.A.
FITS `.fits.fz`	lossless tile compression: GZIP_2 without quantization (floats), RICE_1 (integers)	Siril, astropy, DS9
FITS	uncompressed float32 (or UInt16)	all

XISF: `bounds` sets the range PixInsight maps to [0,1] when opening: nothing to set, same scale for every exposure; the −1,000 lower bound keeps the negative values of short calibrated exposures. Size: 36 % of the original FITS for the T120 (float64), 62 % for IRIS. **Compressed FITS**: measured with astropy 8.0, RICE_1 and HCOMPRESS_1 quantize floats (one-ADU error, even with `quantize_level=0`); only GZIP_1 and GZIP_2 without quantization restore floats bit for bit. **float32 FITS**: in PixInsight, set the FITS input range to 0–65,535 with *Truncate*. For every format except the compatible XISF, the difference to the original FITS stays below half a float32 ULP (checked pixel by pixel).

Compatible XISF (N.I.N.A., Siril). UInt16 = ADU + 1,000 rounded, clipped to [0, 65,535]; the pedestal keeps the negative backgrounds of short calibrated exposures and is written in the header (`PEDESTAL`, PixInsight's convention). Measured on 66 exposures of the bank (`outils/audit2/mesure_perte_uint16`): rounding ≤ 0.5 ADU, background mean changed by at most 0.014 ADU (background noise: 10 to 109 ADU); pixels clipped high: at most 0.012 % (stars above 64,535 ADU); no pixel lost low any more. Without the pedestal, a 10-s T120 exposure with an over-subtracted background (−220 ADU) lost 99.75 % of its pixels to 0 and 6.1 % of the pixels of 1–2-s exposures fell to 0: hence the pedestal. Integer IRIS images stay lossless.

Who reads what (checked)

File	PixInsight	Siril	N.I.N.A.	ASTAP	Note
XISF Float32 (default)	yes	no	no	no	Siril ≥ 1.4 reads it but outside its [0, 1] range; N.I.N.A.: white image
Compatible XISF (UInt16, zlib)	yes	≥ 1.4.0	yes	no	ASTAP: uncompressed XISF only
XISF UInt16 zstd (integer IRIS)	yes	≥ 1.4.0	no (3.2)	no	zstd read by N.I.N.A. since January 2026 (development)
FITS Float32	yes	yes	no	yes	N.I.N.A. assumes [0, 1]; Aladin, astropy: yes
FITS UInt16	yes	yes	yes	yes	
`.fits.fz` Float32 (GZIP)	—	yes	—	no	ASTAP: Rice only
`.fits.fz` UInt16 (Rice)	—	yes	—	≥ 2026.07.16	—: not checked

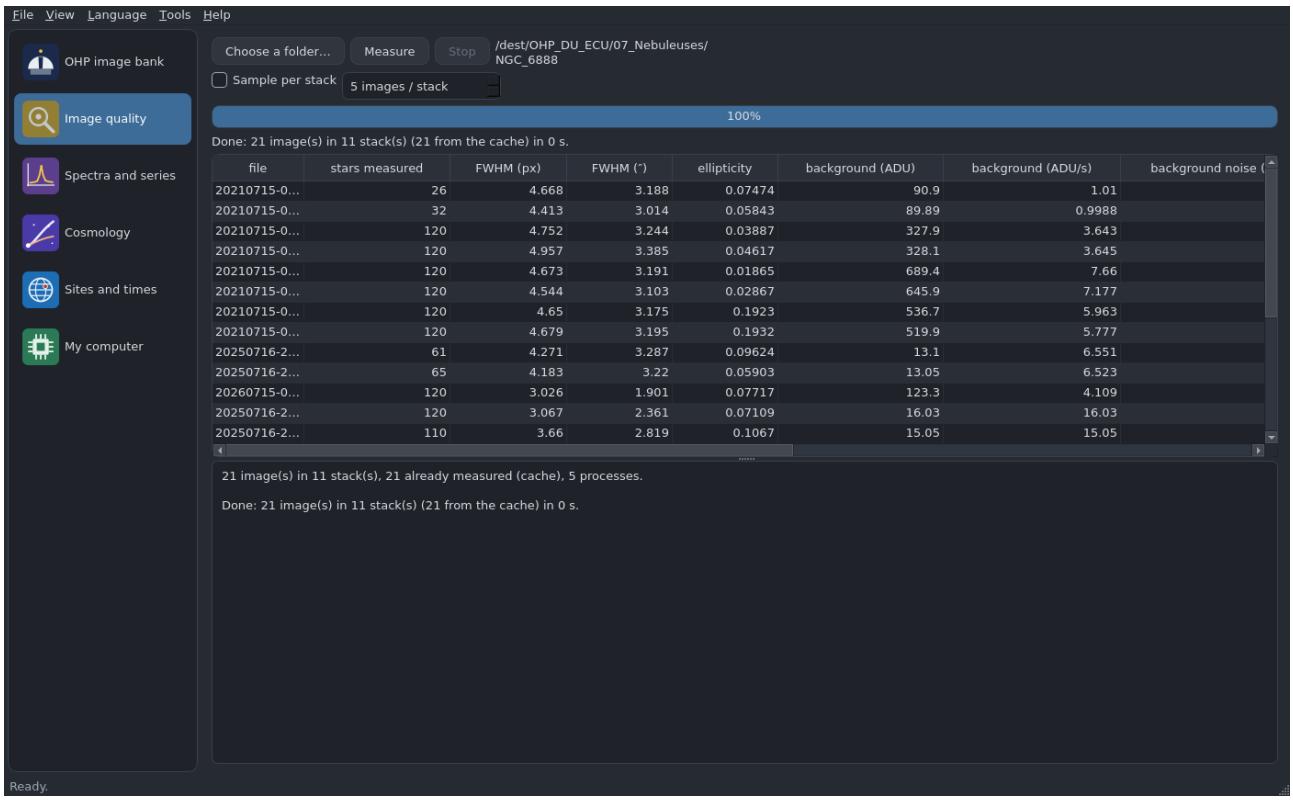
Sources: **PixInsight**, PCL code (`XISF.cpp`: zlib, lz4, lz4hc, zstd codecs, with or without `+sh`; `XISFReader.cpp`: `bounds` mandatory for a float, clipped then mapped to [0, 1]); XISF 1.0 specification revision 1 (September 2026, https://pixinsight.com/doc/docs/XISF-1.0-spec/XISF-1.0-spec.html), XSD schema identical to the one Coupole uses. **Siril**, real test of the official AppImages (siril-cli, pixels read back and compared): 1.2.0 and 1.2.6 do not read XISF (« file type not supported »), 1.4.0 and 1.4.4 read all our codecs bit for bit; but XISF floats stay in ADU there (statistics multiplied by 65,535), whereas float FITS are mapped to [0, 1]; Debian 13 and Ubuntu 24.04 packages (Siril 1.2.x): no XISF. **N.I.N.A.**, code (`Version-3.2` tag, 27/11/2025: `XISF.cs` l. 304–328: lz4, lz4hc, zlib, with or

without `+sh`; `Float32Converter.cs` l. 23: (ushort)($v \times 65535$), `bounds` never read) and real decoding of one of our files. **ASTAP**, official page (« uncompressed » XISF, Rice `.fits.fz` since 2026.07.16, not GZIP). Test files for testers: `tests/donnees/` (synthetic image as default XISF, compatible XISF and FITS, under 300 KB each).

Deviation from revision 1. Coupole writes the centre of the astrometric solution in `Observation:Center:RA/De` (choice of the reference processing, older than the revision); revision 1 keeps these properties for the point the telescope was aimed at and standardizes the solution in the `AstrometricSolution:` namespace (not written: the solution stays in the WCS keywords, which PixInsight reads). Kept as is in 0.1.9 so as not to change files compared with the reference processing.

CHAPTER 9

Image quality (optional)



The module needs the SEP library (Source Extractor in Python, LGPL, about 1 MB); it never runs automatically (*Measure* button, *Check image quality* box of the processing, or `coupole qualite FOLDER -write`). Report per stack: `QUALITE.csv` and `QUALITE.txt`.

Sky frames only. By default only sky frames (lights) are measured. Left out without being opened: Flat(s), Dark(s), Bias, Offset(s), DarkFlat(s), Calibration, Master(s) folders (« FLAT L », « Dark-Bias » included), cosmetized, calibrated, registered, debayered, `_platesolve...` and their French names (Plats, Noirs, Biais); `FLAT_`, `DARK_`, `BIAS_`, `DARKFLAT_` files (N.I.N.A., ASIAIR), masters (`masterDark`, `masterFlat`, `masterLight`, `superbias`) and WBPP outputs (`_c`, `_cc`, `_r`...). As a last resort, a frame whose header (`IMAGETYP`, `FRAME` or the XISF property `Observation:Image:Type`) says flat, dark, bias, offset or master is left out when measured; Light, Light Frame, Science or Object are measured. The summary says « 70 image(s) found, 66 sky frame(s) measured, 0 with an error, 4 calibration file(s) excluded »; *Excluded files...* lists them with the reason. *Also include calibration frames* box (command line: `-with-calibration`): everything is measured.

Other programs' files, errors. XISF files from N.I.N.A. (keywords without a comment), the ASIAIR (no `Metadata`), PixInsight and Siril are read by a tolerant reader: only what really prevents reading the pixels is refused; our own files are still checked strictly. A file that cannot be measured shows in the table, *status* column, with its reason (« error: XISF format not read (...) »); the summary counts the errors and `coupole.log` keeps the stack trace; the command line exits with code 1 when no image could be measured. N.I.N.A. frame from a CMOS camera (9,576 × 6,388, 61 Mpx): 4.2 s, 1,170 MB peak (float32 computation), FWHM 3.5″ at 1.695″/px; the number of processes is reduced from the image size (half of the free memory) and the number of fitted stars grows with the sensor (120 to 400).

Speed, resume, sample. Measurements run in separate processes (count given by the machine plan: cores, memory, economy mode and the Preferences limit; at most 3 when the folder is on a network share, detected and announced, since reading then limits the speed), one image being read while the others are computed; the interface stays free, *Stop* is immediate. Each measured image is kept at once in the cache `_traitement/qualite.sqlite` by (path, size, date) and shown; its stack's `QUALITE.csv` (atomic write) is rewritten at most every 5 s and at the end of the stack or on stop: starting again redoes nothing, closing then reopening resumes. Above 200 images, Coupole offers a **sample** of 5 images per stack (spread in time: first, last, middle…, adjustable in the tab and with `-sample N`; measured first, while you choose), enough to judge a stack, or *Measure everything* with the duration estimated from the images already measured; progress shows the time left and the rate (images/min). On a 6-core processor: 1.21 → 0.33 s per T120 image (vectorised Moffat model, analytic derivatives, 120 stars per image), i.e. the whole bank in about ten minutes on 5 processes.

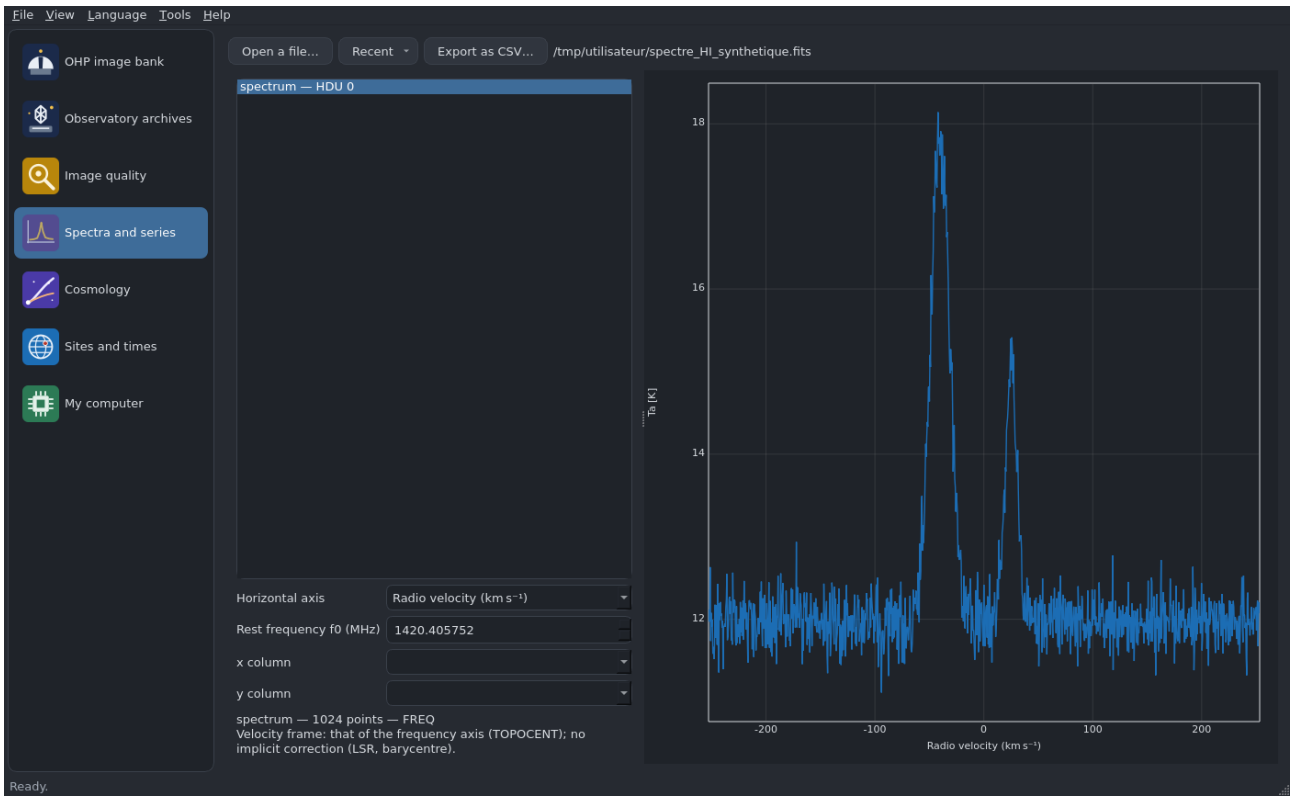
Listing as a stream. Measuring starts with the first stack found, while the listing goes on (« Listing the folder: 1,234 files found in 56 stacks… », *Stop* immediate); the listing opens no file and reads size and date from each folder's listing (at most one `stat` per image). In a Coupole output, only the stack folders are read, from `INDEX_LOTS.csv` or the state database; processing leftovers (`converties`, `telechargements`) are never taken for stacks. On a share, 32 threads read the folders. The cache is **never** written on a network share: Linux's SMB client forwards SQLite's locks to the server, which refuses them (« database is locked » after 30 s, three times before the first measurement in 0.1.8: the 90 s of « Listing the folder… » seen on a real share); it goes to the user's cache folder (an old cache on the share is read). Measured on a real Samba share, 831 XISF of 09_Galaxies: 0.1.8 92.7 s before the first measurement (10 ms latency); 0.1.9 full listing in 0.24–0.57 s (10 ms) and 0.57–1.05 s (20 ms), first result in 0.2–0.6 s.

Measure	Method	Synthetic validation
FWHM	elliptical Moffat with free β fitted to each isolated, unsaturated star (model integrated over the pixel); median	≤ 0.4 % (Gauss and Moffat, 2.5–8 px)
ellipticity	$1 - b/a$ of the same fit	$\pm$ 0.02
3 × 3 map	medians per ninth of the field	order recovered
background, noise	SEP, 64 px mesh	0.5 %; 5 %
gradient, residual	plane fitted to the background map; residual amplitude	gradient and vignetting recovered
saturated	peak above SATURATE/DATAMAX (else 60,000 ADU)	5 out of 5
trails	objects with major axis $\geq$ 12 px and $a/b \geq 6$	1 out of 1, 0 false
SNR	only with the gain (EGAIN/GAIN)	given or absent
sampling	FWHM $<$ 2 px, 2–5 px, $>$ 5 px	—

On 14 IRIS images (12 nights, FWHM from 2.4 to 11.3 px), the measured FWHM and the one written by PinPoint in the header correlate (r = 0.974), median ratio 1.046 (0.85 to 1.16): the two estimators differ (PinPoint's method is not described). Not offered for lack of validation: mag arcsec^{-2} (no zero point), dust-donut detection. A FWHM growing towards an edge may come from the sensor, field curvature, collimation, but also from seeing, tracking or focus: the map alone cannot tell.

CHAPTER 10

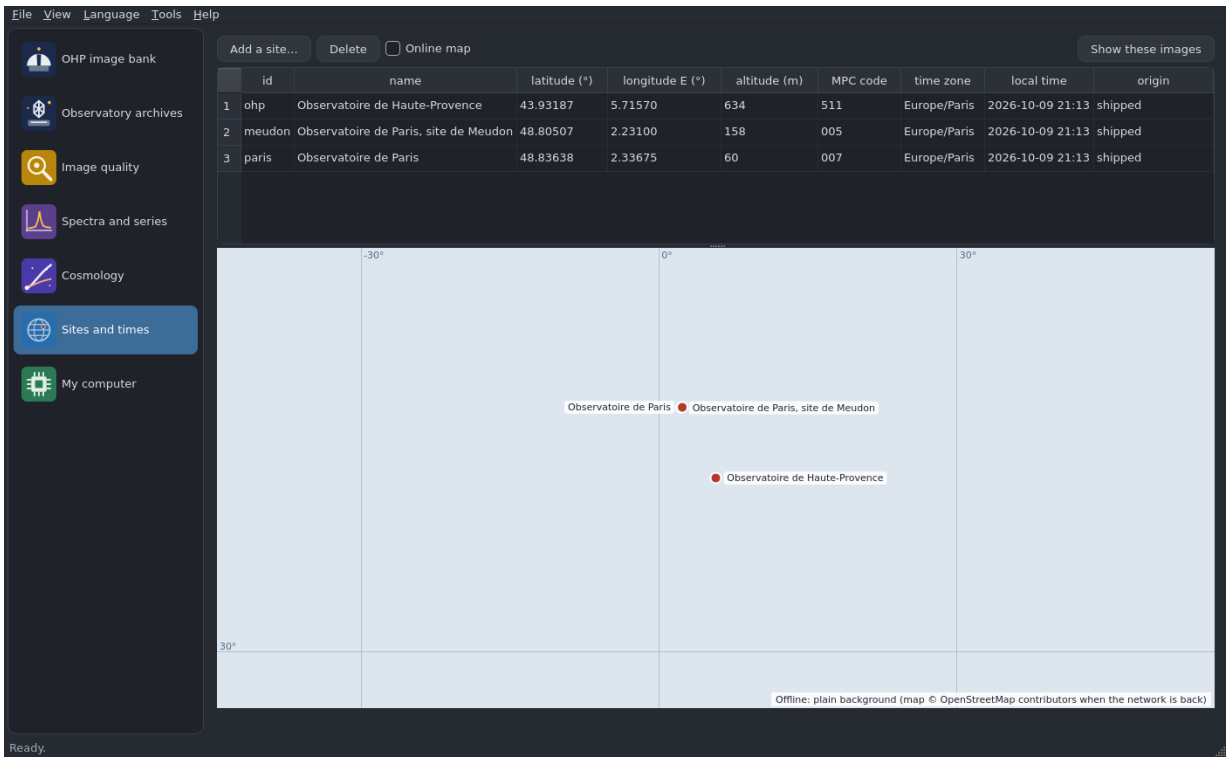
Spectra and series



Formats read: FITS (1D image with a linear `CRVAL/CDELT/CRPIX` WCS axis, 1 × 1 × N radio cubes, binary tables) and CSV or text in columns (optional header, unit in parentheses or brackets). Radio velocity: $v = c(1 - f/f_0)$ (FITS radio convention), f_0 read from `RESTFRQ`, otherwise the H I line at 1,420.405751768 MHz. The frame is that of the frequency axis (`SPECSYS`); no LSR or barycentric correction is made implicitly. Formats of teaching radio antennas are added with a reader (CONTRIBUTING.md, « Adding a format »).

CHAPTER 11

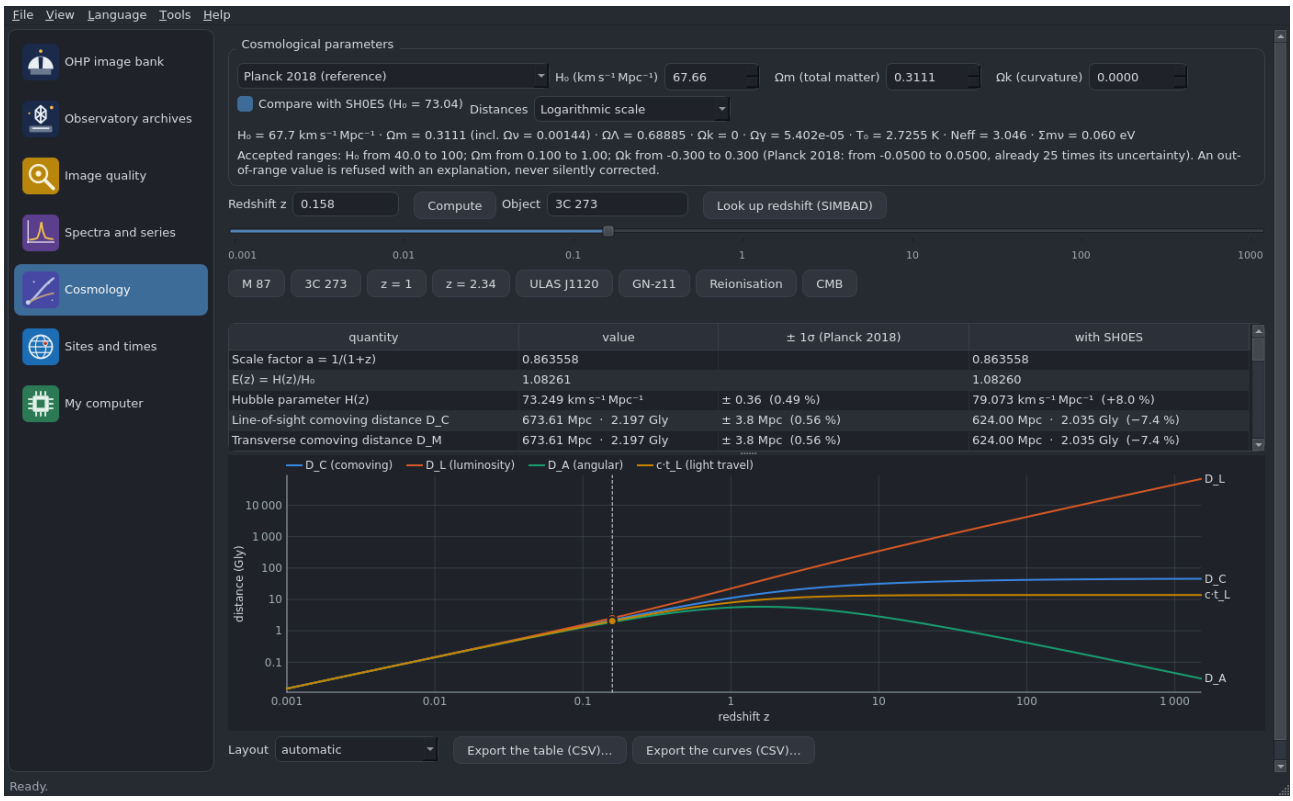
Sites and times



Time is always kept in UTC (aware dates); the site's local time and the user's are shown as well. The « evening date » of a night is the site's local date at the previous noon; daylight is computed from the Sun's altitude at the site (astropy, no IERS download). Shipped sites: OHP (MPC 511), Meudon (005), Paris (007), geodetic coordinates computed from the Minor Planet Center parallax constants. Partner sites are added (IANA time zone, MPC code if any) once their coordinates are known from a reliable source. Map: OpenStreetMap tiles cached on disk, at most two downloads at a time, User-Agent naming the application, attribution shown; offline, plain background and points. On a dense screen (scaling ≥ 150 %, Mac Retina) the map draws the tiles of the next zoom level at half size (four per logical tile, the « @2x » equivalent): it stays sharp instead of being upscaled by the system.

CHAPTER 12

Cosmology



The redshift z of a distant object does not give « one » distance but several, all correct, answering different questions. The module computes them in the ΛCDM model with

$$E(z)^2 = \Omega_r(z)(1+z)^4 + \Omega_m(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda, \qquad D_C = \frac{c}{H_0}\int_0^z \frac{dz'}{E(z')},$$

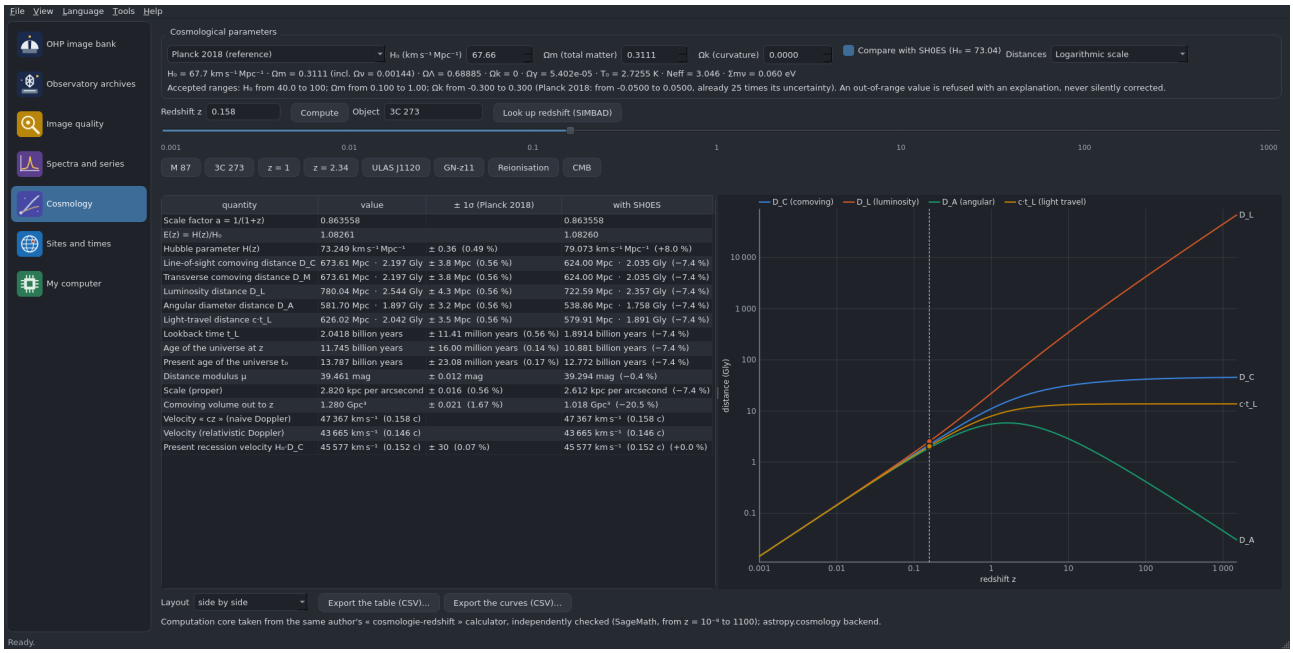
radiation included (CMB photons and neutrinos, one of them massive, 0.06 eV).

Quantity	Use
D_C line-of-sight comoving	present distance of the object; saturates at ≈ 46 Gly (particle horizon)
D_M transverse comoving	equals D_C in a flat universe; enters D_L and D_A
$D_L = (1+z)D_M$	photometry: $F = L/(4\pi D_L^2)$; distance modulus $\mu = 5\log_{10}(D_L/10\text{ pc})$
$D_A = D_M/(1+z)$	sizes: $\theta = \ell/D_A$; scale in kpc per arcsecond; maximum near $z \approx 1.6$
$c\,t_L$	path travelled by the light (the most intuitive, the least physical)
t_L, age at z, t_0	lookback time; check $t_L + t(z) = t_0$
V_C	comoving volume out to z (Hogg 1999, exact in curved universes too)
velocities	cz (naive), relativistic Doppler (inappropriate), $H_0 D_C$ (the only correct one)

Parameters. Planck 2018 by default ($H_0 = 67.66$, $\Omega_m = 0.3111$ incl. $\Omega_\nu = 0.00144$, $T_0 = 2.7255$ K) with the 1σ uncertainty propagated from *correlated* H_0 and Ω_m ($\rho = -0.976$: ignoring it would overestimate the uncertainty) and, optionally, the comparison with SH0ES ($H_0 = 73.04$, Hubble tension). For comparison: Planck 2015, WMAP 9-year, a « textbook » ΛCDM (70 / 0.3 / 0.7 without radiation) and a custom set (H_0, Ω_m, Ω_k, Planck 2018 radiation). Ranges: H_0 from 40 to 100, Ω_m from 0.1 to 1, Ω_k from -0.3 to 0.3 (± 0.05 for Planck 2018); an out-of-range value is *refused with an explanation*, never silently corrected.

Input. A z (comma or point), one of the examples (M 87, 3C 273, $z = 1$, $z = 2.34$, ULAS J1120, GN-z11, reionisation, CMB), or an object name: its redshift is requested from SIMBAD (tolerant naming: « m87 », « ngc4486 »; several candidates: you choose). $z \leq 0$ (approaching object, like M 31) gives no distance; below $z = 0.03$ the object's peculiar velocity dominates (warning); beyond $z \approx 1090$ the universe is opaque. Table and curves export to CSV (« ; », UTF-8; raw values in Mpc, Gyr, Gpc^3).

Slider. Under the z field, a slider spans the whole range on a logarithmic scale (0.001 to 1100, marks at 0.01 / 0.1 / 1 / 10 / 100 / 1000). While dragging, the vertical marker of the curves and the table of values follow live: values are then *interpolated* on the curve grid (300 logarithmic points, typical deviation < 0.5 %, σ and SH0ES columns suspended) and the status line says so; on release the exact computation resumes. Arrows: fine step (1 % of a decade); Page up/down: one decade. The z field and the slider stay in sync both ways.



Layout. Side by side, the table takes the width of its columns (to their content, never truncated) and the curves all the rest; stacked, the table shows its 18 rows when there is room, otherwise the room is shared while keeping at least 300 px for the curves. *View* > *Cosmology layout* (or the module's *Layout* list): automatic (side by side from 1,550 px wide, back to stacked below 1,450 px: no flipping while resizing), always side by side, or always stacked; the choice and the splitter position of each layout are kept.

Units. Negative-exponent notation: H_0 in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, velocities in $\mathrm{km\,s^{-1}}$; in CSV files, unambiguous ASCII (`km s^-1 Mpc^-1`), likewise on a console that cannot show exponents (Windows cp1252).

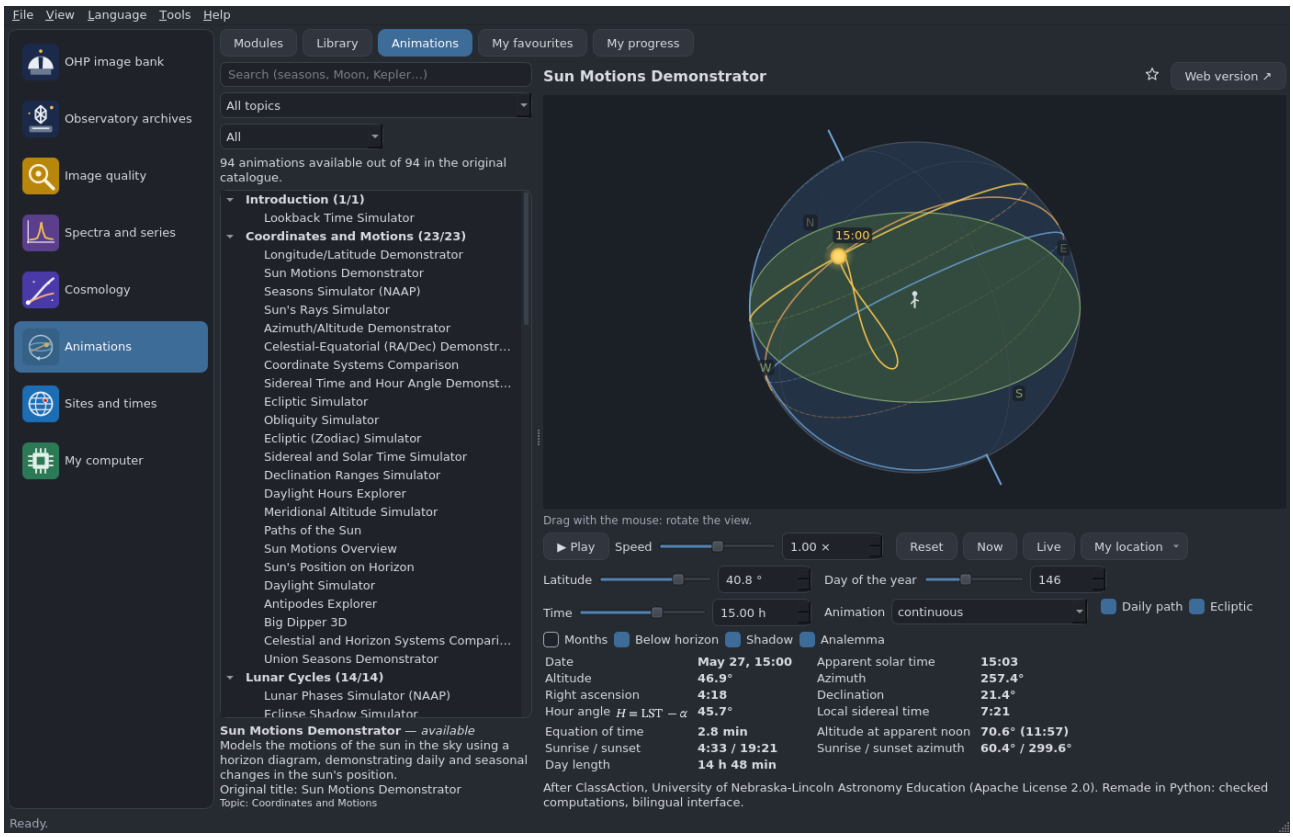
Origin and checks. The computation core is taken from the same author's « cosmologie-redshift » calculator (public repository https://github.com/ARP273-ROSE/cosmologie-redshift), with astropy.cosmology. Before integration it was compared on 8 October 2026 with astropy and with an independent SageMath integration (mpmath, 18 digits, exact Fermi-Dirac function for the neutrinos), from $z = 10^{-8}$ to 1100, for $\Omega_k = 0, \pm 0.01, \pm 0.05$: relative deviation $\leq 3 \times 10^{-6}$ on distances and $\leq 2 \times 10^{-5}$ on ages and $E(z)$ (Komatsu fit of the neutrinos in astropy); no bug. astropy's comoving volume, however, loses all its digits at very small z in curved universes (down to a negative volume): Coupole uses a series expansion there, matched to the closed form. Scripts: `outils/verif_cosmo/`.

Command

```
coupole cosmo 0.158 -shoes
coupole cosmo -object "3C 273"
coupole cosmo 1 2 3 -model wmap9 -csv results.csv
coupole cosmo -curves 0.001 1500 300 -csv curves.csv
```

CHAPTER 13

Animations



The module remakes in Python the *ClassAction* teaching animations of the University of Nebraska-Lincoln (*Astronomy Education*), originally written in Flash and no longer playable since that player was retired. They now run on Windows, macOS and Linux, in French and English, in the Coupole theme, and their computations rely on checked reference formulas. After ClassAction, University of Nebraska-Lincoln Astronomy Education, Apache License 2.0 (`NOTICE` and `LICENCES/Apache-2.0.txt` files).

Catalogue. On the left, the 92 animations of the original catalogue, plus three (the Antikythera Mechanism, section 13.2, Orion in 3D, section 13.3, and Gravity and Orbits, section 13.7), sorted by topic (coordinates and motions, Moon, Renaissance and orbits, stellar properties, light and spectra, extrasolar planets…), with a search (French or English, accents ignored) and filters by topic and status. Topics follow a teaching order, from the introduction and the sky seen from the Earth to cosmology. Each has its name and description in both languages, and its original title. Since version 0.4.0 all 92 are available: those that were only movies (a building's shadow, the melting nail, the hammer thrower, the balloon, the fusion reactions, the traffic jam…) have become simulations, and those that rely on real data (spectra, light curves, exoplanets, planets) cite their source in the `NOTICE` file, updated where it was outdated.

Handling. Dragging a sphere rotates it; dragging a star, the Moon, the Earth or an observer moves it. *Play / Pause* and *Speed* (0.1 to 10 times, logarithmic scale) run the time; *Reset* goes back to the initial state. Each setting has a slider and a numeric field (the exact value is kept: the field only shows a rounded one); computed values are shown below the settings. All settings, the theme and the chosen animation come back at the next launch. Only one animation runs at a time, and only while it is shown; one frame costs a few milliseconds (nothing heavy in the GUI thread). What the mouse does in the shown animation is recalled in small print below the scene (no more tooltip over the drawing). The *Web version* button opens the animation, with its settings, in Coupole Web

(`arp273-rose.github.io/coupole-web`): the same animations in a browser, on a computer, tablet or phone.

13.1 Rendering, formulas and maps (0.4.1)

Bodies. The Sun, the Moon, the Earth and Mars are drawn as spheres: a radial gradient whose focus is shifted towards the light (darkened limb) and, when a Sun is drawn in the same panel, a shaded night side beyond the terminator; the Sun has a darkened limb and a faint halo. Everything stays vector and antialiased; brushes are cached, so a frame still costs a few milliseconds (median 1.5 ms per frame at 720 × 470, as before these changes, measured on the 92 animations). Symbols (icons of the eclipse table, markers) stay flat.

Formulas. Formulas are typeset by a small layout engine written for Coupole (QPainter, without LaTeX or matplotlib): stacked fractions, well-placed superscripts and subscripts, radicals, stretchy parentheses, $\times 10^n$ as a true superscript, one-letter variables in mathematical italic, words (sin, log, LST) upright. The font is STIX Two Math (SIL OFL 1.1 licence, embedded 139 kB subset, `LICENCES/OFL-STIX-Two.txt`). The formulas drawn in the animations (Newton's law calculator, gravity algebra, distance modulus, small angles, luminosity, flux, eccentricity…) and those of descriptions, labels and shown values are all typeset; layouts are cached.

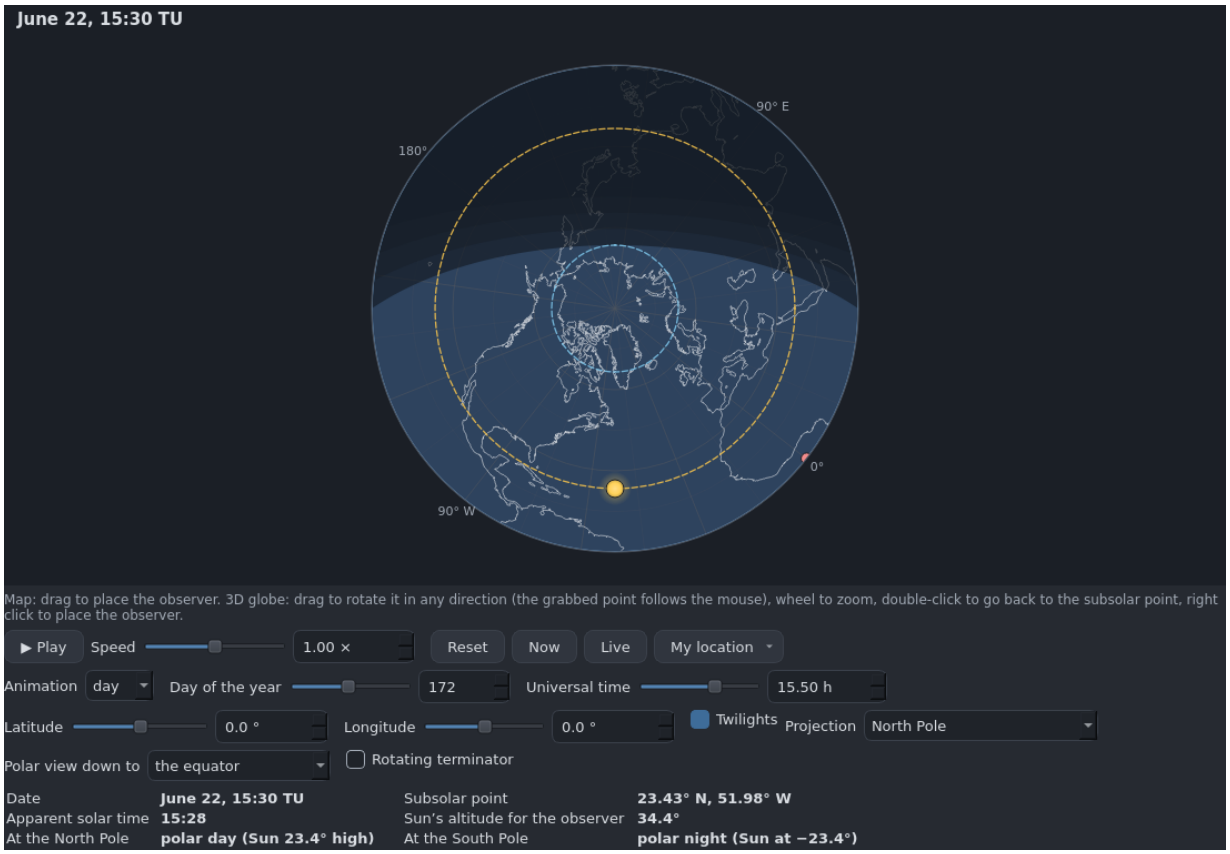
Labels. Every boxed label goes through a common placer: if its place is taken (another label, a small body, text already written, the edge of the image), it tries 24 positions around its object and takes the first free one, linked to the object by a leader line. A test checks, for the 92 animations, four sizes (420 to 2560 pixels wide) and three instants, that no label overlaps another or leaves the image.

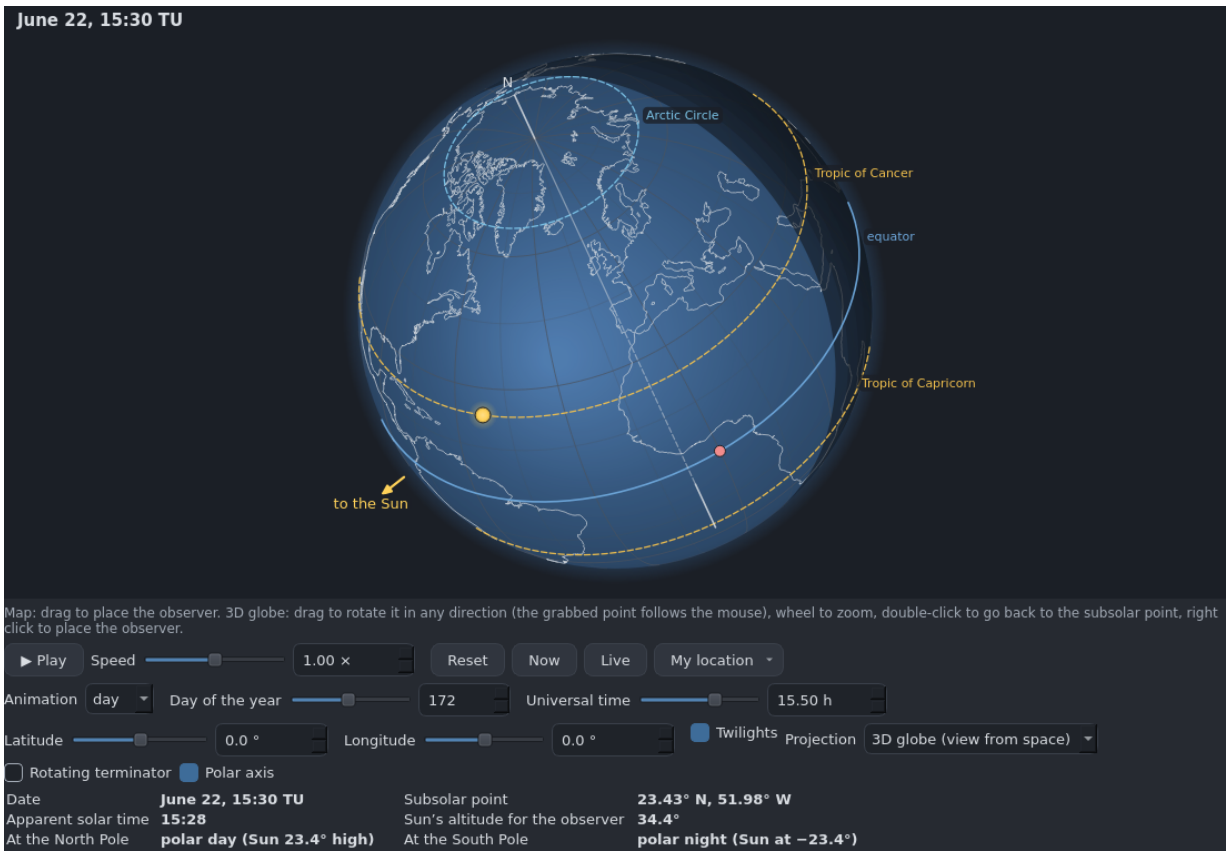
Maps. Coastlines come from Natural Earth 1:50m (public domain), simplified (Douglas-Peucker 0.07°, 65 kB), loaded with the first globe shown. The daylight simulator offers four projections. Equirectangular: linear mapping between the mouse and latitude-longitude. North Pole or South Pole: azimuthal equidistant centred on the pole (true distance to the pole), edge at the equator or 30° beyond to see both tropics; meridians every 15° (30° emphasised), parallels every 15°, tropics and polar circles dashed; by default the Earth turns under a fixed terminator (noon at the bottom), and *Rotating terminator* keeps the map fixed (Greenwich at the bottom in the North). Night and twilights are drawn meridian by meridian, at the exact latitude where the Sun is at 0°, −6°, −12° or −18°: a loop around the pole, with no artefact at the pole (at the solstice it is tangent to the polar circle). 3D globe seen from space (orthographic projection): it turns in any direction with the mouse (trackball rotation: the grabbed point follows the mouse, with a light inertia), the wheel zooms, a double-click goes back to the subsolar point, a right click places the observer; thin atmosphere, sphere lit on the Sun's side, lines cut exactly at the limb, exact night and twilights (visible parts of the spherical caps opposite the Sun), polar axis and a "to the Sun" arrow. Polar day or night at the North and South Poles is shown with the values.

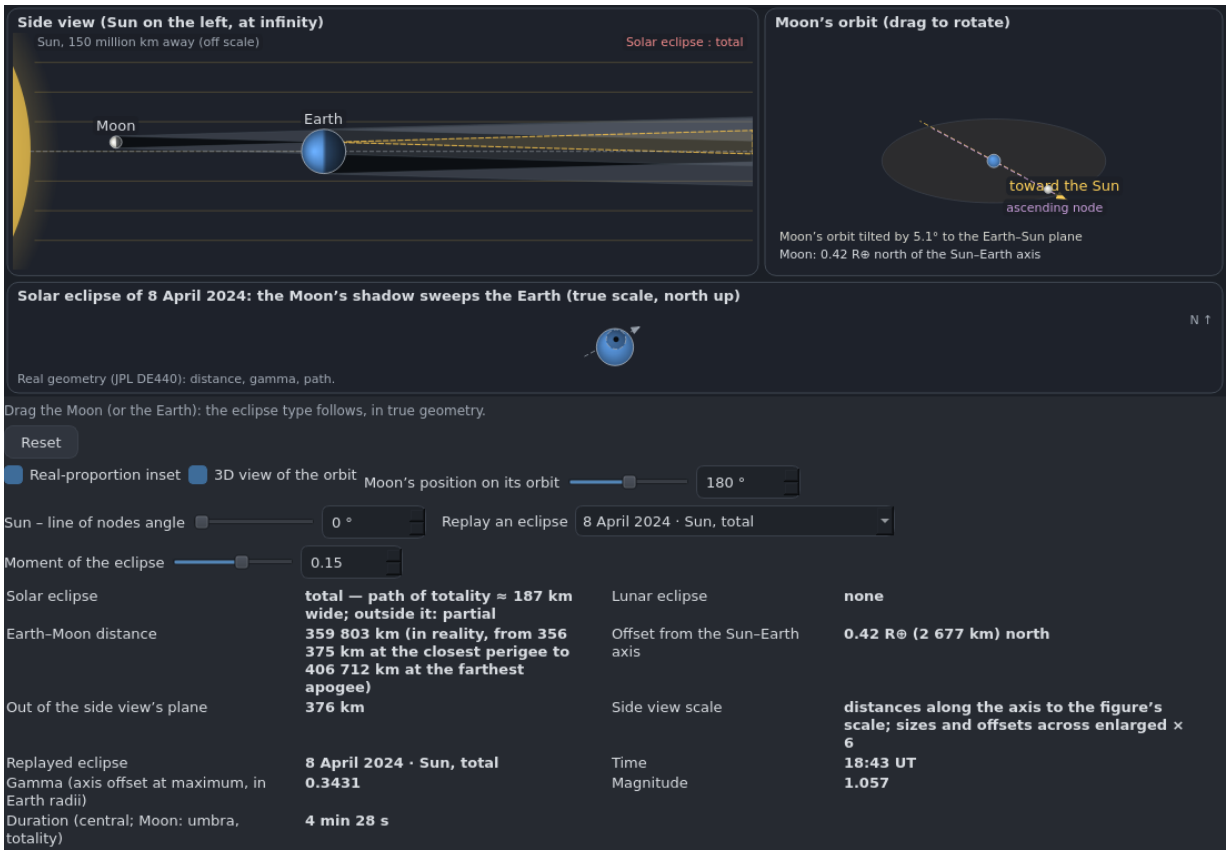
Milky Way. The rotation curve is no longer made of two pieces joined at 5 kpc: it is that of a published mass model, McMillan (2017), computed with galpy (bulge, thin and thick stellar discs, HI and H2 gas, NFW dark-matter halo; components can be shown, v^2 being the sum of the components' v^2), with $R_0 = 8.21$ kpc and $v_0 = 233.1$ km/s. Real measurements are overlaid with their error bars: Eilers et al. (2019) from 5 to 25 kpc and the galkin compilation (Pato & Iocco 2017) from 2 to 5 kpc. The Keplerian curve starts from the mass the model holds within the sphere of radius R. The Galaxy is seen from the North Galactic Pole, to scale: long bar (Wegg et al. 2015), log-periodic spiral arms (Reid et al. 2019), Sun at 8.21 kpc with its direction of rotation, circle of the chosen radius. SageMath recovers the model's halo speed and bulge mass to 1.3×10^{-4} and 5×10^{-5}.

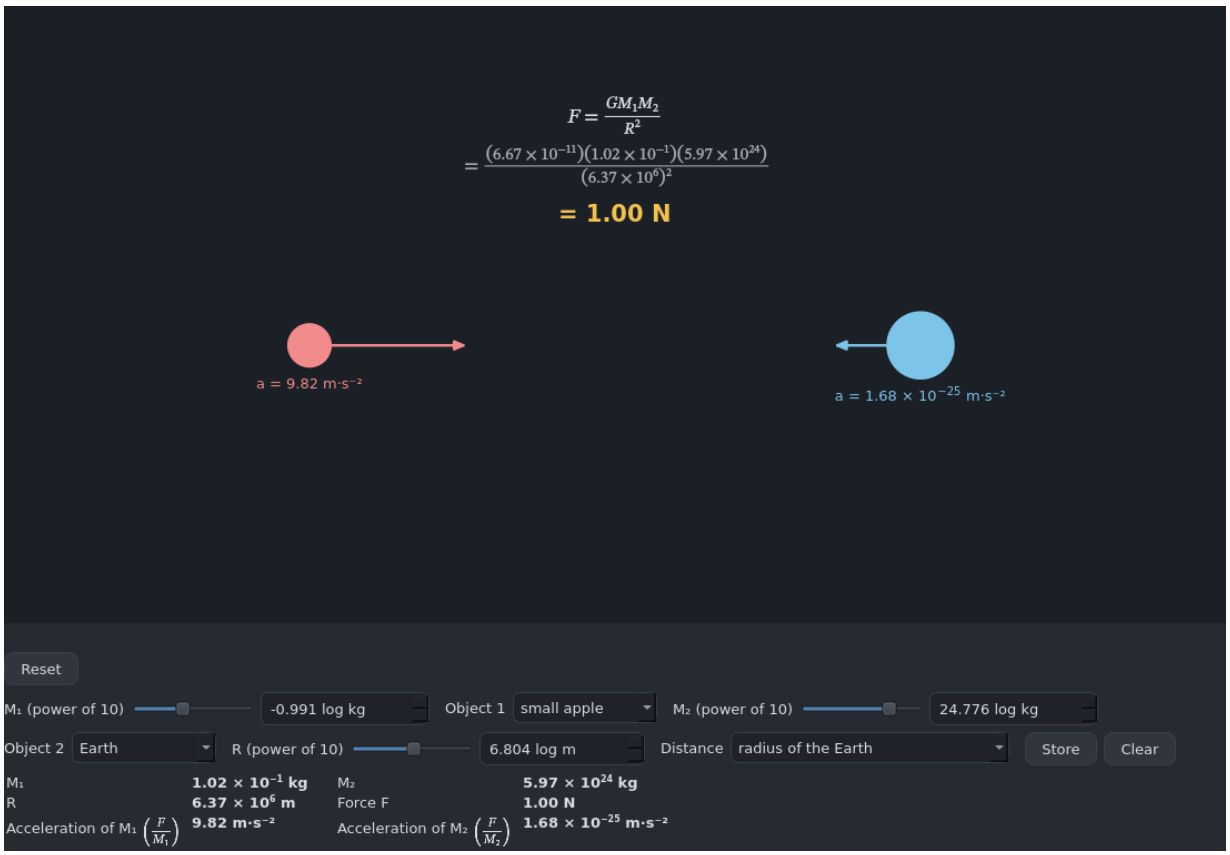
Eclipses. In the eclipse shadow simulator the Sun is drawn whole, and *Replay an eclipse* shows each of the 133 eclipses of the 2015–2044 table to true scale: the Moon crosses the Earth's umbra and penumbra, or the Moon's shadow sweeps the Earth; the offset from the axis comes from the Moon's mean argument of latitude ($y = D \sin 5.145° \sin F$, Meeus), brought into the range allowed by the eclipse type, and a slider moves the instant.

Fixes. The globe of the longitude/latitude demonstrator (and every sphere) turned against the drag: the grabbed point now follows the mouse. The Sun's rays of the seasons simulator stop at the limb and at the panel edge; the star names of the Big Dipper 3D and the hammer thrower labels no longer overlap; a few texts too long for small windows were moved (legends, clocks, hydrogen levels, building shadow, Sun's rays, electromagnetic spectrum).

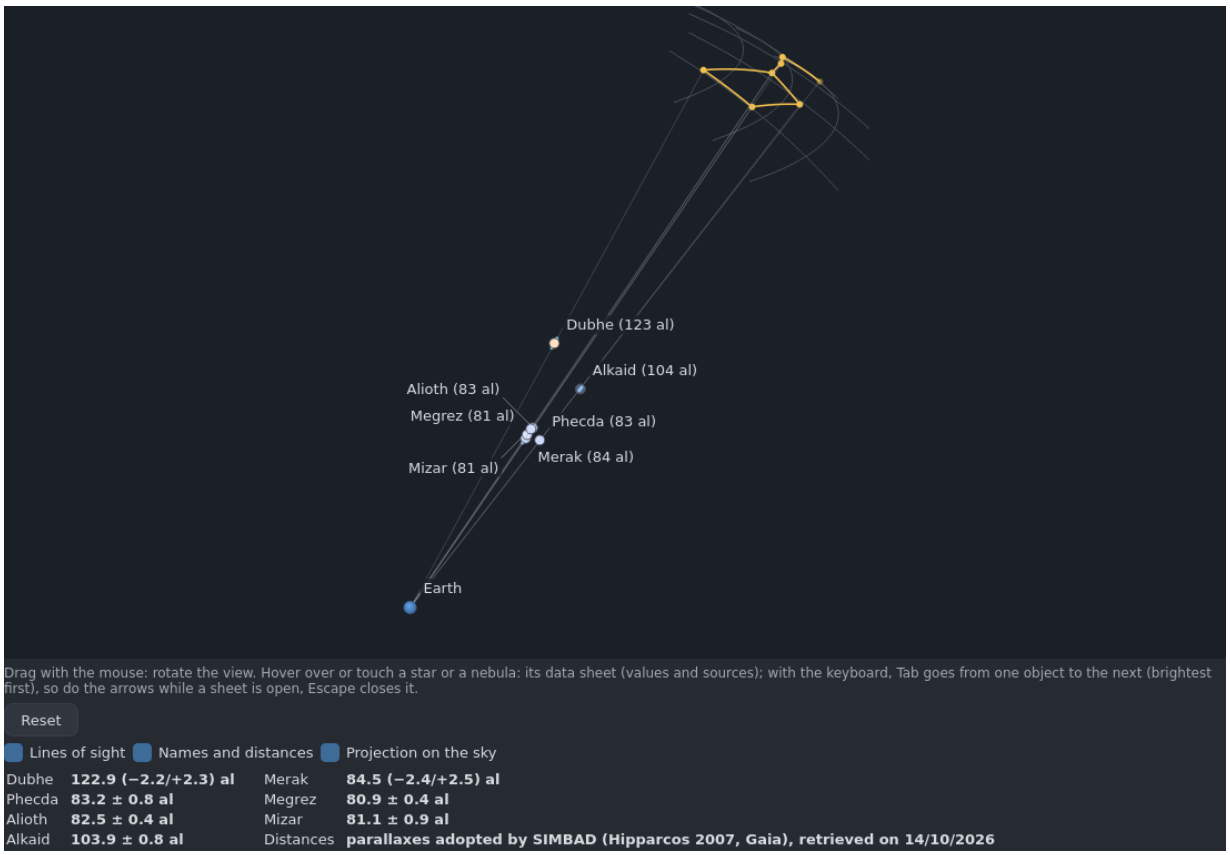



June 22, 15:30 TU
N
Arctic Circle
Tropic of Cancer
equator
Tropic of Capricorn
to the Sun
Map: drag to place the observer. 3D globe: drag to rotate it in any direction (the grabbed point follows the mouse), wheel to zoom, double-click to go back to the subsolar point, right click to place the observer.
▶ Play
Speed
1.00 ×
Reset
Now
Live
My location
Animation
day
Day of the year
172
Universal time
15.50 h
Latitude
0.0 °
Longitude
0.0 °
Twilights
Projection
3D globe (view from space)
Rotating terminator
Polar axis
Date
June 22, 15:30 TU
Subsolar point
23.43° N, 51.98° W
Apparent solar time
15:28
Sun's altitude for the observer
34.4°
At the North Pole
polar day (Sun 23.4° high)
At the South Pole
polar night (Sun at −23.4°)


Side view (Sun on the left, at infinity)
Sun, 150 million km away (off scale)
Solar eclipse : total
Moon
Earth
Moon's orbit (drag to rotate)
toward the Sun
ascending node
Moon's orbit tilted by 5.1° to the Earth–Sun plane
Moon: 0.42 R⊕ north of the Sun–Earth axis
Solar eclipse of 8 April 2024: the Moon's shadow sweeps the Earth (true scale, north up)
N ↑
Real geometry (JPL DE440): distance, gamma, path.
Drag the Moon (or the Earth): the eclipse type follows, in true geometry.
Reset
Real-proportion inset
3D view of the orbit
Moon's position on its orbit
180 °
Sun – line of nodes angle
0 °
Replay an eclipse
8 April 2024 · Sun, total
Moment of the eclipse
0.15
Solar eclipse
total — path of totality ≈ 187 km wide; outside it: partial
Lunar eclipse
none
Earth–Moon distance
359 803 km (in reality, from 356 375 km at the closest perigee to 406 712 km at the farthest apogee)
Offset from the Sun–Earth axis
0.42 R⊕ (2 677 km) north
Out of the side view's plane
376 km
Side view scale
distances along the axis to the figure's scale; sizes and offsets across enlarged × 6
Replayed eclipse
8 April 2024 · Sun, total
Time
18:43 UT
Gamma (axis offset at maximum, in Earth radii)
0.3431
Magnitude
1.057
Duration (central; Moon: umbra, totality)
4 min 28 s


$F = \frac{GM_1M_2}{R^2}$
$= \frac{(6.67 \times 10^{-11})(1.02 \times 10^{-1})(5.97 \times 10^{24})}{(6.37 \times 10^{6})^2}$
= 1.00 N
a = 9.82 m·s⁻²
a = 1.68 × 10⁻²⁵ m·s⁻²
Reset
M₁ (power of 10)
-0.991 log kg
Object 1
small apple
M₂ (power of 10)
24.776 log kg
Object 2
Earth
R (power of 10)
6.804 log m
Distance
radius of the Earth
Store
Clear
M₁ 1.02 × 10⁻¹ kg
R 6.37 × 10⁶ m
Acceleration of M₁ $\left(\frac{F}{M_1}\right)$ 9.82 m·s⁻²
M₂ 5.97 × 10²⁴ kg
Force F 1.00 N
Acceleration of M₂ $\left(\frac{F}{M_2}\right)$ 1.68 × 10⁻²⁵ m·s⁻²


The hammer thrower (seen from above, distance ≈ 2 m)
ball
center of mass
thrower
A star and its planet
planet
center of mass
star
Pause
Speed
1.00 ×
Reset
Thrower mass
110 kg
Hammer mass
7.26 kg
Star / planet mass
5.17
Paths
Thrower's circle (radius) 0.124 m
Ball's circle (radius) 1.876 m
Ratio of radii = inverse mass ratio $\frac{0.124}{1.876} = 0.066$
Star's circle, as a share of the distance 16.2 %



13.2 The Antikythera Mechanism (0.4.2)



A 93rd animation, outside ClassAction, under “Ancient astronomy”: the Greek astronomical calculator of the 2nd century BC, in 3D, adapted from *Anticythere3D*, by the same author (full application: `github.com/ARP273-ROSE/Anticythere3D`). Tooth counts (Freeth et al., *Nature* 444, 2006; *Nature* 454, 2008), trains, exact ratios (fractions), the layout of the 15 arbors (constrained optimisation, 17 exact

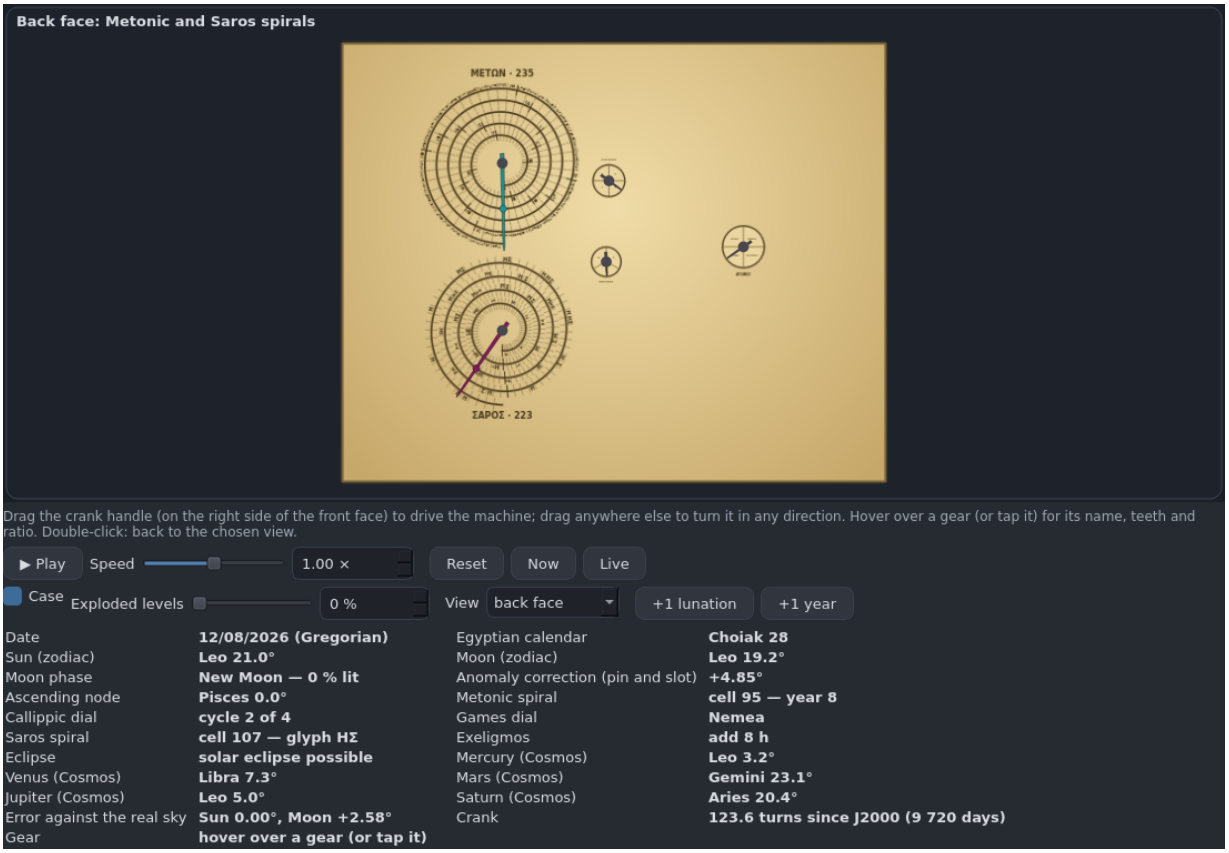
centre distances), the pin and slot, the dials and their graduations are those of Anticythere3D 1.3.1, taken unchanged, with its tests.

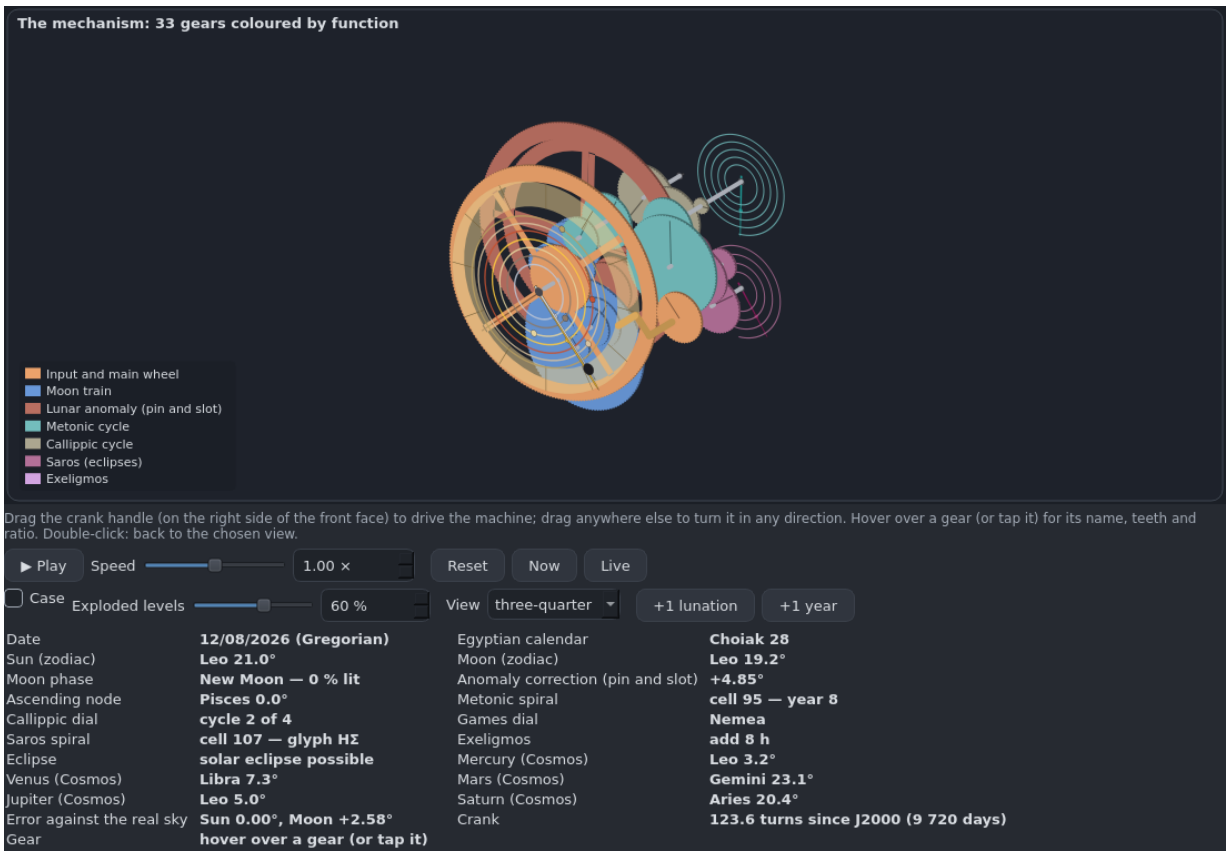
Handling. Grab the crank handle (right side of the front face) and turn it: one turn of the handle is 48/223 of a turn of the great wheel b1, that is 78.6 days. Elsewhere, dragging turns the machine in any direction (quaternion orientation, never invalid); *Play* turns the crank once per second (times the speed); *Today*, *+1 lunation*, *+1 year*. *Case*: closed machine (wooden case, bronze plates engraved in Greek) or open (the 33 gears coloured by function, with a legend); *Exploded levels* spreads the 16 meshing planes along the arbors; *View*: front face, back face or three-quarter. Hovering over a gear (or tapping it) gives its name, teeth, exact ratio in turns per year and function.

What the dials show. Front: zodiac (360 divisions, Greek names, parapegma letters), 365-day Egyptian calendar ring, turned so that the Sun pointer shows the date (era of Nabonassar, 1 Thoth of year 1 = JD 1,448,638, checked with SageMath), Sun and Moon pointers, half-silvered phase ball, Cosmos rings (Freeth et al., *Scientific Reports* 11:5821, 2021) whose small sphere shows the planet's geocentric longitude (period relation and epicycle). Back: Metonic spiral (5 turns, 235 cells, Corinthian months, years in Greek numerals), Saros spiral (4 turns, 223 cells, computed H and Σ glyphs), Callippic, exeligmos (0, 8, 16 h) and Games dials, each on the arbor that carries its pointer. The values: date (Gregorian or Julian), Egyptian date, Sun, Moon, phase, anomaly correction, node, spiral cells, possible eclipse, planets, error against the real sky (Meeus).

Rendering without OpenGL. Orthographic projection and painter's algorithm: all parts are flat and parallel, each is drawn in its plane through an exact affine transform, the planes are sorted along the line of sight, the arbors are cut into segments at the planes of the gears that may cover them; the closed case is convex, only its visible faces are drawn. The engraved dials are drawn as vectors (same strokes as Anticythere3D) and cached in an image whose resolution follows the zoom; teeth smaller than 3 pixels are replaced by the pitch circle. A frame costs 1 to 4 ms closed, 5 to 9 ms open (QPainter, 420 to 720 pixels, loaded NAS); in Coupole Web, 0.4 to 1.2 ms (Chromium, emulated phone) and 1.5 to 5 ms with a CPU slowed down 4 times.

Checks. SageMath (`outils/verif_anim/anticythere_verif.sage`, tooth counts typed again, rational arithmetic): sidereal Moon 254/19, hence $254 - 19 = 235$ lunations in 19 years (235/19, a convergent of the continued fraction of the year in lunations), Metonic 5/19, Callippic 1/76, Saros $940/4237 = (4/223)(235/19)$, exeligmos 235/12711 (54.09 years), carrier 477/4237 (8.8826 years), anomalistic month of the machine 27.5533 d, pin-and-slot amplitude $\arcsin(1.1/9.6) = 6.58°$, 17 centre distances. The tests of Anticythere3D (ratios, carrier, pin and slot, return of the dials, planets, calibration, centre distances) are taken over, with rendering tests (views, aberrant drags while playing, hovered gear, crank); the displayed values are identical, character for character, to those of Coupole Web.

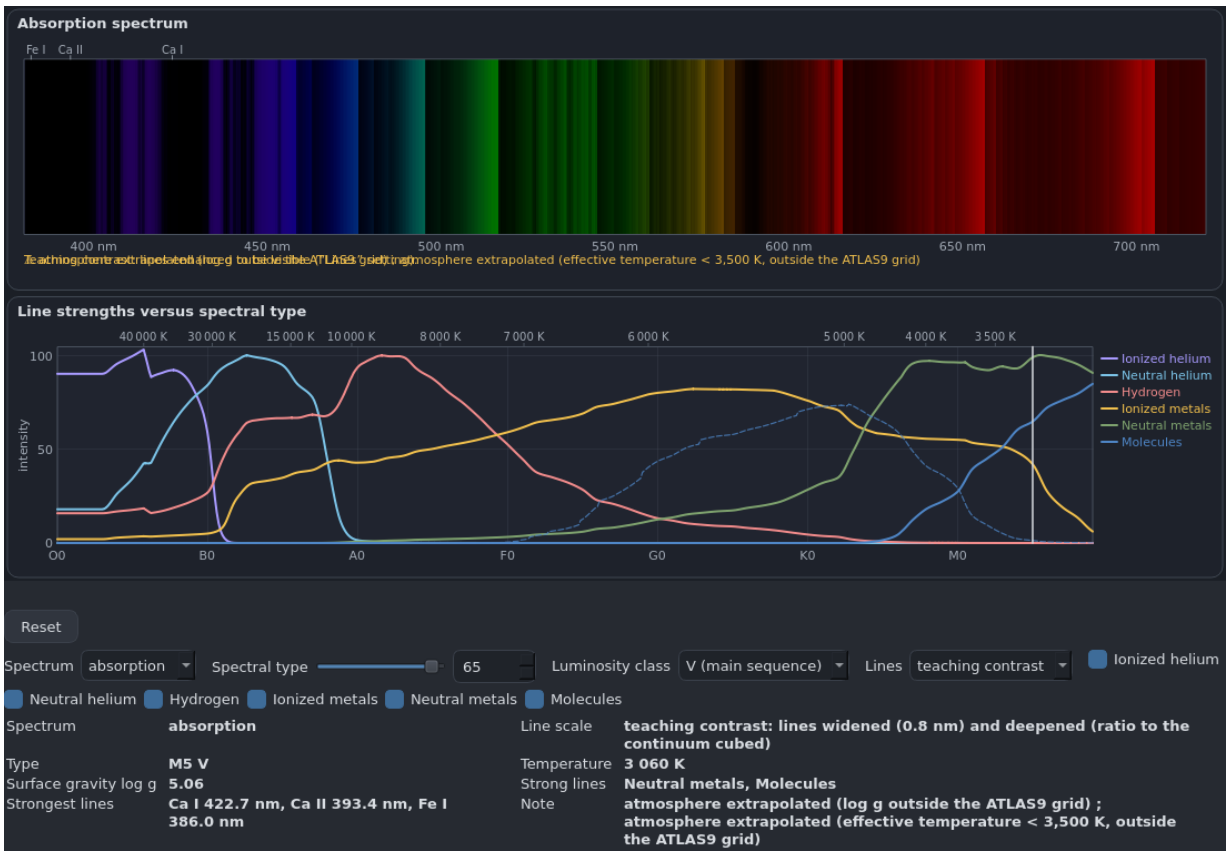

Back face: Metonic and Saros spirals
ΜΕΤΩΝ · 235
ΣΑΡΟΣ · 223
Drag the crank handle (on the right side of the front face) to drive the machine; drag anywhere else to turn it in any direction. Hover over a gear (or tap it) for its name, teeth and ratio. Double-click: back to the chosen view.
▶ Play
Speed
1.00 ×
Reset
Now
Live
Case
Exploded levels
0 %
View
back face
+1 lunation
+1 year
Date 12/08/2026 (Gregorian)
Sun (zodiac) Leo 21.0°
Moon phase New Moon — 0 % lit
Ascending node Pisces 0.0°
Callippic dial cycle 2 of 4
Saros spiral cell 107 — glyph HΣ
Eclipse solar eclipse possible
Venus (Cosmos) Libra 7.3°
Jupiter (Cosmos) Leo 5.0°
Error against the real sky Sun 0.00°, Moon +2.58°
Gear hover over a gear (or tap it)
Egyptian calendar Choiak 28
Moon (zodiac) Leo 19.2°
Anomaly correction (pin and slot) +4.85°
Metonic spiral cell 95 — year 8
Games dial Nemea
Exeligmos add 8 h
Mercury (Cosmos) Leo 3.2°
Mars (Cosmos) Gemini 23.1°
Saturn (Cosmos) Aries 20.4°
Crank 123.6 turns since J2000 (9 720 days)



Other fixes (0.4.2). Panel titles no longer spill onto the neighbouring panel in a narrow window (Phases of Venus, Copernican and Ptolemaic, and 20 other animations): wrapped on two lines, shortened or ended with "…", the full title staying in a tooltip and in the scene's accessible name. Three views of a spectrum: the opaque bulb hides the cloud behind it (continuous spectrum); absorption appears only when the gas lies between the telescope and the bulb. The day-night simulator now opens on the 3D globe.

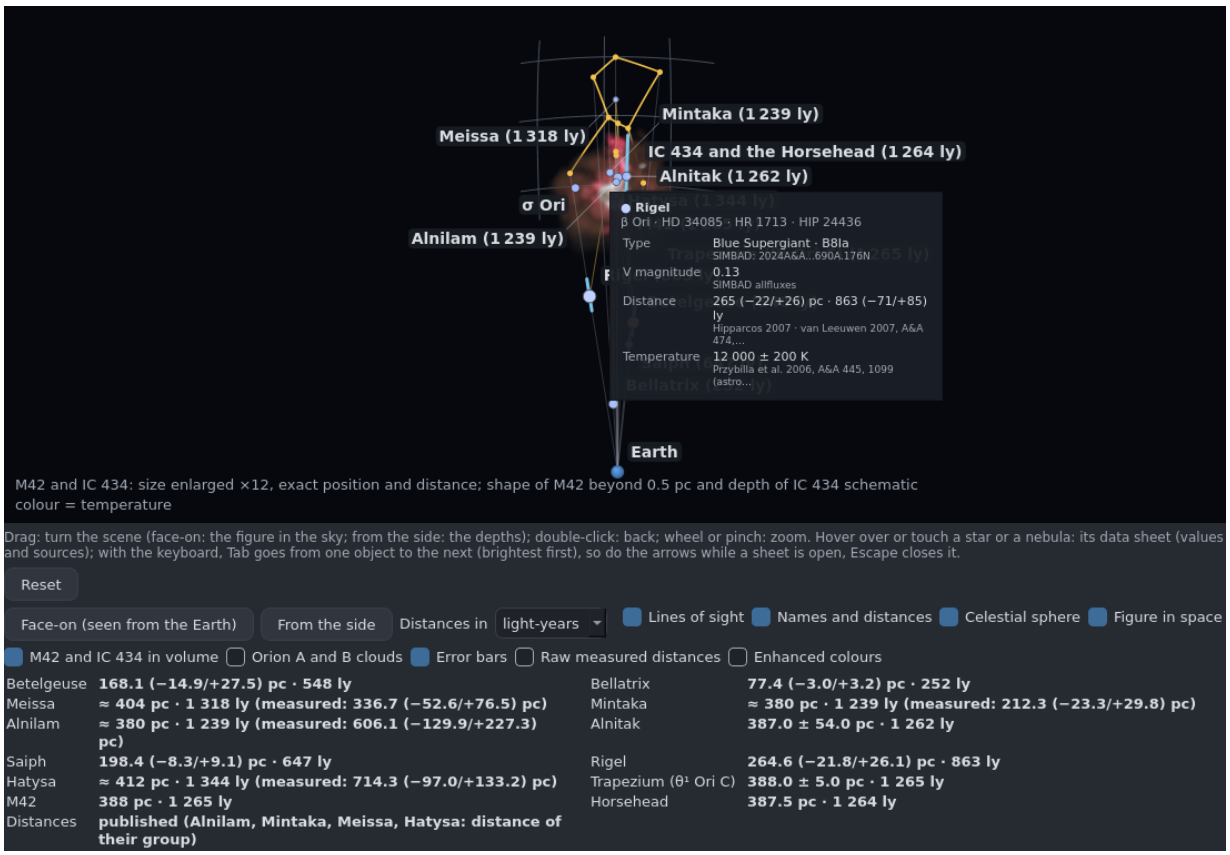
13.3 Physical spectra, Orion in 3D, hover data sheets (0.5.0)

Physically based spectra. The Spectrum Explorer and the Spectroscopic Parallax no longer use the original's straight segments and 37 lines: the strengths of the six line families are computed with Saha–Boltzmann on model atmospheres (Castelli & Kurucz ATLAS9 grid), at optical depth 2/3, for every spectral type and luminosity class (V, III, I); 133 NIST lines (air wavelengths, oscillator strengths), TiO bands and the CH G band degraded to the red, Planck continuum coloured with the CIE 1931 functions. Smooth curves, with the spectral type along the axis and the class temperature on top; names of the strongest lines above the spectrum. G2 V = 5,770 K (Mamajek 2022). Extrapolated cases (atmosphere outside the grid, clamped or interpolated calibration) are flagged below the spectrum and in the values; the Spectroscopic Parallax takes the published absolute magnitude (class V everywhere, III for O stars and from G to M7, I for O stars) and elsewhere the former calibration, flagged too (dotted curve). The engine and its data (copied unchanged, same reference tests as Coupole Web, checked with SageMath) are described in `NOTICE`.



Orion in 3D. A 94th animation, Coupole's own, in a new topic "Interstellar Medium and Star Formation". Orion as we see it (seen from the Sun: the perspective is the gnomonic projection of the sky, north up, east left), then *Unfolding*: every star slides along its line of sight to its measured distance (SIMBAD, Gaia DR3, Hipparcos 2007, dedicated measurements: Betelgeuse, ζ Ori, σ Ori, VLBA Trapezium) and the view turns sideways: Tabit is 8 pc away, Bellatrix 77, Betelgeuse 168, Rigel 265, the Belt around 380–400 pc. Error bars along the line of sight; Alnilam, Mintaka, Meissa and Hatysa, whose own measurement is unreliable, are placed at the published distance of their group (*Raw measured distances* option). Orion A and B molecular clouds (3D skeletons of Zucker et al. 2021), λ Orionis ring, Barnard's Loop on the Orion shell (Foley et al. 2023), the Orion Nebula as a "blister" (bowl-shaped ionisation front, Veil, Trapezium; Wen & O'Dell 1995, O'Dell et al. 2009) and the Horsehead. *Look at*: dive towards M42 or the Horsehead, with ESO (VST, VLT) or Hubble images laid on the sky at their exact place; their credit, verbatim, is shown on the scene and as a link below it. True star colours (black body at their temperature, CIE 1931), *Enhanced colours* option. Everything is to scale except the size of the stars. Dragging turns the scene around the pivot; double-click: back; *Play*: the scene turns slowly. A frame costs 5 to 8 ms (QPainter, 720 pixels, loaded NAS), 0.4 to 1.1 ms in Coupole Web.

Hover data sheets. In Orion in 3D, Big Dipper 3D, the Big Dipper clock, heliacal rising, the coordinate systems comparison and the HR diagram, hovering over a star or a nebula (or touching it, or reaching it with the keyboard: Tab from the brightest to the faintest, arrows, Escape) opens a small sheet laid next to the object, never over it: names and designations, type, magnitudes, distance in parsecs and light-years with its uncertainty ("uncertain measurement" when needed, other measurements), temperature, published radius and luminosity, age and exciting stars of nebulae, and the source of each value. The summary is read by screen readers. On a small canvas the sheet is compact. Big Dipper 3D: updated distances (SIMBAD adopted parallaxes); HR diagram: *Colour = temperature* box.

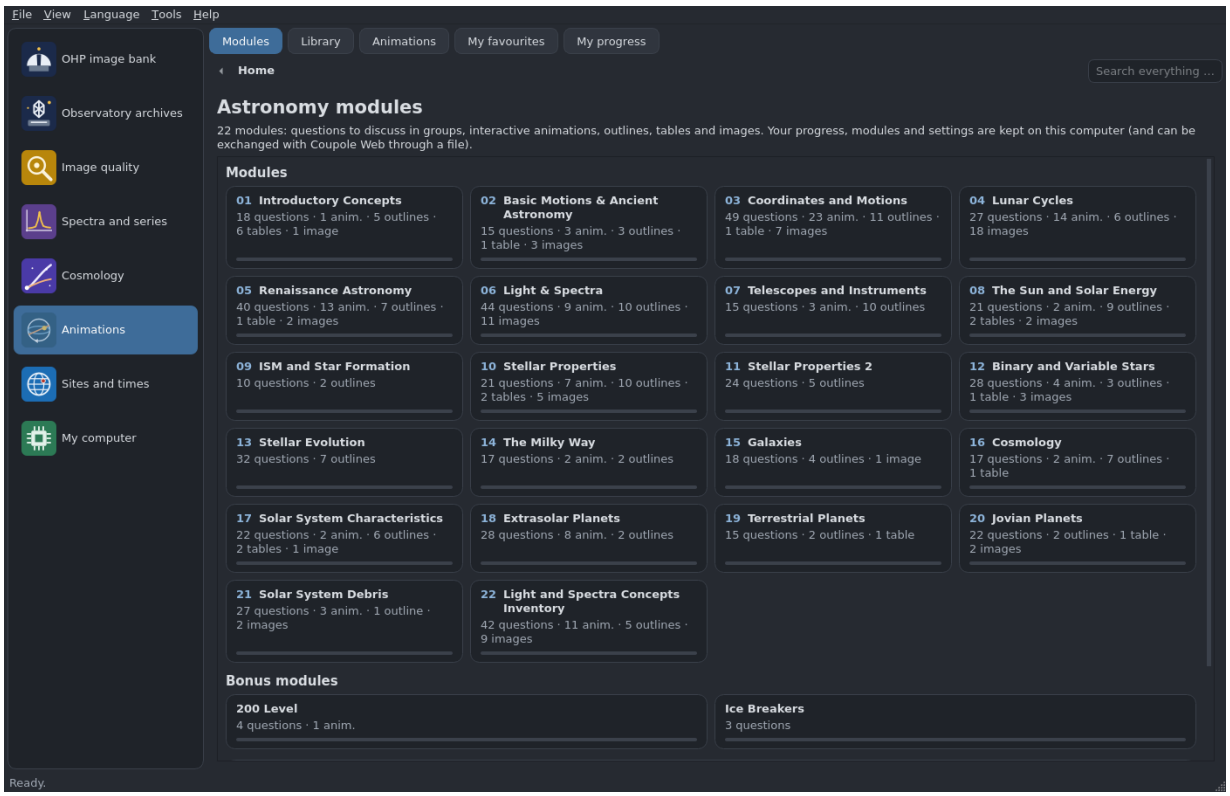


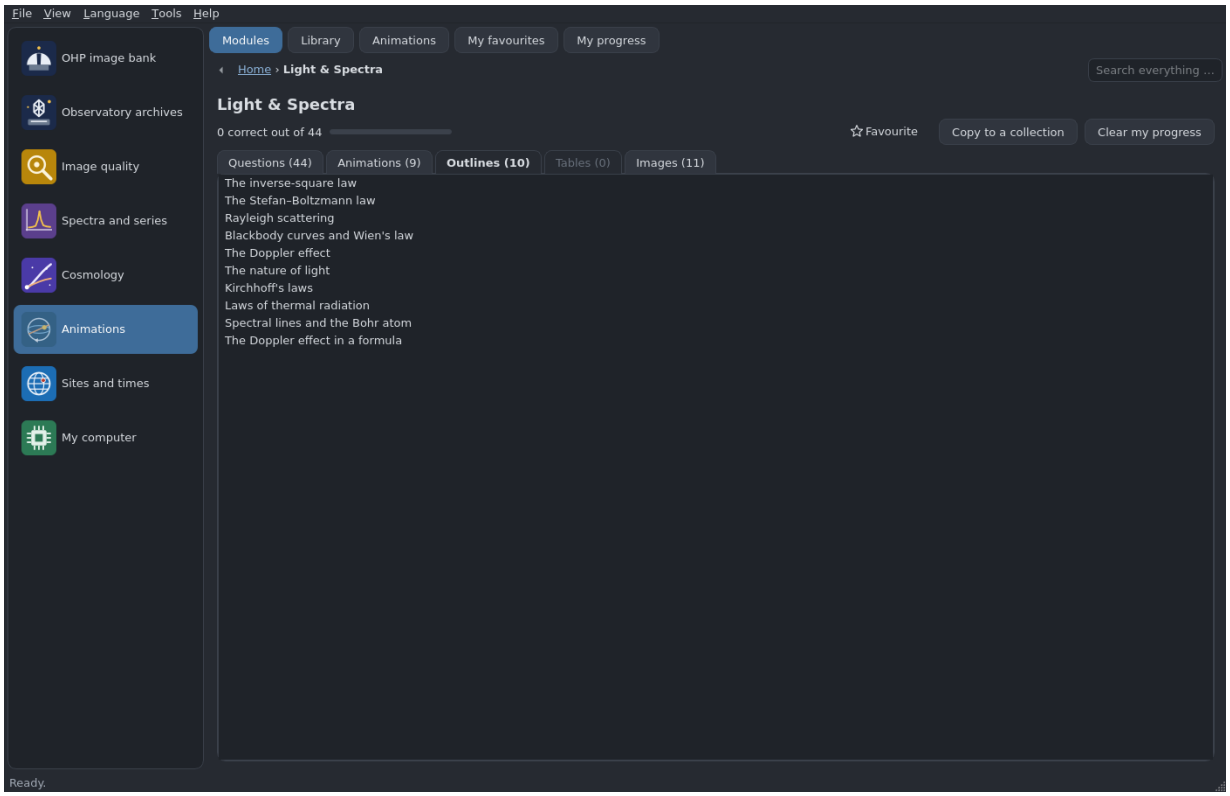
Typography. SI-style units (km·s^{-1}, W·m^{-2}…), true minus sign, thin spaces, decimal comma in French, powers of ten as superscripts, formulas with stretchy parentheses and radicals, multi-letter subscripts upright ($\lambda_{\max}$): in every animation, with a formatter shared by the desktop, the web and the resources, and a test that reviews every displayed text.

Checks. SageMath: parsec → light-year conversion from the IAU definitions, distances and error bars of the 300 data sheets, Trapezium parallax (Gaia DR3, 4 stars: 402 ± 4 pc, shown in the sheet next to the VLBA 388 ± 5 pc), ICRS → galactic matrix against its definition; colours (21 stars) and spectrum model (46 comparisons) checked by the data sets. The numbers of Orion in 3D, the content of the sheets and their placement are identical to Coupole Web's (shared references).

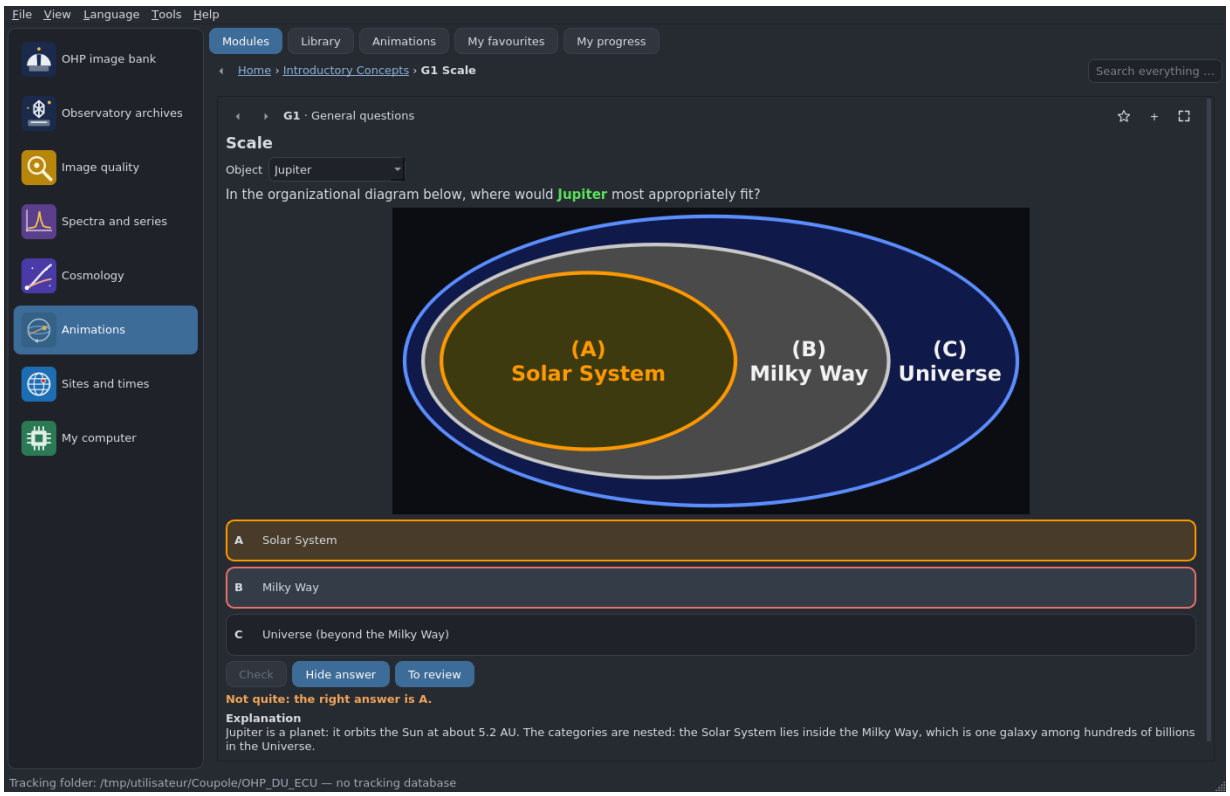
13.4 Modules, questions and resources (0.6.0)

Module navigation. The *Animations* module opens on the home page of the **22 ClassAction modules** (and two bonus modules): *Modules*, *Animations* (the catalogue), *My modules*, *My progress* buttons, a search and a breadcrumb (with back to the previous page). A module page has *Questions* (Warm-up, General, Challenge, Discussion, numbered W1, G1, C1, D1, with their state ○, ●, ✓, ×), *Animations*, *Outlines*, *Tables*, *Images* tabs. Double-click or Enter opens an item; an outline, table or image opens in its own window (several at once, resizable, Escape closes them); hovering over an item shows a preview; right click: *Add to…* a personal module.

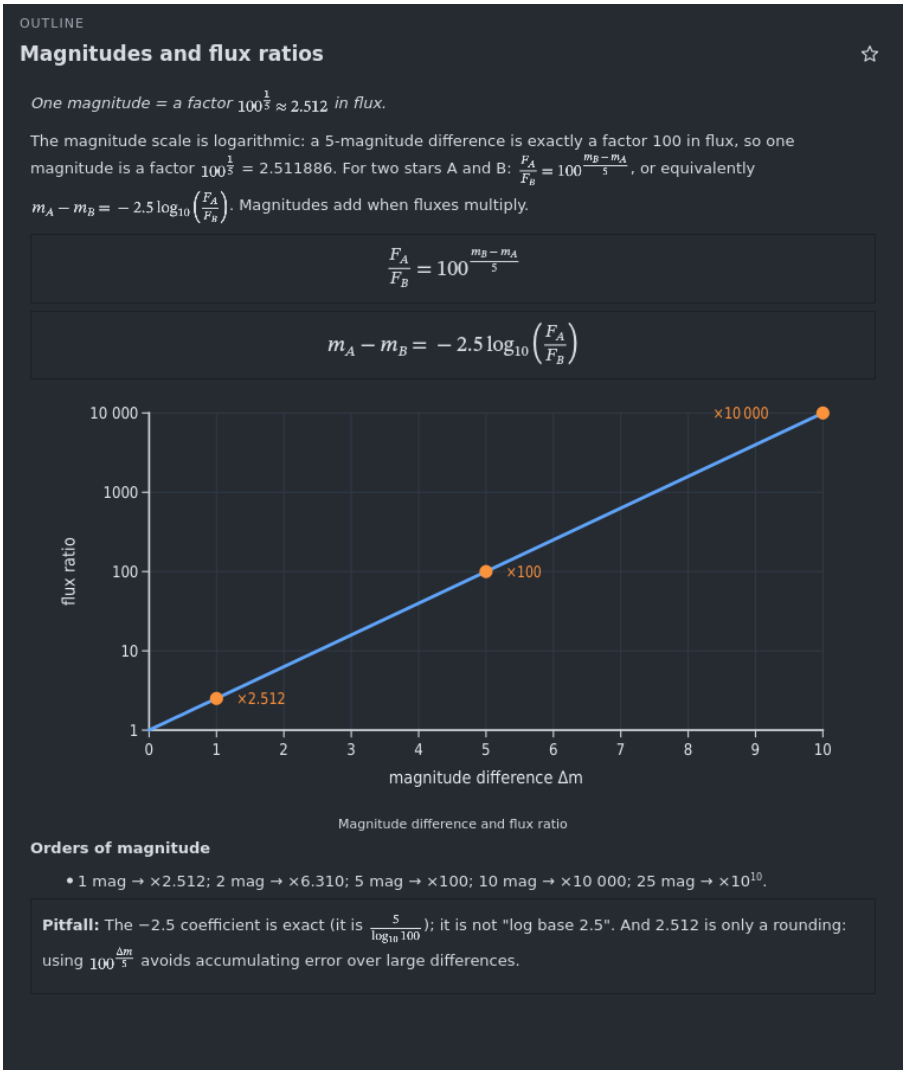




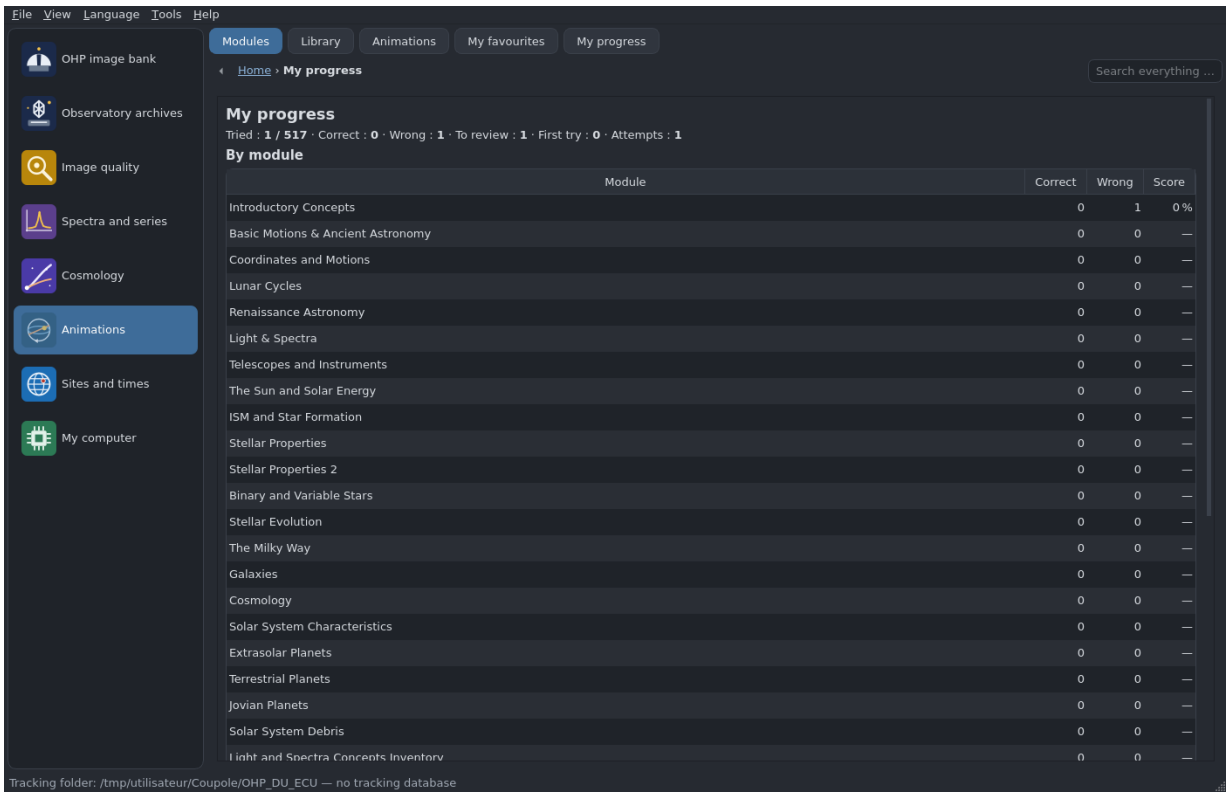
Questions. The 517 ClassAction peer-instruction questions (Apache 2.0) are translated and rewritten as data (the same as in Coupole Web, same engine: the fingerprints of over 8,000 instances are identical). *Adapt question* menus (variants), *New draw*, vector figures, choices A, B, C… (keys A to H), *Check* (Enter) records the attempt, *Show answer* (R) makes the right answer flash orange then stay orange, *Hint* (I), explanation, answer elements for discussion questions, "What changed from the original" panel, related resources ⋆ on the right, previous and next question (arrows, order W → G → C → D, looping), full screen (F).



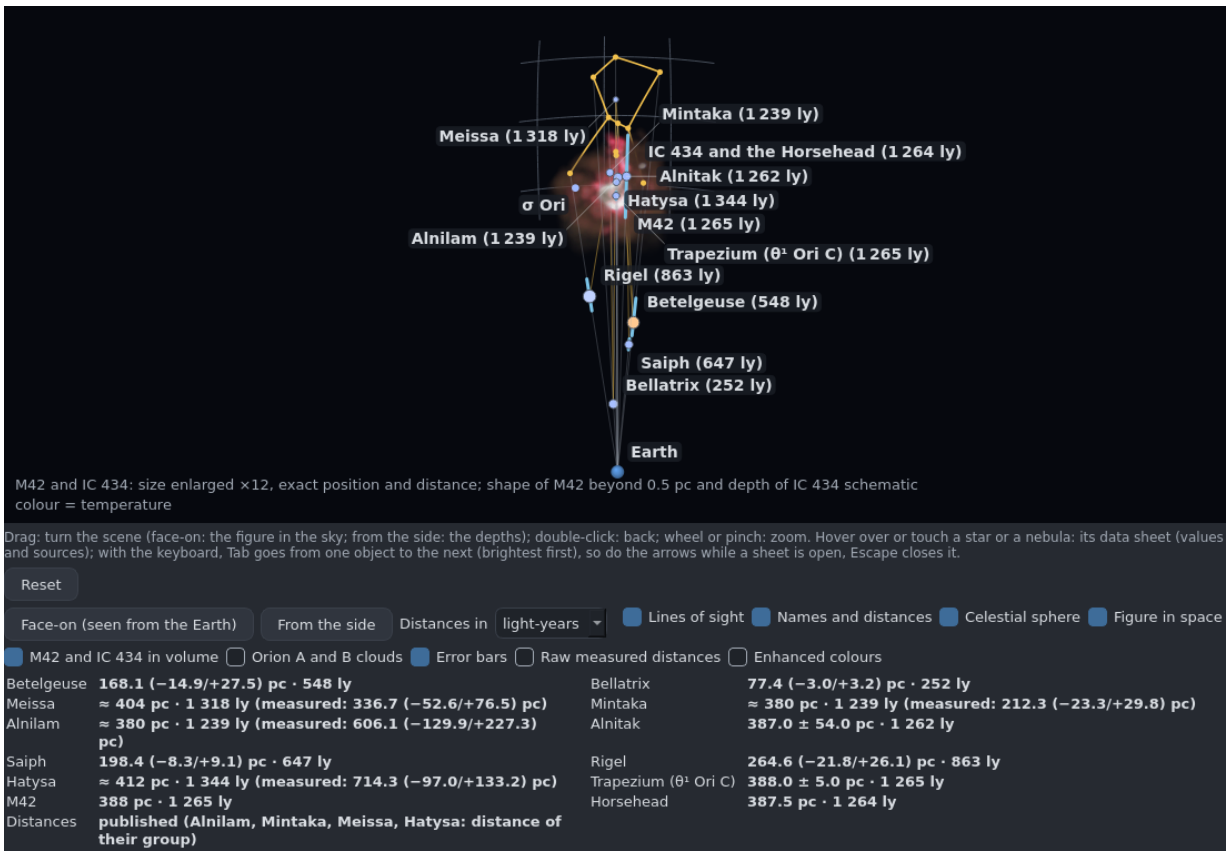
Outlines, tables, images. 110 outlines, 14 tables and 109 diagrams written for Coupole (recomputed, sourced values): text and formulas by Coupole's formula engine, diagrams drawn by a QPainter engine that reads their description (real text width through `QFontMetricsF` for margins, overlap check), tables with right-aligned numeric columns (source in the tooltip), free photographs with their credit and licence (full list in `NOTICE`).



My modules and backup. *My modules*: create, copy (including a ClassAction module), rename, delete (with *Undo*), add or remove any question or resource, reorder (*Organise* tab: drag, or Alt+↑ and Alt+↓). Everything is kept in `sauvegarde.json` in the configuration folder (atomic write, copy of the previous version): each animation's settings, answers, scores and attempts, progress, My modules, preferences, in the same format as Coupole Web. *My progress*: summary per module, questions to review, *Export* and *Import* (merge or replace; Coupole Web files accepted), *Reset* all or part (one animation, one module, settings, answers, My modules, preferences, everything), with confirmation, a copy kept in `sauvegardes/` and *Undo*.



Orion in 3D, remade. After the Big Dipper 3D: the Earth, the main stars (Betelgeuse, Bellatrix, Meissa, the Belt, Saiph, Rigel, the sword with Hatysa and the Trapezium) at their distances, the lines of sight and the figure on the celestial sphere; drag to turn the scene, *Face-on (seen from the Earth)* and *From the side* buttons. M42 (blister hollowed into the OMC-1 cloud, emitting layer, Trapezium, Veil, Bar) and IC 434 (red front lit by σ Ori, Horsehead pillar, edge of Orion B, NGC 2023) are drawn in volume after their published geometry, at their exact position, size enlarged $\times 12$ (written in the scene). Still scene: nothing is redrawn while nothing moves.



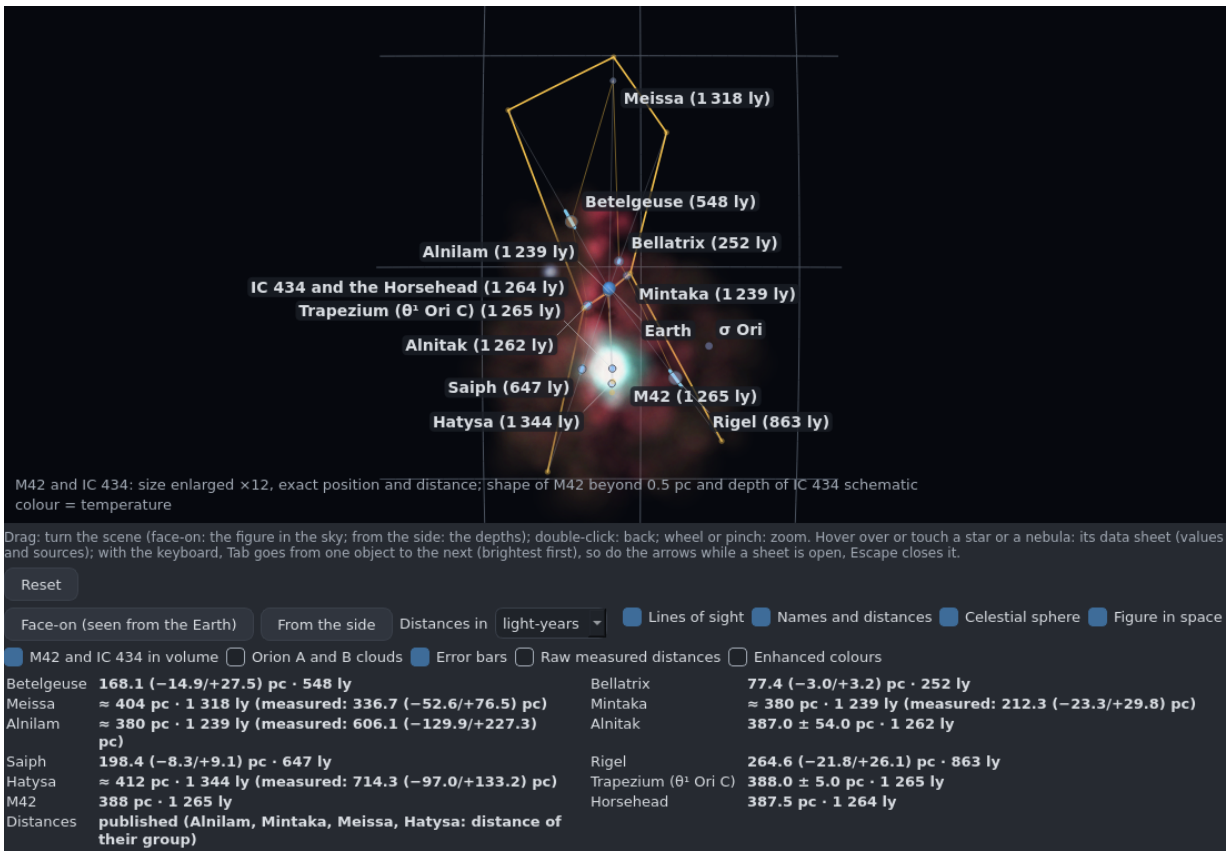
Drag: turn the scene (face-on: the figure in the sky; from the side: the depths); double-click: back; wheel or pinch: zoom. Hover over or touch a star or a nebula: its data sheet (values and sources); with the keyboard, Tab goes from one object to the next (brightest first), so do the arrows while a sheet is open, Escape closes it.

Reset

Face-on (seen from the Earth) | From the side | Distances in light-years | Lines of sight | Names and distances | Celestial sphere | Figure in space

M42 and IC 434 in volume | Orion A and B clouds | Error bars | Raw measured distances | Enhanced colours

Betelgeuse	**168.1 (−14.9/+27.5) pc · 548 ly**	Bellatrix	**77.4 (−3.0/+3.2) pc · 252 ly**
Meissa	**≈ 404 pc · 1 318 ly (measured: 336.7 (−52.6/+76.5) pc)**	Mintaka	**≈ 380 pc · 1 239 ly (measured: 212.3 (−23.3/+29.8) pc)**
Alnilam	**≈ 380 pc · 1 239 ly (measured: 606.1 (−129.9/+227.3) pc)**	Alnitak	**387.0 ± 54.0 pc · 1 262 ly**
Saiph	**198.4 (−8.3/+9.1) pc · 647 ly**	Rigel	**264.6 (−21.8/+26.1) pc · 863 ly**
Hatysa	**≈ 412 pc · 1 344 ly (measured: 714.3 (−97.0/+133.2) pc)**	Trapezium (θ¹ Ori C)	**388.0 ± 5.0 pc · 1 265 ly**
M42	**388 pc · 1 265 ly**	Horsehead	**387.5 pc · 1 264 ly**
Distances	**published (Alnilam, Mintaka, Meissa, Hatysa: distance of their group)**		

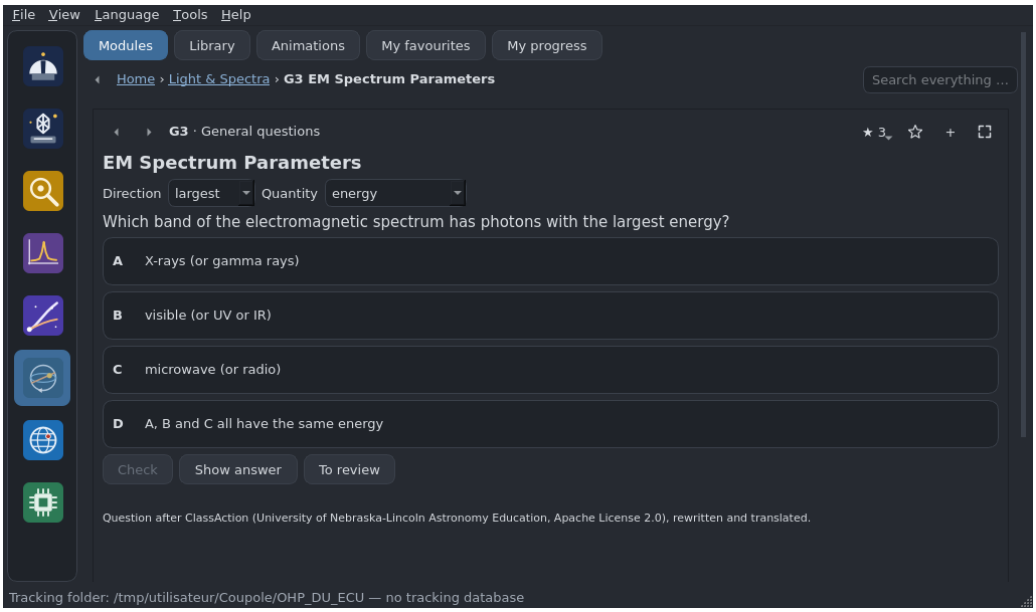


Spectra. The line-family intensities are computed every eighth of a type (same physics as the model, calibration interpolated monotonically and smoothly, PCHIP): no more plateaus or kinks. The spectrum band has a *teaching contrast* by default (lines widened and deepened, stated below the band) or a *realistic scale*.



13.5 Look and checks (0.7.0)

Look. The home page shows the modules as equal-width cards (number, title, contents, thin progress bar); the current view is highlighted in the *Modules*, *Animations*, *My modules*, *My progress* bar. On a question, related resources are on the right in a wide window and in the « ⋆ n » menu of the header in a narrow one; the figure takes at most 45 % of the height, so the choices stay visible at 1024 × 600. Texts of figures and diagrams stay readable in both themes and never overlap, whatever the system font. The interface adapts from 1024 × 600 to 4K and from 100 to 200 % scaling; in a small window, an animation page scrolls instead of squeezing the scene.

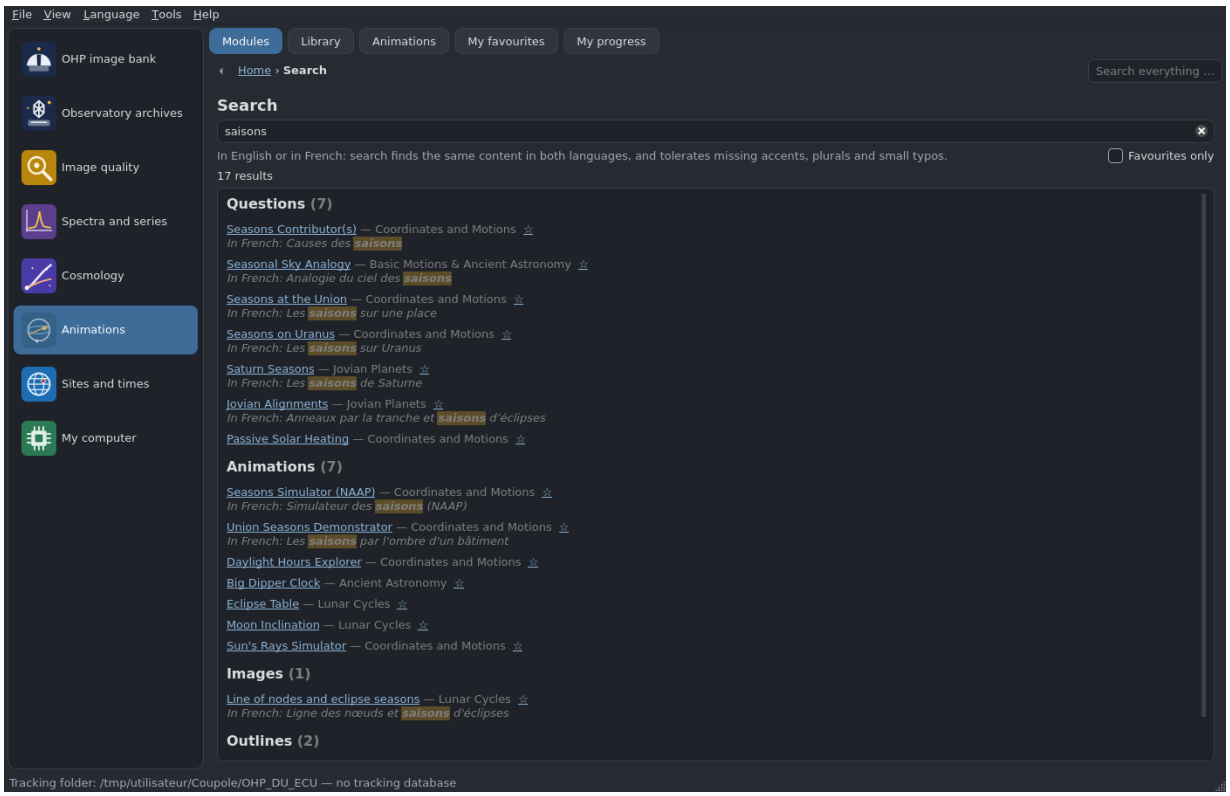


Automatic checks. The 517 questions are scored one by one (right and wrong answer), the 95 animations are handled (every control at its limits, every button, random drags) without exception and without freezing the GUI thread, the backup makes the round trip export → erase → import identically (and accepts a Coupole Web export), language and theme change live, settings come back on reopening. Every built package (Windows, macOS, Linux) passes a headless « smoke » test: `python -m coupole.essai_fumee`.

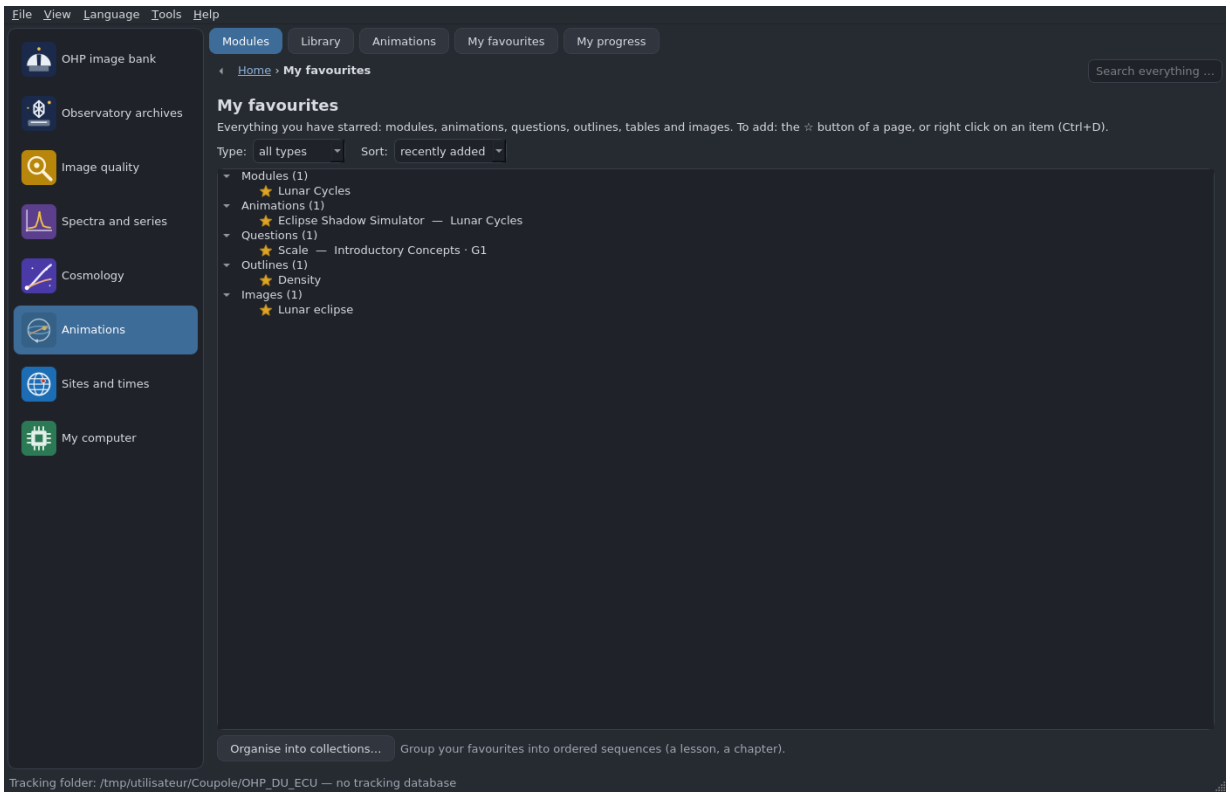
Common kit. Coupole now follows the author's standard for desktop applications: updates and incident reports are the common modules `updater.py` and `reporting.py`, identical across applications and configured by `kit.json`; settings, reports and data live outside the installation folder (consent given to earlier versions is carried over); no computer name is sent (random identifier). `coupole --check` checks an installation without a screen (window, modules, animations, data, manuals, logo, style) and returns code 0 when all is well. The Windows installer is bilingual; the latest installer has a permanent link (`releases/latest/download/Coupole-Setup.exe`); on Linux, the `Coupole-X-linux-x86_64.tar.gz` (or `-arm64`) archive installs into the user profile through `installer.sh` (`coupole` command), and the file dialog is Qt's complete one by default.

13.6 Search, favourites, library, "Now" (0.7.1)

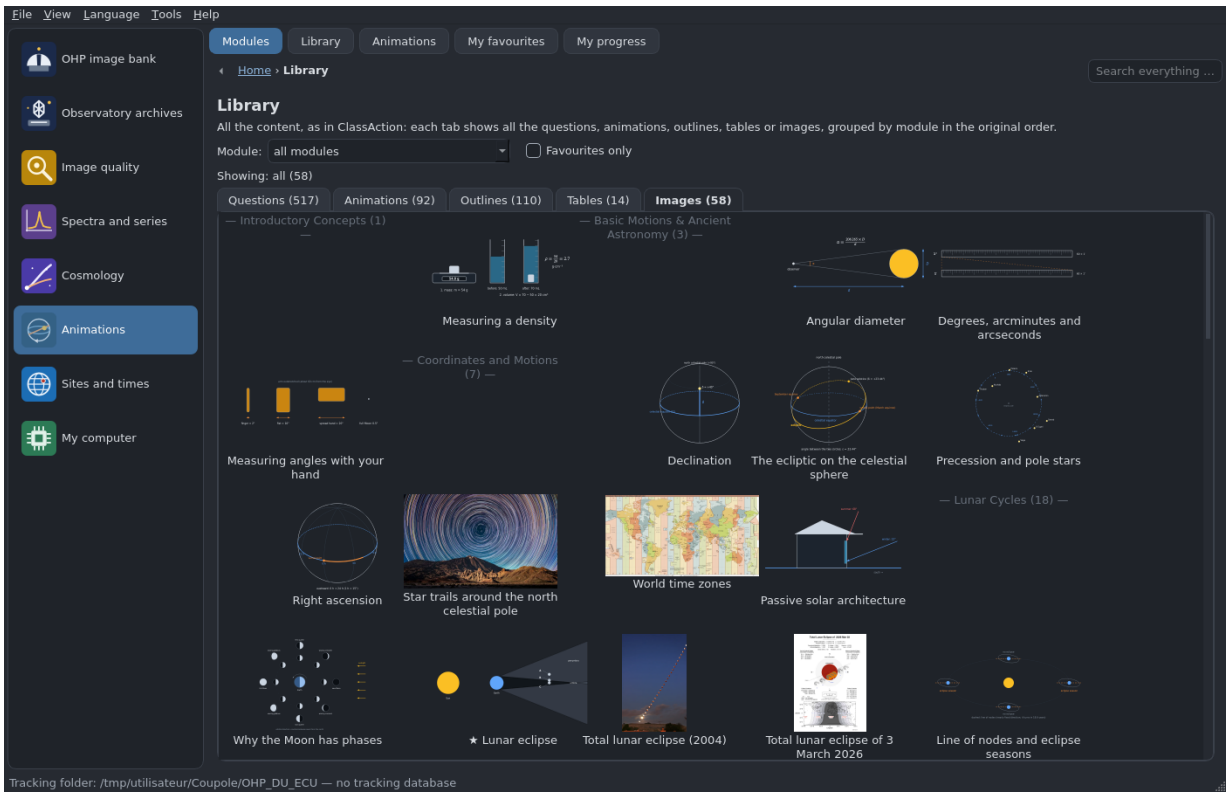
Bilingual search. In English or in French, whatever the interface language, search finds the same content (modules, animations, questions, outlines, tables, images) and shows it in the interface language. The index gathers titles, keywords, texts and the original ClassAction titles in both languages; it ignores accents and case, reduces simple plurals and tolerates one typo (two for long words). Results are grouped by type, matches in bold; when only the other language's title matches, it is recalled. The engine is the same, line for line, as Coupole Web's.



Favourites and collections. The ⋆ star of a page (module, question, outline, table, image, animation), right click or Ctrl+D in a list mark a favourite; “My favourites” (top bar and home page) groups them by type, with a filter and a sort. Collections (the former “My modules”: your own sequences in the chosen order) move to the background: “Organise into collections…” button. Favourites and collections are in the shared backup (format version 2), exchanged with Coupole Web.



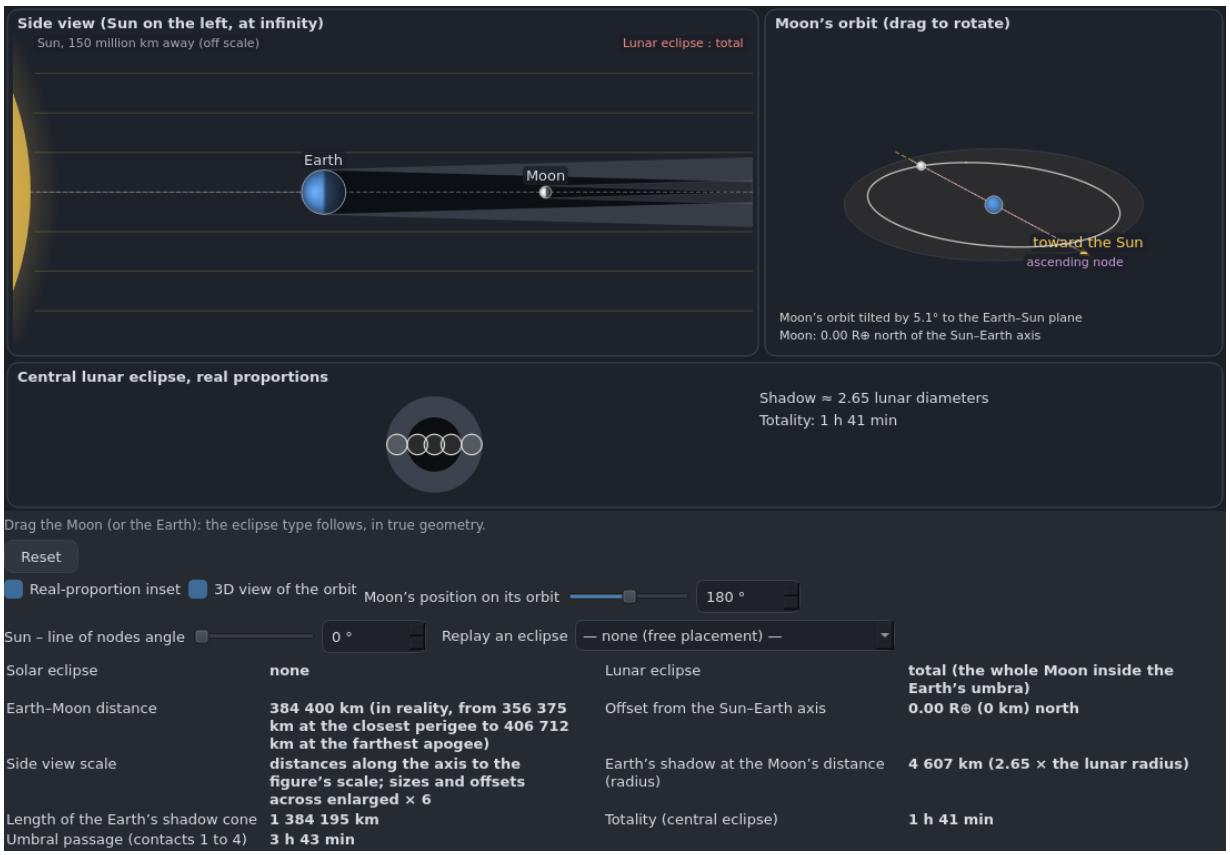
Library. ClassAction's original presentation: one tab per type (Questions, Animations, Outlines, Tables, Images) shows all the content of that type, grouped by module in the original order, with thumbnails; module and "favourites only" filters.



Now and live. The 19 date- or time-dependent animations have the *Now* (the computer's present instant, in universal time, shown in your time zone, no daylight-saving error), *Live* (the time follows the clock; stops as soon as the date is changed or playback restarted) and *My location* (one of the known sites) buttons. An animation set in solar time receives the mean solar time at the time zone meridian (without daylight saving), or at the chosen site's longitude.



Eclipse shadows in true geometry. The eclipse type is computed with real sizes and the Moon's distance as placed: bringing it closer, well aligned, goes from no eclipse to a partial one (penumbra only), an annular one (antumbra), then a total one (umbra), with the path width; the distance is compared with the extreme perigee and apogee (356,375 and 406,712 km, DE440 ephemeris, 1900–2100). The side view draws the cones of the Earth and the Moon (sizes and offsets enlarged ×6, stated); the 3D view shows the orbit tilted by 5.1° and the line of nodes. "Replay an eclipse" places the Moon with the real geometry (JPL DE440); the 133 table eclipses get their type back.

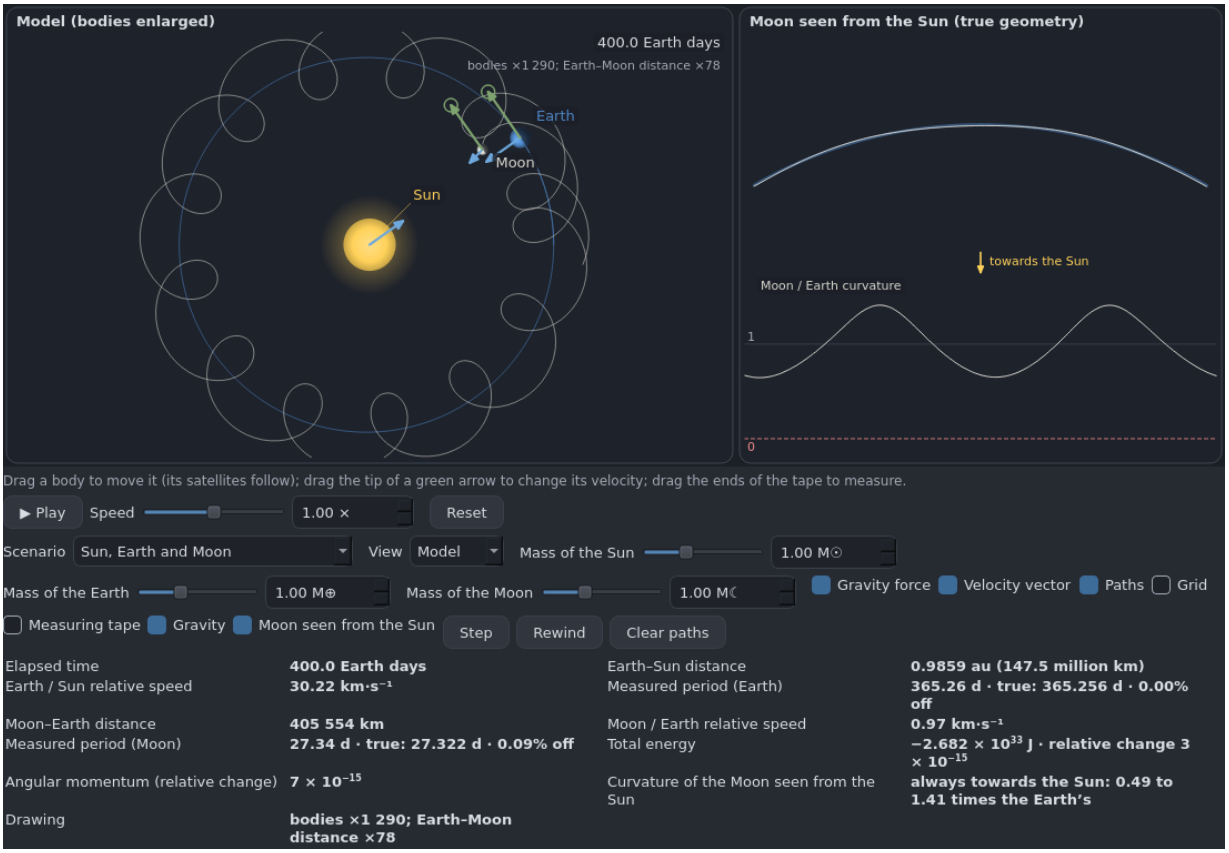


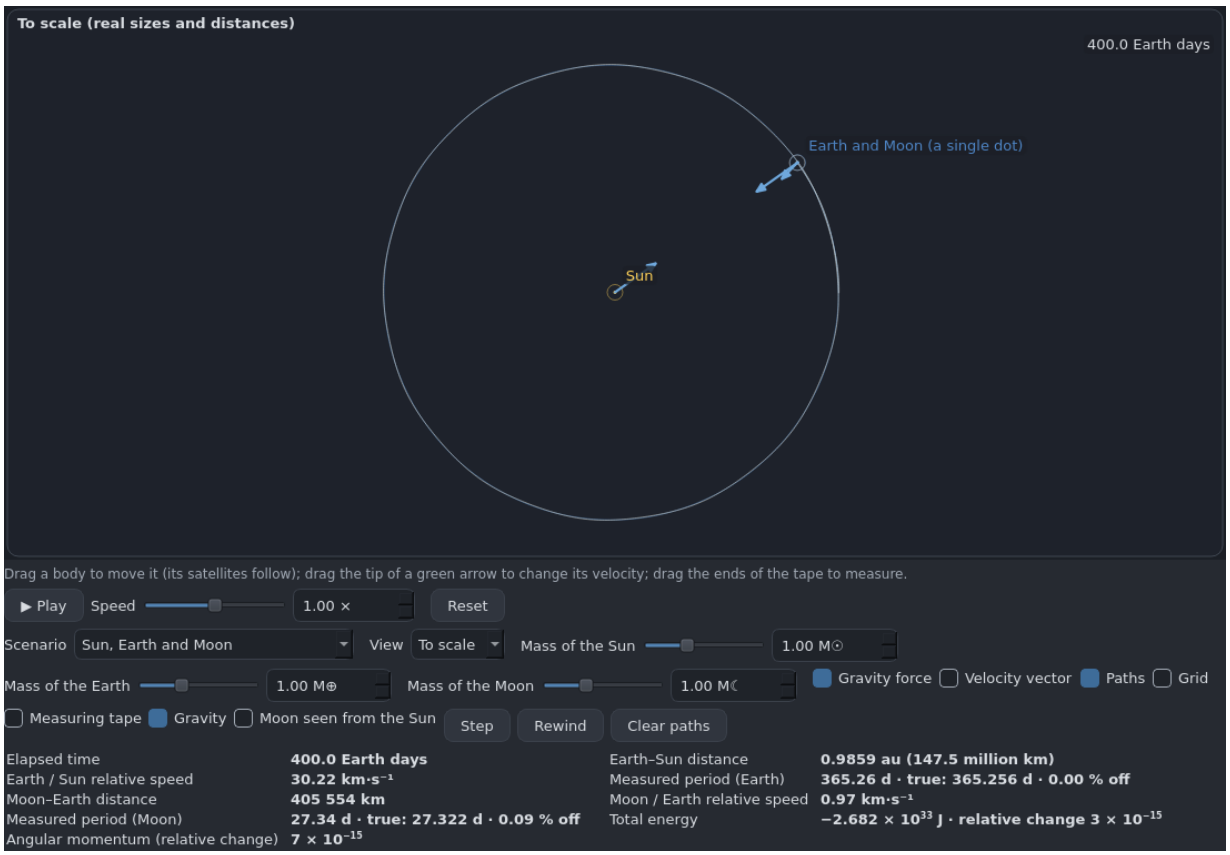
13.7 Gravity and Orbits (0.7.2)

A 95th animation, in the "Renaissance" topic and the "Renaissance Astronomy" module, after the PhET simulation "Gravity and Orbits" of the PhET Interactive Simulations project at the University of Colorado Boulder, shown in class. Only the idea and the list of features are taken up: the code, drawings and texts are Coupole's own, no PhET file or logo is used (PhET licences checked on 15 October 2026 and cited in `NOTICE`).

Scenarios. Sun and Earth; Sun, Earth and Moon; Earth and Moon; Earth and space station (ISS); free (Sun, Earth, Moon and a comet with a 2.4 au aphelion). Pairs start on circular orbits; the Sun, the Earth and the Moon together start from their *real* positions and velocities on 1 January 2026 (JPL DE440 ephemeris, each orbit brought into the ecliptic by a rotation about its line of nodes): a Moon launched on a Keplerian circle would, in the Sun's tidal field, take a distorted orbit whose period depends on the starting phase (27.05 to 27.81 d).

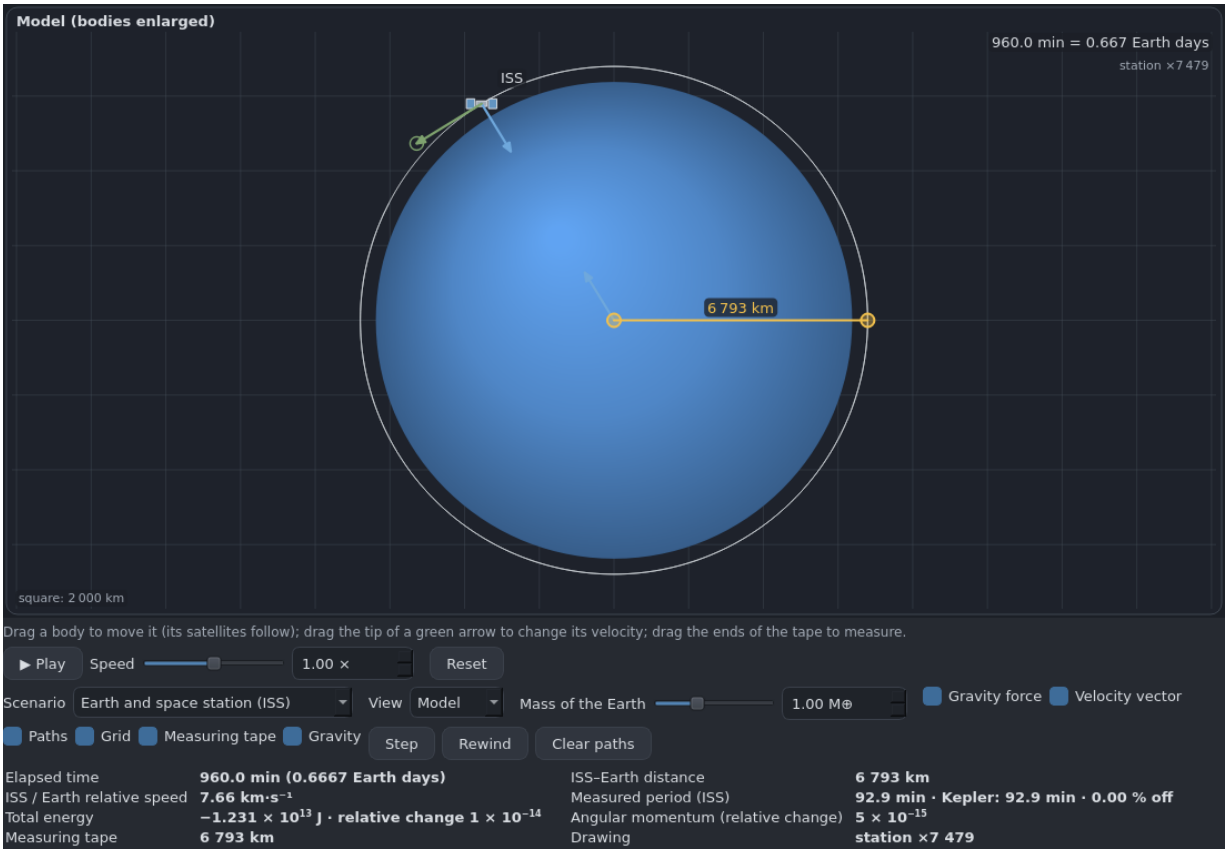
Two views. "Model": bodies enlarged and the Earth–Moon distance enlarged 78 times, to be readable (the ratios are written on the canvas). "To scale": true sizes and distances; seen from the Sun, the Earth and the Moon are a single dot (circled by a faint ring), the Sun is only a small disc, and the ISS skims the Earth. The view moves smoothly from one to the other (radii and gap interpolated on a log scale, 0.9 s), then nothing is redrawn.


Model (bodies enlarged)
400.0 Earth days
bodies ×1 290; Earth–Moon distance ×78
Earth
Moon
Sun
Moon seen from the Sun (true geometry)
towards the Sun
Moon / Earth curvature
1
0
Drag a body to move it (its satellites follow); drag the tip of a green arrow to change its velocity; drag the ends of the tape to measure.
▶ Play
Speed
1.00 ×
Reset
Scenario
Sun, Earth and Moon
View
Model
Mass of the Sun
1.00 M☉
Mass of the Earth
1.00 M⊕
Mass of the Moon
1.00 M☾
Gravity force
Velocity vector
Paths
Grid
Measuring tape
Gravity
Moon seen from the Sun
Step
Rewind
Clear paths
Elapsed time
400.0 Earth days
Earth / Sun relative speed
30.22 km·s⁻¹
Moon–Earth distance
405 554 km
Measured period (Moon)
27.34 d · true: 27.322 d · 0.09% off
Angular momentum (relative change)
7 × 10⁻¹⁵
Drawing
bodies ×1 290; Earth-Moon distance ×78
Earth–Sun distance
0.9859 au (147.5 million km)
Measured period (Earth)
365.26 d · true: 365.256 d · 0.00% off
Moon / Earth relative speed
0.97 km·s⁻¹
Total energy
−2.682 × 10³³ J · relative change 3 × 10⁻¹⁵
Curvature of the Moon seen from the Sun
always towards the Sun: 0.49 to 1.41 times the Earth's



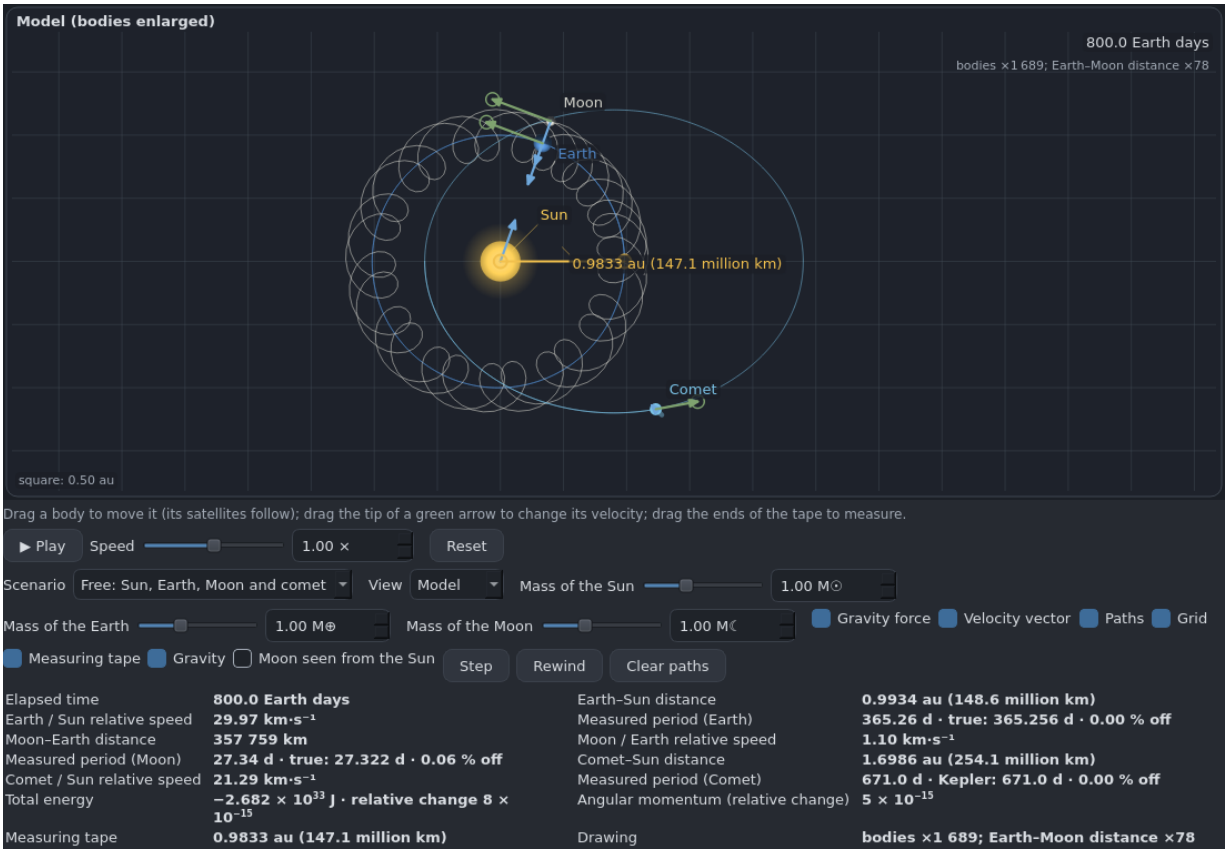
Computation. N bodies under Newtonian gravity, 4th-order Yoshida symplectic scheme (three composed velocity-Verlet steps), adaptive step ($0.01\sqrt{r^3/G(m_1+m_2)}$ for the tightest pair); constants: G (CODATA 2018), nominal $GM_\odot$ and $GM_\oplus$ (IAU 2015), GM_{Moon} (DE440), radii (IAU 2015, NASA), astronomical unit (IAU 2012), ISS (NASA: 419,725 kg at 415 km). Collisions at the real radii: the lighter body crashes, momentum is kept. The numbers are the same, bit for bit, as in Coupole Web.

Settings and mouse. Boxes: gravity force (blue arrows, length relative to the start for each body), velocity vector (green arrows, same scale for all), paths, grid (square size written), measuring tape (real distance), gravity (without it, each body moves off in a straight line), Moon seen from the Sun; masses of the Sun, the Earth and the Moon (0.5 to 2 times the real ones); *Step*, *Rewind* (positions and velocities of the last start), *Clear paths* buttons. Dragging a body moves it (its satellites follow; time is paused while it is held), pulling the tip of a green arrow changes the velocity, dragging the ends of the tape measures.



Moon seen from the Sun. A panel, beside the scene (wide window) or below it (tall window), never over the motion, shows the Moon's real path around the Sun over the last 60 days, at the same scale in width and height: it merges with the Earth's and never loops (the loops of the "Model" view come from the enlarged Earth–Moon distance). Below, the Moon / Earth curvature ratio always stays positive (0.49 to 1.41): the Moon's path always curves towards the Sun, which pulls it 2.2 times harder than the Earth ($GM_\odot/d_\odot^2$ versus $GM_\oplus/d_{\text{Moon}}^2$).

Values and checks. Distances, relative speeds ($\text{km} \cdot \text{s}^{-1}$), measured periods (successive passages through the starting direction seen from the central body) compared with the true ones or with Kepler, total energy and angular momentum with their relative change (about 10^{-14}). Measurements: Earth 365.26 d (sidereal year 365.256 d), Moon 27.34 d (sidereal month 27.322 d), ISS 92.87 min, comet 671.0 d; an independent SageMath integration (DOP853, tolerance 10^{-13}, same initial conditions) gives the same periods to within 2×10^{-6} (`outils/verif_anim/gravite_orbites_verif.sage`).


Model (bodies enlarged)
800.0 Earth days
bodies ×1 689; Earth–Moon distance ×78
Moon
Earth
Sun
0.9833 au (147.1 million km)
Comet
square: 0.50 au
Drag a body to move it (its satellites follow); drag the tip of a green arrow to change its velocity; drag the ends of the tape to measure.
▶ Play
Speed
1.00 ×
Reset
Scenario
Free: Sun, Earth, Moon and comet
View
Model
Mass of the Sun
1.00 M☉
Mass of the Earth
1.00 M⊕
Mass of the Moon
1.00 M☾
Gravity force
Velocity vector
Paths
Grid
Measuring tape
Gravity
Moon seen from the Sun
Step
Rewind
Clear paths
Elapsed time 800.0 Earth days
Earth / Sun relative speed 29.97 km·s⁻¹
Moon–Earth distance 357 759 km
Measured period (Moon) 27.34 d · true: 27.322 d · 0.06 % off
Comet / Sun relative speed 21.29 km·s⁻¹
Total energy −2.682 × 10³³ J · relative change 8 × 10⁻¹⁵
Measuring tape 0.9833 au (147.1 million km)
Earth–Sun distance 0.9934 au (148.6 million km)
Measured period (Earth) 365.26 d · true: 365.256 d · 0.00 % off
Moon / Earth relative speed 1.10 km·s⁻¹
Comet–Sun distance 1.6986 au (254.1 million km)
Measured period (Comet) 671.0 d · Kepler: 671.0 d · 0.00 % off
Angular momentum (relative change) 5 × 10⁻¹⁵
Drawing bodies ×1 689; Earth–Moon distance ×78

13.8 Available animations

Animation	What it shows, what it computes
Longitude / latitude	globe (coastlines, cities, equator, Greenwich meridian, date line); a point to move, coordinates in decimal or sexagesimal
Sun motions	local sphere: daily path, culmination, shadow, months, analemma; altitude, azimuth, α, δ, hour angle, sidereal time, equation of time, sunrise and sunset ($h_0 = -0.833°$), day length
Seasons (NAAP)	orbit (axis fixed in space), Earth from the side or from the Sun, subsolar point, noon altitude $90° - \lvert\varphi - \delta\rvert$, beam spread $1/\sin h$
Lunar phases (NAAP)	orbit seen from above, phase seen from the Earth, observer's sky; illuminated fraction $(1 - \cos\phi)/2$, age (29.53-day lunation), steps of ±1 d, 1 h, 1 min
Sun's rays	parallel rays, latitude of the most direct ray ($\delta_\odot$), tropics and polar circles, 2026 equinoxes and solstices
Azimuth / altitude	horizon system: azimuth arc (from north toward east) and altitude arc; draggable star
Equatorial coordinates	celestial sphere, Earth at the center, α arc (from the 0h hour circle) and δ arc, ecliptic
Coordinate systems comparison	celestial sphere and horizon diagram side by side; circumpolar, rise and set, never rise regions; star trails; Big Dipper, Orion, Southern Cross
Sidereal time and hour angle	$H = \mathrm{LST} - \alpha$ brought between −12 and +12 h, hour angle arc, the sky turns with the LST
Ecliptic	(α, δ) map seen from inside or outside, ecliptic $\tan\delta = \tan\varepsilon \sin\alpha$, path of the Sun between two dates
Obliquity	definition of the angle; Earth (IAU 2006, J2000 and 2026), Mars, Uranus
Ecliptic (zodiac)	Sun in front of the 13 constellations crossed (official IAU boundaries, Ophiuchus included); midnight constellation; 30° signs
Sidereal and solar time	orbit and two 24-hour clocks; sidereal day $86\,400\,N/(N+1)$ s for a year of N days (365, 365.25 or 365.2422)
Declination ranges	cut of the Earth and celestial sphere; limits $\delta = \pm(90° - \lvert\varphi\rvert)$, pole altitude $= \lvert\varphi\rvert$
Eclipse shadows	draggable Earth and Moon, umbra and penumbra from the common tangents, type of eclipse; real-proportion inset (Aristarchus' method)
Ptolemaic Mars	eccentric deferent, equant point, epicycle run in one year, Sun; loops of the path
Epicycles	three bodies, one held still: heliocentric and geocentric give the same relative positions
Retrograde motion	Earth and Mars on their orbits, lines of sight, Mars against the stars (loop with the 1.85° inclination); length and size of the retrograde arc
Planetary orbit (NAAP)	Kepler's laws: ellipse and foci, swept areas, $P^2 = a^3$, speed from the vis-viva relation; planets' orbits
Parallax	$d = 1/p$; arcsecond, parsec, light-year, AU conversions; Proxima, Sirius, Vega
Distance modulus	$m - M = 5\log_{10} d - 5$, one quantity locked, the star on a 1 to 1,000 pc scale
Daylight hours	day length over the year for a latitude, day and night arcs of the parallel, yearly average; geometric horizon or refraction
Meridional altitude	meridian cut: $h = 90° - \varphi + \delta$ (or $180° - h$ north of the zenith), Sun (±23.44°) and Moon (±28.6°) ranges
Paths of the Sun	daily paths at the solstices, equinoxes and on the chosen day, rising and setting points, day lengths
Sun motions overview	four-step construction: poles, celestial equator, equinox path, solstice paths
Position on the horizon	sunrise (or sunset) point sliding along the horizon over the year, spread between the solstices
Daylight simulator	flat map: terminator, subsolar point (latitude $\delta_\odot$, apparent noon), civil, nautical, astronomical twilights
Antipodes	a point and its antipode $(-\varphi, \lambda \pm 180°)$ on the globe and the map, constrained dragging, 5° steps
Big Dipper 3D	the seven stars at their distances (81 to 123 ly, SIMBAD adopted parallaxes) in true colour, lines of sight, projection on the celestial sphere; hover data sheet
Celestial and horizon frames	switching frames by a rotation of $90° - \varphi$ about the east-west axis

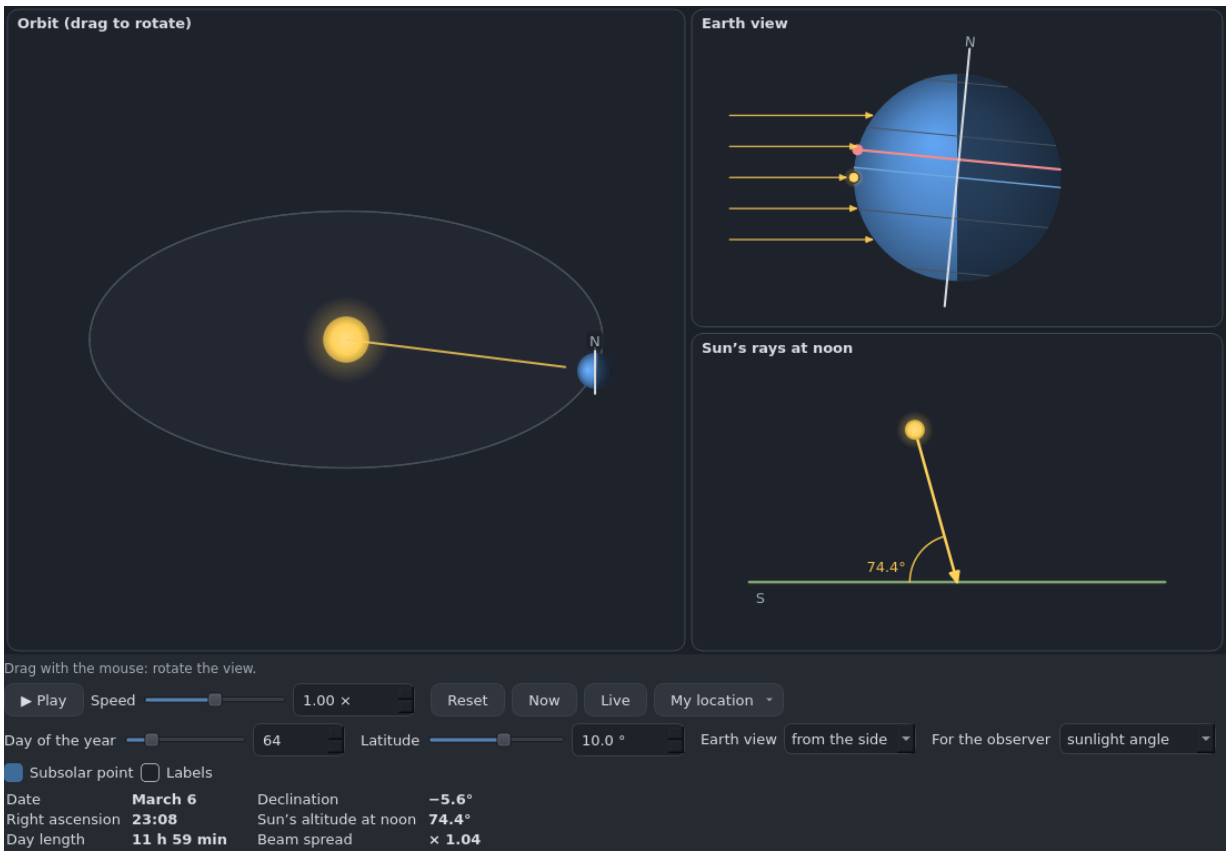
Animation	What it shows, what it computes
Lunar phase vocabulary	the Moon at each of the eight named phases, animated lunation (29.53 s on screen: one second per day), age and days elapsed
Basketball phases	lamp, ball and eye seen from above; what the eye sees, phase angle i, fraction $(1+\cos i)/2$, equivalent lunar phase
Three views	the Moon seen from the Earth, from the Sun (always full) and from a freely placed eye
Phase positions	two bodies to place around the Sun, each seen from the other; very close, one becomes the moon of the other
Phases and the horizon diagram	Sun and Moon on eight positions of the celestial equator: time, phase and position in the sky
Lunar phase quizzer	the phase, the Moon or the Sun is hidden; answer with the slider or the mouse, verdict (same named phase)
Bisecting planes	planes perpendicular to the Sun and to the Earth–Moon line: the phase is the lit part of the half facing us
Full moon declination	opposite the Sun within $\pm 5.145°$; yearly curve, midnight altitude $90° - \varphi + \delta$
Synodic lag	sidereal and solar day: $1/S = 1/P_{\text{rot}} - 1/\text{year}$ (real rotation or slowed down to 240 h or 1,200 h)
Tides	equilibrium tide of the Moon and the Sun $\zeta = (M/M_\oplus)(R/d)^3 R\,(3\cos^2\psi - 1)/2$, tidal force, spring and neap tides, level at an observer
Moon inclination	orbit tilted by 5.145°, draconic month 27.212 d, draconic year 346.62 d; to-scale view from the Sun, umbra and penumbra
Eclipse table	the 133 eclipses from 2015 to 2044 in slanted bands; eclipse seasons (Sun at a node), check with Meeus' mean phases

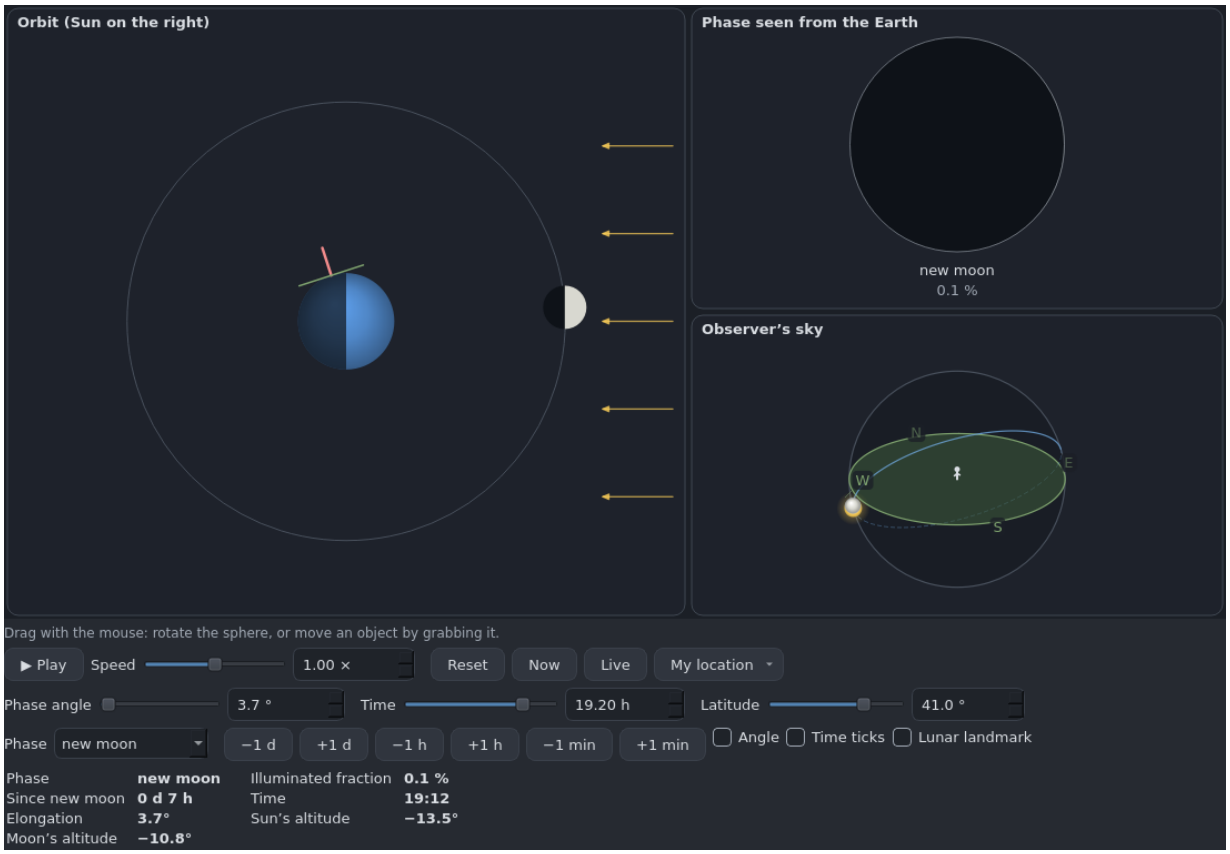
Animation	What it shows, what it computes
Phases of Venus	Copernican system: elongation, phase, apparent diameter (10 to 60″), greatest elongation $\arcsin a$
Ptolemaic Venus	epicycle on the Earth–Sun line: never more than a crescent
Planetary configurations	conjunctions, oppositions, quadratures, greatest elongations; synodic period, stop or pause at each configuration
Synodic period	$1/S = 1/E - 1/P$ (superior) or $1/P - 1/E$ (inferior), E sidereal year (365.256 d)
Kepler's third law	$P^2 = a^3/M$, logarithmic plot, the planets on the line
Eccentricity	$e = c/a$, $b = \sqrt{a^2 - c^2}$, sum of the distances to the foci $= 2a$
Gravity algebra	factors $\frac{1}{3}$ to 3 on M_1, M_2, d: force multiplied by ab/c^2 (exact fraction)
Law of gravity	$F = GM_1M_2/R^2$ from the electron to the Milky Way, accelerations, presets, stored results
Earth orbits	$P = 2\pi\sqrt{r^3/GM}$: Space Station, GPS, geostationary (42,164 km), Moon
Gravity and Orbits	after PhET: Sun, Earth, Moon, ISS, comet under gravity (4th-order Yoshida, real DE440 state on 1/01/2026); model and to-scale views, forces, velocities, paths, tape, gravity on/off, measured periods, Moon seen from the Sun
Small angles	$\alpha \approx 206\,265''\, d/D$ against the exact value $2\arcsin(d/2D)$
Big Dipper clock	northern sky in stereographic projection; one turn per sidereal day, about 1° per day at a fixed time
Heliacal rising	first morning the star is up at dawn (Sun at −7°); whole year, Sirius, Vega…
Shadow of a building	instead of the original movie: the cast shadow day after day, at a fixed time or at true noon

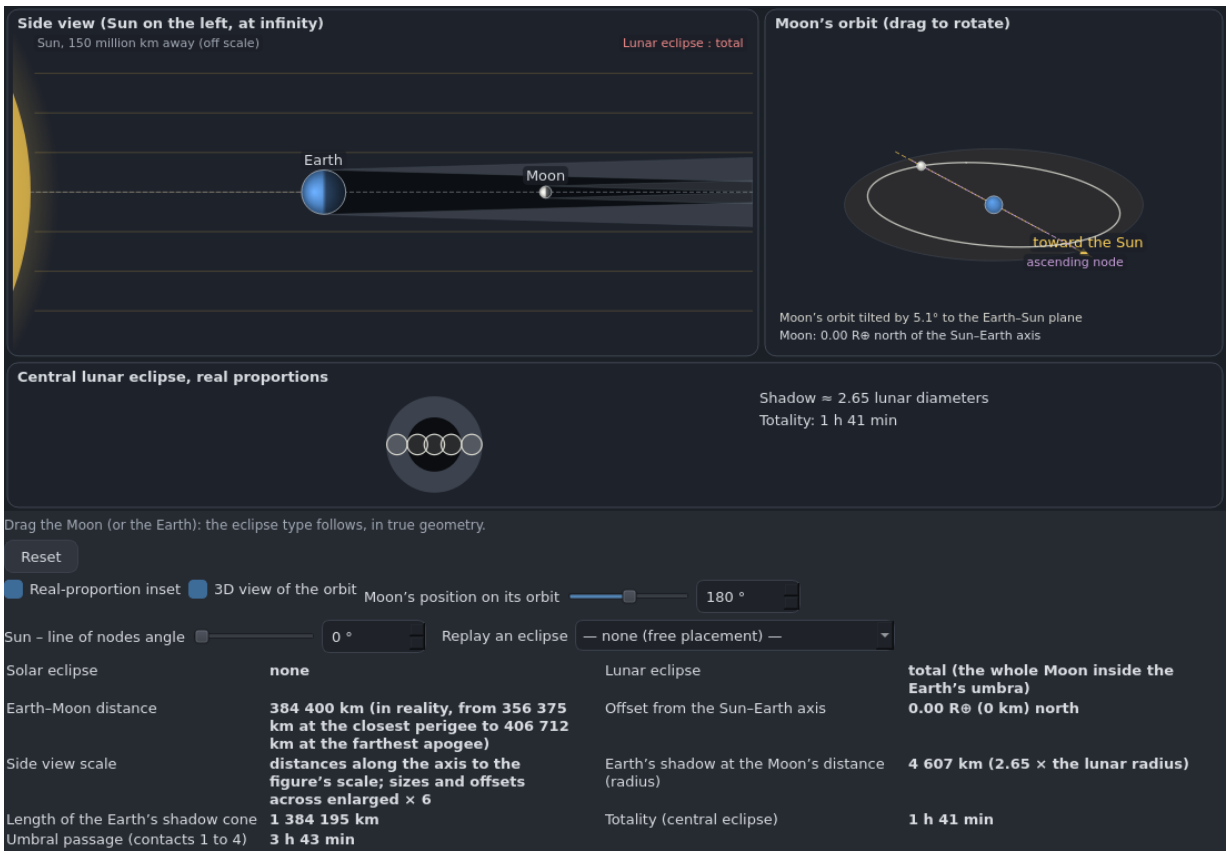
Animation	What it shows, what it computes
Stellar luminosity	$L/L_\odot = (R/R_\odot)^2(T/T_\odot)^4$, $L = 4\pi R^2\sigma T^4$
Flux	a bulb and a detector: $F = P/4\pi d^2$
Stellar velocity	$v_r = c\,\Delta\lambda/\lambda_0$ (Hα), $v_t = 4.74\,\mu d$, $v = \sqrt{v_r^2 + v_t^2}$
Spectroscopic parallax	type and class → published M_V (or former calibration, flagged); $d = 10^{(m-M+5)/5}$ pc; computed spectrum
HR diagram (NAAP)	nearest and brightest stars, sequences, lines of equal radius, $R/R_\odot = \sqrt{L/L_\odot}/(T/T_\odot)^2$

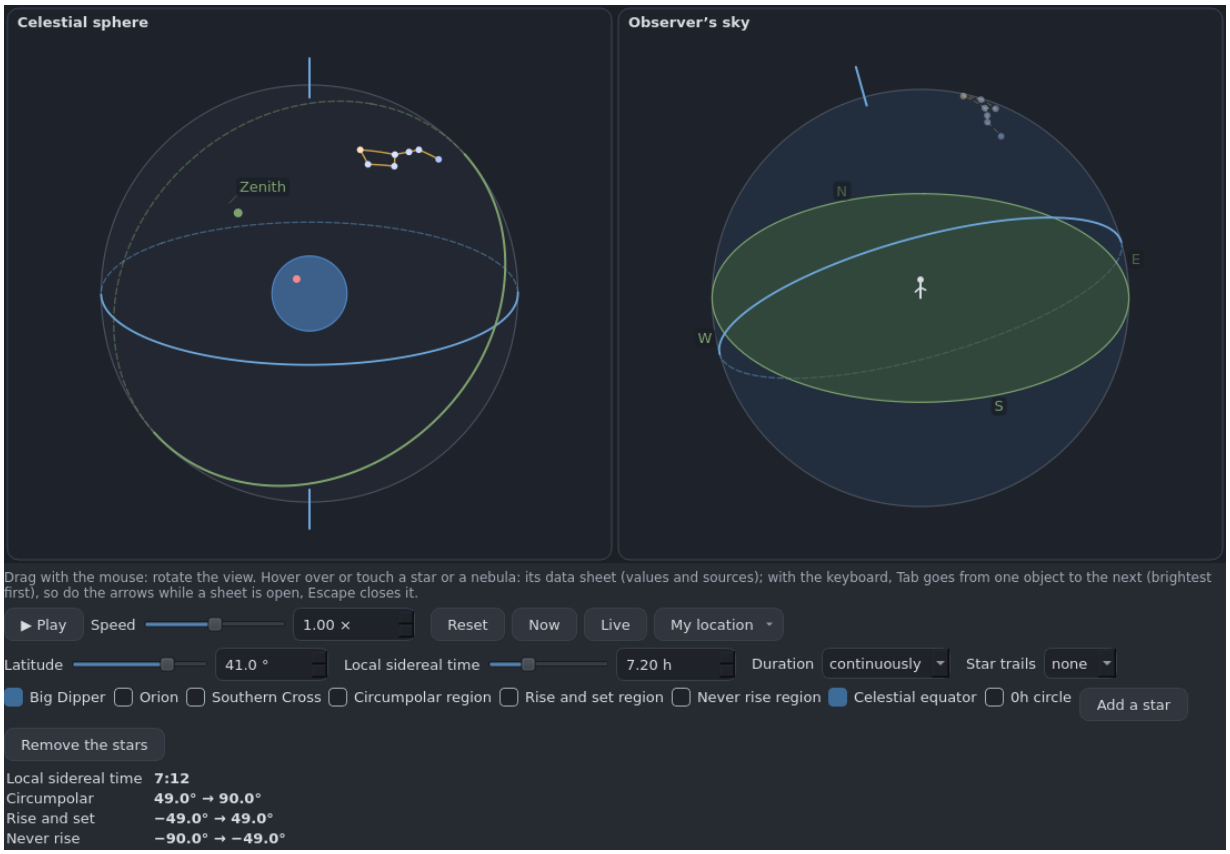
Animation	What it shows, what it computes
Melting nail	instead of the movie: colour and glow of the nail from Planck's law and the CIE colour matching functions (invisible below ≈ 800 K); Planck curve, $\lambda_{\max} = b/T$, share radiated in the visible
Blackbody curves (NAAP)	up to five curves, peak (Wien), area $\sigma T^4/\pi$, perceived colour; U, B, V, R magnitudes and colour indices
Spectrum explorer	continuous, emission, absorption; six families of lines (He II, He I, H, ionized and neutral metals, molecules) computed with Saha–Boltzmann versus spectral type and class; 133 NIST lines, TiO and CH bands
Electromagnetic spectrum	$\nu = c/\lambda$, $E = h\nu$, blackbody peaking there, band; the wave drawn next to an object of the same size
Doppler effect	crests emitted by a moving source, signal received through the retarded time $\lvert\vec{r}_{\text{obs}} - \vec{r}_s(\tau)\rvert = c(t-\tau)$: $f = f_0/(1 - v_r/c)$; draggable observer
Filters (NAAP)	source (blackbody, Gaussian, Sun) × filters (U, B, V, R, atmosphere, Gaussian) = received spectrum; perceived colour (CIE, sRGB)
Hydrogen atom (NAAP)	photon absorbed if $E = 13.598\,(1/n^2 - 1/m^2)$ eV, ionization beyond, re-emission; energy levels, log
Three views of spectra (Kirchhoff)	light bulb, gas cloud, draggable telescope: continuous, emission or absorption at the same wavelengths
Hammer thrower	instead of the movie: athlete and hammer (7.26 kg), star and planet, around the center of mass; $r_1/r_2 = m_2/m_1$
Radial velocity graph	Sun and Jupiter ($K = 12.5$ m/s, 11.86 yr), Sun and Earth, 51 Pegasi b; $K = (2\pi G/P)^{1/3} M_p \sin i/(M_* + M_p)^{2/3}$
The Sun and the planets	Sun's center around the barycenter (true positions of the planets, Standish elements, DE440 masses, 1800–2050); up to 2.2 $R_\odot$
Pulsar period	pulses emitted along an orbit: arrival intervals $P(1 \pm v/c)$ (exact travel time)
Spectral line shifts	star and planet seen from above; spectral lines shifted by $\Delta\lambda/\lambda = v/c$ (magnified), real shift of Hα
Transits (NAAP)	depth $(R_p/R_*)^2$ (exact overlap area), duration, impact parameter; 9 real planets (NASA Exoplanet Archive); noisy measurements
Radial velocities (NAAP)	$v_r = K[\cos(\nu + \omega) + e\cos\omega]$, eccentric and inclined orbits, $M_p \sin i$; real planets, noisy measurements

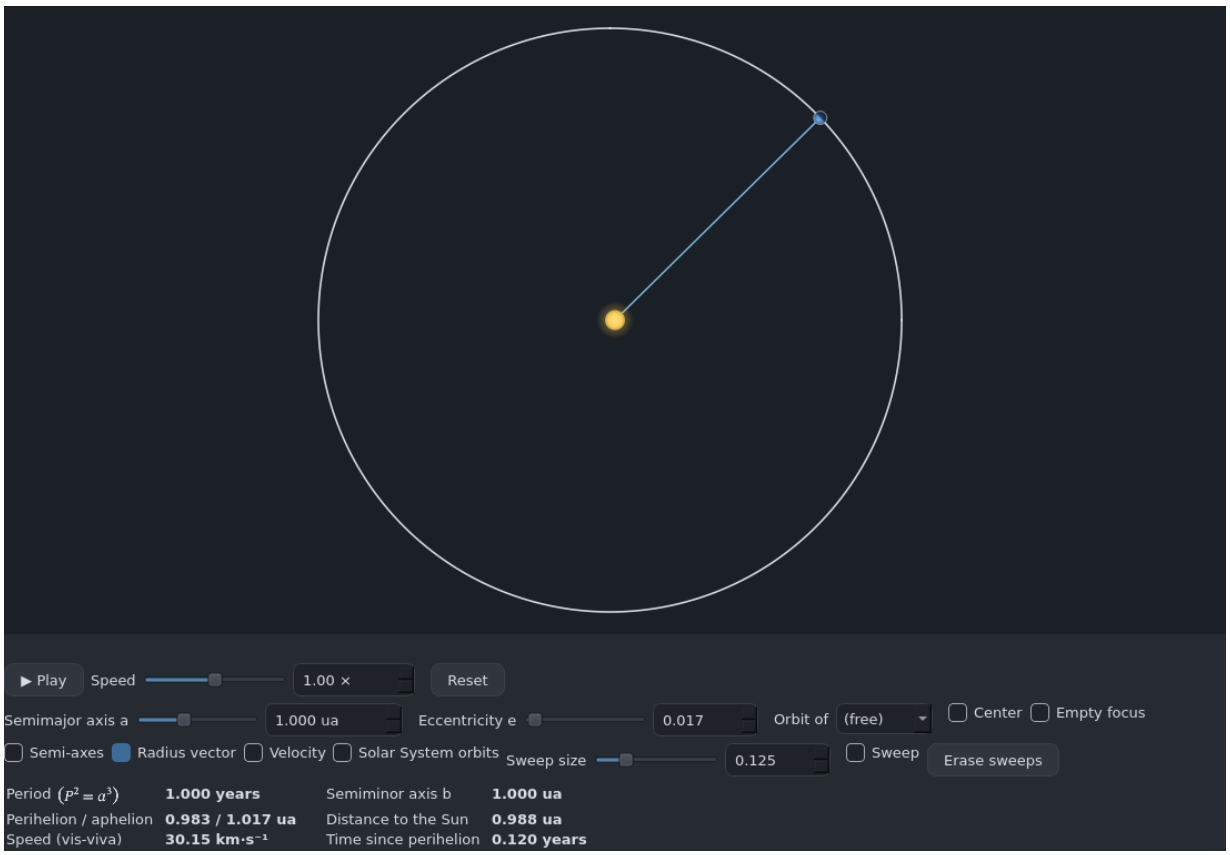
Animation	What it shows, what it computes
Eclipsing binaries (NAAP)	two stars and the light curve (V filter, surface brightness from Planck's law); 46 real binaries and their measurements (CALEB), 7 of them « to complete »; model–data deviation
Light curve	a pair with a 3.65 d period observed measurement after measurement, each point plotted at its phase
Center of mass	$r_1 = d\,m_2/(m_1+m_2)$, $m_1 r_1 = m_2 r_2$; the pair turns around it
Aliasing (wagon wheel)	wheel seen frame by frame: apparent rotation folded into $]-180°/n, 180°/n]$, wheel stopped or backwards; same trap for a variable star
Planet properties	histograms: distance, period, mass, radius, moons (Jupiter 95, Saturn 274), rotation, density (NASA, 2026)
Formation temperatures	$T = 600\ \mathrm{K}\,(d/1\ \mathrm{au})^{-0.747}$; state (gas or solid) of nine substances, snow line
Driving through snow	the flakes seem to burst from one point: radiant at $\arctan(v_{\mathrm{snow}}/v_{\mathrm{car}})$, like a meteor shower
Proton-proton chain, CNO cycle	instead of the movies: each reaction, energy released (AME 2020), net 26.731 MeV, share of the neutrinos
Orion in 3D	the constellation unfolded to its true distances, error bars, Orion A and B clouds, M42 as a "blister", Horsehead, λ Ori, Barnard's Loop; dive with ESO and Hubble images; hover data sheets
Milky Way rotation	McMillan (2017) model and its components, Eilers et al. (2019) and galkin measurements, $M = v^2 R/G$, Keplerian curve; Galaxy seen from the pole (bar, Reid et al. 2019 arms)
Density waves	tractor and cars: $n_{\mathrm{jam}}/n = (v - v_t)/(v_{\mathrm{slow}} - v_t)$, analogy of the spiral arms
Balloon universe	instead of the movie: galaxies on an inflating balloon, $v = Hd$, $H = \dot{a}/a$; the balloon turns with the mouse
Galactic redshift	spectrum of M32 shifted by $1+z$, lines, light received in U, B, V, R; speed from the relativistic Doppler effect
Earth's atmosphere	airmass (sec z, Kasten & Young, Pickering), extinction $\Delta m = kX$ and reddening, atmospheric windows (ESO SkyCalc), refraction, flattened Sun, earlier rising, dispersion
Telescope	magnification F/f, true field, resolving power $1.22\lambda/D$, light-gathering power, focus; Moon, Saturn, globular cluster
CCD sensor	exposure (random photons, Poisson law, saturation) then readout row by row and pixel by pixel
Snell's law	$n_1 \sin\theta_1 = n_2 \sin\theta_2$, total internal reflection, share reflected (Fresnel), Brewster angle
Lookback time	the explosion of a star d light-years away seen d years later; supernovae of 1054 and 1006

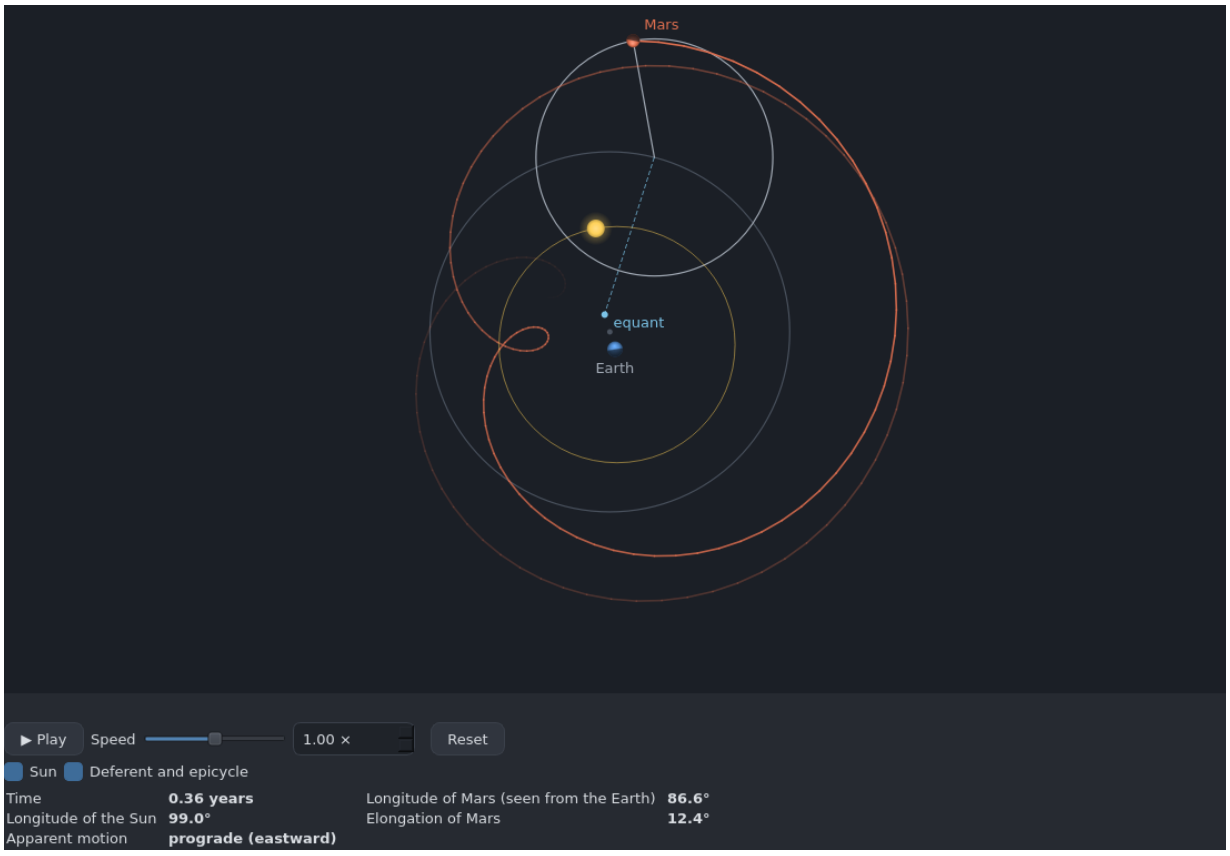

Orbit (drag to rotate)
N
Earth view
N
Sun's rays at noon
74.4°
S
Drag with the mouse: rotate the view.
▶ Play
Speed
1.00 ×
Reset
Now
Live
My location
Day of the year
64
Latitude
10.0 °
Earth view
from the side
For the observer
sunlight angle
Subsolar point
Labels
Date March 6
Right ascension 23:08
Day length 11 h 59 min
Declination −5.6°
Sun's altitude at noon 74.4°
Beam spread × 1.04

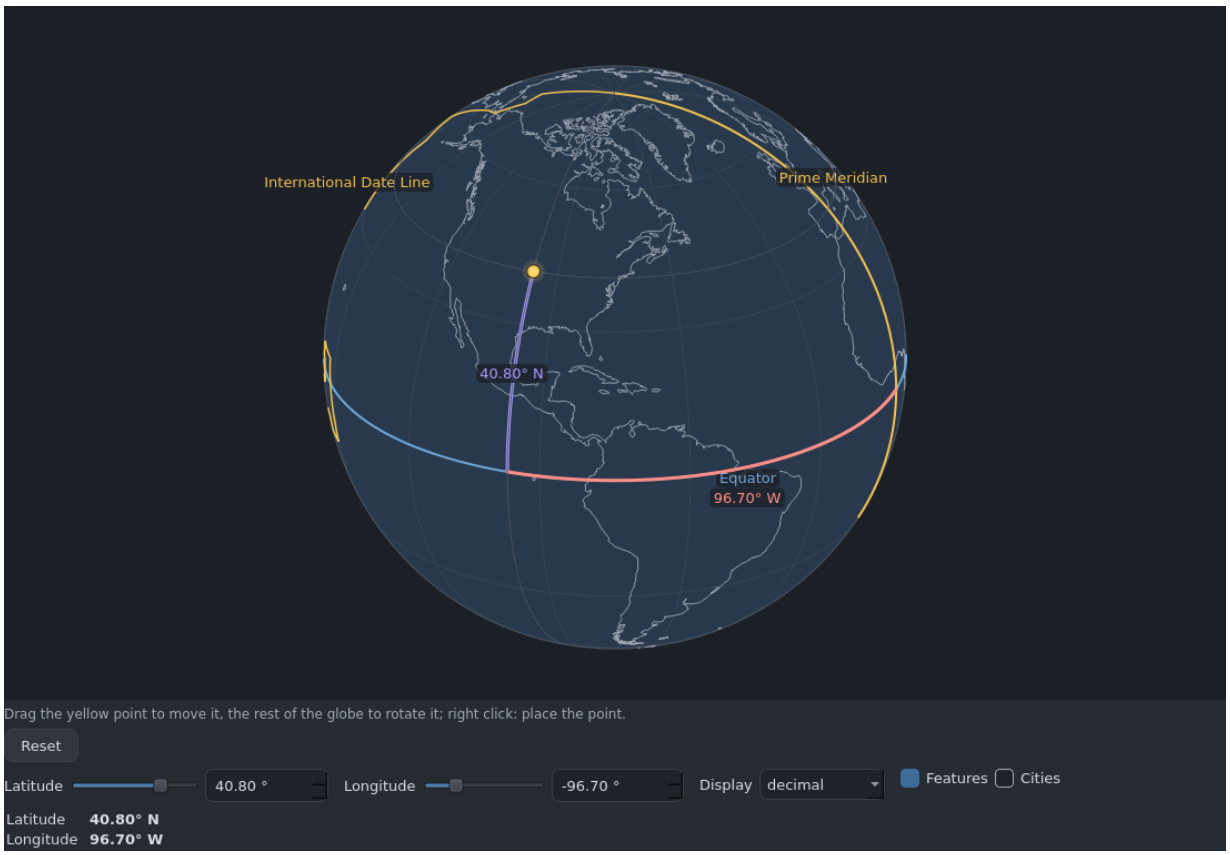

Orbit (Sun on the right)
Phase seen from the Earth
new moon
0.1 %
Observer's sky
N
E
W
S
Drag with the mouse: rotate the sphere, or move an object by grabbing it.
▶ Play
Speed
1.00 ×
Reset
Now
Live
My location
Phase angle
3.7 °
Time
19.20 h
Latitude
41.0 °
Phase
new moon
−1 d
+1 d
−1 h
+1 h
−1 min
+1 min
Angle
Time ticks
Lunar landmark
Phase new moon
Since new moon 0 d 7 h
Elongation 3.7°
Moon's altitude −10.8°
Illuminated fraction 0.1 %
Time 19:12
Sun's altitude −13.5°

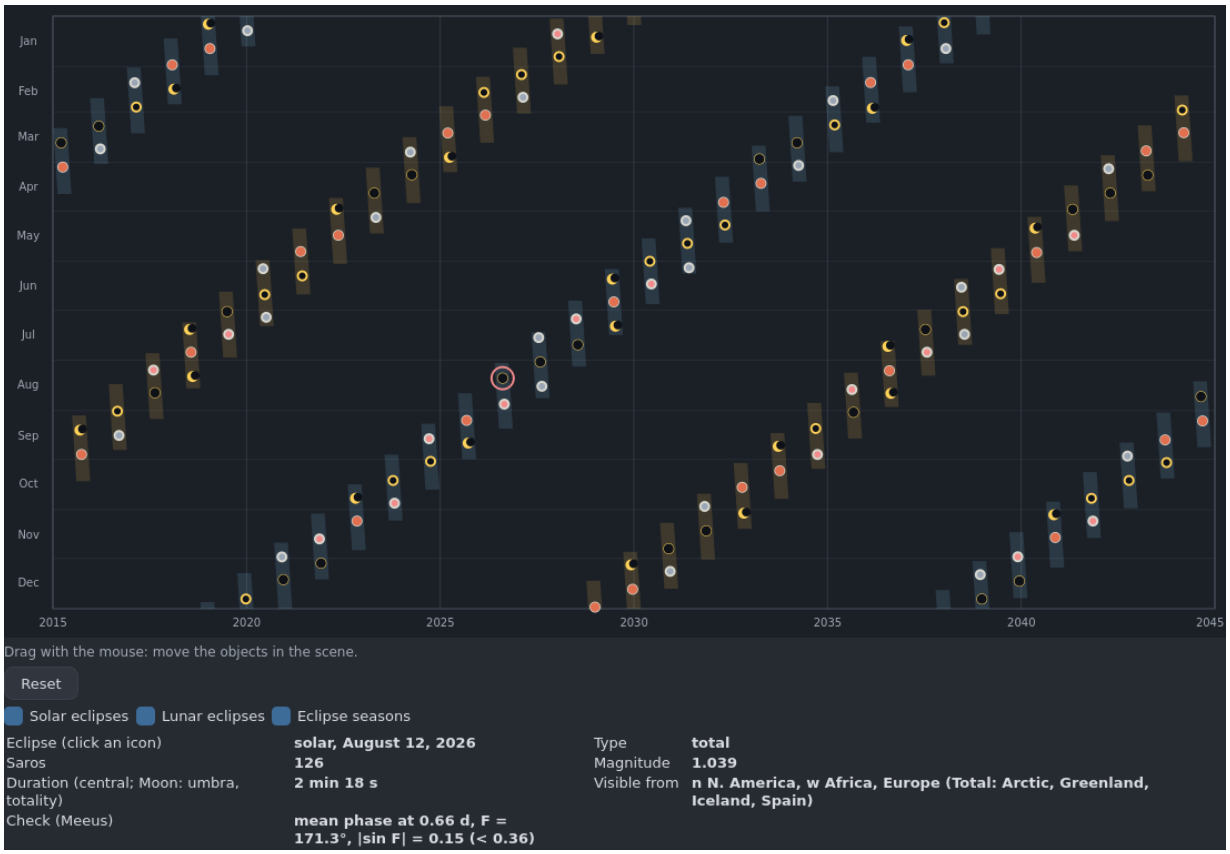

Side view (Sun on the left, at infinity)
Sun, 150 million km away (off scale)
Lunar eclipse : total
Earth
Moon
Moon's orbit (drag to rotate)
toward the Sun
ascending node
Moon's orbit tilted by 5.1° to the Earth–Sun plane
Moon: 0.00 R⊕ north of the Sun–Earth axis
Central lunar eclipse, real proportions
Shadow ≈ 2.65 lunar diameters
Totality: 1 h 41 min
Drag the Moon (or the Earth): the eclipse type follows, in true geometry.
Reset
Real-proportion inset
3D view of the orbit
Moon's position on its orbit
180 °
Sun – line of nodes angle
0 °
Replay an eclipse
— none (free placement) —
Solar eclipse
none
Lunar eclipse
total (the whole Moon inside the Earth's umbra)
Earth–Moon distance
384 400 km (in reality, from 356 375 km at the closest perigee to 406 712 km at the farthest apogee)
Offset from the Sun–Earth axis
0.00 R⊕ (0 km) north
Side view scale
distances along the axis to the figure's scale; sizes and offsets across enlarged × 6
Earth's shadow at the Moon's distance (radius)
4 607 km (2.65 × the lunar radius)
Length of the Earth's shadow cone
1 384 195 km
Totality (central eclipse)
1 h 41 min
Umbral passage (contacts 1 to 4)
3 h 43 min

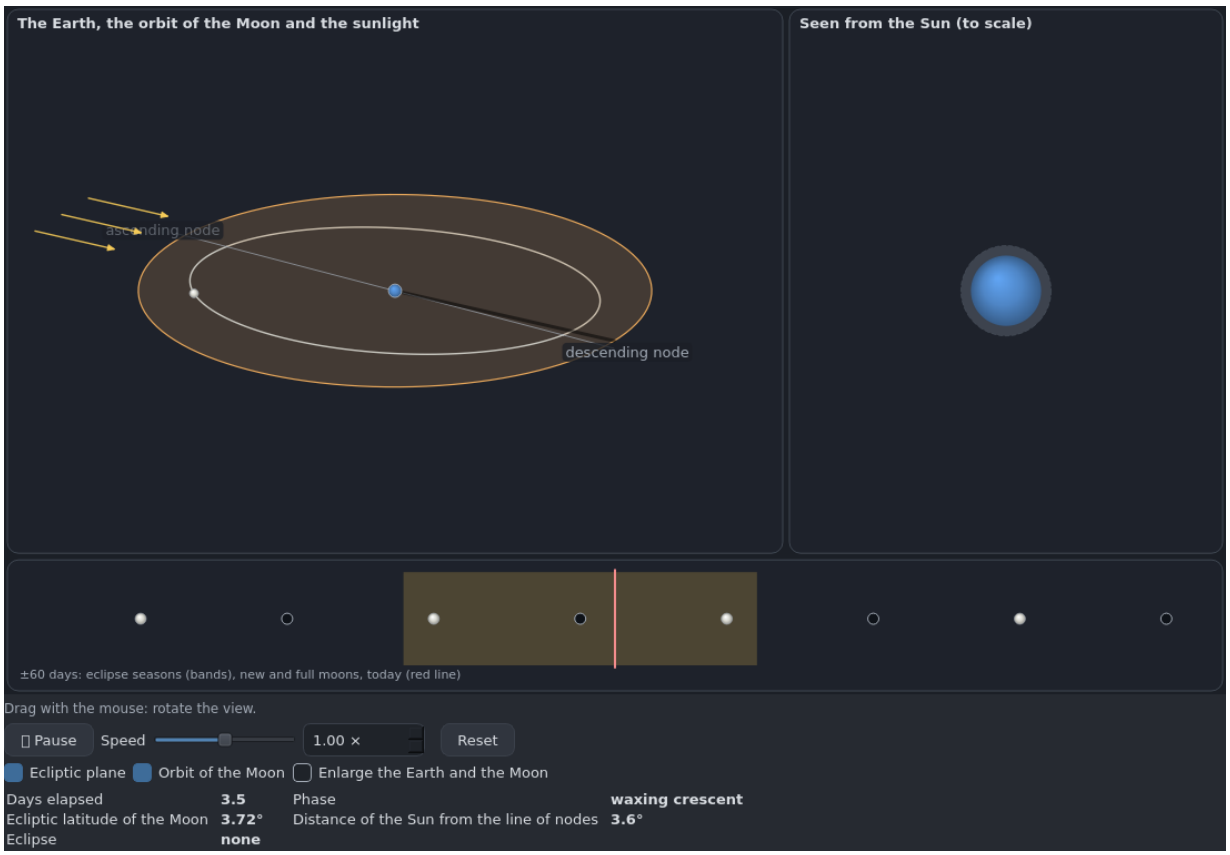

Celestial sphere
Zenith
Observer's sky
N
E
W
S
Drag with the mouse: rotate the view. Hover over or touch a star or a nebula: its data sheet (values and sources); with the keyboard, Tab goes from one object to the next (brightest first), so do the arrows while a sheet is open, Escape closes it.
▶ Play
Speed
1.00 ×
Reset
Now
Live
My location
Latitude
41.0 °
Local sidereal time
7.20 h
Duration
continuously
Star trails
none
Big Dipper
Orion
Southern Cross
Circumpolar region
Rise and set region
Never rise region
Celestial equator
0h circle
Add a star
Remove the stars
Local sidereal time 7:12
Circumpolar 49.0° → 90.0°
Rise and set −49.0° → 49.0°
Never rise −90.0° → −49.0°


▶ Play
Speed
1.00 ×
Reset
Semimajor axis a
1.000 ua
Eccentricity e
0.017
Orbit of
(free)
Center
Empty focus
Semi-axes
Radius vector
Velocity
Solar System orbits
Sweep size
0.125
Sweep
Erase sweeps
Period $(P^2 = a^3)$ 1.000 years
Semiminor axis b 1.000 ua
Perihelion / aphelion 0.983 / 1.017 ua
Distance to the Sun 0.988 ua
Speed (vis-viva) 30.15 km·s⁻¹
Time since perihelion 0.120 years

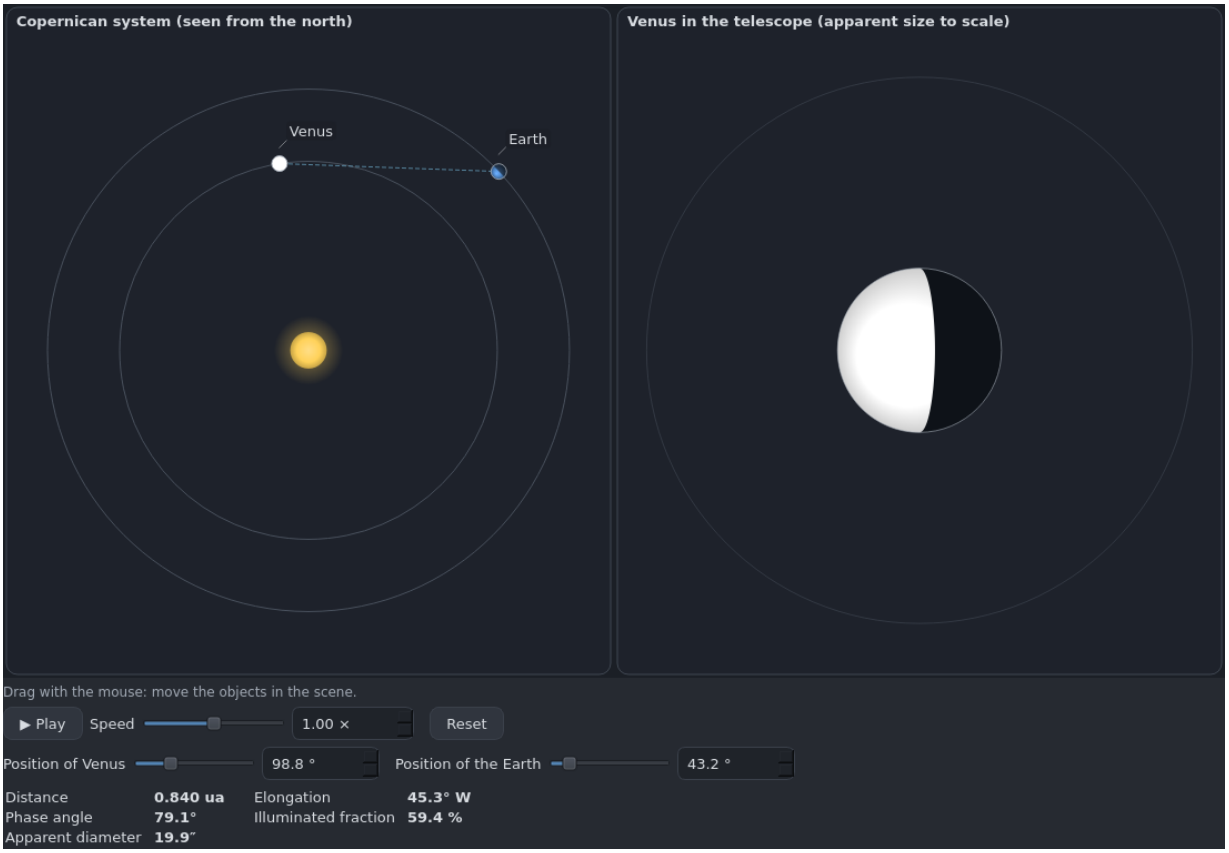

Mars
equant
Earth
▶ Play
Speed
1.00 ×
Reset
Sun
Deferent and epicycle
Time 0.36 years
Longitude of the Sun 99.0°
Apparent motion prograde (eastward)
Longitude of Mars (seen from the Earth) 86.6°
Elongation of Mars 12.4°

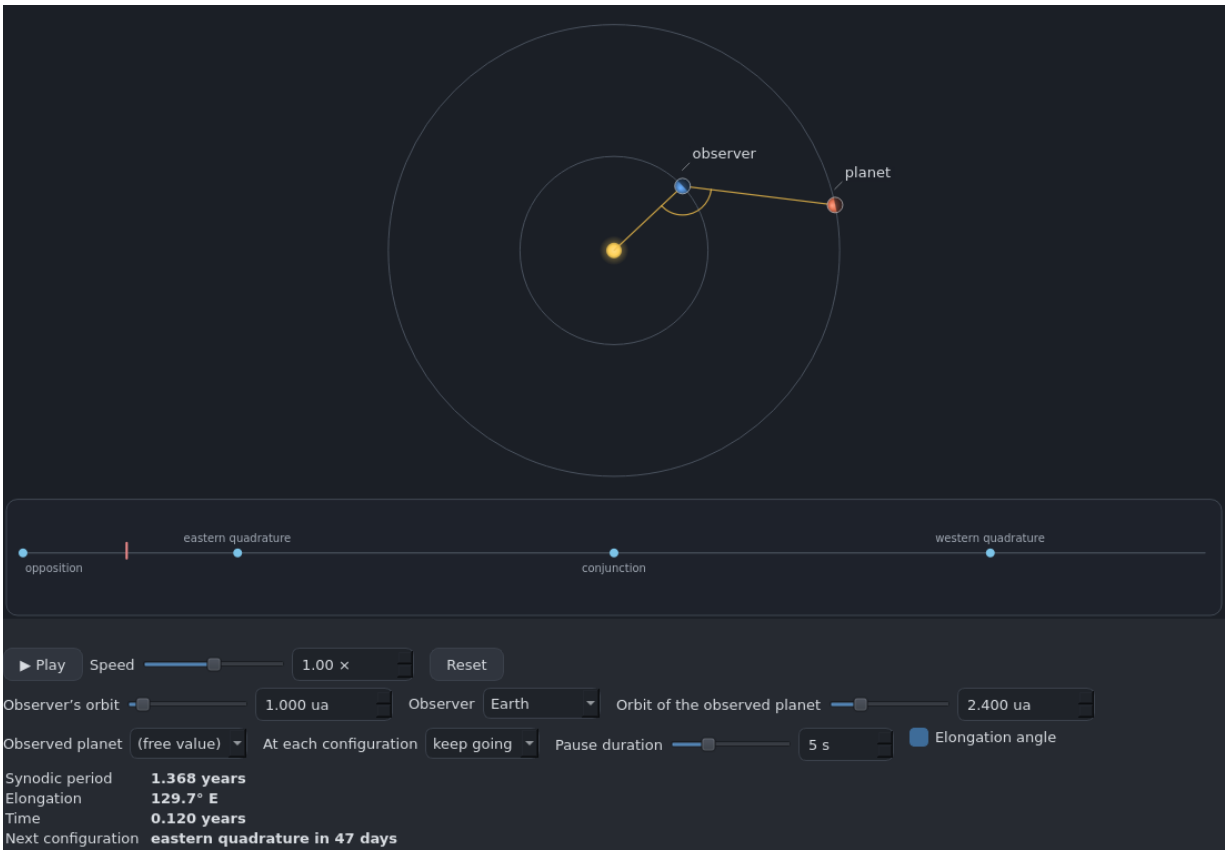

International Date Line
Prime Meridian
40.80° N
Equator
96.70° W
Drag the yellow point to move it, the rest of the globe to rotate it; right click: place the point.
Reset
Latitude
40.80 °
Longitude
-96.70 °
Display
decimal
Features
Cities
Latitude 40.80° N
Longitude 96.70° W

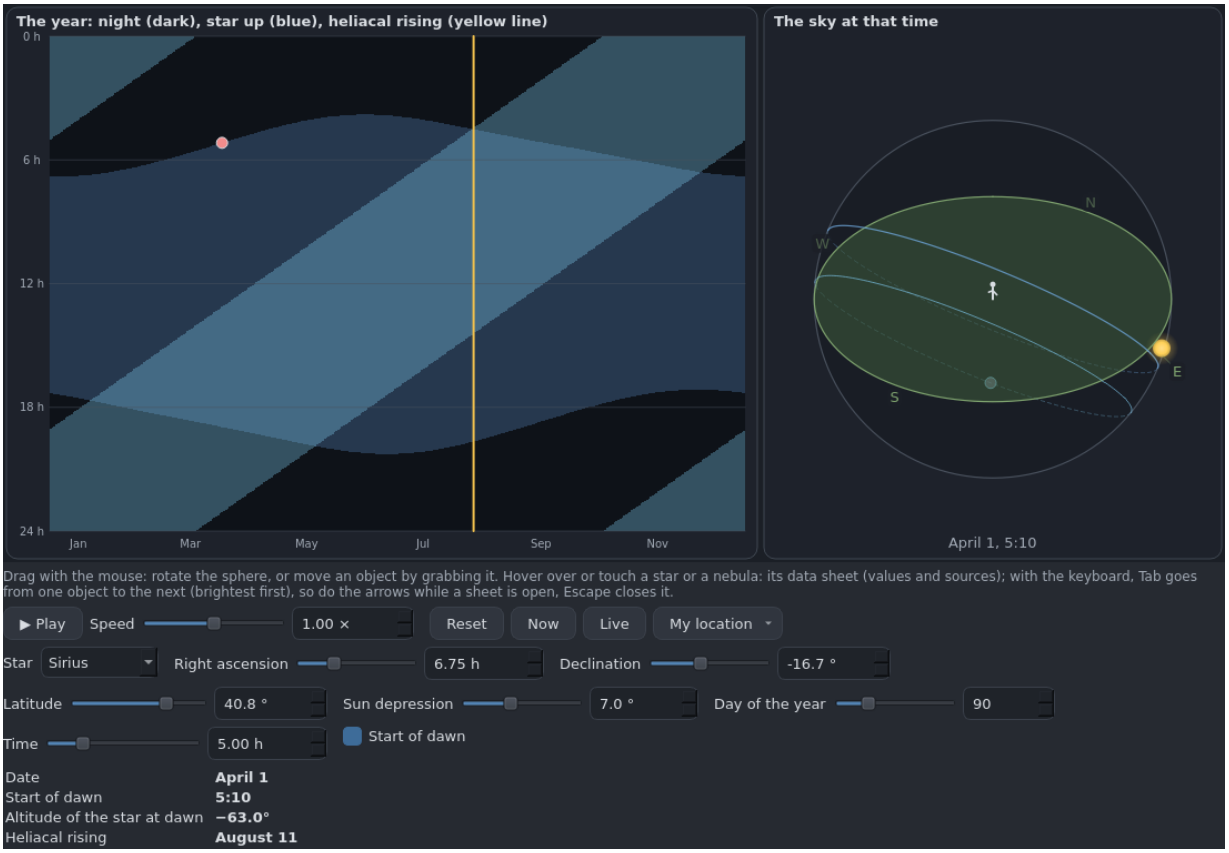

Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
2015
2020
2025
2030
2035
2040
2045
Drag with the mouse: move the objects in the scene.
Reset
Solar eclipses
Lunar eclipses
Eclipse seasons
Eclipse (click an icon) solar, August 12, 2026
Saros 126
Duration (central; Moon: umbra, totality) 2 min 18 s
Check (Meeus) mean phase at 0.66 d, F = 171.3°, |sin F| = 0.15 (< 0.36)
Type total
Magnitude 1.039
Visible from n N. America, w Africa, Europe (Total: Arctic, Greenland, Iceland, Spain)

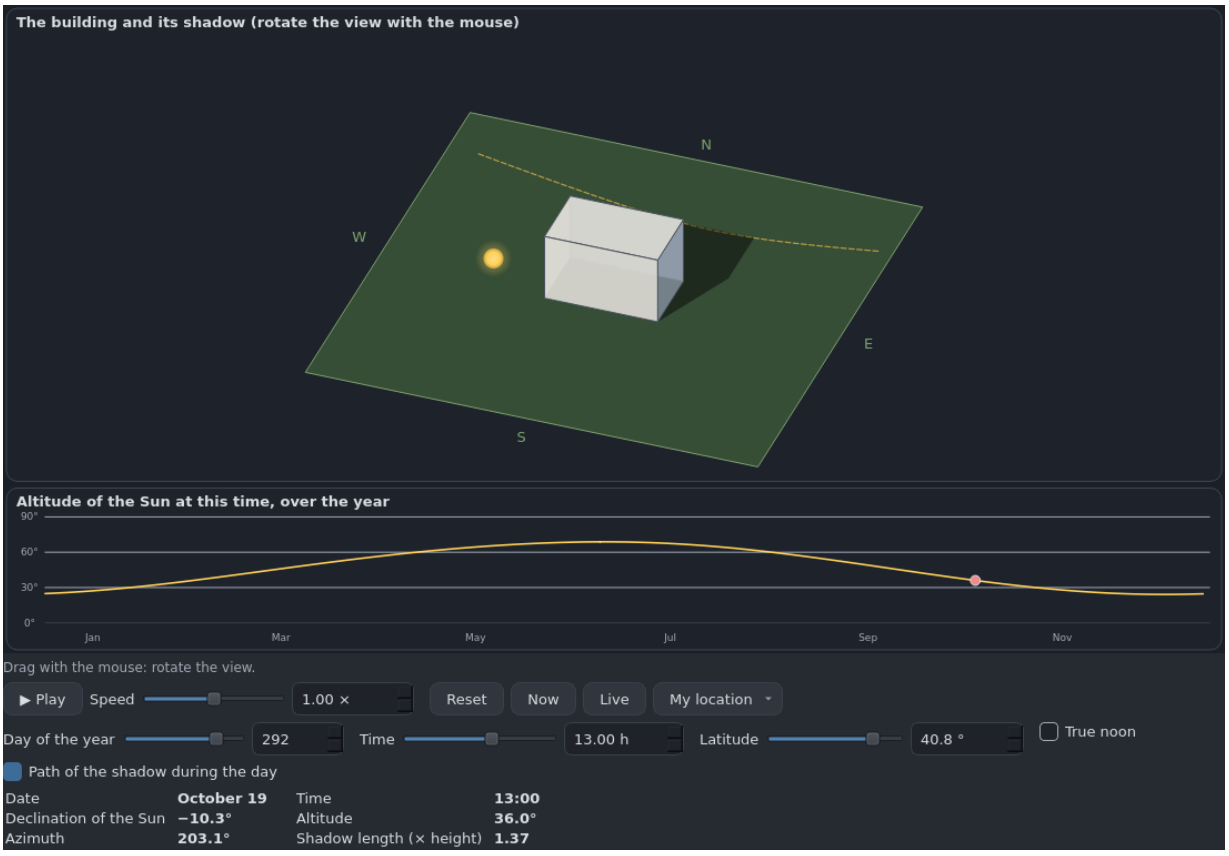

The Earth, the orbit of the Moon and the sunlight
ascending node
descending node
Seen from the Sun (to scale)
±60 days: eclipse seasons (bands), new and full moons, today (red line)
Drag with the mouse: rotate the view.
Pause
Speed
1.00 ×
Reset
Ecliptic plane
Orbit of the Moon
Enlarge the Earth and the Moon
Days elapsed 3.5
Phase waxing crescent
Ecliptic latitude of the Moon 3.72°
Distance of the Sun from the line of nodes 3.6°
Eclipse none

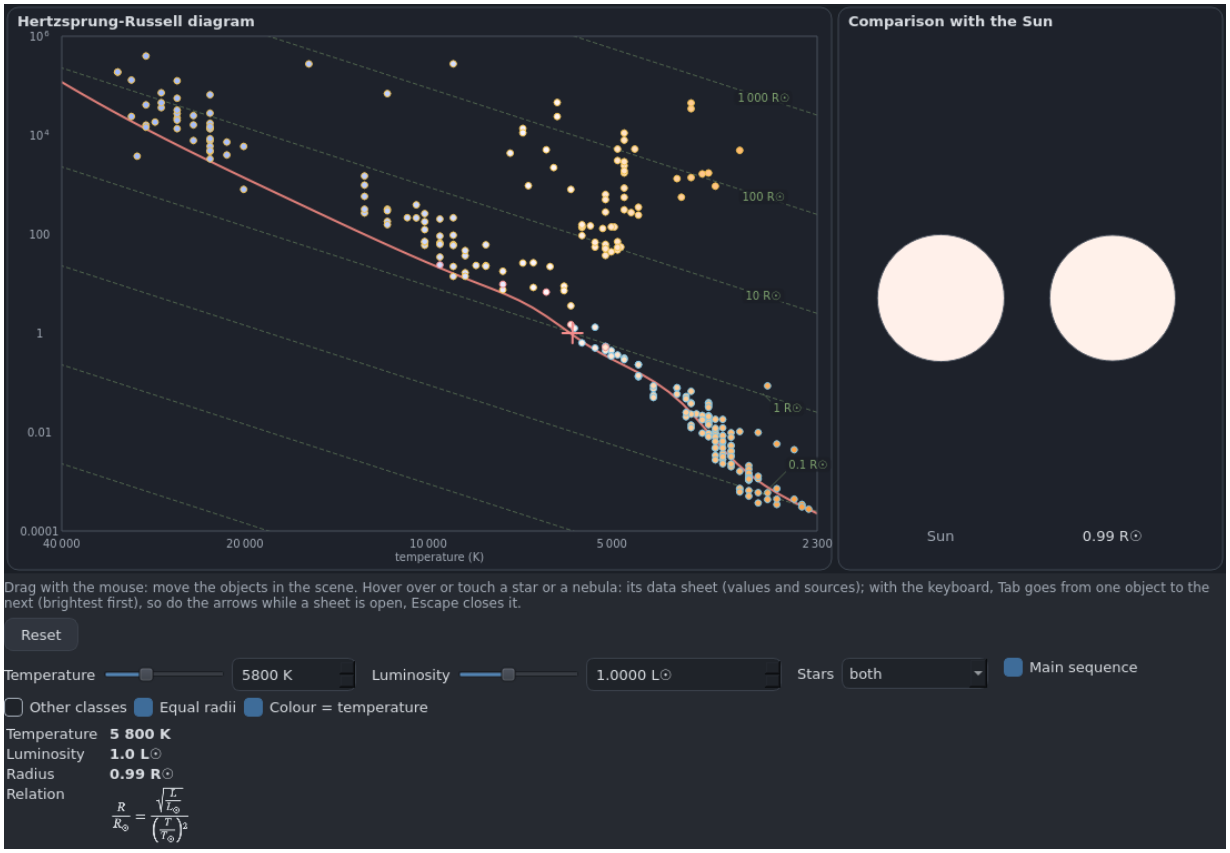

Sea level at the observer (red dot), ±1.5 day
m
0.5
−0.5
▶ Play
Speed
1.00 ×
Reset
With the Sun
Tidal force
Bulge offset
0 °
Exaggeration
4.0 ×
Time 7:12
Phase new moon
Level at the observer −14.6 cm
Tidal range from the Moon alone 53.8 cm
Sun / Moon ratio 0.459


Copernican system (seen from the north)
Venus
Earth
Venus in the telescope (apparent size to scale)
Drag with the mouse: move the objects in the scene.
▶ Play
Speed
1.00 ×
Reset
Position of Venus
98.8 °
Position of the Earth
43.2 °
Distance 0.840 ua
Phase angle 79.1°
Apparent diameter 19.9″
Elongation 45.3° W
Illuminated fraction 59.4 %

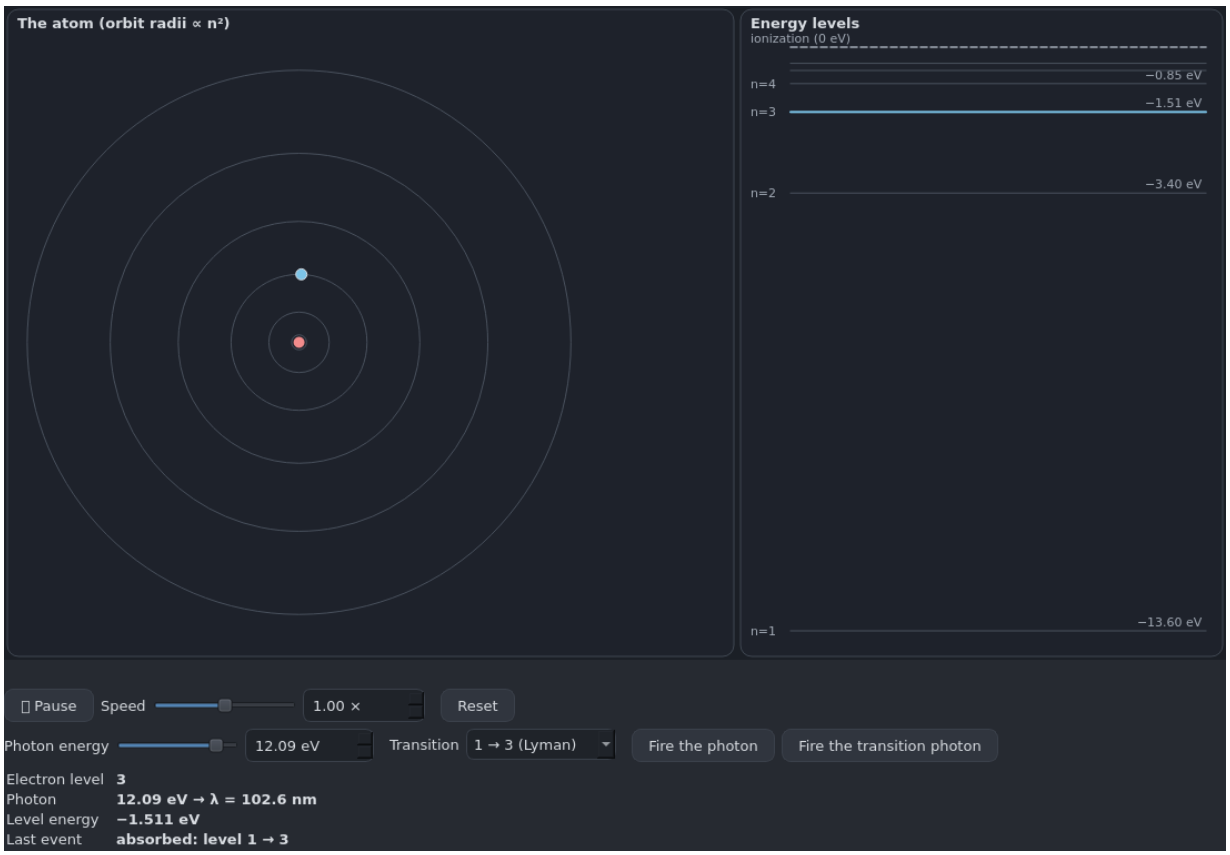

observer
planet
opposition
eastern quadrature
conjunction
western quadrature
▶ Play
Speed
1.00 ×
Reset
Observer's orbit
1.000 ua
Observer
Earth
Orbit of the observed planet
2.400 ua
Observed planet
(free value)
At each configuration
keep going
Pause duration
5 s
Elongation angle
Synodic period 1.368 years
Elongation 129.7° E
Time 0.120 years
Next configuration eastern quadrature in 47 days

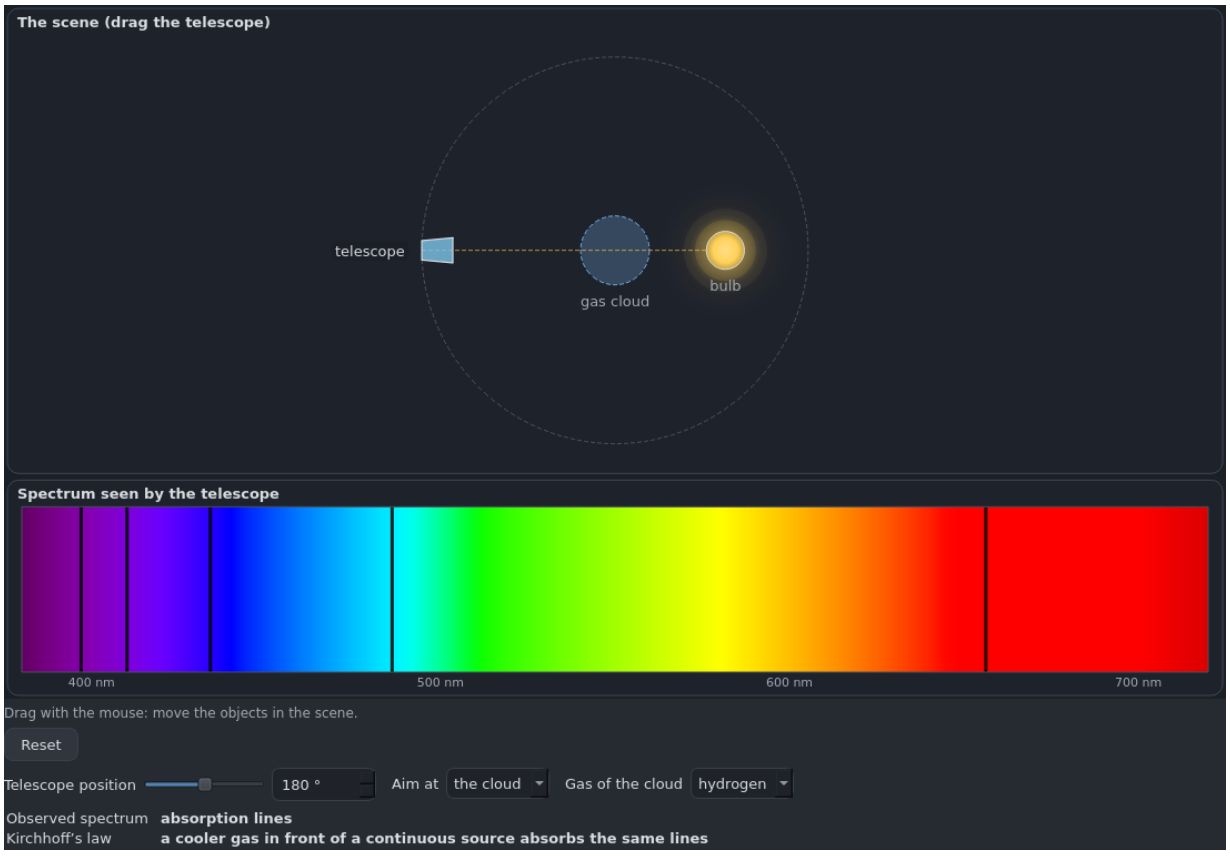

The year: night (dark), star up (blue), heliacal rising (yellow line)
0 h
6 h
12 h
18 h
24 h
Jan
Mar
May
Jul
Sep
Nov
The sky at that time
N
W
E
S
April 1, 5:10
Drag with the mouse: rotate the sphere, or move an object by grabbing it. Hover over or touch a star or a nebula: its data sheet (values and sources); with the keyboard, Tab goes from one object to the next (brightest first), so do the arrows while a sheet is open, Escape closes it.
▶ Play
Speed
1.00 ×
Reset
Now
Live
My location
Star
Sirius
Right ascension
6.75 h
Declination
-16.7 °
Latitude
40.8 °
Sun depression
7.0 °
Day of the year
90
Time
5.00 h
Start of dawn
Date April 1
Start of dawn 5:10
Altitude of the star at dawn −63.0°
Heliacal rising August 11

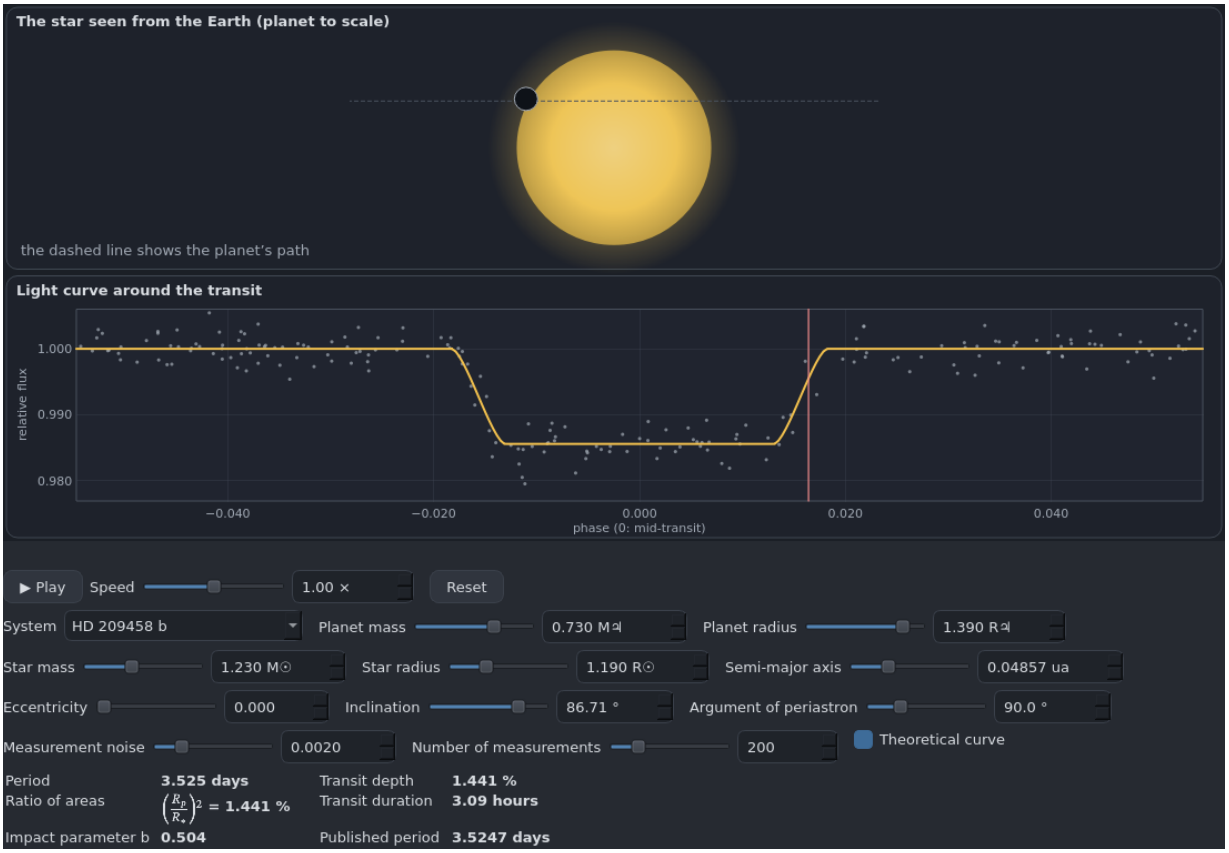

The building and its shadow (rotate the view with the mouse)
N
W
E
S
Altitude of the Sun at this time, over the year
90°
60°
30°
0°
Jan
Mar
May
Jul
Sep
Nov
Drag with the mouse: rotate the view.
▶ Play
Speed
1.00 ×
Reset
Now
Live
My location
Day of the year
292
Time
13.00 h
Latitude
40.8 °
True noon
Path of the shadow during the day
Date October 19
Time 13:00
Declination of the Sun −10.3°
Altitude 36.0°
Azimuth 203.1°
Shadow length (× height) 1.37

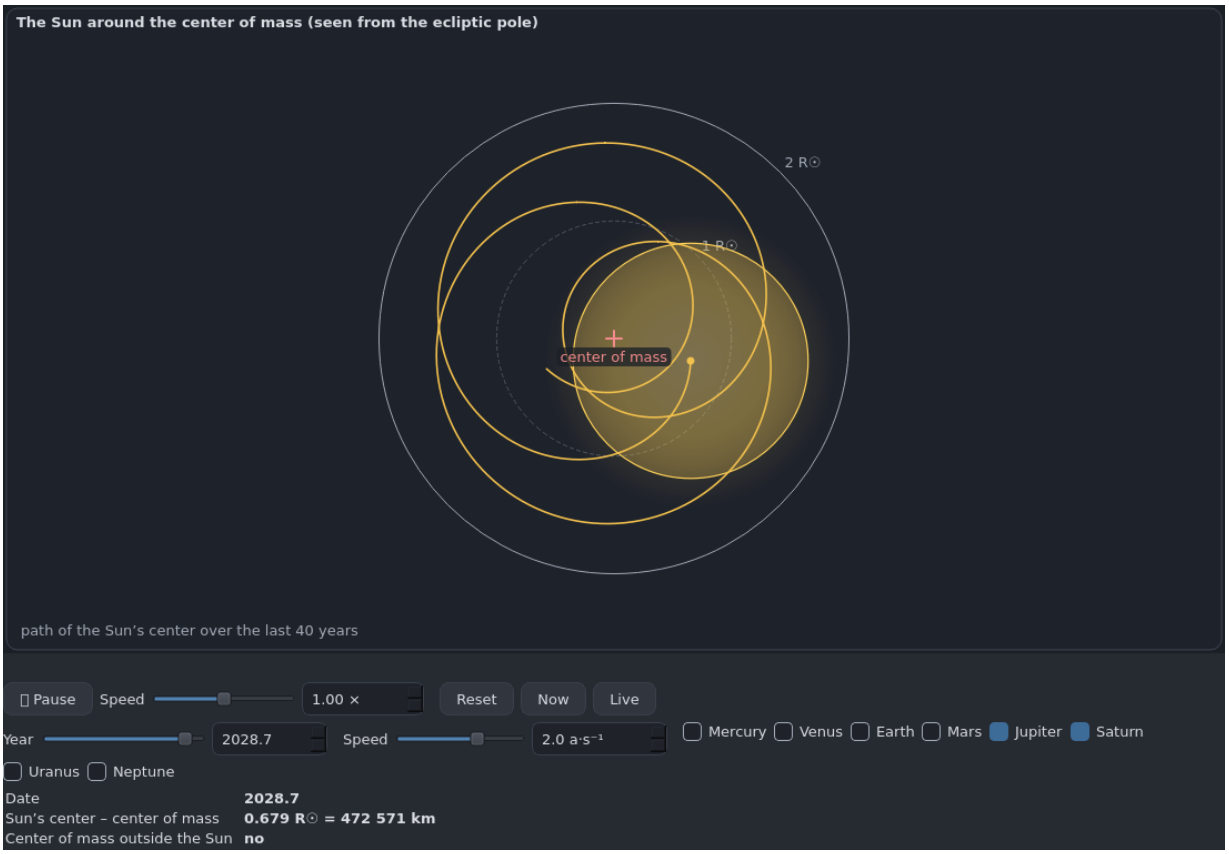

Hertzsprung-Russell diagram
10⁸
10⁴
100
1
0.01
0.0001
40 000
20 000
10 000
5 000
2 300
temperature (K)
1 000 R☉
100 R☉
10 R☉
1 R☉
0.1 R☉
Comparison with the Sun
Sun
0.99 R☉
Drag with the mouse: move the objects in the scene. Hover over or touch a star or a nebula: its data sheet (values and sources); with the keyboard, Tab goes from one object to the next (brightest first), so do the arrows while a sheet is open, Escape closes it.
Reset
Temperature
5800 K
Luminosity
1.0000 L☉
Stars
both
Main sequence
Other classes
Equal radii
Colour = temperature
Temperature 5 800 K
Luminosity 1.0 L☉
Radius 0.99 R☉
Relation
$\frac{R}{R_\odot} = \frac{\sqrt{\frac{L}{L_\odot}}}{\left(\frac{T}{T_\odot}\right)^2}$


Blackbody curves
peak at 322.0 nm
6 000 K
4 000 K
9 000 K
radiance (TW m⁻² sr⁻¹ m⁻¹)
200
100
0
0
200
400
600
800
1 000
wavelength (nm)
Filters
0.38
0.30
0.25
0.22
U
B
V
R
magnitudes
Reset
Curve
curve 3
Temperature
9000 K
Add a curve
Remove the curve
Axis up to
1000 nm
Vertical scale
all curves
Area under the curve
Peak
U, B, V, R filters
Temperature (selected curve) 9 000 K
Power per m² (σT⁴) 3.72 × 10⁸ W·m⁻²
U − B 0.07
V − R 0.03
Peak (Wien's law) 322.0 nm
Power per m², relative to the Sun × 5.91
B − V 0.06

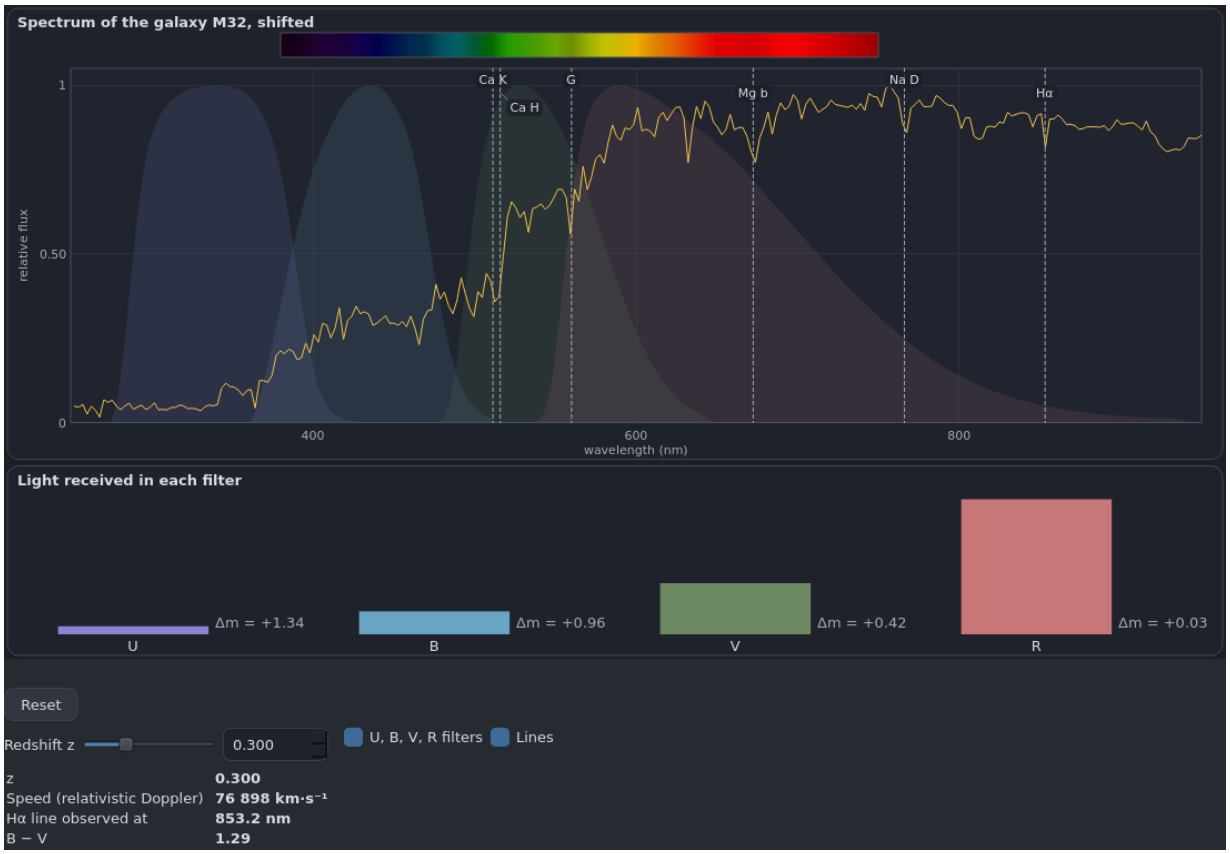

The atom (orbit radii ∝ n²)
Energy levels
ionization (0 eV)
n=4
−0.85 eV
n=3
−1.51 eV
n=2
−3.40 eV
n=1
−13.60 eV
Pause
Speed
1.00 ×
Reset
Photon energy
12.09 eV
Transition
1 → 3 (Lyman)
Fire the photon
Fire the transition photon
Electron level 3
Photon 12.09 eV → λ = 102.6 nm
Level energy −1.511 eV
Last event absorbed: level 1 → 3

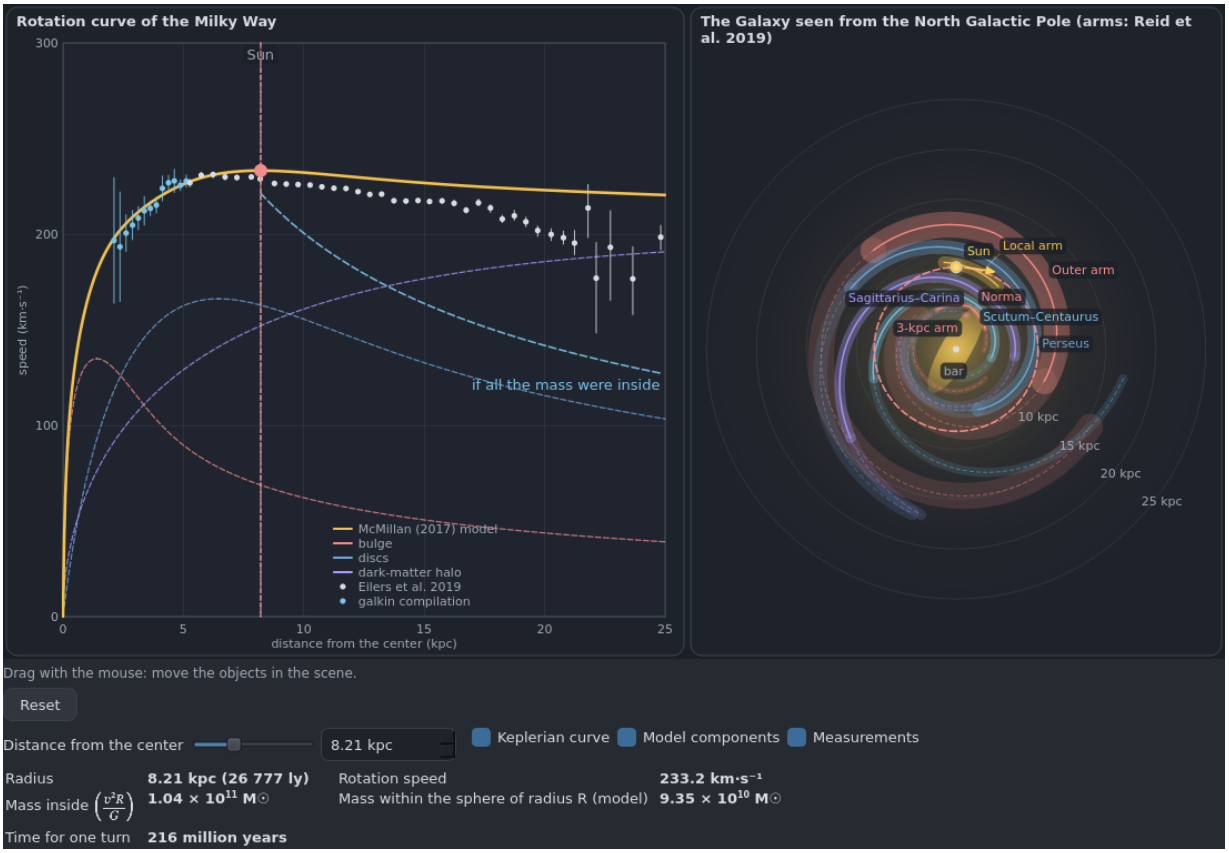

The scene (drag the telescope)
telescope
gas cloud
bulb
Spectrum seen by the telescope
400 nm
500 nm
600 nm
700 nm
Drag with the mouse: move the objects in the scene.
Reset
Telescope position
180 °
Aim at
the cloud
Gas of the cloud
hydrogen
Observed spectrum absorption lines
Kirchhoff's law a cooler gas in front of a continuous source absorbs the same lines

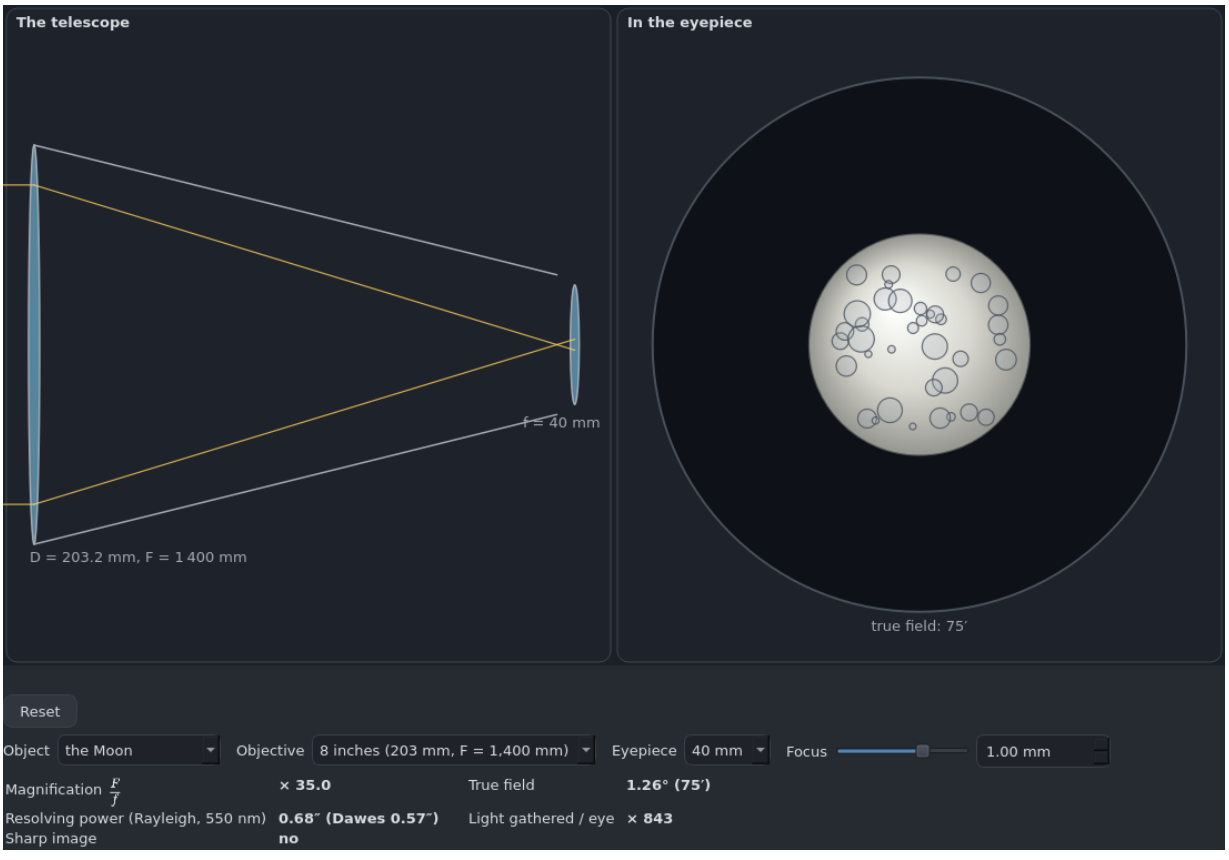

The star seen from the Earth (planet to scale)
the dashed line shows the planet's path
Light curve around the transit
relative flux
1.000
0.990
0.980
−0.040
−0.020
0.000
0.020
0.040
phase (0: mid-transit)
▶ Play
Speed
1.00 ×
Reset
System
HD 209458 b
Planet mass
0.730 M♃
Planet radius
1.390 R♃
Star mass
1.230 M☉
Star radius
1.190 R☉
Semi-major axis
0.04857 ua
Eccentricity
0.000
Inclination
86.71 °
Argument of periastron
90.0 °
Measurement noise
0.0020
Number of measurements
200
Theoretical curve
Period 3.525 days
Transit depth 1.441 %
Ratio of areas $\left(\frac{R_p}{R_*}\right)^2$ = 1.441 %
Transit duration 3.09 hours
Impact parameter b 0.504
Published period 3.5247 days

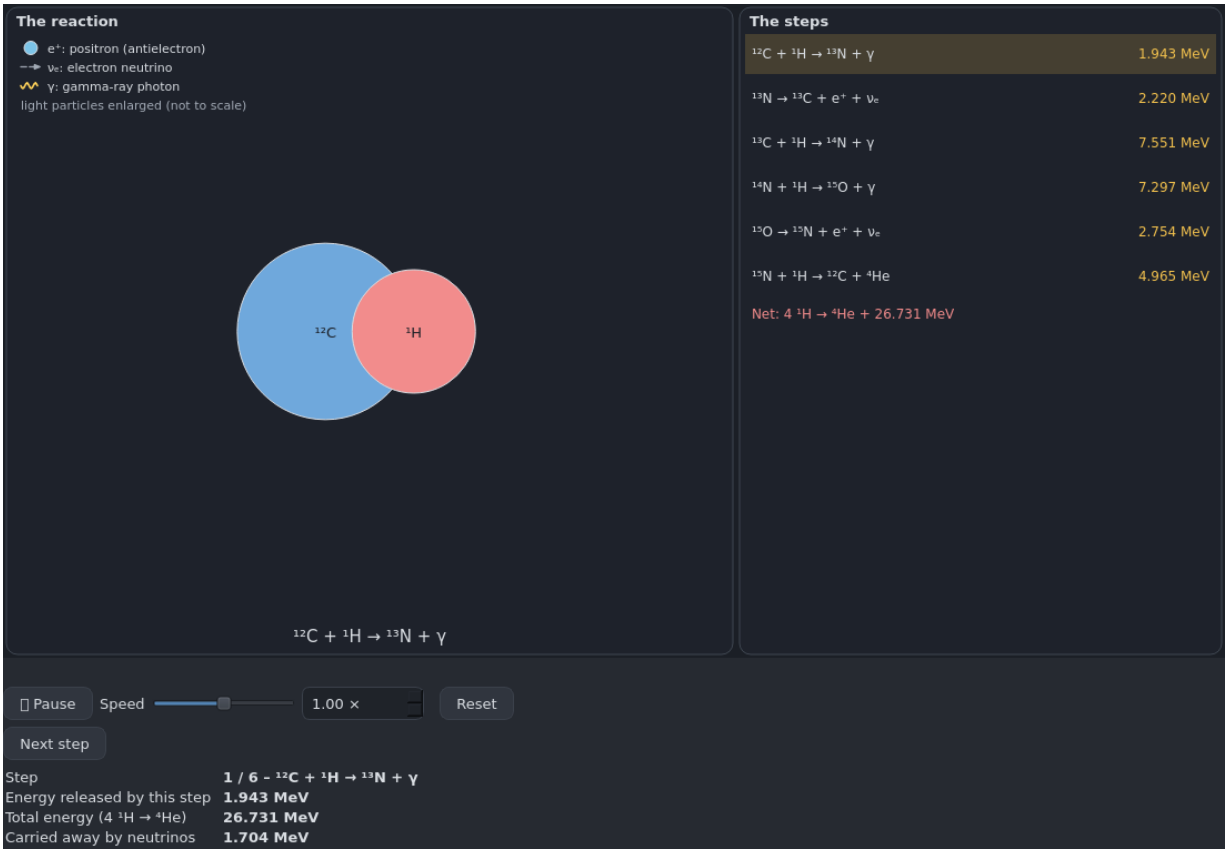

The Sun around the center of mass (seen from the ecliptic pole)
2 R☉
1 R☉
center of mass
path of the Sun's center over the last 40 years
Pause
Speed
1.00 ×
Reset
Now
Live
Year
2028.7
Speed
2.0 a·s⁻¹
Mercury
Venus
Earth
Mars
Jupiter
Saturn
Uranus
Neptune
Date 2028.7
Sun's center – center of mass 0.679 R☉ = 472 571 km
Center of mass outside the Sun no


The two stars seen from the Earth
2
1
Light curve (V filter)
relative flux
1.00
0.80
0.60
0.40
0.0
0.2
0.4
0.6
0.8
1.0
Phase
▶ Play
Speed
1.00 ×
Reset
System
EK Cep
Mass 1
2.02 M☉
Mass 2
1.11 M☉
Radius 1
1.575 R☉
Radius 2
1.311 R☉
Temperature 1
8995 K
Temperature 2
5689 K
Semi-major axis
16.58 R☉
Eccentricity
0.109
Inclination
89.16 °
Argument of periastron
49.8 °
Phase shift
0.000
Vertical axis
flux
Measurements
Period 4.420 days
Deepest eclipse 62.0 %
Model – data deviation 0.0442
Spectral types A1.5V/?


Spectrum of the galaxy M32, shifted
Ca K
G
Ca H
Mg b
Na D
Hα
relative flux
1
0.50
0
400
600
800
wavelength (nm)
Light received in each filter
Δm = +1.34
U
Δm = +0.96
B
Δm = +0.42
V
Δm = +0.03
R
Reset
Redshift z
0.300
U, B, V, R filters
Lines
z 0.300
Speed (relativistic Doppler) 76 898 km·s⁻¹
Hα line observed at 853.2 nm
B − V 1.29


Rotation curve of the Milky Way
Sun
300
200
100
0
speed (km·s⁻¹)
0
5
10
15
20
25
distance from the center (kpc)
if all the mass were inside
McMillan (2017) model
bulge
discs
dark-matter halo
Eilers et al. 2019
galkin compilation
The Galaxy seen from the North Galactic Pole (arms: Reid et al. 2019)
Sun
Local arm
Outer arm
Sagittarius-Carina
Norma
Scutum-Centaurus
3-kpc arm
Perseus
bar
10 kpc
15 kpc
20 kpc
25 kpc
Drag with the mouse: move the objects in the scene.
Reset
Distance from the center
8.21 kpc
Keplerian curve
Model components
Measurements
Radius
8.21 kpc (26 777 ly)
Rotation speed
233.2 km·s⁻¹
Mass inside (v²R/G)
1.04 × 10¹¹ M⊙
Mass within the sphere of radius R (model)
9.35 × 10¹⁰ M⊙
Time for one turn
216 million years


The telescope
f = 40 mm
D = 203.2 mm, F = 1 400 mm
In the eyepiece
true field: 75′
Reset
Object
the Moon
Objective
8 inches (203 mm, F = 1,400 mm)
Eyepiece
40 mm
Focus
1.00 mm
Magnification F/f
× 35.0
True field
1.26° (75′)
Resolving power (Rayleigh, 550 nm)
0.68″ (Dawes 0.57″)
Light gathered / eye
× 843
Sharp image
no



13.9 Computations and checks

The original animations used convenient approximations (23.5° obliquity, circular orbit, Fourier fit for the Sun, 3.26 ly per parsec). Coupole replaces them with:

- Julian day and Greenwich mean sidereal time (IAU 1982, Meeus eq. 12.4); mean obliquity P03 / IAU 2006 ($\varepsilon_{J2000} = 84\,381.406''$);
- apparent position of the Sun and equation of time (Meeus ch. 25 and 28, accuracy $\approx 0.01°$);
- coordinate changes by spherical trigonometry (azimuth measured from north toward east) and rotation matrices for drawing;
- Kepler's equation (Newton), vis-viva relation ($GM_\odot = 1.327\,124\,400\,18 \times 10^{20}$ m^3 s^{-2}), 3rd law $P^2 = a^3$;
- exact IAU parsec and light-year (1 pc = 3.261 564 ly); IAU nominal radii.
- mean lunar phases and argument of latitude F (Meeus ch. 49 and 54), node of the lunar orbit (ch. 47); rigorous IAU 1976 precession (ch. 21); equilibrium tide ($M_\oplus/M_{\text{Moon}} = 81.300\,568$);
- synodic periods, $G = 6.674\,30 \times 10^{-11}$ (CODATA 2018), $GM_\oplus = 3.986\,004\,418 \times 10^{14}$; Stefan-Boltzmann ($\sigma$ exact) and IAU nominal solar constants ($R_\odot$, $L_\odot$, hence $T_\odot = 5\,772$ K; $M_{\text{bol},\odot} = 4.74$ and Pogson's factor 2.5, the original wrote 2.51189). The empirical calibrations of the original (spectral type, bolometric corrections, HR diagram sequences, line strengths) are kept as they are;
- (batch 3) exact SI constants; Planck's law, Wien's constant from $x = 5(1-e^{-x})$ ($b = 2.897\,772$ mm K; the original: 2.897 768), σ; hydrogen with the reduced mass (13.598 eV, $hc = 1\,239.84$ eV nm; the original: 13.6 and 1,240); Snell's law and Fresnel; Doppler effect through the retarded time; perceived colour from the CIE 1931 functions (Judd-Vos) and the standard sRGB transform; magnitudes in Landolt's filters;

- (batch 3) two bodies: $P = 2\pi\sqrt{a^3/G(M_1 + M_2)}$ (0.1158 d for $a = 1$ $R_\odot$, the original: 0.1155), Kepler's equation, K and v_r, exact overlap area of two discs (transits, eclipses); masses from the IAU and DE440 GM values, barycenter of the Solar System from Standish's elements; energy budgets of the fusion reactions from the AME 2020 mass excesses; G in kpc $(\mathrm{km/s})^2/M_\odot$.

The script `outils/verif_anim/verif_anim.sage` recomputes each quantity under SageMath with 100 bits, by a different route than the code (rotations instead of closed forms, numerical roots, similar triangles); its values are frozen in `tests/anim_reference_sage.json` and `tests/test_anim_calculs.py` compares Coupole with them (differences $\leq 10^{-9}$), with Meeus' published examples (sidereal time of 10 April 1987, Sun of 13 October 1992) and with astropy (apparent Sun better than 0.01°, constellation boundaries within 0.05°). A few results: central lunar eclipse, Earth's shadow 4,607 km in radius at the Moon's distance, i.e. 2.65 lunar radii, totality of 1 h 41 min; retrograde motion of Mars (circular orbits) lasting 73 days over 15.9°; sidereal day of 23 h 56 min 4.091 s.

Batch 2: the 133 eclipses of the table fall within 1.01 day of a mean phase, with $|\sin F| \leq 0.28$, and all within an eclipse season; in 2026 the Sun passes the nodes on 27 February and 22 August; heliacal rising of Sirius at 40.8° N on 11 August (Vega on 14 November, Aldebaran on 19 June); geostationary orbit of 42,164 km; lunar equilibrium tide of 53.8 cm total range, Sun 0.459 times the Moon; precession within 10^{-4} degree of astropy.

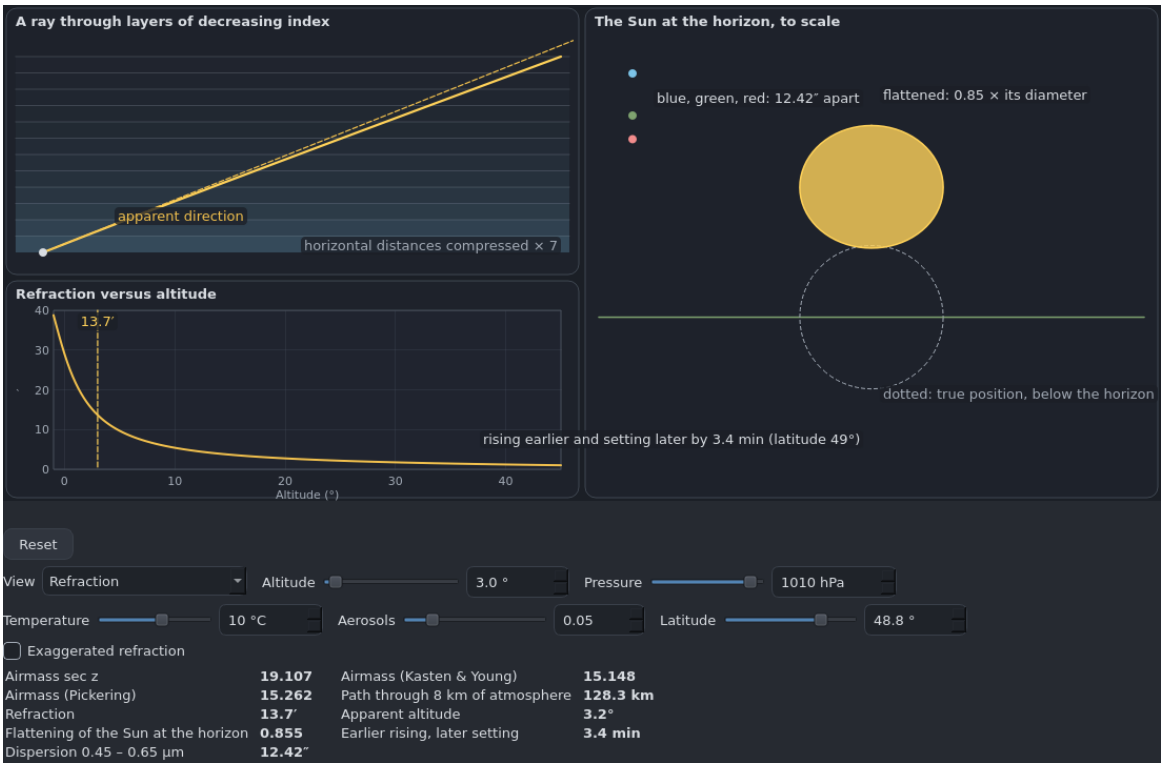
Batch 3: Wien's constant, σ, Planck's law within 10^{-11} of SageMath, and σT^4 equal to the numerical integral of πB_λ; Hα at 656.47 nm in vacuum; Fresnel identical to the sine and tangent laws, r_p zero at Brewster's angle; Doppler frequency equal to $f_0/(1 - v_r/c)$ within 10^{-5}; Sun around the barycenter at 1.65 $R_\odot$ on 1 January 2000, better than 5×10^{-5} AU from ERFA; Jupiter moves the Sun by 12.5 m/s; HD 209458 b: transit of 1.44 % lasting 3 h 05; the 39 complete binaries follow their measurements within 0.037 (median, relative flux), and DI Her « to complete » recovers $e = 0.49$ by fitting; fusion 26.731 MeV; mass of the Galaxy inside the Sun's orbit 9.9×10^{10} $M_\odot$; the jam of the density waves reaches the predicted density (×5). The light curves of the binaries, drawn as images in the original, were digitized by `outils/verif_anim/donnees_lot3.py`.

Command

```
coupole anim liste
coupole anim liste -faites -json
coupole anim info sunmotions
```

Earth's atmosphere (0.8.0). Coupole's own animation (Telescopes theme), four views: thickness crossed (airmass: $\sec z$, Kasten & Young 1989, Pickering 2002, 8 km spherical shell, ≈ 38 at the horizon), extinction and reddening ($\Delta m = kX$, Rayleigh $\propto \lambda^{-4}$, aerosols, ozone), atmospheric windows (ESO sky model from 0.3 to 30 µm, sketch from γ rays to radio) and refraction (Bennett, Sæmundsson, flattened Sun to scale, earlier rising, dispersion after Ciddor 1996, pressure and temperature). Formulas checked with SageMath (`outils/verif_atmosphere/`).



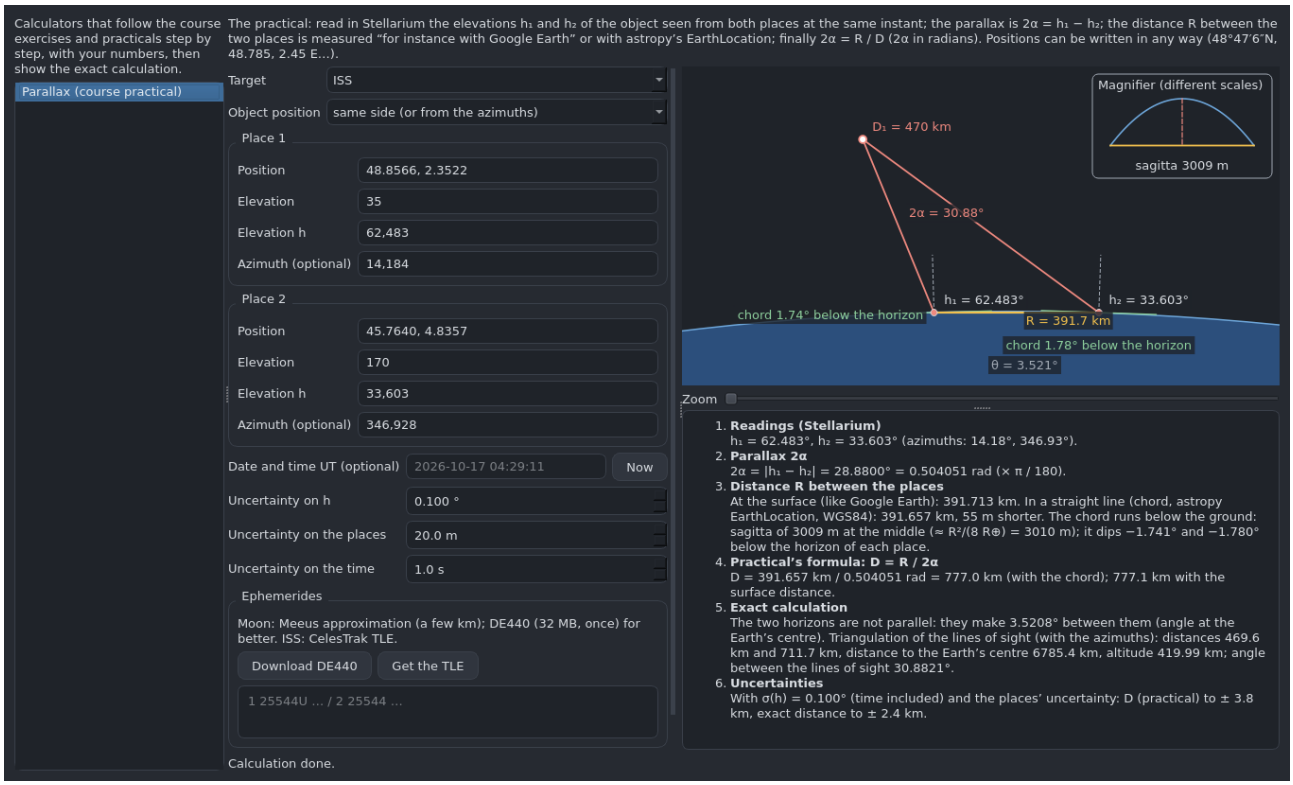


Sidereal time and hour angle: the southern hemisphere. The hour angle arc is shown by default (a choice already made is kept). The "What to look for" card has an *In the southern hemisphere* box: $H = \mathrm{LST} - \alpha$ does not change (counted westward from the meridian, $H < 0$ before culmination), but most objects culminate toward the north ($\delta > \varphi$); facing north, the sky moves from right to left; the south celestial pole has no bright star (σ Octantis, magnitude 5.4, about 1° away); a star is circumpolar if $\delta < -(90° - |\varphi|)$. These statements are checked by SageMath (`outils/verif_anim/temps_sideral_sud_verif.sage`).

CHAPTER 14

Course calculators

Calculators that follow the approach of a course exercise or practical *step by step*, with your numbers, then show the exact calculation and what changes. The first: **Parallax (course practical)**, after the official practicals (ISS altitude, Earth–Moon distance).

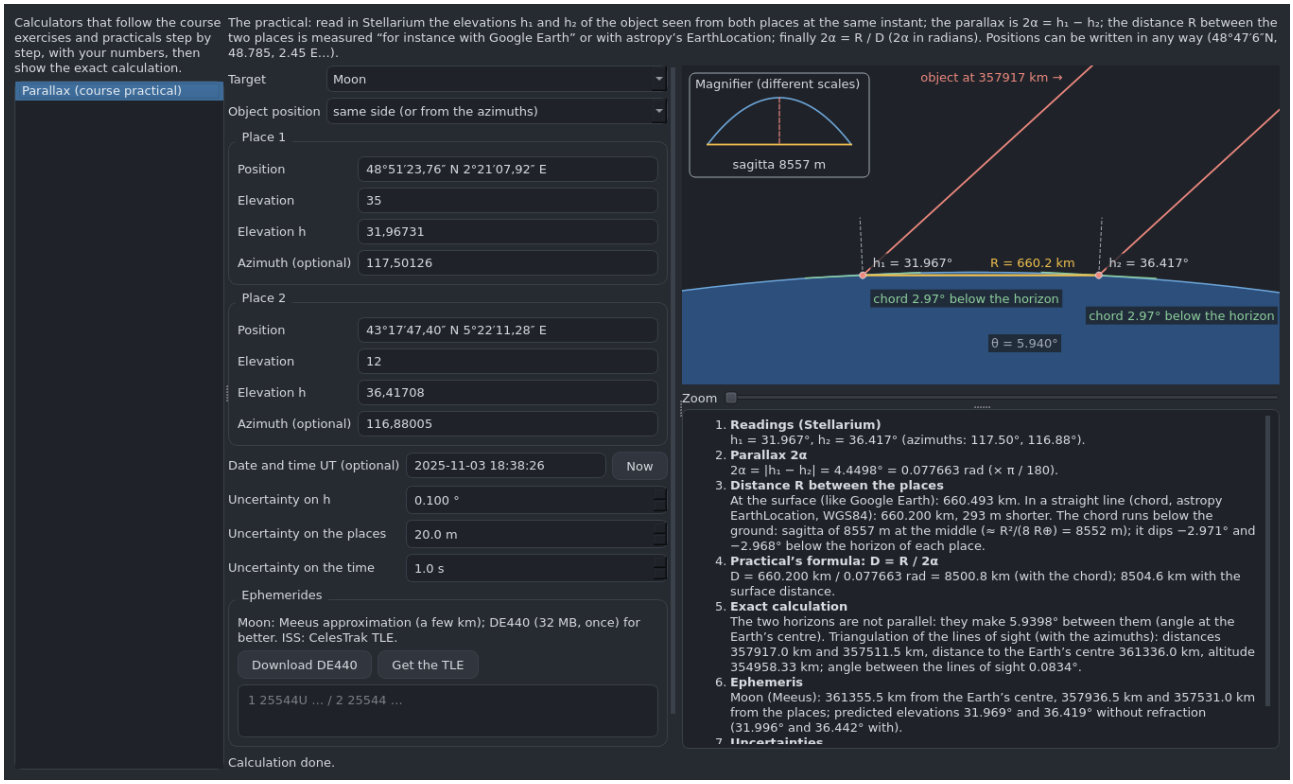


The practical's approach.

1. read in Stellarium the elevations h_1 and h_2 of the object seen from both places at the same instant (and, if possible, the azimuths) — for the Moon, the practical's Moon–star angular distance φ works too;
2. parallax $2\alpha = h_1 - h_2$ (object seen on the same side by both); if the object is *between* the places, the lines of sight cross above the baseline and $2\alpha = 180° - h_1 - h_2$;
3. distance R between the places "for instance with Google Earth" (at the surface) or with astropy's `EarthLocation` (in a straight line: the chord);
4. $2\alpha = R/D$, 2α in radians, hence D.

Positions can be written *in any way*: decimal with point or comma, signed or with N/S/E/O/W, degrees-minutes-seconds with ° ′ ″ ’ " or spaces ("48°47′6″N", "48 47 06 N", "2,45° E", "-2.35", "48d47m6s"), latitude and longitude in any order if the letters say so (without letters: latitude then longitude). An impossible latitude ($> 90°$) is corrected and reported; a huge baseline that would become reasonable by swapping latitude and longitude of one place is reported. Minutes or seconds ≥ 60, contradictory sign and letter: the error is stated, nothing is guessed.

Chord, surface and sagitta. The chord $R = |\vec{r}_1 - \vec{r}_2|$ joins the WGS84 geocentric positions (like `EarthLocation.from_geodetic`); the surface distance is the ellipsoid geodesic (Vincenty). The chord runs *below* the ground: its **sagitta** (gap at the middle between chord and surface) is $\approx R^2/(8R_\oplus)$ — 3 km for 400 km — and the calculator gives it exactly on the ellipsoid, with the dip of the chord below each place's horizon; a magnifier makes it visible in the drawing.



The exact calculation. The two horizons are not parallel: they make the angle θ at the Earth's centre (5.9° for 660 km). With both azimuths, the two lines of sight are triangulated in space (closest point of the two lines): distances to both places, distance to the Earth's centre, altitude above the ellipsoid; without azimuths, the object is assumed in the vertical plane of both places. The Moon example shows it: $h_1 - h_2 = 4.45°$ is almost entirely due to θ, the practical's formula applied to elevations gives $\approx$ 8,500 km, the exact calculation 357,917 km — that is why the Moon practical measures the Moon–star distance.

Ephemerides and uncertainties. With the time (*Now* button): Moon from JPL DE440 (32 MB, downloaded once, then offline) or Meeus otherwise — topocentric and geocentric distances, predicted elevations without and with refraction; ISS from the CelesTrak TLE (the closest to the instant among those kept in cache, date shown, warning beyond 3 days; TLE history is not freely available without a Space-Track account: for an old pass, paste a TLE of that time) and SGP4. Uncertainties propagated to D (elevations, positions, time multiplied by the object's speed in elevation). *Example* fills in consistent fictitious data; *Export the details* writes a CSV for the "Calculs" sheet of the share.

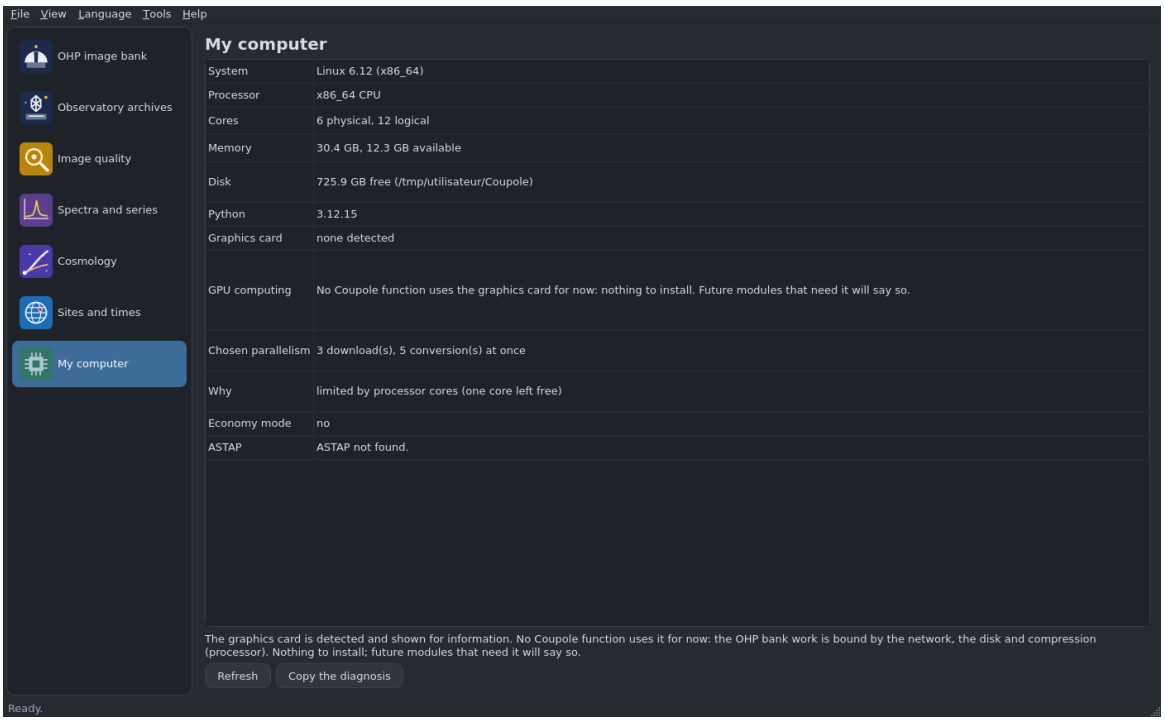
Verification. Coordinate reader: a battery of cases shared with Coupole Web (`tests/td_coordonnees_cas.json`). Geometry and calculations against SageMath at 200 bits (`outils/verif_td/verif_td.sage`): chord, surface, sagitta, dips, central angle within 1 mm and 10^{-9} degree, triangulation, the practical's formula.

Command

```
coupole td parallax -p1 "48.8566, 2.3522" -h1 62.483 -az1 14.184 -alt1 35
    -p2 "45°45′50″N 4°50′09″E" -h2 33.603 -az2 346.928 -alt2 170
```

CHAPTER 15

Computer and performance



Detection without mandatory dependency: system, architecture, processor, physical and logical cores (bounded by affinity and container quota), total and available memory, free space, graphics cards (nvidia-smi, `/sys/class/drm`, system_profiler, CIM). Parallelism: 3 downloads (at most 4), conversions = physical cores − 1 bounded by available memory at 500 MB per conversion (peak measured on an IRIS 4096^2 image, at most half of the free memory); economy mode (1 conversion, 2 downloads) automatic below 4 GB or 2 cores; manual limits, also bounded by memory. Examples: 2 cores and 2 GB → 2 downloads, 1 conversion; 6 cores and 20 GB free → 3 and 5; 32 cores and 60 GB free → 3 and 16 (cap). The graphics card is not used by the OHP bank (work bound by network, disk and Zstandard); `coupole.core.calcul` offers CuPy to future modules.

Nothing heavy runs on the GUI thread: astropy, scipy and SEP are loaded only when first needed, the machine probes and the ASTAP detection run in the background, and closing the window stops then waits for every thread, task and process (see `docs/AUDIT_2026-10.md`). Large lists stay smooth at 80,000 rows: cells are computed only for the rows shown, sorting uses one key per row (not pairwise comparisons), selecting no longer makes every drawing walk all rows, and changing the theme only recomputes colours (see `docs/AUDIT2_2026-10.md`).

Measurement (6-core processor, NVMe; Debian container)	
`coupole -version`	
window shown (astropy, scipy, SEP not loaded)	0.42 s (1.5 s be
inventory loaded (background thread)	3.3 s at first launch, (
full chain, 24 T120 images of 8 MB, 8 MB/s cap	24.5 s = duration of the downloa
same chain without a cap (local server)	2.3 s, 86 MB/s of FIT
conversion of one T120 image (one process)	0.13 s (zstd 14 ms, reading 18 ms, writing and readi
memory peak of a T120 / IRIS 4096^2 conversion (dedicated process)	163 MB / 365 MB (17
interface during 3 downloads (28 MB/s)	max gap of a 20 ms
Second audit (0.1.4): real bank of 7,625 XISF; network share simulated at 2 ms per access	
final sorting with nothing to move, on the share	1.5 s
48 images to a destination on the share	4.9 s
quality, starting again on the already measured bank (share)	1.9
reorganise the bank (reading the 7,625 headers)	57 s
select all, 80,000 images (10 × the bank)	0.2 s (3.1 s before; 0.5 s → 0.17
sort 80,000 images	0.1–0.2 s (3

CHAPTER 16

Command line

Everything the interface does is available on the command line; `-json` for scripts. Without argument, `coupole` opens the interface. Commands have a French name and an English alias.

Command

```
coupole -lang en ohp estimate "NGC 6888" -night 2025-07-16
coupole -lang en ohp process "NGC 6888" -night 2025-07-16 -filter B -filter V -filter R
-dest ~/OHP -yes
coupole -lang en ohp download Pluto -no-doubtful-dates -dest ~/FITS
coupole -lang en qualite ~/OHP -write
```

coupole	
`-version`	show the version and exit
`-check, -verifier`	check the installation without a screen (window, modules, animations, shipped data) and exit: code 0 if all is well
`-lang, -langue {auto,fr,en}`	message language (auto: system language)
`-v, -verbeux, -verbose`	detailed log
`-reinitialiser-interface, -reset-interface`	erase the saved layout (window, columns, filters, file-dialog folders) and exit; settings do not change

coupole astap
Look for ASTAP and its star catalogue, show what to install for this system, or record their location. Coupole works without ASTAP.

`-guide`	show the installation guide for this system
`-definir, -set PATH`	path of the ASTAP executable (or its folder)
`-catalogue, -catalog FOLDER`	star catalogue folder (D80, D50...)
`-oublier, -forget`	forget recorded paths (back to detection)
`-json`	JSON output (for scripts)

coupole rapports (reports)
Nothing is sent without consent. Without consent, reports stay in a local folder.

`-activer, -enable`	allow sending (anonymous) reports
`-desactiver, -disable`	forbid sending reports

coupole maj (update)
Query the releases published on GitHub. The standalone package updates itself; a pip/pipx installation shows the command to run.

`-appliquer, -apply`	download and install the update (standalone package)

coupole sources
Without option: list the addresses and their origin (shipped default, remote file, forced value).

`-forcer, -set KEY VALUE`	force a value (never overwritten by an update)
`-defaut, -default KEY`	go back to the default value for this key
`-tout-reinitialiser, -reset-all`	go back to the default values for every key
`-tester, -test KEY`	test the connection to this address
`-distant, -remote`	fetch the sources file published in the repository now

coupole ohp inventaire (inventory)
Bank inventory from the public TAP service (http://tap-ufe.obspm.fr/tap, table ivoa.obscore), kept in a local cache.

`-rafraichir, -refresh`	query the Observatory's TAP service again (one request)
`-manquantes, -missing`	list what remains to download compared with the output folder (–dest)
`-dest FOLDER`	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)
`-json`	JSON output (for scripts)

coupole ohp catalogue (catalog)
The 194 database names grouped into objects, sorted by type.

`-type TYPE`	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
`-telescope {T120,IRIS}`	telescope (repeatable)
`-chercher, -search TEXT`	keep objects whose name contains this text
`-json`	JSON output (for scripts)

coupole ohp images (list)
One line per image: date, telescope, filter, exposure, flags (duplicate, shared date, daytime).

`OBJECT`	object(s): « NGC 6888 », « Pluto », « 1998 AG6 », or a database name
`-type TYPE`	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
`-telescope {T120,IRIS}`	telescope (repeatable)
`-filtre, -filter F`	filter: B, V, R (Johnson and Cousins), Bc, Vc, Rc, Ha, OIII, g, r, i, z, or the exact name (« Johnson R »); repeatable or comma-separated
`-nuit, -night YYYY-MM-DD`	night (evening date); repeatable
`-sans-dates-douteuses, -no-doubtful-dates`	leave out exposures with a shared DATE-OBS or dated in daytime
`-tout, -all`	the whole bank (≈ 78 GB)
`-csv FILE`	also write the list to this CSV file
`-dest FOLDER`	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)

coupole ohp estimer (estimate)
Estimate before downloading, with the free space at the destination.

`OBJECT`	object(s): « NGC 6888 », « Pluto », « 1998 AG6 », or a database name
`-type TYPE`	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
`-telescope {T120,IRIS}`	telescope (repeatable)
`-filtre, -filter F`	filter: B, V, R (Johnson and Cousins), Bc, Vc, Rc, Ha, OIII, g, r, i, z, or the exact name (« Johnson R »); repeatable or comma-separated
`-nuit, -night YYYY-MM-DD`	night (evening date); repeatable
`-sans-dates-douteuses, -no-doubtful-dates`	leave out exposures with a shared DATE-OBS or dated in daytime
`-tout, -all`	the whole bank (≈ 78 GB)
`-format {xisf,xisf16,fz,fits}`	output format: xisf (PixInsight, default), xisf16 (XISF compatible with N.I.N.A. and Siril: 16-bit integers, zlib), fz (lossless compressed FITS), fits (float32)
`-dest FOLDER`	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)

coupole ohp traiter (process)

Full processing of a selection, resumable: running the same command again resumes where it stopped.

`OBJECT`	object(s): « NGC 6888 », « Pluto », « 1998 AG6 », or a database name
`-type TYPE`	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
`-telescope {T120,IRIS}`	telescope (repeatable)
`-filtre, -filter F`	filter: B, V, R (Johnson and Cousins), Bc, Vc, Rc, Ha, OIII, g, r, i, z, or the exact name (« Johnson R »); repeatable or comma-separated
`-nuit, -night YYYY-MM-DD`	night (evening date); repeatable
`-sans-dates-douteuses, -no-doubtful-dates`	leave out exposures with a shared DATE-OBS or dated in daytime
`-tout, -all`	the whole bank (≈ 78 GB)
`-dest FOLDER`	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)
`-format {xisf,xisf16,fz,fits}`	output format: xisf (PixInsight, default), xisf16 (XISF compatible with N.I.N.A. and Siril: 16-bit integers, zlib), fz (lossless compressed FITS), fits (float32)
`-noms, -names {fr,en}`	language of folder names, object names and headers (default: interface language)
`-astap {auto,tous,suspectes,jamais, all,suspicious,never}`	ASTAP check: auto/all (every image when ASTAP is present), suspicious, never
`-telechargements, -downloads N`	simultaneous downloads (automatic by default; at most 4)
`-conversions N`	simultaneous conversions (default: from cores and memory)
`-econome, -eco`	economy mode: 1 conversion, 2 downloads
`-debit, -rate MB/s`	download rate cap in MB/s (default: setting; no cap, 0 = unlimited)
`-garder-fits, -keep-fits`	also keep the original FITS (in _traitement/fits_origine)
`-garder-doublons, -keep-duplicates`	also process duplicates (inventory and identical pixels) instead of leaving them out
`-oui, -yes`	accept a selection over 5 GB without confirmation

coupole ohp tout (all)

The whole bank (≈ 8,000 images, ≈ 78 GB of FITS → ≈ 30 GB as XISF), with a volume and time estimate before confirmation; running the same command again resumes where it stopped.

`-dest FOLDER`	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)
`-format {xisf,xisf16,fz,fits}`	output format: xisf (PixInsight, default), xisf16 (XISF compatible with N.I.N.A. and Siril: 16-bit integers, zlib), fz (lossless compressed FITS), fits (float32)
`-noms, -names {fr,en}`	language of folder names, object names and headers (default: interface language)
`-astap {auto,tous,suspectes,jamais, all,suspicious,never}`	ASTAP check: auto/all (every image when ASTAP is present), suspicious, never
`-telechargements, -downloads N`	simultaneous downloads (automatic by default; at most 4)
`-conversions N`	simultaneous conversions (default: from cores and memory)
`-econome, -eco`	economy mode: 1 conversion, 2 downloads
`-debit, -rate MB/s`	download rate cap in MB/s (default: setting; no cap, 0 = unlimited)
`-garder-fits, -keep-fits`	also keep the original FITS (in _traitement/fits_origine)
`-garder-doublons, -keep-duplicates`	also process duplicates (inventory and identical pixels) instead of leaving them out
`-oui, -yes`	accept a selection over 5 GB without confirmation

coupole ohp nouveautes (new)
Query the TAP service again, compare with the local copy (output folder) and list what is new; –download processes them into the same tree (never without this option).

`-telecharger, -download`	download and convert the listed new images
`-dest FOLDER`	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)
`-format {xisf,xisf16,fz,fits}`	output format: xisf (PixInsight, default), xisf16 (XISF compatible with N.I.N.A. and Siril: 16-bit integers, zlib), fz (lossless compressed FITS), fits (float32)
`-noms, -names {fr,en}`	language of folder names, object names and headers (default: interface language)
`-astap {auto,tous,suspectes,jamais, all,suspicious,never}`	ASTAP check: auto/all (every image when ASTAP is present), suspicious, never
`-telechargements, -downloads N`	simultaneous downloads (automatic by default; at most 4)
`-conversions N`	simultaneous conversions (default: from cores and memory)
`-econome, -eco`	economy mode: 1 conversion, 2 downloads
`-debit, -rate MB/s`	download rate cap in MB/s (default: setting; no cap, 0 = unlimited)
`-garder-fits, -keep-fits`	also keep the original FITS (in _traitement/fits_origine)
`-garder-doublons, -keep-duplicates`	also process duplicates (inventory and identical pixels) instead of leaving them out
`-oui, -yes`	accept a selection over 5 GB without confirmation

coupole ohp reorganiser (reorganise, reorganize)
Walk a folder, recognise files produced by Coupole (header) and move them into the stack tree of –dest; name conflict → suffix and a log line; nothing is overwritten.

`FOLDER`	folder holding the files to sort
`-dest FOLDER`	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)

coupole ohp telecharger (download)
Layout: <folder>/<object>/<night>_<telescope>/<original file>.

`OBJECT`	object(s): « NGC 6888 », « Pluto », « 1998 AG6 », or a database name
`-type TYPE`	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
`-telescope {T120,IRIS}`	telescope (repeatable)
`-filtre, -filter F`	filter: B, V, R (Johnson and Cousins), Bc, Vc, Rc, Ha, OIII, g, r, i, z, or the exact name (« Johnson R »); repeatable or comma-separated
`-nuit, -night YYYY-MM-DD`	night (evening date); repeatable
`-sans-dates-douteuses, -no-doubtful-dates`	leave out exposures with a shared DATE-OBS or dated in daytime
`-tout, -all`	the whole bank (≈ 78 GB)
`-dest FOLDER`	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)
`-telechargements, -downloads N`	simultaneous downloads (automatic by default; at most 4)
`-debit, -rate MB/s`	download rate cap in MB/s (default: setting; no cap, 0 = unlimited)
`-oui, -yes`	accept a selection over 5 GB without confirmation

coupole ohp anomalies (anomalies-report)
Duplicates (same file, daytime copy, identical pixels), shared or daytime dates, inconsistent field, classification to check, unknown instrument. Nothing is deleted.

`-dest FOLDER`	add the pixel duplicates found in this processing folder
`-csv FILE`	also write the list to this CSV file
`-nouveaux, -new`	only images new since the last refresh

coupole ohp classer (classify)
Without argument: list the names to check. With a database name and –object: record the correction (merge into an existing object if –type is omitted). –online: query JPL SBDB and CDS Sesame.

`DB_NAME`	target name as it appears in the database
`-objet, -object OBJECT`	canonical object to associate (existing one: merge)
`-type {ast,neocp,com,pla,tno,pn, neb,amas,gal,eto,autre}`	category: ast, neocp, com, pla, tno, pn, neb, amas, gal, eto (repeatable)
`-oublier, -forget`	forget the correction of this name
`-en-ligne, -online`	resolve unknown names online (JPL SBDB, CDS Sesame), with cache

coupole ohp ranger (sort)
Useful after an interruption, or to merge several runs in the same folder.

`-dest FOLDER`	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)

coupole ohp bilan (status)
progress of a processing folder

`-dest FOLDER`	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)
`-json`	JSON output (for scripts)

coupole ohp fusionner (merge)
merge the state database on the share with the local working database (after a divergence)

`-dest FOLDER`	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)

coupole ohp reconnaitre (recognise, recognize)
Reads the header of each converted image of the output folder (OHP:Source:URL or HISTORY) and records it as owned in the tracking database. Downloads nothing, moves nothing, rewrites no file.

`-dest FOLDER`	destination folder (default: ~/Coupole/OHP_DU_ECU or the one in the settings)

coupole ohp metadonnees (metadata)
For each XISF/FITS of the folder: FOCALLEN matched to the measured scale (old value in HISTORY) and the XISF properties Instrument:Sensor:XPixelSize, Instrument:Camera:XBinning, Instrument:Telescope:FocalLength… Pixels are not touched (read back and compared), atomic writing, log in _traitement/metadonnees.csv. Without –rewrite: dry run.

`FOLDER`	output folder (or one stack)
`-reecrire, -rewrite`	really write (otherwise: only say what would change); also records each file's location, relative to the output folder, in the state database (copy from an older processing run, another computer)
`-noms, -names {fr,en}`	language of folder names, object names and headers (default: interface language)

coupole partages ajouter (add)
Checks the share link, shows the rights (warns if deletion is allowed) and remembers it under a free name.

`NAME`	name given to the share
`LINK`	share link (https://…/s/…) or WebDAV address
`-mot-de-passe, -password PASSWORD`	share password, if any

coupole partages liste (list)
remembered shares

`-json`	JSON output (for scripts)

coupole partages oublier (forget)
forget a share (nothing is touched on the server)

`NAME`	name given to the share

coupole partages explorer (explore)
Reads the whole tree and states what was recognised.

`NAME`	name given to the share
`-fichiers, -files`	list each file as well
`-vus, -seen`	mark the current content as seen
`-json`	JSON output (for scripts)

coupole partages tableau (table)
Normalises each row and reports anomalies (never changes anything).

`NAME`	name given to the share
`PATH`	table to read (all by default)
`-entetes, -headers`	read the FITS headers first (places and times, safer checks)
`-json`	JSON output (for scripts)

coupole partages entetes (headers)
read the FITS headers (place, UT time, object)

`NAME`	name given to the share
`PATH`	share folder (everything by default)
`-json`	JSON output (for scripts)

coupole partages paires (pairs)
Looks for shots of the same target by two distant observers, at almost the same time.

`NAME`	name given to the share
`-iss S`	maximum difference for the ISS (s)
`-lune, -moon S`	maximum difference for the Moon (s)
`-base N`	minimum baseline (km)
`-entetes, -headers`	read the FITS headers first (places and times, safer checks)
`-sans-corrections, -no-corrections`	do not apply the proposed corrections
`-json`	JSON output (for scripts)

coupole partages telecharger (download)
Same engine as the OHP image bank; FITS prepared as XISF or FITS; sorted by date, target and observer.

`NAME`	name given to the share
`PATH`	paths in the share (files or folders)
`-dossier, -folder FOLDER`	download folder
`-format {xisf,fits}`	format of the prepared FITS
`-astrometrie, -astrometry {wcs, auto,astap,astrometry.net,nova}`	astrometric solution when the header has none
`-en-ligne, -online`	agree to send the images without a solution to nova.astrometry.net (API key from the Preferences)
`-sans-originaux, -no-originals`	do not keep the original FITS

coupole partages mesurer (measure)
Centre of the lunar disc (or ISS trail) and its position (α, δ) from the astrometric solution of the image.

`FILE`	images (FITS with an astrometric solution)
`-cible, -target {lune,moon,iss}`	measured target
`-json`	JSON output (for scripts)

coupole partages calculer (compute)

JSON file: {"cible": "lune"|"iss", "methode": "deux"|"diurne"|"vitesse"|"orbite", "visees": [{"nom", "lat", "lon", "alt_m", "points": [{"instant": "2026-10-21T04:33:44Z", "ra": 96.2, "dec": 23.7}]}], "tle": [line 1, line 2]}.

`FILE`	JSON sightings file
`-html FILE`	also write the HTML report
`-pdf FILE`	also write the PDF report
`-json`	JSON output (for scripts)

coupole partages deposer (upload)

Without –yes, only shows what would be sent. Never overwrites: free names, conditional uploads, local copy of the table.

`NAME`	name given to the share
`FILE`	files to upload
`-dossier, -folder PATH`	share folder to upload to
`-tableau, -table PATH`	table to add the row to
`-valeur, -value ROLE=VALUE`	row value: auteur=…, cible=…, date=…, latitude=…, longitude=…, altitude=…
`-oui, -yes`	really send

coupole archives liste (list)

list the known archives (stage, missions, status)

`-json`	JSON output (for scripts)

coupole archives chercher (search)

Search MAST, ESO and IRSA (by default) for public final products around an object; –archive opus or pds with –body for planetary probes.

`OBJECT`	object (name or coordinates)
`-rayon, -radius ARCMIN`	search radius in arcminutes (default: 3)
`-archive A`	archive to query (repeatable): mast, eso, irsa, noirlab, koa, sdss, goa, smoka, opus, pds
`-mission M`	mission (repeatable): JWST, HST, GALEX, ESO, Spitzer, WISE, 2MASS, NOIRLab, Keck, SDSS, Voyager, Cassini, Juno
`-instrument I`	instrument (part of the name, repeatable)
`-filtre, -filter F`	filter (exact name, repeatable)
`-debut, -from YYYY-MM-DD`	observations on or after this date
`-fin, -to YYYY-MM-DD`	observations up to this date
`-tous-niveaux, -all-levels`	also the calibrated single frames and the raw data
`-non-publiques, -proprietary`	also the still proprietary data (for information: not downloadable)
`-corps, -body BODY`	Solar System body (Voyager, Cassini, Juno)
`-limite, -limit N`	maximum number of rows per archive (default: 2000)
`-id ID`	keep only this identifier (repeatable)
`-variantes, -variants`	also show variants (other products of the same exposures) and detection images; by default, one product per filter
`-programme, -program N`	keep only the products of this programme (repeatable), e.g. 9978
`-json`	JSON output (for scripts)
`-csv FILE`	also write the results to this CSV file

coupole archives estimer (estimate)
Like « search », then add up the sizes (announced or measured) without downloading anything.

`OBJECT`	object (name or coordinates)
`-rayon, -radius ARCMIN`	search radius in arcminutes (default: 3)
`-archive A`	archive to query (repeatable): mast, eso, irsa, noirlab, koa, sdss, goa, smoka, opus, pds
`-mission M`	mission (repeatable): JWST, HST, GALEX, ESO, Spitzer, WISE, 2MASS, NOIRLab, Keck, SDSS, Voyager, Cassini, Juno
`-instrument I`	instrument (part of the name, repeatable)
`-filtre, -filter F`	filter (exact name, repeatable)
`-debut, -from YYYY-MM-DD`	observations on or after this date
`-fin, -to YYYY-MM-DD`	observations up to this date
`-tous-niveaux, -all-levels`	also the calibrated single frames and the raw data
`-non-publiques, -proprietary`	also the still proprietary data (for information: not downloadable)
`-corps, -body BODY`	Solar System body (Voyager, Cassini, Juno)
`-limite, -limit N`	maximum number of rows per archive (default: 2000)
`-id ID`	keep only this identifier (repeatable)
`-variantes, -variants`	also show variants (other products of the same exposures) and detection images; by default, one product per filter
`-programme, -program N`	keep only the products of this programme (repeatable), e.g. 9978

coupole archives telecharger (download)
Like « search », then download (confirmation beyond the threshold: –yes), extract the science image and file it in <dest>/Archives.

`OBJECT`	object (name or coordinates)
`-rayon, -radius ARCMIN`	search radius in arcminutes (default: 3)
`-archive A`	archive to query (repeatable): mast, eso, irsa, noirlab, koa, sdss, goa, smoka, opus, pds
`-mission M`	mission (repeatable): JWST, HST, GALEX, ESO, Spitzer, WISE, 2MASS, NOIRLab, Keck, SDSS, Voyager, Cassini, Juno
`-instrument I`	instrument (part of the name, repeatable)
`-filtre, -filter F`	filter (exact name, repeatable)
`-debut, -from YYYY-MM-DD`	observations on or after this date
`-fin, -to YYYY-MM-DD`	observations up to this date
`-tous-niveaux, -all-levels`	also the calibrated single frames and the raw data
`-non-publiques, -proprietary`	also the still proprietary data (for information: not downloadable)
`-corps, -body BODY`	Solar System body (Voyager, Cassini, Juno)
`-limite, -limit N`	maximum number of rows per archive (default: 2000)
`-id ID`	keep only this identifier (repeatable)
`-variantes, -variants`	also show variants (other products of the same exposures) and detection images; by default, one product per filter
`-programme, -program N`	keep only the products of this programme (repeatable), e.g. 9978
`-dest FOLDER`	output folder (default: the one of the Preferences)
`-format {xisf,fits}`	format of the prepared images (default: xisf)
`-sans-original, -no-original`	(default since 0.2.2) delete the original file once the image is prepared
`-garder-original, -keep-original`	also keep the original file (in <field>/_originaux); by default only the prepared product remains
`-masque, -mask`	also write the mask of pixels without data (compressed, after cropping)
`-debit, -rate MB/S`	rate cap in MB/s (default: setting; no cap, 0 = unlimited)
`-paralleles, -parallel N`	simultaneous downloads (default: 2, at most 6)
`-max N`	at most N files (the closest)
`-oui, -yes`	confirm a download beyond the threshold

coupole archives preparer (prepare)
Extract the science image of a multi-extension FITS, or convert a planetary product, to XISF or FITS Float32 next to the file.

`FILE`	files to prepare
`-format {xisf,fits}`	format of the prepared images (default: xisf)
`-etiquette, -label FILE`	detached PDS3 (.LBL) or PDS4 (.xml) label

coupole archives aligner (align)
Resample the images (WCS) onto the grid of one of them or an optimal grid; write the aligned files, the common mask, the colour preview and composition.txt.

`FILE`	prepared images (FITS or XISF with WCS)
`-dest FOLDER`	folder of the aligned files
`-reference N`	index of the reference image (default: 0, the first)
`-optimale, -optimal`	optimal grid covering every image
`-methode, -method {bilineaire, adaptative,exacte,proche}`	resampling: bilineaire (bilinear), adaptative (adaptive), exacte (exact, reproject), proche (nearest)
`-echelle, -scale X`	output pixel size factor (default: 1)
`-sans-recadrage, -no-crop`	do not crop to the common area
`-format {xisf,fits}`	format of the prepared images (default: xisf)

coupole archives catalogue (catalog)
Lists an archive's targets, most observed first (aggregate query, 30-day cache), or the famous programmes. Lists: celebres, hst, jwst, eso, voyager, cassini, juno.

`liste {celebres,famous,hst,jwst, eso,voyager,cassini,juno}`	list to show (celebres by default)
`-type T`	object type: galaxie, amas_galaxies, amas_stellaire, nebuleuse, etoile, systeme_solaire, champ, autre
`-filtre, -filter OBJECT`	keep only targets whose name contains this text
`-page N`	page to show (100 targets per page)
`-actualiser, -refresh`	read the list online again (ignore the cache)
`-json`	JSON output (for scripts)

coupole archives reorganiser (reorganise)
Renames the files already downloaded (<target>_<instrument>_<filter>_<date>_<type>_<n>x<exp>s), files them by field, crops them to their useful area, deletes full-size masks and moves variants and detection images to "_variantes". Never overwrites; log in _etat/JOURNAL.txt.

`-dest FOLDER`	output folder (default: the one of the Preferences)
`-simuler, -dry-run`	show what would be done, changing nothing
`-garder-masques, -keep-masks`	rewrite the masks (compressed) instead of deleting them
`-sans-recadrage, -no-crop`	do not crop the images

coupole archives instruments
guide to instruments and filters (or one filter)

`FILTER`	filter name (F656N, F444W, Ks…): wavelength, width, use, amateur equivalent
`-instrument I`	instrument (part of the name, repeatable)

coupole archives hudf
Lists the HUDF high-level products (XDF, HUDF09, HUDF12); –download fetches the "3-filter colour" preset (XDF F435W, F775W, F160W, 3 × 110 MB) or the filters given by –filter.

`-telecharger, -download`	download and prepare (otherwise: list only)
`-filtre, -filter F`	XDF filter to download (repeatable)
`-dest FOLDER`	output folder (default: the one of the Preferences)
`-format {xisf,fits}`	format of the prepared images (default: xisf)
`-oui, -yes`	confirm a download beyond the threshold

coupole archives bilan (status)
what is already downloaded and prepared

`-dest FOLDER`	output folder (default: the one of the Preferences)
`-json`	JSON output (for scripts)

coupole qualite
Optional: background, noise, FWHM and ellipticity (3×3 map), background gradient, saturation, trails, sampling

`FOLDER`	folder of stacks (or one image file)
`-ecrire, -write`	write QUALITE.csv and QUALITE.txt in each stack
`-echantillon, -sample N`	measure N images per stack (spread in time) instead of all
`-tout, -all`	measure everything even above 200 images (otherwise: sample of 5 per stack, announced)
`-processus, -processes N`	number of measuring processes (default: machine plan)
`-avec-calibration, -with-calibration`	also measure flats, darks, bias, masters and intermediate files (excluded by default)

coupole donnees lire (read)
describe the contents of a file (FITS, CSV)

`FILE`	file to read
`-json`	JSON output (for scripts)

coupole donnees exporter (export)
export a spectrum or a series as CSV

`FILE`	file to read
`-csv FILE`	export a spectrum or a series as CSV
`-hdu HDU`	index of the dataset in the file (default: the first 1D one)
`-vitesse, -velocity`	add the radio velocity (frequency spectrum)
`-f0 MHz`	rest frequency in MHz (default: RESTFRQ, else H I)

coupole cosmo
From redshift to distances: comoving, luminosity, angular, lookback time, age, volume, distance modulus, scale (Planck 2018 and other parameter sets)

`Z`	redshift(s) to compute
`-objet, -object NAME`	object name: redshift requested from SIMBAD
`-modele, -model {planck18,planck15, wmap9,simple,perso}`	parameter set (default: planck18)
`-h0 H0`	H_0 in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (custom model)
`-om OM`	total Ωm (custom model)
`-ok OK`	Ωk, curvature (planck18 or perso)
`-shoes`	add the SH0ES comparison (planck18)
`-sans-incertitudes, -no-uncertainties`	without uncertainties (faster)
`-csv FILE`	write the results to this CSV file
`-courbes, -curves ZMIN ZMAX N`	write the curves (z from ZMIN to ZMAX, N points) to the CSV file given by –csv
`-json`	JSON output (for scripts)

coupole anim liste

`-faites`	only the available animations
`-json`	JSON output (for scripts)

coupole anim info

`ID`	ClassAction id (e.g. sunmotions)
`-json`	JSON output (for scripts)

coupole td parallaxe (parallax)
"Parallax (course practical)" calculator

`-p1 TEXT`	position of place 1 (any format)
`-h1 TEXT`	elevation measured at place 1
`-az1 TEXT`	azimuth measured at place 1 (optional)
`-alt1 TEXT`	elevation of place 1
`-p2 TEXT`	position of place 2 (any format)
`-h2 TEXT`	elevation measured at place 2
`-az2 TEXT`	azimuth measured at place 2 (optional)
`-alt2 TEXT`	elevation of place 2
`-cible, -target {iss,lune,moon}`	target: iss or moon
`-instant TEXT`	date and time UT (ephemeris comparison)
`-json`	JSON output (for scripts)

coupole sites
Observing sites (OpenStreetMap map), time zones, UTC and local time

`-ajouter, -add X`	add a site: ID NAME LAT LON ALT TIMEZONE [MPC]
`-supprimer, -delete ID`	delete an added site
`-heure, -time ISO_DATE`	show a UTC date (ISO) in site time and evening date
`-site ID`	site identifier (default: ohp)

coupole machine
Hardware diagnosis, chosen parallelism, graphics card, ASTAP

`-json`	JSON output (for scripts)

CHAPTER 17

Evolution

Sources. Addresses (TAP, collection, portal, download URL rewriting, resolvers, collection point, repository, map tiles) are in `coupole/donnees/sources.json`. Three layers: shipped values; file published in the repository, fetched at startup when newer and passing the shape check (known keys, types, domains from an allow-list; no cryptographic signature); values forced by the user, never overwritten. Preferences > Sources: edit, back to default, test. If the image server moves, `ohp.telechargement.de/vers` rewrites the URLs.

Classification. Shipped table `data/classement.json` (194 checked names); published aliases; user corrections (*Correct the classification*, `coupole ohp classify`); name-shape rules; optional online resolution (JPL SBDB, CDS Sesame), cached; grouping by position on a known fixed object; otherwise « unclassified ». Every automatic classification is marked « to check ».

Database. Extra ObsCore columns ignored, missing ones replaced by a default, unknown instruments and filters stored apart, non-image data never flagged « daytime »; tested on simulated inventories.

Security. No secret in the repository; remote sources file and collection point over HTTPS only, bounded size, domain allow-list; update archive downloaded from GitHub only and checked before extraction (absolute paths or « .. » refused, symbolic links refused, unpacked size and member count bounded), replacement with rollback on failure; XISF and FITS read with size bounds; folder names built from the database made safe (ASCII, forbidden characters, Windows reserved names, length); ASTAP launched without a shell; no `eval` nor `pickle` on external data.

CHAPTER 18

Incident reports and privacy

At first launch Coupole asks whether it may send anonymous reports: installation, sign of life (once a day), crash, native crash, frozen interface, written report. Content: version, system, processor, memory, graphics card, a random installation identifier, error trace with the home folder replaced by « ~», the user name and the computer name masked and deep paths shortened; a report never exceeds 64 kB. Never a computer name, a user name or an image. Without consent nothing is sent: reports stay in `_rapports` in the settings folder. The choice can be changed (Help menu, `coupole reports -enable|-disable`).

CHAPTER 19

Writing a module

A module is a Python package with a `manifest.json` (identifier, version, French and English names, icon, texts, graphical and command-line entry points), discovered in `coupole/modules/`, through the `coupole.modules` entry point, or in the `modules/` folder of the settings. Texts through `tr()`, mandatory tooltips, long work off the GUI thread: see CONTRIBUTING.md.

CHAPTER 20

Troubleshooting

Symptom	What to do
Nothing starts (Windows)	Start menu > Coupole > Diagnostic: the error stays on screen.
« Graphical interface unavailable »	missing Qt libraries (Linux: `libxcb-cursor0`, `libegl1`); the command line still works.
Download fails	run again (resume); check the connection; Preferences > Sources > Test; server moved: URL rewriting.
« ASTAP not found »	ASTAP assistant; or `coupole astap -set PATH -catalog FOLDER`.
Incomplete catalogue	reinstall the catalogue (1,476 tiles expected).
Not enough space	choose another output folder; the space needed includes pending FITS.
Slow computer	economy mode, or fewer simultaneous conversions.
Unknown target name	Catalogue tab, « To check » box, *Correct the classification*.
Quality unavailable	`pip install sep` (included in the packages).
Settings back to defaults	the `reglages.json` file was unreadable: it is kept alongside with the suffix `.corrompu-...`.
Output folder not writable	clear message at start; choose another folder (Preferences or the Folder field).
File dialog without the NAS, or generic	Linux: install `xdg-desktop-portal` and the desktop backend (`-kde`, `-gnome` or `-gtk`), restart; KDE: `kio-fuse`; otherwise type the path of the mounted share into the field (section 3.1).
System dialog does not open	0.1.10 and earlier, Spectra and series on KDE: update (nameless filter refused by the portal); otherwise Preferences > File dialogs: Qt; each opening is logged in `coupole.log`.
Owned images but nothing to open (copy from an older run)	update (0.1.11): locations are found through `journal.csv` (section 4.11).
Quality: empty columns on your own frames	update (0.1.11): tolerant XISF reader; the *status* column gives the reason.
`~/.local/bin/coupole` not found	add `~/.local/bin` to the `PATH` (line given by the installer).
`.part` or `.tmp` file in `_traitement`	left by a cut: resuming reuses it (`.part`) or replaces it; nothing to do.
WBPP does not find focal length and pixel	enter the values of the *For PixInsight / N.I.N.A.* box (Stacks tab) in the *Astrometric solution* panel; older files: `coupole ohp metadata FOLDER -rewrite`.
White image in N.I.N.A. or Siril	compatible XISF format (download options), section 8.
« database is locked » towards a NAS (0.1.9 and earlier)	update: the local working database solves it (section 4.6).
« Two versions of the state database »	two computers (or the NAS) processed into the same folder: accept the merge, or `coupole ohp merge -dest FOLDER`; both original databases are kept.
Everything « to download » on Windows although the folder (network drive) is complete	0.2.0 and earlier: update (section 4.6); the « Tracking folder » line says what is read.
« The tracking database of this folder could not be read »	the cause follows the message (and `coupole.log`): damaged database, share unplugged, permissions; do not start a full download again.
No `_traitement/` but sorted images	*Recognise existing files* (banner), or `coupole ohp recognise -dest FOLDER`.
« Could not copy the state database to the share »	share unplugged or database open elsewhere: nothing is lost, retried on the next pass and at the next start.
Archives: black image except a strip, 300 MB mask, 17 files for 3 filters (0.2.1)	update then *Reorganise downloads*: cropping, masks deleted, variants aside (section 6).
Archives: “jupiter”, “saturne” not found, nothing for Cassini, no preview (0.2.1)	update: bodies searched by target name, shared limit, thumbnails read in full.
Archives: black image when opened	normal: electrons/s or MJy/sr; stretch it (STF, autostretch).

Sources

« OHP student observations » bank, Observatoire de Paris / PADC (Paris Astronomical Data Centre), public TAP service, Etalab 2.0 licence; official guide of the database (PADC-BDD-DU-ECU). XISF 1.0 specification (Pleiades Astrophoto). ASTAP: https://www.hnsky.org/astap.htm. Observatory codes: Minor Planet Center. SEP: Barbary (2016), Bertin & Arnouts (1996). Adaptive moments: Bernstein & Jarvis (2002). Map background: © OpenStreetMap contributors. SIMBAD and Sesame: CDS, Strasbourg (Wenger et al. 2000); Aladin Lite (CDS); NED (NASA/IPAC); JPL Small-Body Database (NASA/JPL). Cosmology: Planck Collaboration (2020, A&A 641, A6); Hinshaw et al. (2013, WMAP 9); Riess et al. (2022, SH0ES); Hogg (1999, astro-ph/9905116); Komatsu et al. (2011) for the massive-neutrino density; astropy.cosmology; « cosmologie-redshift » calculator (same author). Archives: MAST (STScI), ESO Science Archive Facility, IRSA (NASA/IPAC), NOIRLab Astro Data Archive, Keck Observatory Archive (NExScI), SDSS, Planetary Data System (OPUS of the Ring-Moon Systems Node, Imaging Node); reproject (Astropy); Rector et al. (2007, AJ 133, 598) for chromatic ordering; amateur practices: see `docs/archives_methode.md`. Animations: ClassAction, University of Nebraska-Lincoln Astronomy Education (Apache License 2.0); Meeus (1998, *Astronomical Algorithms*, 2nd ed.); P03 / IAU 2006 obliquity; IAU constellation boundaries (data shipped with astropy); Natural Earth 1:50m coastlines (public domain); STIX Two Math font (SIL OFL 1.1); NASA Exoplanet Archive, NASA Planetary Fact Sheet, JPL DE440 ephemeris and Standish elements, AME 2020, Landolt filters (1992), spectrum of M32 (Santos et al. 2002), Catalog and Atlas of Eclipsing Binaries, rotation curve: McMillan (2017) model through galpy, Eilers et al. (2019) measurements and galkin compilation (Pato & Iocco 2017), Reid et al. (2019) arms, Wegg et al. (2015) bar; atmosphere: Kasten & Young (1989), Pickering (2002), Hansen & Travis (1974), Ångström (1929), Ciddor (1996), ESO sky model (Noll et al. 2012; Jones et al. 2013); DU shared folders and course calculators: the course's "ISS parallax" and "Moon parallax" practicals (Observatoire de Paris); JPL DE440 (Park et al. 2021) read by jplephem (MIT); ERFA (BSD, derived from SOFA, IAU); SGP4 (Vallado et al. 2006, sgp4 package, MIT) and CelesTrak TLEs; WGS84 (NIMA TR8350.2); Vincenty (1975); Kåsa (1976); Bennett (1982) and Saemundsson (1986) for refraction; skyfield (Rhodes, MIT) for the references; astrometry.net and nova.astrometry.net (Lang et al. 2010, AJ 139, 1782); WebDAV (RFC 4918) and Nextcloud public shares; Office Open XML (ECMA-376); Antikythera Mechanism: adapted from Anticythere3D (same author), after Freeth et al. (*Nature* 444, 2006; *Nature* 454, 2008; *Scientific Reports* 11:5821, 2021); physical spectra: CODATA 2022, NIST ASD, Mamajek (2022), Malkov et al. (2020), van Belle et al. (2021), Martins et al. (2005), Castelli & Kurucz (2003, ATLAS9), NIST-JANAF, Allard et al. (2000), CIE 1931; Gravity and Orbits: after the PhET simulation "Gravity and Orbits" (PhET Interactive Simulations, University of Colorado Boulder; nothing reused), CODATA 2018, IAU 2015 (B3) and 2012 (B2), DE440 (Park et al. 2021), NASA (Earth, Moon, ISS fact sheets), Yoshida (1990); Orion in 3D and hover data sheets: SIMBAD (CDS, Strasbourg), Gaia DR3 (ESA/DPAC), Hipparcos (van Leeuwen 2007), Zucker et al. (2021, CC0), Großschedl et al. (2018), Kounkel et al. (2017, 2018), Foley et al. (2023), O'Dell et al. (2009), Stellarium (figure, CC BY-SA 4.0); images ESO/G. Beccari, ESO, and NASA, ESA, M. Robberto (STScI/ESA) and the Hubble Space Telescope Orion Treasury Project Team, NASA, ESA, and the Hubble Heritage Team (AURA/STScI), under the CC BY 4.0 licence (details in `NOTICE`).